

ESSENTIAL OF
INTERNAL
MEDICINE

Block Medicine I (**MED – 421**)



Nephrology, Hematology & Endocrinology

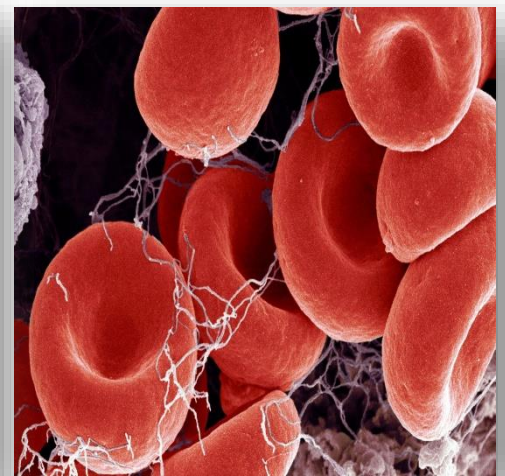
2026-2027



**Sohag University
Faculty of Medicine
Department of Internal Medicine**

Block Medicine I (MED – 421)

(Nephrology, Hematology & Endocrinology)



**Prepared By
Internal Medicine Department Staff Members
Head of Department of Internal Medicine
Prof. Nayel Abd El-Hameed Zaki
2026-2027**

Tables of contents

No.	Titles	Page
1	Renal disease introduction	6
2	Glomerular Diseases (introduction)	20
3	Glomerular Diseases (primary and secondary glomerulonephritis)	25
4	Acute Kidney injury	39
5	Chronic kidney diseases	46
6	Management of CKD and Renal Replacement therapies	52
7	Tubulointerstitial diseases	57
8	Fluid and electrolyte disorders	62
9	Renal cystic diseases	67
10	Infectious kidney diseases	71
11	Anemia (Iron deficiency anemia and megaloblastic anemia)	75
12	Aplastic anemia	93
13	Hemolytic Anemia	100
14	Myeloproliferative disorders	118
15	Lymphadenopathy and splenomegaly (Hodgkin and non Hodgkin lymphoma)	124
16	Blood Transfusion	138
17	Acute leukemia	145
18	Chronic leukemia	155
19	Bleeding Tendency	161
20	Introduction and hypothalamus (diabetes insipidus and SIADH)	182
21	Growth hormone (acromegaly, gigantism and short stature)	192
22	Pan-hypopituitarism	202
23	Hyperthyroidism	205
24	Hypothyroidism	220
25	Disorders of parathyroid glands (Hyperparathyroidism and Hypoparathyroidism)	229
26	Diseases of the adrenal gland (Cushing, Addison and clinical use of corticosteroids)	240
27	Diseases of the adrenal gland (hyperaldosteronism and pheochromocytoma)	247

Nephrology



Contents

No.	Titles
1	Renal disease introduction
2	Glomerular Diseases (introduction)
3	Glomerular Diseases (Glomerulonephritis)
4	Acute Kidney injury
5	Chronic kidney diseases
6	Management of CKD and Renal Replacement therapies
7	Tubulointersitial diseases
8	Fluid and electrolyte disorders
9	Renal cystic diseases
10	Infectious kidney diseases

1

Renal disease introduction

□ Intended Learning Outcomes:-

Objective 1

- Identify anatomical parts of nephrology, their physiology and their basic symptoms and signs about basics.

Objective 2

- Describe, clinical presentations, types, differential diagnosis investigation regarding renal diseases.

• It includes:-

Overview of the urinary system

Kidneys and its physiology

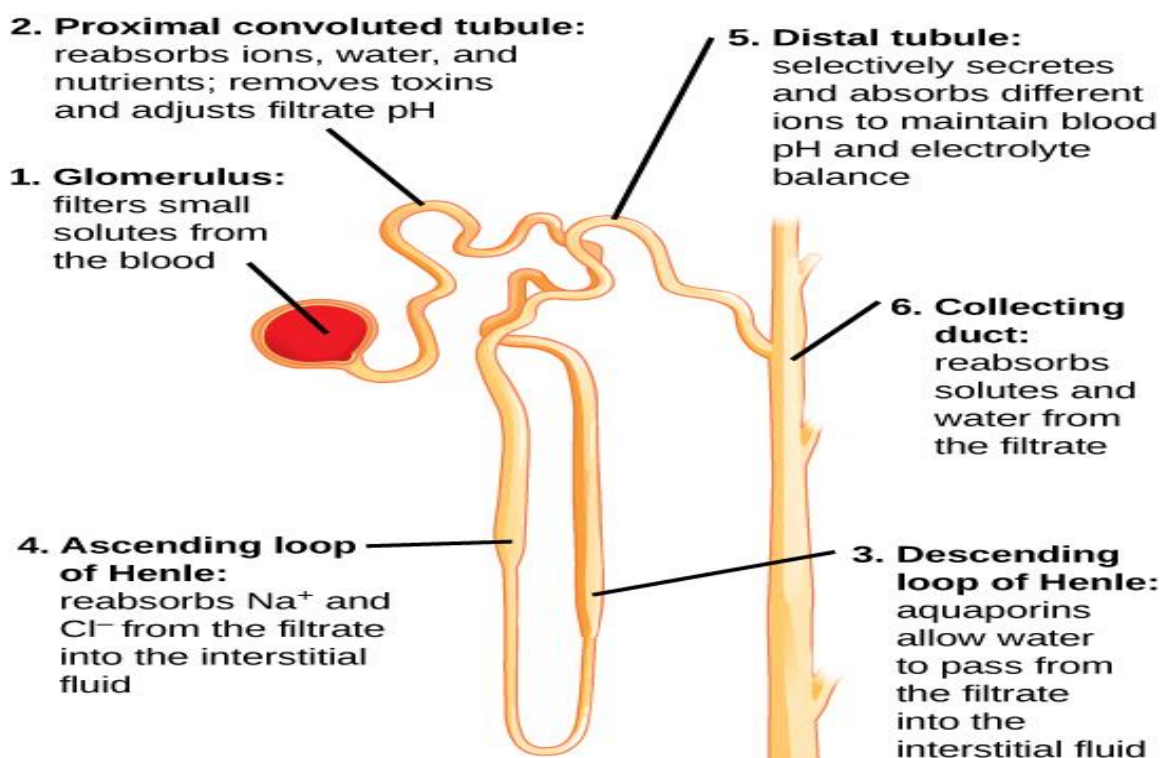
Diagnostic evaluation

Symptoms and signs

Definitions

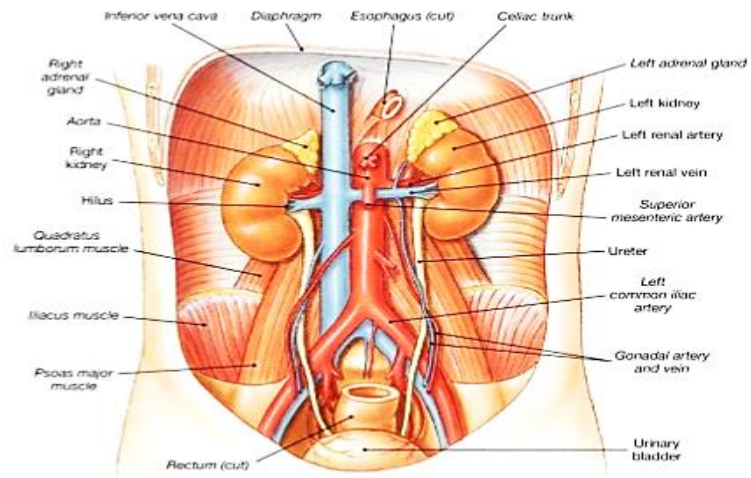
Nephrons

are the functional units of the kidneys. They are composed of a renal corpuscle (the glomerulus and the Bowman capsule) and a renal tubule (the proximal convoluted tubule, the loop of Henle, the distal convoluted tubule, the collecting tubule, and the collecting ducts).



Anatomy of the kidney

- ❑ A retroperitoneal organ
- ❑ T11-L3
- ❑ Normal size: 11-15cm in adults.
- ❑ Right kidney usually shorter than the left (upper limit of variation in length between right & left 1.5 cm)



A Kidney

Blood, waste and water enter here through the Renal Artery

Blood without waste or excess water leave here through the Renal Vein

Excess water and Toxic Waste in the form of Urine leaves here via the Ureter



Function of the kidneys

1- Excretory function:

- a. metabolites
- b. drugs
- c. toxins

2- Homeostatic function:

- a. Maintenance of water balance.
- b. Maintenance of electrolyte balance.
- c. Maintenance of acid-base balance.

3- Endocrine (hormonal) secretory function:

- a) Renin by the juxtaglomerular cells (JG)
- b) Erythropoietin hormone by endothelial cells of peritubular capillaries of renal cortex
- c) Prostaglandins.

4- Endocrine (hormonal) metabolic function:

- The kidney converts vitamin D3 → active 1,25-dihydroxycholecalciferol by alpha one hydroxylase enzyme in cells of PCT under effect of PTH.

Presenting features of renal disease

Dysuria

It means Pain in passing urine (prior to, during or following micturition usually in the urethra or suprapubic region).

Causes

- Urethritis and cystitis.
- Inflammation of vagina and penis.

Polyuria and nocturia

-> Urine output exceeding 3 liters/day in adults and 2 liter/day in children.

Causes

- Diabetes Mellitus.
- Diabetes Insipidus.
- Psychogenic polydipsia.
- Chronic renal failure.
- Hypercalcemia.

Oliguria

- Urine output is less than 400 ml per day.

Causes

- Dehydration.
- Shock.
- Hemorrhage.
- Acute kidney injury.
- hypotension, hypovolaemia.
- urinary tract obstruction

Anuria

- Urine output less than 100 ml per day.
- A person may first experience oliguria or low output of the urine and then progress to anuria, which must be differentiated from retention of urine.

Causes

- Shock or sever hypotension.
- Complete obstruction of both ureters.
- Bilateral cortical necrosis

Hematuria

- Blood in the urine, arise anywhere in renal tract.

Types

- ▶ **Gross hematuria:** when a person can see the blood in urine.
- ▶ **Microscopic hematuria:** Red blood cells seen under the microscope.

Causes

- Infection in the kidney, bladder, prostate.
- Renal calculi
- Trauma.
- Bladder and kidney cancer.
- Polycystic kidney disease.
- Bleeding disorders such as hemophilia

Frothy urine

- Urine looks foamy commonly due to proteinuria.

Causes

Proteinuria: Presence of excess protein in the urine.

Types

1- Transient (intermittent) proteinuria:

- The most common form of proteinuria.
- Transient proteinuria usually resolves without treatment.
- Stresses such as fever and heavy exercise may cause transient proteinuria.

2-Orthostatic proteinuria:

- Related to sitting/standing or lying down.
- Orthostatic proteinuria occurs when one loses protein in the urine while in an upright position but not when lying down.
- It occurs in 2 to 5 percent of adolescents but is unusual in people over the age of 30 years.
- The cause of orthostatic proteinuria is not known.
- Orthostatic proteinuria is not harmful, does not require treatment, and typically disappears with age.

3- Persistent proteinuria

- In contrast to transient and orthostatic proteinuria, persistent proteinuria occurs in people with underlying kidney disease or other medical problems.

Causes:

- Glomerular disease.
- Tubular disease.

Renal pain

Types

- Suprapubic pain: Cystitis
- Perineal pain or rectal pain: Prostatitis
- Flank pain, hypochondrium or iliac fossa: Renal pain

Causes

- Acute obstruction of ureters due to stone or blood clot – Colicky pain radiates from loin to groin and to the testicle in the male.
- Chronic obstruction – less severe pain and sometimes no pain.

Disturbance in the act of micturition

1. Urinary Incontinence

involuntary leakage of urine from the bladder.

Types

- **Stress incontinence:**
 - Involuntary leakage of urine with exertion (cough, sneezing, laughing, exercising or lifting something heavy).
- **Urge incontinence:**
 - Sudden intense urge to urinate followed by an involuntary loss of urine.
- **Overflow incontinence:**
 - Dribbling of small amount of urine (too full bladder either with or without the presence of the desire to urinate).

Causes

- Cerebrovascular lesions.
- Paraplegia. Multiple sclerosis.
- Prolapsed uterus with cystocele.

2- Urine retention

Types

- **Acute:** sudden inability to urinate.
- **Chronic:** gradual inability to completely empty the bladder.

Causes

- Obstructive lesions of the urethra.
- Disease of the spinal cord.
- Coma.

3-Hesitancy

Difficulty in starting or maintaining the flow of urine during the act of urination.

Causes

- Bladder muscle disorders.
- Lower urinary tract infections.
- Bladder outlet obstruction.
- Enlarged prostate.
- Psychological issues.

4- Strangury

Painful, frequent urination of small volume that are expelled slowly only by straining and despite a severe sense of urgency.

Causes

- Cystitis.
- Bladder tumor.

5-Nocturnal enuresis

Involuntary urination while asleep after the age at which staying dry at night.

Causes

- High fluid intake.
- Urinary tract infection.
- Sleep disorders

Outlines of renal disorders

Features of common renal diseases

Acquired

- Glomerulopathies (GN)
- Urinary tract infection
- Urinary tract obstruction
- Renal failure

Congenital

- Polycystic Kidney Disease
- Others

Glomerulopathies (GN)

Glomerulopathies are the third most common cause of endstage renal disease.

Glomerulopathy is a general term for a group of disorders in which:

- The kidneys are involved symmetrically.
- There is primarily an immunologically mediated injury to glomeruli.
- May be part of a generalized disease eg:SLE

Classification of glomerulopathies

Nephrotic syndrome

Nephritic syndrome

Nephrotic syndrome

- 1 .Proteinuria (>3.5 g/day)
- 2 .Hypoalbuminaemia(<30 g/L).
- 3.Oedema.
- 4 .Hyperlipidaemia ($\uparrow$ LDL & cholesterol)

Nephritic syndrome

- 1-Haematuria
- 2-Hypertension
- 3-Oliguria
- 4-Uraemia

Urinary tract infection

- Presence of pure growth of >100000 colony forming units/ml in urine with pyuria.
- UTI sites: bladder (cystitis), prostate (prostatitis), kidney (pyelonephritis)

Cystitis: frequency, dysuria, urgency, haematuria, suprapubic pain

Pyelonephritis: fever, rigors, vomiting, loin pain, tenderness, oliguria (if ARF)

Prostatitis: flu-like symptoms, low backache, swollen and tender prostate

Renal calculi

Consist mainly of crystal aggregates, form in the Collecting Duct and deposited anywhere in UT

Manifestations

- Asymptomatic.
- Loin pain, renal colic (radiates from loin to groin).
- Haematuria, sterile pyuria, calculus anuria

Acute Renal failure

Definition

Significant deterioration in renal function occur over hrs or days.

-Reversible over days/weeks (injury to kidney is short term and potentially reversible)

Clinically no symptoms or sign but oliguria <400ml/day) common.

-No long term complication seen in CKD e.g.: renal anemia, renal bone disease.

Clinical feature:

- Early stage asymptomatic

-ARF does not produce a classic set of symptoms. The most common symptom is decreased urine output, which occurs in 70% of patients.

Chronic Renal failure

Definition

-CKD implies long-standing, and usually progressive, impairment in renal function. -In many instances, no effective means are available to reverse the primary disease process.

Clinical feature:

Produces symptoms when renal function-which is measured as the glomerular filtration rate (GFR)-falls below 30 milliliters per minute (<30 mL/min). This is approximately 30% of the normal value.

- When GFR slows to below 30 mL/min, signs of uremia (high blood level of protein by-products, such as urea) may become noticeable. When the GFR falls below 15 mL/min most people become increasingly symptomatic.

Polycystic renal disease

- Autosomal dominant disease with formation of multiple cysts associated with extrarenal abnormalities-hepatic and CVS. May present as:
- Loin pain or haematuria due to haemorrhage into a cyst, cyst infection or urinary tract stone formation
- Loin or abdominal discomfort
- Hypertension

Different Investigations

kidney Function Tests

1-Blood urea (N= 20 -40 mg/dl)

- The level of urea depends both on the G.F.R & its production rate. Also, it is heavily influenced by protein intake, increased tissue catabolism and GIT bleeding (↓ production). So it ↑↑ with protein diet and tissue breakdown (provided that there is renal insufficiency).
- Also, it ↓↓ with chronic advanced liver disease (as it is produced in the liver) and with reduced catabolism in old age (↓production).

2- BUN (blood urea nitrogen) (N= 10-20 mg/dl)

- It is similar to blood urea as regard the influencing factors affecting its blood level.
- ∴ $Urea = BUN \times 2$

3- Serum creatinine (N= 0.7-1.4 mg/dl)

- The level of the serum creatinine is much less dependent on the diet but is more related to the age, sex & muscular mass.
- So, S. creatinine is a better guide to (GFR) than urea.
- It starts elevation when kidney Function (GFR) is reduced > 70%

4- Creatinine Clearance:

- It is a method to measure GFR approximately.
- It is the volume of plasma cleared from creatinine/min. = $\frac{U \times V}{P}$
 U = urine creatinine mg/dl.
 V = volume of urine/min
 P = plasma creatinine mg/dl.

- Normal Value(90-140 mU/min for males, 80-125 mU/min for females)

Note

Serum creatinine starts elevation when creatinine clearance < 25%-30%.

- The glomerular filtration rate (GFR) varies with age and sex but is approximately 120-130 ml/min/1.73 m² surface area. It can be measured by renal scan using Diethylene triamine penta-acetic acid labeled with technetium (99mTc-DTPA), it is more accurate than creatinine clearance. GFR 35-50% of normal is sufficient to maintain the patient symptom free, at GFR 20-35% azotemia occurs with initial manifestations of renal failure.

Urine analysis

Physical Examination:

A. Color

- Normally it is amber yellow due to urochrome pigment which is a break down product of Hb related to bile pigments found in the urine.

Abnormal colors:

1. *Colorless = polyurea*
 - D.M. – D.I
 - Compulsive H₂O intake.
2. *Reddish*
 - Haematuria. - Haemoglobinuria.
 - Myoglobinuria. " ". Beetroot, Rhubarb.
 - Drugs e.g. Rifampicin (orange-red).
3. *Brownish*
 - Bilirubin. -:Nitrofurantoin.
 - Flagyl.
4. *Bluish:*
 - Amitriptyline (Tryptizole).
5. *Greenish?! P*
 - pseudomonas;"

B- Aspect

- Cloudy or turbid urine provides a warning of possible abnormality as presence of pyuria, hematuria, excessive epithelial cells, excessive amorphous urate or mucus.

C- Oduor

Fruity or sweet odor may be present in cases of ketonuria.

D.PH

(Normally acidic, pH= 5-7)

Acidic urine

- Ketosis
 - diabetes mellitus
 - Starvation
 - fever
- Urinary tract infection by *Escherichia coli*
- High protein diet.

Alkaline urine

- Severe vomiting
- Old ammoniacal urine sample
- Chronic renal failure
- Urinary tract infection by bacteria that split urea to ammonia - *proteus* or *pseudomonas*
- Vegetarian diet

E- Specific gravity

- Measurement of urine osmolality is more accurate.
- 1005 - 1025 (or more)
- After H₂O deprivation for 12 hours it reaches 1025 or more.
- After plenty fluid intake it reaches 1005 or less.

Decreased specific gravity

- diabetes insipidus
- tubulointerstitial disease
- compulsive water intake

Increased specific gravity

- diabetes mellitus
- increased proteins
- water depletion

Chemical examination:

It includes:

- Proteins
- Glucose and ketones
- Bilirubin and urobilinogen

Urinary proteins

- Normally, there is a trace of protein.
- It is a glycoprotein secreted by the cells of the ascending limb of the loop of Henle called + trace of albumin,
- (normal urine protein) = 150-200 mg/24 hr.
- Slightly higher values up to 300 mg daily may be excreted by adolescents.

Causes of proteinuria (albuminuria)

1. **Heavy proteinuria** > 3.5gm/24 hr urine
= Nephrotic \$ (glomerular Disease).
2. **Mild proteinuria** < 3.5gm
 - Nephritic \$.
 - Tubulointerstitial disease.
 - Orthostatic.
 - Exercise.
 - Febrile albuminuria.
 - Excessive RBCs or pus cells may give False albuminuria.
 - Congestive heart failure → Congested kidney → escape of protein through the glomerulus.

Microscopic examination of urine sediment

A- Pus cells = dead WBCs (normally up to 4/ H.P.F)

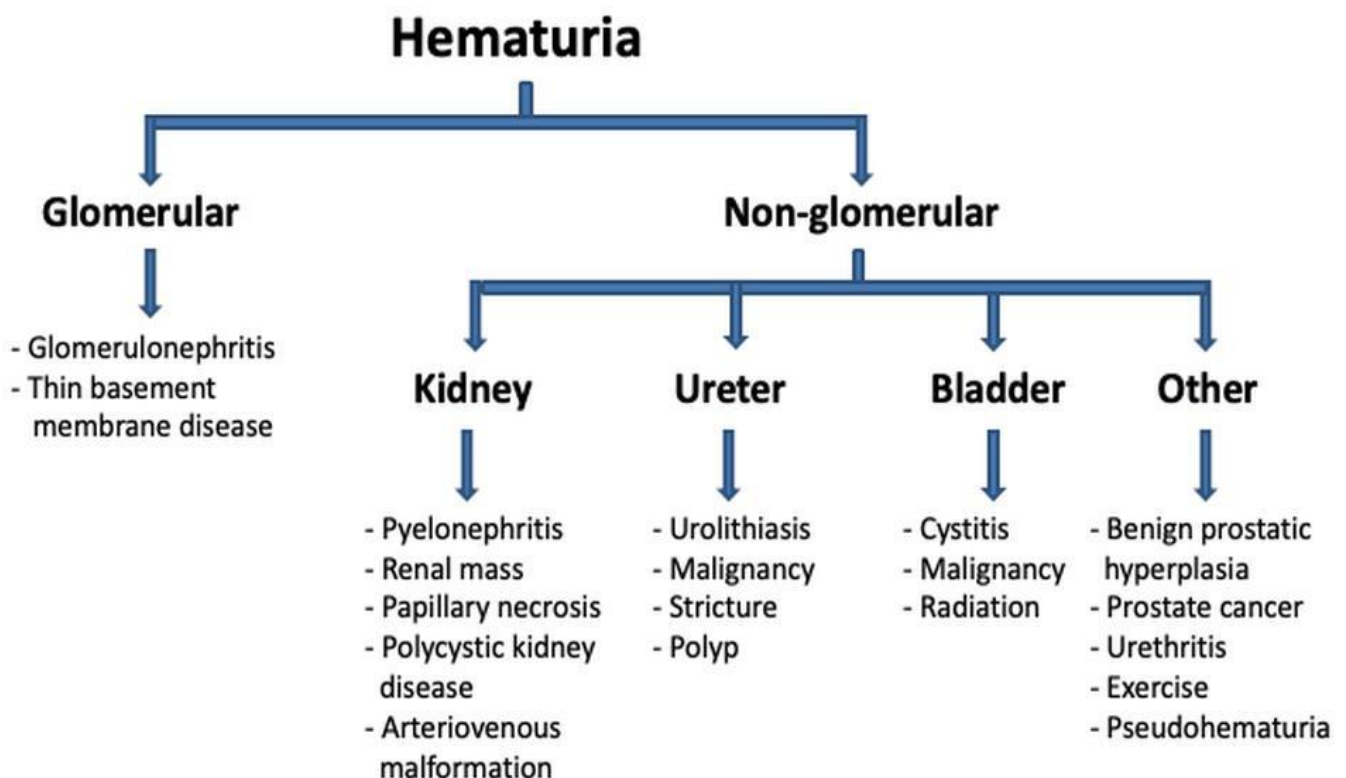
- If $> 4/ H.P.F$ = pyuria, it is significant if $> 10/ H.P.F$.
- Pyuria indicates an inflammatory reaction within the urinary tract.
- So if pyuria + ve, Culture of mid-stream urine sample is indicated.

Causes of pyuria with negative culture = sterile pyuria

- 1- Urinary tract infection in patient under antibiotic.
- 2- TB.
- 3- Renal stones.
- 4- Graft rejection.
- 5- Glomerulonephritis.
- 6- Drug induced interstitial nephritis.
- 7- Non gonococcal urethritis caused by mycoplasma and Chlamydia.

B- RBCs Normally up to 5/ H.P.F.

- may be overt, with bloody urine, or microscopic and found only on chemical testing.
- A positive Stix test must always be followed by microscopy of fresh urine (with the exception of menstruating women) to confirm the presence of red cells and so exclude the relatively rare conditions of haemoglobinuria or myoglobinuria.



C. Casts:

- Microscopic cylindrical structures due to coagulation of tubular protein (Tamm-Horsfall proteins) + albumin $\pm$ cellular element formed within kidney tubules.
- Because they are formed inside the tubules, they indicate upper urinary tract lesion (inside the kidneys.) Le. within the nephrons.
- Any excessive cells or increase albumin in urine $\rightarrow$ coagulation of tubular Protein $\rightarrow$ cast formation.
- The name of the cast may refer to its cellular element.
- The presence of red or white blood cells in the casts gives evidence of inflammatory parenchymal renal diseases.
- Casts may contain cellular material (RBCs, WBCs, tubular 'cells'), lipids, fibrin, bile and crystals.

D- Urine Crystals

- (In the absence of specific symptoms crystals of calcium oxalate and uric acid are of little clinical significance)
 - 1- Uric acid present in acidic urine, acute uric acid nephropathy and hyperuricosuria.
 - 2- Calcium phosphate, in alkaline urine.
 - 3- Calcium oxalate, in acidic urine and hyperoxaluria.
 - 4- Cystine, in cystinuria.

Urinary tract imaging

Abdominal x-ray

- Look at kidneys, ureters and bladder Abnormal calcification, eg. calculi (80%)

Ultrasound

- Characterizing renal masses as cystic or solid.
- Diagnosing polycystic kidney disease. -Detecting intrarenal and/or perinephric fluid (e.g. pus, blood)
- Demonstrating renal arterial perfusion or detecting renal vein thrombosis using Doppler.

IV pyelography

- has largely been replaced by ultrasonography and CT scanning.

Retrograde pyelography

- Reasons for performing a retrograde pyelogram include identification of filling defects (e.g. stones or tumors), as an adjunct during the placement of ureteral stents or ureteroscopy, or to delineate renal anatomy in preparation for surgery.

Antegrade pyelography/percutaneous nephrostomy

- involves percutaneous puncture of a pelvicalyceal system with a needle and the injection of contrast medium to outline the pelvicalyceal system and ureter to the level of obstruction.

CT scan

- Renal calculi, characterization of masses, renal trauma, retroperitoneal lesion
- CT+ IV contrast can reveal kidney function

Renal angiography

- Assessment of renal artery stenosis

MRI

- to characterize renal masses, demonstrate the renal arteries and cancer staging.

Radionuclide

- Dynamic scintigraphy: investigation of obstruction, RBF & GFR -Static scintigraphy: Kidney visualization, localization of infection, renal function.

Renal biopsy

- Renal biopsy is carried out under ultrasound control in specialized centers and requires interpretation by an experienced pathologist.
- Renal biopsy is helpful in the investigation of the nephritic and nephrotic syndromes, acute and chronic renal failure, hematuria after urological investigations and renal graft dysfunction.

Indications

- ☐ Microscopic hematuria .
- ☐ Proteinuria
- ☐ Diabetic nephropathy only if rapid deterioration couldn't be explained
- ☐ Acute nephritic syndrome (RPGN)
- ☐ Acute kidney injury

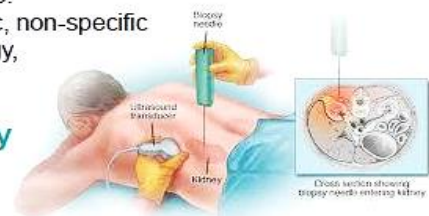
Unexplained AKI, particularly with an active urinary sediment (i.e. possible RPGN), or AKI that does not behave as expected, e.g. failing to recover.

- ☐ Chronic kidney disease

The most important determinants of the appropriateness of a biopsy are:

- Renal size. If <9cm, a biopsy histology is more likely to show chronic, non-specific
- Clinical context. A high pre-test probability the need to obtain histology,
- Degree of proteinuria.

- ☐ Systemic disease potentially involving the kidney



2

Glomerulopathies (GN)

Intended Learning Outcomes

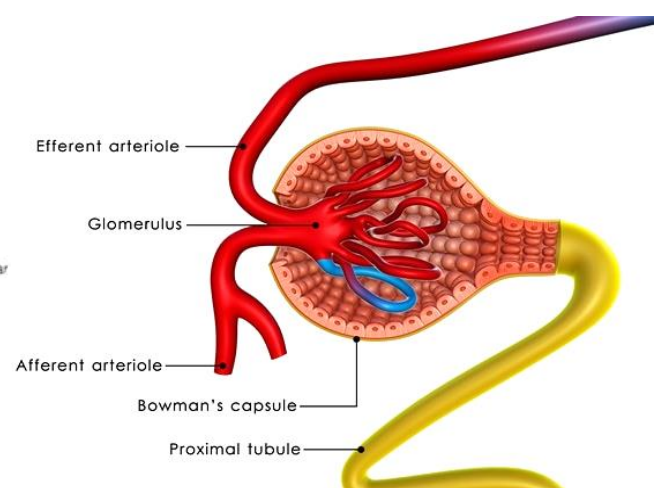
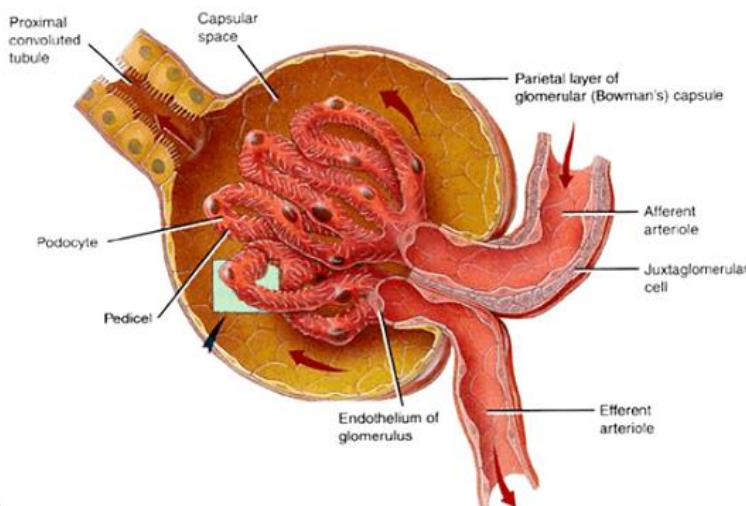
- 1- Define glomerular diseases
- 2- Describe definition, terminology, overview, Nephrotic vs Nephritic and their clinical features, diagnosis, differential diagnosis and treatment of different glomerular diseases.

Contents

- 1-Summary
- 2-Definition
- 3-Terminology of glomerular diseases
- 4-Pathophysiology
- 5-Overview of glomerular diseases
- 6-Nephrotic vs. nephritic syndrome
- 7-Definition of both Nephrotic & nephritic syndromes
- 8-Overview of both Nephrotic & nephritic syndrome
- 9-Pathophysiology
- 10-Clinical features
- 11.Diagnostics
- 12-Differential diagnosis
- 13-treatment.

Introduction

- Renal glomeruli excrete urinary substances and excess water as an ultrafiltrate into the urine .
- Any damage to the glomeruli disrupts the filtration process and results in the appearance of blood components (proteins and red blood cells) in the urine.
- Glomerular damage is commonly caused by immune-mediated processes, which often lead to glomerulonephritis. Non-inflammatory causes, such as metabolic disease (e.g., diabetes, amyloidosis), can also result in significant damage to the glomeruli.



GLOMERULOPATHIES (GLOMERULAR DISEASES).

Glomerulopathies are the **third** most common cause of end-stage renal disease (after diabetes and hypertension)

- Glomerulonephritis is an inflammation of the glomerulus, while glomerulopathy is a term for disorder affecting this structure.
- Glomerulopathy (GN) is a general term for a group of disorders in which: there is primarily an immunologically mediated injury to glomeruli, although renal interstitial damage is a regular accompaniment
- Most of the glomerular diseases are immunologically mediated, whereas tubular and interstitial disorders are frequently caused by toxic or infectious agents

Terminology of glomerular diseases

- **Primary:** a kidney disease specifically affecting the glomeruli (e.g., minimal change glomerulonephritis)
- **Secondary:** a disease affecting the glomeruli in the context of a systemic disease (e.g., lupus nephritis in SLE) or a disease affecting another organ (e.g., diabetic nephropathy)
- **Diffuse:** >50% of glomeruli affected (e.g., diffuse proliferative glomerulonephritis)
- **Focal:** < 50% of glomeruli affected (e.g., focal segmental glomerulosclerosis)
- **Global:** entire glomerulus is affected
- **Segmental:** only part of the glomerulus is affected
- **Proliferative:** an increased number of cells in the glomerulus
- **Membranous:** thickening of the glomerular basement membrane (e.g., membranous nephropathy)
- **Sclerosing:** scarring of the glomerulus
- **Necrotizing:** cell death within the glomerulus
- **Crescentic:** accumulation of cells such as macrophages, fibroblasts, and epithelial cells in Bowman space

Etiology

- Glomerular diseases may be caused by primary glomerular disorders (80–90% of cases) and/or systemic diseases and toxic exposures (10–20% of cases).

1 Primary

- The following types of nephrotic syndrome are commonly associated with other conditions.
 - **Minimal change disease**
 - **Focal segmental glomerulosclerosis**
 - **Membranous nephropathy**
 - **Membranoproliferative glomerulonephritis** (can manifest as nephrotic or nephritic syndrome)

2 Secondary

- **Diabetic nephropathy**
- **Amyloid nephropathy:** can be associated with multiple myeloma (AL amyloidosis) or chronic inflammatory disease such as rheumatoid arthritis (AA amyloidosis)
- **Lupus nephritis** (can manifest as nephrotic or nephritic syndrome)

Pathophysiology

The glomerular filtration barrier consists of 3 parts :

- **Initial segment:** Fenestrated glomerular capillary endothelium prevents large proteins from passing through.
- **Second segment:** The glomerular basement membrane (GBM) contains a negative charge produced by heparan sulfate.
- **Final segment:** Visceral epithelial cells produce/maintain the GBM and contain intercellular junctions created by podocytes that prevent further protein loss.
- **Damage to the glomeruli** → disruption of the glomerular filtration barrier → can lead to nephritic syndrome (low-level proteinuria, microhematuria, oliguria, and hypertension) or nephrotic syndrome (high-level proteinuria and generalized edema).

Clinical presentation of glomerular diseases:

The clinical manifestations of glomerular disease are clustered into the **five** major glomerular syndromes summarized in Table .

BKMC The Glomerular Syndromes

Syndrome	Manifestations
Nephritic syndrome	Hematuria, azotemia, variable proteinuria, oliguria, edema, and hypertension
Rapidly progressive glomerulonephritis	Acute nephritis, proteinuria, and acute renal failure
Nephrotic syndrome	>3.5 gm/day proteinuria, hypoalbuminemia, hyperlipidemia, lipiduria
Chronic renal failure	Azotemia → uremia progressing for months to years
Isolated urinary abnormalities	Glomerular hematuria and/or subnephrotic proteinuria

Nephrotic vs. nephritic syndrome

- Nephritic syndrome and nephrotic syndrome are both common clinical manifestations of glomerular diseases.
- Both syndromes are composed of characteristic clinical (e.g., edema, hypertension) and laboratory findings (e.g., glomerular hematuria, massive proteinuria), which result from damage to the glomeruli.
- Glomerular diseases are usually categorized by the syndrome they cause, which is either nephritic or nephrotic.
- Some diseases that manifest with nephritic syndrome can simultaneously cause nephrotic-range proteinuria (> 3.5 g/day), the main feature of nephrotic syndrome. When the criteria for both syndromes are fulfilled, the findings are referred to as mixed nephritic-nephrotic syndrome.

Definition of both Nephrotic & nephritic syndromes

1 Nephritic syndrome

- A syndrome characterized by glomerular damage, leading to
 - hematuria, occasionally pyuria
 - water retention,
 - hypertension
 - edema.
- Can be caused by a variety of conditions, including autoimmune, hereditary, and infectious diseases.

2 Nephrotic syndrome

- A combination of symptoms seen in various renal diseases that are associated with dysfunction of the renal filtration system. Characterized by
 - massive renal loss of protein (> 3.5 g/day)
 - resulting in edema
 - and hypoalbuminemia,
 - hyperlipidemia,
 - hypercoagulability (antithrombin III deficiency),
 - and an increased risk of infection (loss of immunoglobulins).

	Nephritic syndrome	Nephrotic syndrome
Presentat ion	- Proteinuria (< 3.5 g/day) (can be in nephrotic range in severe cases) - Hematuria with acanthocytes - RBC casts in urine - Mild to moderate edema - Oliguria - Azotemia - Hypertension - Sterile pyuria	- Heavy proteinuria (> 3.5 g/day) - Hypoalbuminemia - Generalized edema - Hyperlipidemia and fatty casts in urine → frothy urine - Hypertension ↑ Risk of thromboembolism: (via loss of antithrombin III) ↑ Risk of infection (via loss of IgG and tissue edema which compromises the local blood supply and immune response)
Pathophysiology	Inflammatory response within glomeruli → GBM disruption → loss of renally excreted RBCs (acanthocytes) and ↓ GFR → hematuria, oliguria, azotemia, and ↑ renin → edema and hypertension	Damage to <u>podocytes</u> → structural damage of <u>glomerular filtration barrier</u> → massive renal loss of protein

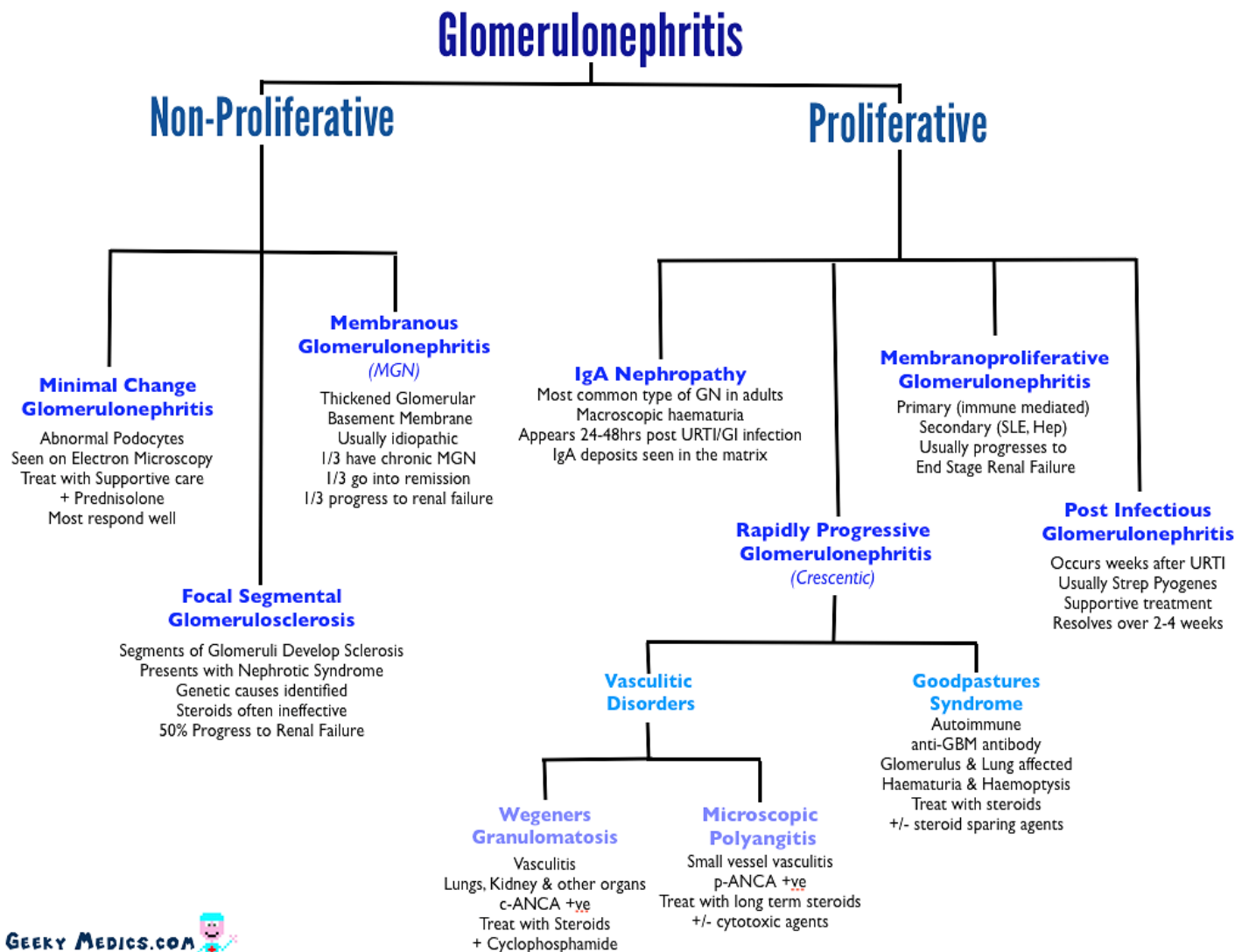
	Nephritic syndrome	Nephrotic syndrome
Causes	- Poststreptococcal glomerulonephritis - IgA nephropathy (Berger disease) - Granulomatosis with polyangiitis - Microscopic polyangiitis - Eosinophilic granulomatosis with polyangiitis - Goodpasture syndrome (anti-GBM disease) - Alport syndrome (hereditary nephritis) - Thin basement membrane disease - Rapidly progressive glomerulonephritis (RPGN) - Lupus nephritis - Most common causes of nephritic-nephrotic syndrome: - Membranoproliferative glomerulonephritis - Diffuse proliferative glomerulonephritis	- Due to primary or secondary podocyte damage - Minimal change disease - Focal segmental glomerulosclerosis - Membranous nephropathy - Due to secondary podocyte damage - Diabetic nephropathy - Amyloid light-chain (AL) amyloidosis, light chain deposition disease - Lupus nephritis

Note

All glomerular diseases can lead to acute and chronic kidney failure

3

Glomerulonephritis (GN)



GEEKY MEDICS.COM

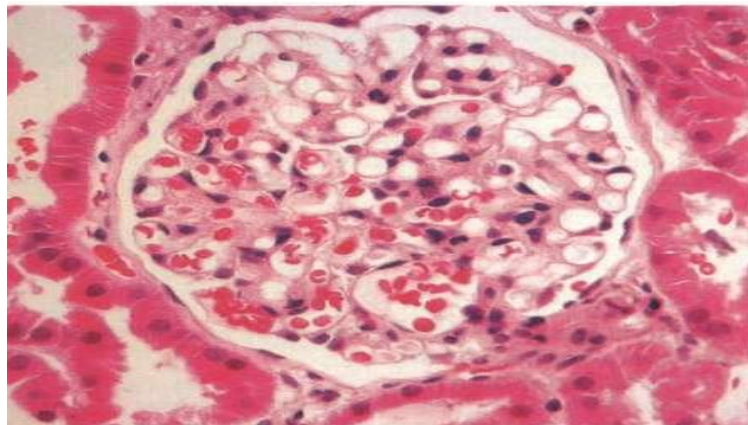
Non-proliferative GN

1

Minimal-change GN (minimal change nephropathy)

- Accounts for 80% of all cases of the idiopathic nephrotic syndrome in children.
- Majority of cases seen in 3 to 4 year age groups.
- Male predominance of 2.5:1 in children but no difference seen in adults.
- 80-90% idiopathic.
- -Associated with infectious disease, recent immunization, ingestion of heavy metals.
- In adults related to use of NSAIDS. A full-blown nephrotic syndrome with heavy proteinuria often of selective type is the most common presentation.
- In this condition the glomeruli appear normal on light microscopy.
- The only abnormality seen on electron microscopy is fusion of the foot processes of epithelial cells (podocytes).

Normal glomerulus on light microscopy in minimal change disease.



Clinical features

- Minimal-change nephropathy is most common in children, particularly males, accounting for the large majority of cases of nephrotic syndrome (proteinuria is usually highly selective) in childhood.
- Edema is present and in children this may be facial
- The condition accounts for 20-25% of cases of adult nephrotic syndrome.

Management

- **High-dose corticosteroid** therapy with prednisolone 60 mg/m² daily (up to a maximum of 80 mg/day) for a maximum of 4-6 weeks followed by 40 mg/m² every other day for a further 4-6 weeks corrects the urinary protein leak in more than 95% of children.
- In children, one-third subsequently do not relapse, but further courses of corticosteroids are indicated in the remainder.
- **In relapsing cases:** a course of cyclophosphamide 1.5-2.0 mg/kg daily is given for 8-12 weeks with concomitant prednisolone 7.5-15 mg/day. This increases the likelihood of long-term remission.
- **In Steroid unresponsive patients** may also respond to cyclophosphamide.
- **An alternative treatment** : to cyclophosphamide is cyclosporine 3-5 mg/kg/day, which is effective but must be continued long term to prevent relapse on stopping treatment.

2 Focal segmental glomerulosclerosis

- This is a disease of unknown etiology.
- It usually presents as massive proteinuria, haematuria, hypertension and renal impairment.
- Patients with nephrotic syndrome are often resistant to steroid therapy.
- All age groups are affected. It usually recurs in transplanted kidneys

CLASSIFICATION

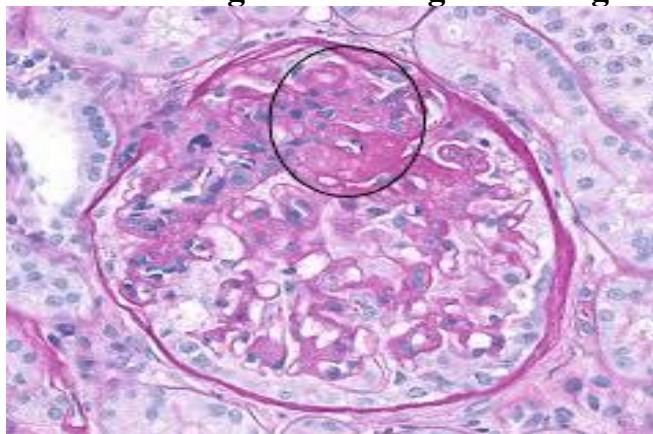
FSGS may be primary (idiopathic) or secondary to various etiologies.

PRIMARY	SECONDARY
Typical FSGS	Unilateral renal agenesis
Collapsing glomerulopathy	HIV infection
Glomerular tip lesion	Heroin addiction
	Morbid obesity
	Sickle cell disease
	Congenital heart disease
	Glycogen storage disease
	Hypertensive nephropathy

Pathology

- This glomerulopathy is defined primarily by its appearance on light microscopy.
- **Segmental** glomerulosclerosis is seen, which later progresses to global sclerosis.
- **Focal** tubular atrophy and interstitial fibrosis are invariably present.
- Immunofluorescence may show deposits of C3 and IgM in affected portions of the glomerulus.

Classic FSGS showing sclerotic segments in glomerulus.



Management

- **Prednisolone 0.5-2 mg/kg/day** is used in most patients and continued for 6 months before declaring the patient resistant to therapy.
- Plus The use of **cyclosporine**
- **Relapse** after reducing or stopping cyclosporine is very common.
- Long-term use may be required to maintain remission.
- **Cyclophosphamide, or azathioprine** is used for **second-line therapy** in adults.
- About 50% of patients progress to end-stage renal failure within 10 years of diagnosis, particularly those who are resistant to therapy.

3

Membranous glomerulopathy

- This condition occurs mainly in adults, predominantly in males.
- Patients are present with asymptomatic proteinuria or frank nephrotic syndrome.
- Microscopic haematuria, hypertension and/or renal impairment may accompany the nephrotic syndrome.
- Like all nephritides, hypertension and the degree of renal impairment are poor prognostic signs.
- Younger patients, females and those with asymptomatic proteinuria of modest degree at the time of presentation do best.

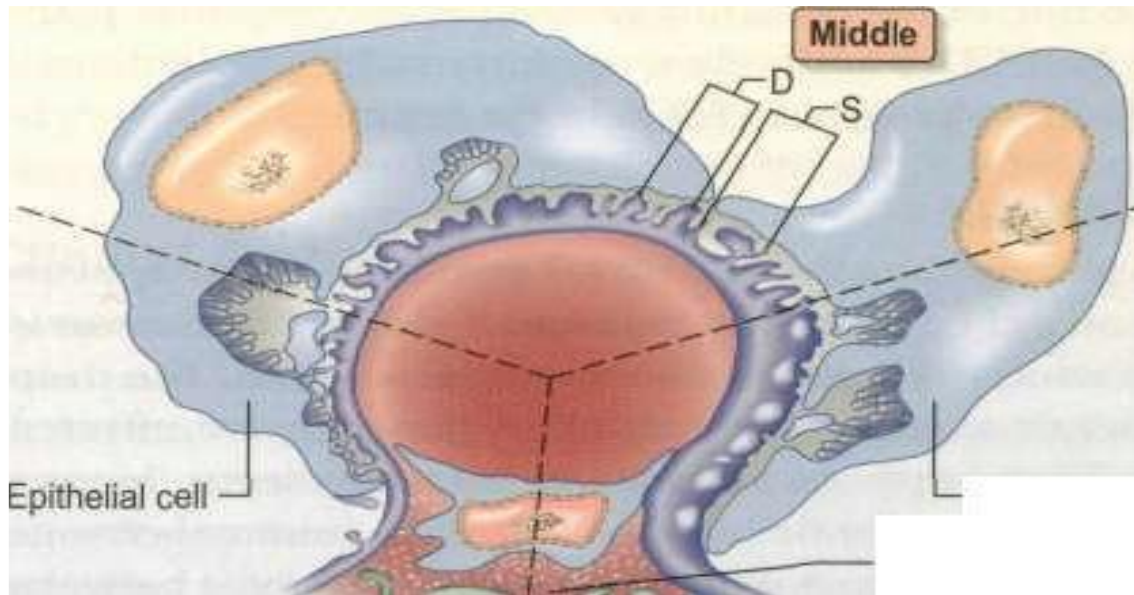
Etiopathogenesis

➤ Types:

1- **primary** or idiopathic form (which comprises 75% of the cases)

2- **secondary to**

- drugs (e.g. penicillamine, gold, NSAIDs, probenecid, mercury, captopril),
 - autoimmune disease (e.g. SLE, thyroiditis),
 - infectious disease (e.g. hepatitis B, hepatitis C, schistosomiasis, Plasmodium malariae),
 - neoplasia (e.g. carcinoma of lung, colon, stomach, breast and lymphoma) .
 - and other causes (e.g. sarcoidosis, kidney transplantation, sickle cell disease).
- At all stages, immunofluorescence shows the presence of uniform granular capillary wall deposits of IgG and complement C3.



Management

- **Oral high-dose corticosteroids** are **ineffective** in producing either a sustained remission of nephrotic syndrome or preserving renal function in patients with membranous GN.
- The use of **azathioprine** is **not** associated with any significant benefits.
- **cyclophosphamide**, and **cyclosporine** and **mycophenolate** are **effective** in the management of membranous GN.
- Because of growing concern about long-term toxicity, these drugs should be reserved for patients who exhibit clinical features, such as
 - severe or prolonged nephrosis (i.e. proteinuria > 6 g/day or > 6 months),
 - renal insufficiency and
 - hypertension
- **Anti-CD20 antibodies** (rituximab, which ablates B lymphocytes), has been shown to improve renal function, reduce proteinuria and increase the serum albumin; no significant adverse effects have been shown in the short term.
-
- **Treatment of Hypercoagulability:**
- **Prophylactic anticoagulation** is indicated if
 - Serum albumin level: Prophylaxis is usually considered when serum albumin is < 2–2.5 g/dL.
 - Coexistence of other risk factors for thromboembolism
- **Options**
- Unfractionated heparin
- Low molecular weight heparin (e.g., enoxaparin)
- Oral warfarin (target INR 2.0–3.0)

Proliferative GN

1 IgA nephropathy

- ❑ This disease has replaced post-streptococcal glomerulonephritis as the commonest form of glomerulonephritis world-wide.
- ❑ There is a focal and segmental proliferative glomerulonephritis
- ❑ The disease may be a result of an exaggerated bone marrow and tonsillar IgA immune response to viral or other antigens
 - Up to 50% of patients exhibit elevated serum IgA (polyclonal) concentration.
 - Superimposed crescent formation is frequent, particularly following macroscopic haematuria due to upper respiratory tract infection.

Clinical features

- IgA nephropathy tends to occur in children and young males.
- They present with asymptomatic microscopic haematuria or recurrent macroscopic haematuria sometimes following an upper respiratory or gastrointestinal viral infection.
- Proteinuria occurs and 5% can be nephrotic.
- The prognosis is usually good, especially in those with normal blood pressure, normal renal function and absence of proteinuria at presentation.
- The risk of eventual development of end-stage renal failure is about 25% in those with proteinuria of more than 1 g per day, elevated serum creatinine, hypertension, and tubulointerstitial fibrosis on renal biopsy.

Management

- Patients with proteinuria over 1-3 g/day, mild glomerular changes only and preserved renal function should be treated with **steroids**.
- **Steroids** reduce proteinuria and stabilize renal function.
- **In patients with progressive disease** (creatinine clearance less than 70 mL/min),
 - Fish oil or prednisolone with cyclophosphamide for 3 months followed by maintenance with prednisolone and azathioprine may be tried.
 - A tonsillectomy can reduce proteinuria and haematuria in those patients with recurrent tonsillitis.
- **All patients, with or without hypertension and proteinuria**, should receive either of ACE inhibitor and angiotensin II receptor antagonist

Diseases commonly associated with acute **nephritic syndrome**

- Post-streptococcal glomerulonephritis
- Non-streptococcal post-infectious glomerulonephritis, e.g. Staphylococcus, pneumococcus, Legionella, syphilis, mumps, varicella, hepatitis B and C, malaria, schistosomiasis,
- Infective endocarditis
- Shunt nephritis
- Visceral abscess
- Systemic lupus erythematosus
- Cryoglobulinaemia

A)

Acute glomerulonephritis **(Acute nephritic syndrome)**

This comprises:

- **Hematuria** (macroscopic or microscopic) - red-cell casts are typically seen on urine microscopy
- **Proteinuria**
- **Hypertension**
- **Edema** (periorbital, leg or sacral)
- temporarily **Oliguria and uraemia**.
- The histological pattern is characterized by cellular proliferation (mesangial and endothelial) and inflammatory cell infiltration (neutrophils, macrophages).

Post-streptococcal glomerulonephritis (PSGN) **(Post infectious GN)**

Pathogenesis

- Immune complex mediated
- Antibodies elicited by streptococcal infection react with normal glomerular antigens leading to immune complex formation and glomerular deposition
- Complement activation by directly deposited streptococcal antigens (serum C3 decreased)

Clinical features

- Most common in 5-12 years old group
- The patient, usually a child, suffers a streptococcal infection 1-3 weeks before the onset of the acute nephritic syndrome.
- Severity of kidney involvement ranging from asymptomatic microscopic hematuria with normal renal function to gross hematuria with ARF

- Oliguria and Hypervolemia
- Hypertension and may lead to hypertensive encephalopathy
- Edema and eye puffiness
- Heart failure secondary to hypervolemia and HTN may occur

Diagnosis of APSGN

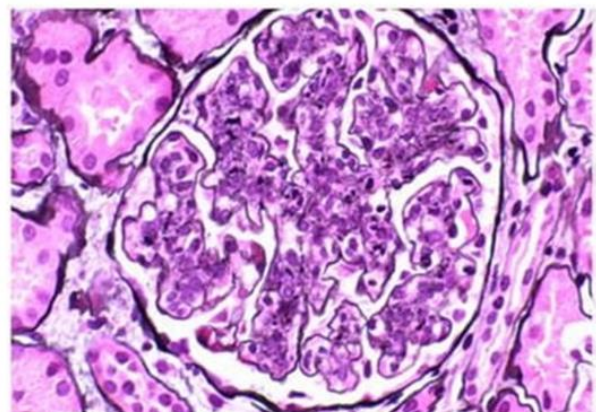
Sr. No	Name of investigation	finding
1.	Urine routine & microscopoy	RBC, RBC cast, PMN leucocytes, proteinuria
2.	CBC	Anemia (d/t hemodilution and low grade hemolysis)
3.	Serum C3, CH50	Low
4.	Throat culture	If positive, support diagnosis or might represent carrier state
5.	Antibody titre to streptococcal antigen (i) Antistreptolysin O (ASO) titre (ii) Anti DNAase B (iii) Streptozyme screen	- Increased after pharyngitis but rarely increase after skin infection - Best for cutaneous streptococcal infection - Measures multiple antibodies to different streptococcal Ag
6.	Chest X ray	For pulmonary edema
7.	MRI (if neurological symptoms)	PRES in parieto-occipital area

Indications of renal biopsy:

- Presence of persistent ARF
- Presence of nephrotic syndrome
- Absence of evidence of streptococcal infection
- Normal complement level
- Hematuria, proteinuria, impaired renal function or low C3 level persist more than 2 months after the onset

Renal biopsy finding:-

- Glomeruli- enlarged and relatively bloodless and show diffuse mesengial cell proliferation with increase in mesengial matrix.



Management

○ The acute phase

- **Antibiotics:** to limit the spread of the organism
- **Treatment of hypervolemia:** by salt and fluid restriction and diuretic therapy
- **Antihypertensives :** for high blood pressure, calcium channel blockers (amlodipine) and B-blockers (bisoprolol and labetalol) is preferred.

Note

If recovery is slow, corticosteroids may be helpful.

Prognosis

- The prognosis is usually good in children
- A small number of adults develop hypertension and/or renal impairment later in life.
- Therefore, in older patients, an annual blood pressure check, and an estimation of serum creatinine, is a reasonable precaution, even after apparent complete recovery.
- Evidence in support of long - term penicillin prophylaxis after the development of glomerulonephritis is lacking.
- In non-streptococcal post-infectious glomerulonephritis, prognosis is equally good if the underlying infection is eradicated.

Dialysis indication

1. Anuria/ oliguria
2. Volume overload with evidence of HTn and / or pulmonary edema refractory to diuretic therapy.
3. Persistent hyperkalemia
4. Severe metabolic acidosis unresponsive to medical management.
5. Uremia (encephalopathy, pericarditis, neuropathy)
6. BUN >100-150 mg/dl (or lower if rising rapidly)

B) Secondary glomerular disease

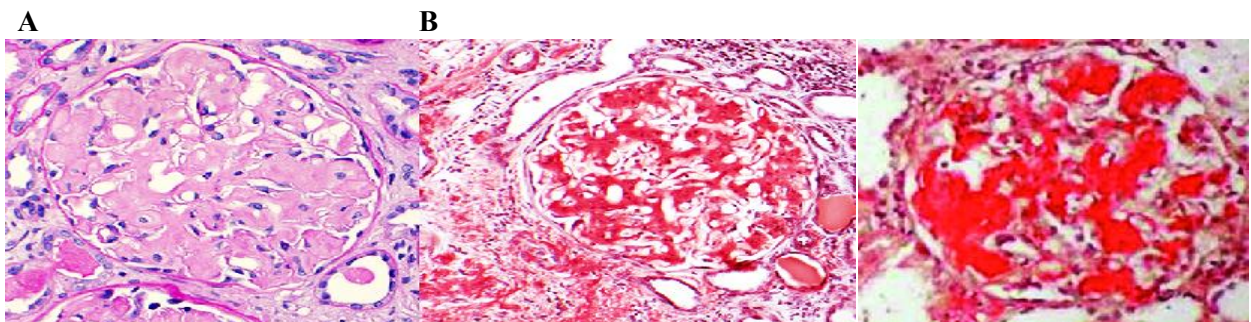
1

Amyloidosis

- Amyloidosis is an **acquired or inherited** disorder of protein folding, in which normally soluble proteins or fragments are deposited extracellularly as abnormal insoluble fibrils.

Pathology

- On light microscopy, eosinophilic deposits are seen in the myocangium, capillary loops and arteriolar walls.
- Staining with Congo red renders these deposits pink and they show green birefringence under polarized light. Immunofluorescence is unhelpful, but on Electron microscopy the characteristic fibrils of amyloid can be seen.



- Amyloid, (a) Light microscopy of eosinophilic amyloid deposits (arrow), (b) Congo red stain under polarized light showing apple green refringence.

Diagnosis

- The diagnosis can often be made clinically when features of amyloidosis are present elsewhere.
- On imaging, the kidneys are often **large**.
- Scintigraphy** with radiolabelled serum amyloid P (SAP), a technique for quantitatively imaging amyloid deposits in vivo, is desirable because of its ability to detect the rate of regression or progression of amyloidosis over a period of time.
- Renal biopsy** is necessary in all suspected cases of renal involvement.

Treatment

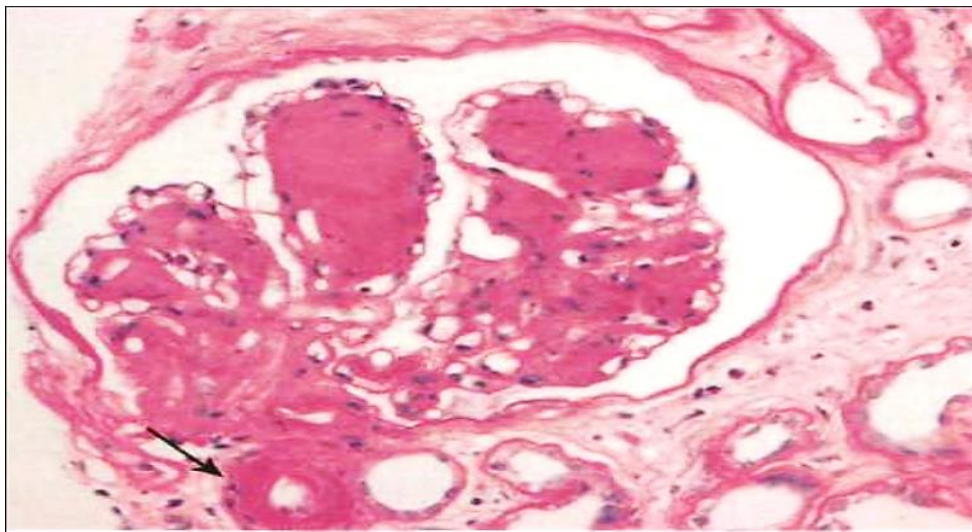
- Treatment of the underlying cause should be undertaken.
- Renoprotective measures should be started.
- The success of dialysis and kidney transplantation is dependent upon the extent of amyloid deposition in extrarenal sites, especially the heart.

Diabetic nephropathy

- Diabetic renal disease is the leading cause of end-stage renal failure in the western world.
- Type 1 and type 2 diabetic patients have equivalent rates of proteinuria, azotaemia, and ultimately end-stage renal failure.
- Both types of diabetes show strong similarities in their rate of renal functional deterioration, and onset of co-morbid complications.

Pathology

- The kidneys enlarge initially and there is glomerular hyperfiltration (GFR > 150 mL/min).
- The major early histological lesions seen are glomerular basement membrane thickening and mesangial expansion.
- Later, glomerulosclerosis develops with nodules (Kimmelstiel- Wilson lesion) and hyaline deposits in the glomerular arterioles).
- These later changes are associated with heavy proteinuria. The lesions seen in type 1 are also seen in type 2.



Diabetes mellitus. Advanced diabetic glomerulopathy and arteriolar sclerosis (arrow).

Treatment

- Lifestyle changes (cessation of smoking and increase in exercise), hypertension, poor metabolic regulation, and hyperlipidaemia should be addressed in every diabetic.
- Microalbuminuria is a reason to start treatment with ACE inhibitors or an angiotensin II receptor antagonist in either type of diabetes, regardless of blood pressure elevation.
- Despite intensified metabolic control and antihypertension treatment in diabetic patients, a substantial number still go on to develop end-stage renal failure.

Systemic lupus erythematosus (lupus GN)

- Overt renal disease occurs in at least one-third of SLE patients and, of these, 25% reach end-stage renal failure within 10 years.
- Histologically almost all patients will have changes which are classified by the World Health Organization (modified with added clinical manifestations)

Histopathological pattern of Lupus nephritis

- **Type I** - Asymptomatic (Mesangial immune deposits (normal glomeruli) on light microscopy).
- **Type II** - Mesangial immune deposits with mesangial cell hypercellularity and matrix expansion. Clinically there is mild renal disease.
- **Type III** - Focal proliferative glomerulonephritis (involving < 50% of the total number of glomeruli on renal biopsy) with subdivisions for active or sclerotic lesions. Haematuria and proteinuria are seen in almost all.
- **Type IV** - Diffuse proliferative glomerulonephritis (involving > 50% of total number of glomeruli on renal biopsy) either with segmental (class IV-S) or global (class IV-G) involvement, Clinically, as in type III, but the nephrotic syndrome, hypertension and renal insufficiency are often seen.
- **Type V** - Membranous GN. Heavy proteinuria is seen, with signs of the nephrotic syndrome, haematuria and hypertension.
- **Type VI** - Advanced sclerosing lesions (> 90 glomeruli are sclerosed). Progressive renal failure is seen.

Note

The prognosis is better in patients with types **I, II and V**.

Note

Immune deposits in the glomeruli and mesangium are characteristic of SLE and stain positive for **IgG, IgM, IgA** and the complement components **C3, C1q and C4** on immunofluorescence.

Management

- Initial treatment depends on the clinical presentation, but hypertension and oedema should be treated.
- A definite histopathological diagnosis is required.
- Type I requires no treatment.
- Type II usually runs a benign course but some are treated with steroids.
- There have been a number of clinical trials with immunosuppressive agents in types III, IV and V.
- At present, steroids and cyclophosphamide are usually used for induction, with azathioprine and mycophenolate mofetil for maintenance therapy.
- B cell depletion with rituximab (anti-CD20) has been shown to be effective in a small study.

Prognosis

- Treatment leading to the normalization of proteinuria,
- hypertension and renal dysfunction indicates a good prognosis.
- Glomerulosclerosis (type VI) usually predicts end-stage renal disease.

Diagnostic Approach for glomerulonephritis

- 1-Confirm nephrotic or nephrotic-range **proteinuria**.
- 2-Assess for potential concomitant and underlying conditions.
- 3-Assess for complications.
- 4-Consider **kidney biopsy** to determine renal pathology.

➤ Confirmation of proteinuria

- Qualitative assessment by **urine dipstick** (commonly used for screening)
 - Usually shows $\geq 3+$ **proteins**
 - **Hematuria** may indicate concomitant glomerulonephritis.
 - Quantitative assessment of **urine** protein excretion
 - 24-hour **urine** protein (test of choice): > 3.5 g/24 hours
 - Spot urine protein/creatinine ratio: > 3.5 g/g

➤ **Urine sediment microscopy**

☐ Nephrotic sediment

- ✓ Lipiduria, fatty casts with Maltese cross appearance under polarized light
- ✓ Renal tubular epithelial cell casts
- ✓ Hematuria with acanthocytes and/or RBC casts may indicate concomitant glomerulonephritis (see “Diagnostics” in “Nephritic syndrome”).

☐ Nephritic sediment

☐ Hematuria (either microhematuria or intermittent macrohematuria)

☐ Acanthocytes

☐ Red blood cell casts: RBC casts form through the congregation of proteins and RBCs inside the tubules.

☐ Mild to moderate proteinuria of $>150\text{mg}/24\text{h}$ but < 3.5 g/24h (nonselective glomerular proteinuria)

☐ Sterile pyuria and sometimes WBC casts.

➤ Additional laboratory studies

- CBC: $\uparrow$ Hb/Hct may indicate hemoconcentration
- $\uparrow$ Cr and/or $\uparrow$ BUN may be seen; $\downarrow$ Na is commonly seen.
- Serum protein: $\downarrow$ total protein, $\downarrow$ albumin (< 3 g/dL)
- Coagulation factors: $\downarrow$ ATIII, $\downarrow$ protein S, $\downarrow$ plasminogen
- Lipid profile: Hyperlipidemia ($\uparrow$ LDL, $\uparrow$ triglycerides) may be present.
- Vitamin D levels: $\downarrow$ 25-OH Vit-D
- Inflammatory markers: $\uparrow$ ESR, $\uparrow$ CRP may suggest underlying infection, inflammatory condition, or vasculitis.
- **Imaging**: Renal ultrasound: to assess kidney size and shape and rule out obstruction

➤ Renal biopsy

Indication: to confirm the diagnosis when the etiology of nephrotic syndrome is unclear and/or to guide management Often required in adult patients to guide management.

- Sometimes deferred when a likely cause of nephrotic symptoms is evident
- Consult nephrology to determine the necessity and timing of biopsy.

General Treatment Approach for glomerulonephritis

1. Reduce proteinuria / intraglomerular pressure

- **ACE inhibitor or ARB** — first-line when albuminuria/proteinuria is present.
 - Titrate to the **maximum tolerated dose**.
 - Do **not** combine ACEi + ARB.
 - Monitor creatinine and potassium after initiation/dose increase.
- **SGLT2 inhibitor** — dapagliflozin or empagliflozin in appropriate patients with CKD and albuminuria, including many patients without diabetes.
- **Finerenone** — mainly for **CKD associated with type 2 diabetes and albuminuria**, despite optimized ACEi/ARB; monitor K⁺.

2. Edema

- **Loop diuretic**: furosemide/torsemide.
- If resistant edema: combine with a **thiazide/thiazide-like diuretic** (e.g., metolazone) with careful monitoring.
- Dietary **sodium restriction** is important.

3. Blood pressure

- Generally target **SBP <120 mmHg when tolerated and when standardized office BP measurement is used** in appropriate CKD patients.
- ACEi/ARB is preferred when significant albuminuria is present.

4. Dyslipidemia / cardiovascular protection

- **Statin therapy** according to CKD age/risk profile.
- Avoid routine treatment of isolated hyperlipidemia solely because of nephrotic syndrome without considering overall CV risk.

5. Thrombosis in nephrotic syndrome

- **Anticoagulation is not routine for every patient.**
- Consider prophylactic anticoagulation in selected patients with severe hypoalbuminemia and high thrombotic risk after balancing bleeding risk.

6. Other supportive measures

- Avoid **NSAIDs** and other nephrotoxins where possible.
- Correct metabolic complications of CKD.
- Vaccination and infection prevention, particularly before immunosuppression.
- Dietary protein intake should generally be around **0.8 g/kg/day in adults with CKD G3–G5**, rather than high-protein diets.

4

Acute Renal failure (Acute kidney injury)

1

Definition of acute kidney

- Acute kidney injury (AKI) or acute renal failure (ARF), was previously defined as a rapid deterioration of renal functions resulting in the accumulation of nitrogenous wastes such as urea and creatinine. However, immediately after a kidney injury, BUN or creatinine levels may be normal, and the only sign of a kidney injury may be decreased urine production.

►

- **KDIGO** is the most recent and most commonly used tool. According to KDIGO, AKI is the presence of any of the following:
 - Increase in serum creatinine by 0.3 mg/dL or more (26.5 micromoles/L or more) within 48 hours
 - Increase in serum creatinine to 1.5 times or more baseline within the prior seven days
 - Urine volume less than 0.5 mL/kg/h for at least 6 hours.

2

Staging of AKI

Patients that meet the definition for AKI can be staged according to whichever criteria (serum creatinine or urine output) gives them the highest stage.

Stage	Serum creatinine	Urine output
1	1.5–1.9 times baseline OR ≥0.3 mg/dl (≥26.5 μmol/l) increase	<0.5 ml/kg/h for 6–12 hours
2	2.0–2.9 times baseline	<0.5 ml/kg/h for ≥12 hours
3	3.0 times baseline OR Increase in serum creatinine to ≥4.0 mg/dl (≥353.6 μmol/l) OR Initiation of renal replacement therapy OR, In patients <18 years, decrease in eGFR to <35 ml/min per 1.73 m ²	<0.3 ml/kg/h for ≥24 hours OR Anuria for ≥12 hours

3

Signs and symptoms:

- Decreased urine output, although occasionally urine output remains normal
- Fluid retention, causing swelling in your legs, ankles or feet
- Shortness of breath
- Fatigue
- Confusion
- Nausea
- Weaknesses
- Irregular heartbeat
- Chest pain or pressure
- Seizures or coma in severe cases

4

Causes of AKI:

1.Impaired blood flow to the kidneys (pre renal) causes:

- Blood or fluid loss
- Low Blood pressure
- Heart attack
- Heart disease
- Infection
- Liver failure
- Use of aspirin, ibuprofen (Advil, Motrin IB, others), naproxen sodium (Aleve, others) or related drugs
- Severe allergic reaction (anaphylaxis)
- Severe burns
- Severe dehydration

2.Damage to the kidneys (renal causes):

- Blood clots in the veins and arteries in and around the kidneys
- Hemolytic uremic syndrome,
- Infection
- Lupus, an immune system disorder causing glomerulonephritis
- Medications, such as certain chemotherapy drugs, antibiotics and dyes used during imaging tests
- Scleroderma
- Thrombotic thrombocytopenic purpura
- Toxins, such as alcohol, heavy metals and cocaine
- Rhabdomyolysis
- Tumor lysis syndrome

3.Urine blockage in the kidneys (post renal causes)

- Bladder cancer
- Blood clots in the urinary tract
- Cervical cancer
- Colon cancer
- Enlarged prostate
- Kidney stones
- Nerve damage involving the nerves that control the bladder
- Prostate cancer

5

Risk factors of AKI:

- Being hospitalized, especially for a serious condition that requires intensive care
- Advanced age
- Blockages in the blood vessels in your arms or legs (peripheral artery disease)
- Diabetes
- High blood pressure
- Heart failure
- Kidney diseases
- Liver diseases
- Certain cancers and their treatments

Acute tubular necrosis

Causes :

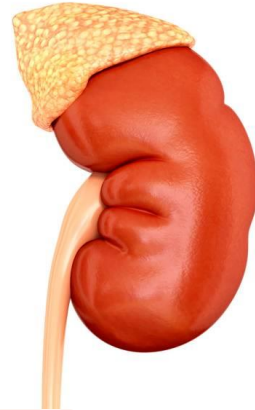
- **Ischemic causes** (Prerenal factors if not corrected rapidly)
 - Prolonged pre-renal cause for ARF → Tubular ischemia & necrosis
- **Toxic causes :**
 - Endogenous :
 - Hemoglobin (Hemolytic anemia).
 - Myoglobin (Rhabdomyolysis).
 - Bilirubin Nephropathy in obstructive jaundice.
- **Exogenous :**
 - Antibiotics: Aminoglycosides, Amphotercin B
 - NSAIDs
 - Radio-contrast dye
 - Heavy metals: Arsenic, Mercury.

Pathophysiology : 3 stages

1-Oliguric phase

2-Diuretic phase

3-Post diuretic phase



1. Oliguric phase : Lasts few hours to weeks:

- Oliguria ($\downarrow$ urine output < 400 cc/d) due to :
- Obstruction of tubules by epithelial casts.
- Afferent vasoconstriction due to excess renin activation.
- Na : decreased (dilutional hyponatremia)
- K : Hyperkalemia.
- Phosphorous : increased
- Ca : $\downarrow$ or may still normal.
- H^+ ions, urea & creatinine : $\uparrow\uparrow$

2- Diuretic phase (3-7 days):

- Polyuria in this phase is due to partial recovery of the tubules (incomplete power of the tubules to reabsorb the filtered water)
- The kidney tubules still unable to concentrate urine
- The urine output increases (up to 10 L/d) but it may take several days before the creatinine starts to drop.
- Dehydration & hypokalemia may occur

3-Post diuretic phase:

- Convalescent phase
- may takes weeks or months
- there is complete recovery with improvement of the general health
- normalization of the chemistry occurs

Clinical picture:

Oliguric phase : (2-4 weeks)

- 1- Oliguria: ↓ urine output < 400 cc/d.
- 2- Manifestations of hypervolemia: in cases of acute renal failure due to pre renal causes e.g. marked reduction of blood pressure with oliguria, decreased skin turgor, reduced jugular venous pressure and dry mucous membranes.
- 3- Manifestations of hypervolemia: in cases of acute renal failure due to intrinsic renal disease
 - Headache.
 - Congested neck vein.
 - Pulmonary edema.
 - Hypertension.
 - Edema lower limb.
- 4- Manifestations of hyponatremia: Convulsion, Confusion, Coma.
- 5- Manifestations of hyperkalemia:
 - Skeletal muscle: weakness, flaccid paralysis, areflexia.
 - Smooth muscle: Intestinal colic, nausea, vomiting.
 - Cardiac muscle: Arrhythmias & can progress to cardiac arrest.
- 6- Manifestations of uremia:
 - Anorexia, nausea, vomiting.
 - Pericarditis, Bleeding.
- 7- Convulsion, Confusion, Coma

Diuretic phase :

- Improvement of general condition.
- Urine output 3-5 L /d
- Dehydration & hypokalemia may occur

Post diuretic phase: Convalescent phase

- Improvement of the general health & normalization of chemistry
- It may take weeks or months.
- Urine output becomes normal volume, concentrated.

Investigations:

1- To diagnose ARF

- 1) **Urine examination:** Oliguria with low fixed specific gravity
- 2) **Biochemical changes:**
 - Azotemia: ↑ Urea & creatinine.
 - Creatinine clearance: ↓
 - Na: ↓ (dilutional hyponatremia)
 - K: ↑
 - PO4: ↑ & ↓ **Ca**
 - ↓ Hco3
- 3) **Ultrasound:** Normal diameter.
- 4) **Urinary Indices :** Fractional Excretion of Sodium

$$FE_{Na} = 100 \times \frac{\text{sodium}_{\text{urinary}} \times \text{creatinine}_{\text{plasma}}}{\text{sodium}_{\text{plasma}} \times \text{creatinine}_{\text{urinary}}}$$

- **FENa < 1** indicates an intact tubular function.
- **FENa > 1** indicates euvoolemia or tubular dysfunction.

2- To Diagnose the cause :

☐ in renal causes:

- Muddy-brown granular cast: in ATN.
- Pigmented cast in myoglobinuria.
- White cells cast in AIN.
- Red cell casts: in AGN
- Pus cells: in acute pyelonephritis

☐ In post renal causes:

- to detect the cause of obstruction
- Retrograde pyelography & angiography.
- Plain film of the abdomen: to detect the stones.

Indications of Renal Biopsy in AKI

- 1-Clinical and laboratory findings suggesting Glomerular or Interstitial nephritis
- 2-Unknown cause
- 3-Prolonged ATN

Treatment

1.Oliguric phase:

- The aim of management is to keep the patient alive until spontaneous recovery.
- Urinary catheter : to assess urine volume & to exclude post-renal cause.
- Central venous catheter : to assess the blood volume.
- Immediate volume infusion to prevent ATN.
- Fluid balance i.e, daily fluid intake should equal urine output plus losses from fistulae and from vomiting plus an allowance of 500 ml daily for insensible loss.
- Acidosis is treated with I.V sodium bicarbonate.
- Control blood pressure.
- Diet:
 - Sodium and potassium restriction with rare exception.
 - Protein restriction to 40 gm/day if it is hoped to avoid dialysis.
 - Patients treated by dialysis must receive 70 gm/day protein.
 - Hypercatabolic patients will require an even higher nitrogen intake to prevent negative nitrogen balance.
- **Hyperkalemia:**
 - Acidosis correction: NaHCO_3 .
 - Ca gluconate (slowly infusion)
 - Insulin + glucose ↑ intracellular shift of K.
 - Dialysis.
- **Indication of dialysis in patients with ARF**
 - Bad general condition (severe symptoms)
 - Marked acidosis
 - Serum K > 7 or if not controlled by conservative measures
 - Pulmonary edema
 - pericarditis

2.Diuretic phase

- Adequate fluid to avoid dehydration.
- Na and K supplements

How to differentiate Acute from chronic renal failure?

1. A rapid rate of change of serum urea & creatinine suggesting an acute process.
2. Small kidneys with increased echogenicity diagnostic of chronic process.
3. Evidence of renal osteodystrophy is indicative of chronic disease.
4. Measurement of carbamylated Hb (a product of non-enzymatic reaction between urea & Hb) if high suggesting a chronic disease.
5. ↓Hb, ↓ Ca, ↑ P may suggest a chronic disease.
6. Low fixed specific gravity of urine (1010) and presence of broad casts in urine analysis indicates chronic renal failure.

5

Chronic Kidney Disease

1

Definition

Structural or functional abnormalities of the kidneys for ≥ 3 months, as manifested by

1. Kidney damage, with or without decreased GFR, as defined by

Pathologic abnormalities

Markers of kidney damage

Urinary abnormalities (proteinuria)

Blood abnormalities (renal tubular syndromes)

Imaging abnormalities

Kidney transplantation

2. $\text{GFR} < 60 \text{ ml/min/1.73 m}^2$, with or without kidney damage

CKD, chronic kidney disease; GFR, glomerular filtration rate; KDIGO, Kidney Disease Improving Global Outcomes.

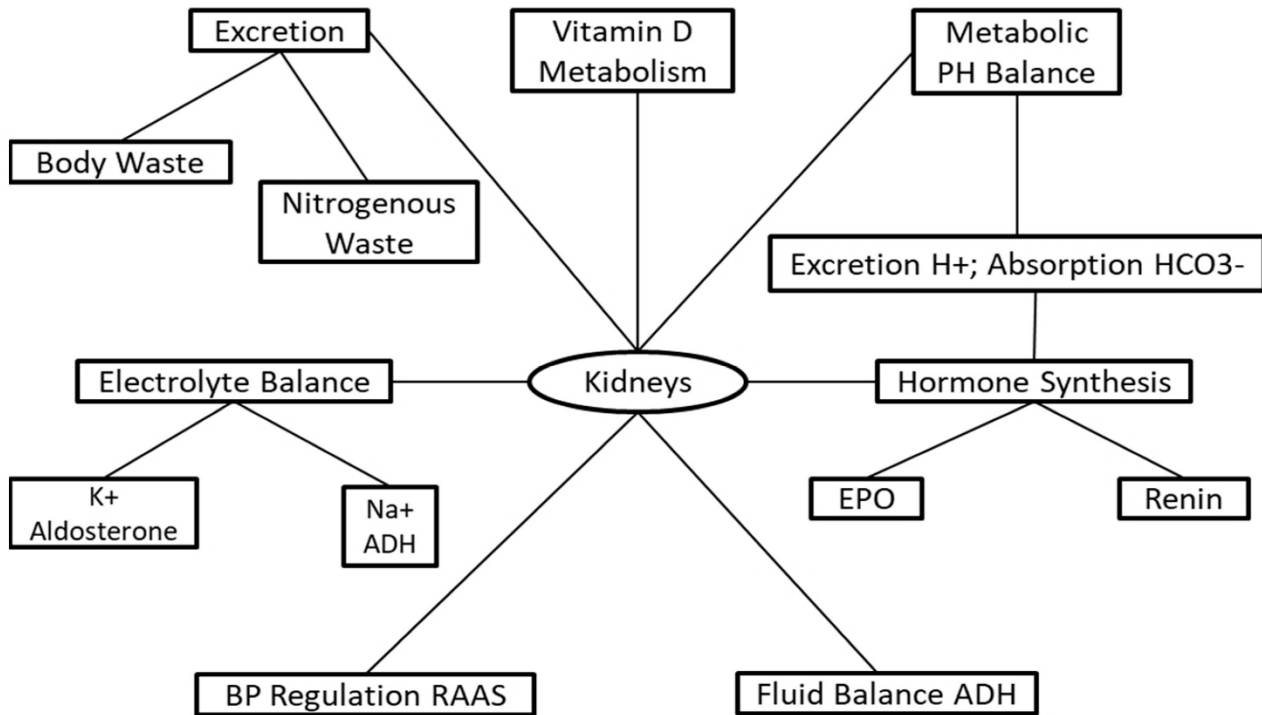
2

Stages of CKD.

CKD-stage	eGFR _{CKD-EPI} (ml/min/1.73m ²)	Albuminuria (ACR)			Total
		A1 ($< 30 \text{ mg/g}$)	A2 ($30 - 300 \text{ mg/g}$)	A3 ($> 300 \text{ mg/g}$)	
G1	≥ 90	79 15.2%	9 1.7%	2 0.4%	90 17.3%
G2	60 – 89	257 49.4%	36 6.9%	10 1.9%	303 58.3%
G3a	45 – 59	76 14.6%	17 3.3%	3 0.6%	96 18.5%
G3b	30 – 44	15 2.9%	5 1.0%	2 0.4%	22 4.2%
G4	15 – 29	3 0.6%	2 0.4%	3 0.6%	8 1.5%
G5	< 15	0	0	1 0.2%	1 0.2%
total		430 82.7%	69 13.3%	21 4.0%	520 100%

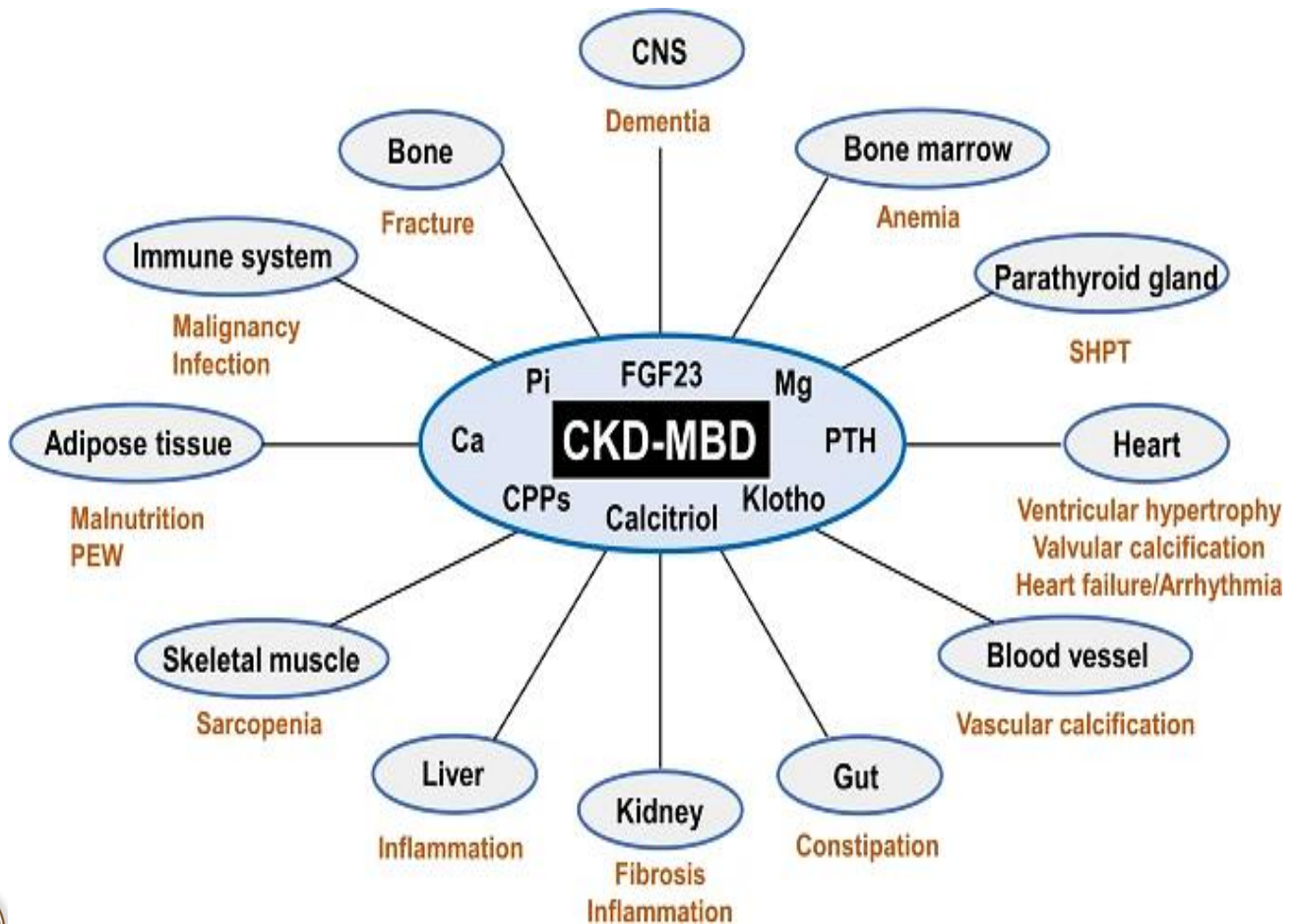
3

Functions:



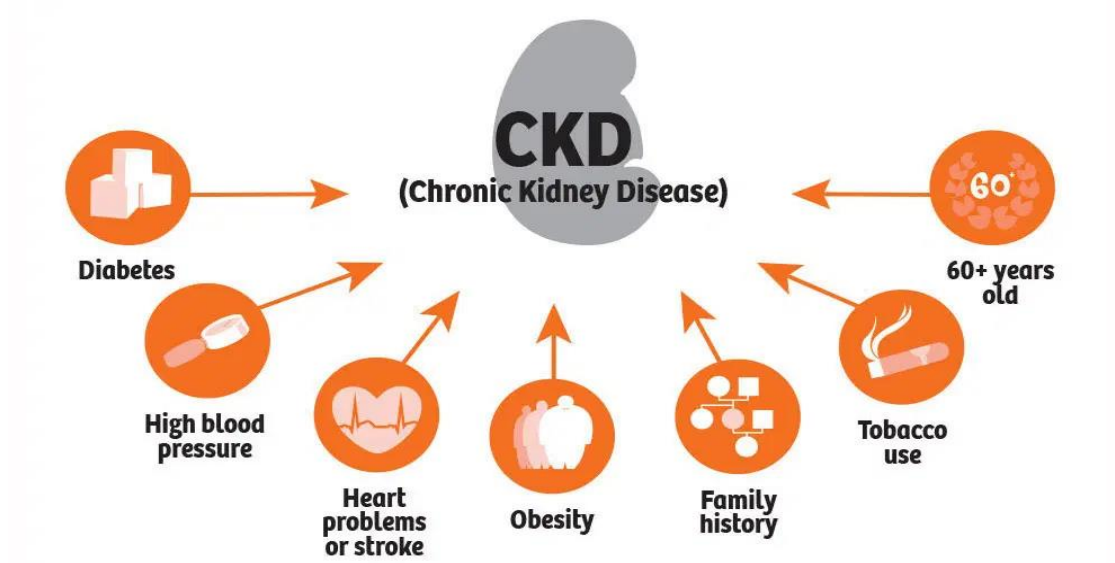
4

Complications of CKD:



5

Risk factors



5

Etiology

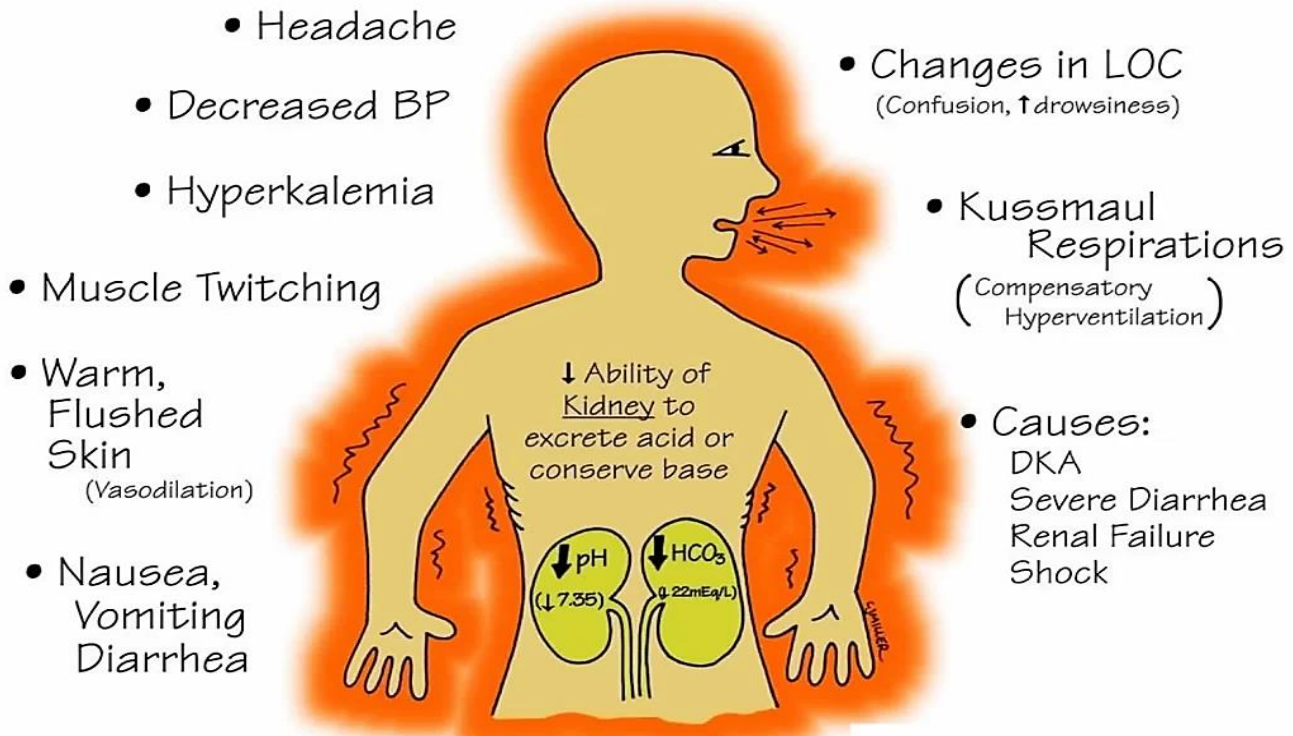
- Diabetes mellitus type 2 (30% to 50%)
- Diabetes mellitus type 1(3.9%)
- Hypertension(27.2%)
- Primary glomerulonephritis(8.2%)
- Chronic Tubulointerstitial nephritis(3.6%)
- Hereditary or cystic diseases(3.1%)
- Secondary glomerulonephritis or vasculitis(2.1%)
- Plasma cell dyscrasias or neoplasm(2.1%)
- Sickle Cell Nephropathy (SCN) which accounts for less than 1% of ESRD patients in the United States

5

Symptoms of CKD

- Most of KD patients are asymptomatic and usually find it through accidental screening tests or until they become unwell due to advanced CKD
- Inpatients presenting symptoms depending on the cause , the following symptoms may vary
 - ✓ Pallor
 - ✓ Breathlessness
 - ✓ Altered cognition
 - ✓ Taste disturbances
 - ✓ Uremic breathing
 - ✓ Polyuria
 - ✓ Proteinuria
 - ✓ Nocturia
 - ✓ Hematuria
 - ✓ Peripheral oedema

METABOLIC ACIDOSIS



5

Laboratory Measurement for ascertainment of CKD

For all patients at increased risk for CKD:

- Serum creatinine to estimate GFR;
- Albumin-to-creatinine or protein-to-creatinine ratio in a first-morning or random untimed "spot" urine specimen;
- Examination of the urine sediment or dipstick for red blood cells and white blood cells.

For patients found to have CKD:

- Imaging of the kidneys, usually by ultrasound;
- Serum electrolytes (sodium, potassium, chloride and bicarbonate).

Chronic kidney disease

Definition

Abnormality of kidney structure or function
> 3 months as $eGFR < 60 \text{ mL/min/1.73 m}^2$

Etiology

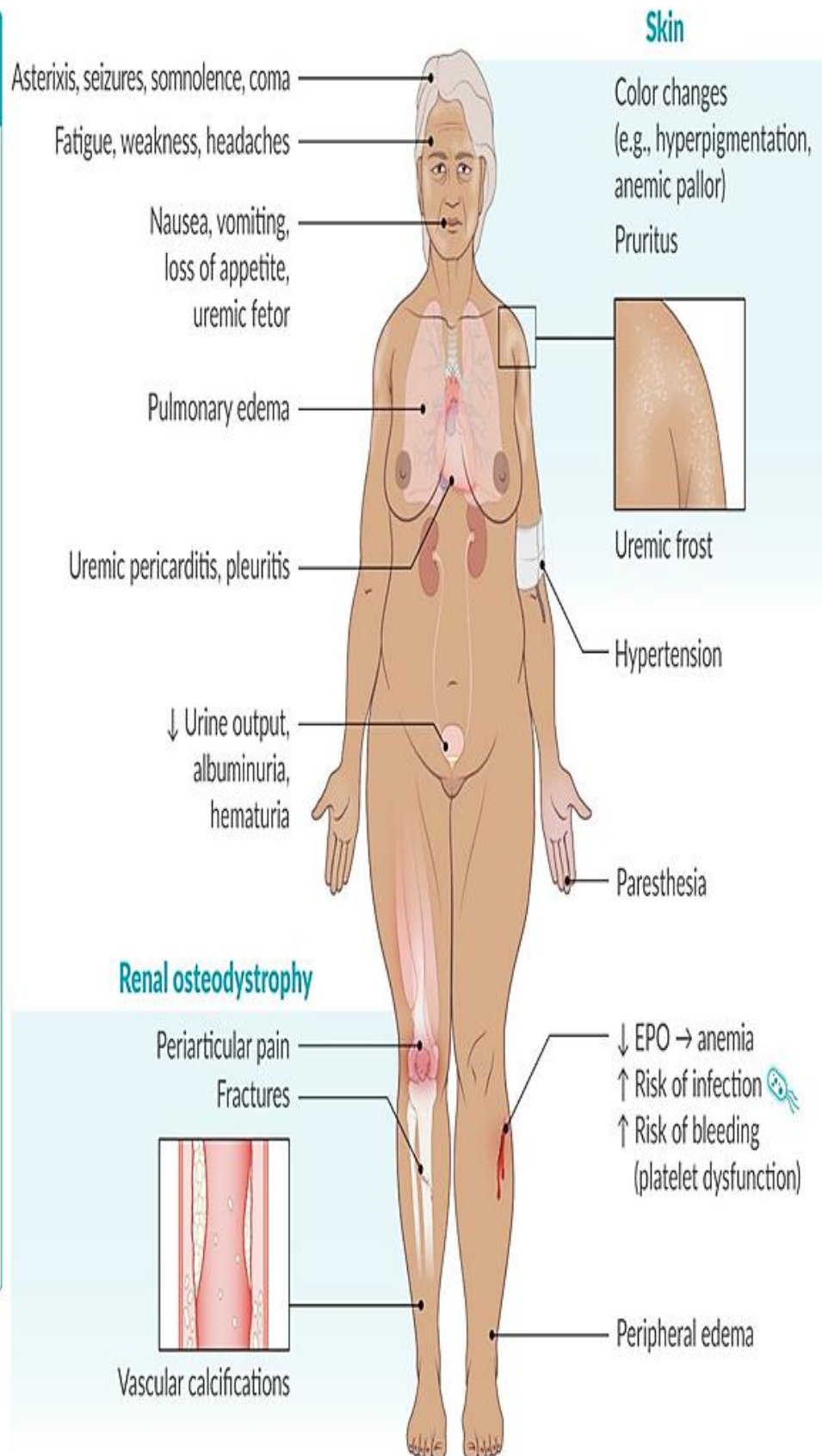
- Diabetic nephropathy
- Hypertensive nephropathy
- Glomerulonephritis
- Idiopathic

Risk factors

- Diabetes, hypertension, obesity
- Age > 60 years of age
- Recreational drug use
- AKI
- Family history of CKD
- African American race or Hispanic ethnicity

Laboratory findings

- $\uparrow$ Creatinine and BUN, $\downarrow$ eGFR
- Hyperkalemia
- Metabolic acidosis
- $\downarrow$ Hb, normal MCV



Acute versus chronic renal disease

History, physical examination, and diagnostic findings often indicate whether a patient has acute or chronic renal disease (**Table**). With acute-on-chronic renal disease, new injury occurs in patients with background renal insufficiency. Findings suggestive of CKD include:

- **History:** In preceding weeks to months, polyuria/polydipsia, weight loss, and decreased appetite
- **Physical examination:** Small, irregular kidneys that are nonpainful
- **Diagnostics:** Degree of azotemia more severe than clinical signs; normocytic, normochromic nonregenerative anemia typical

TABLE. Comparison of Acute and Chronic Renal Disease

	ACUTE	CHRONIC
Degree of illness	Clinical signs more severe than azotemia	Azotemia more severe than clinical signs
History	Lethargy, vomiting	Decreased appetite, PU/PD, weight loss
Kidney size/shape	Normal/large ± pain	Small, irregular, BKLK
Packed cell volume	Normal/regenerative/acute hemorrhage	Normocytic/normochromic/nonregenerative anemia
Potassium	Normal to high	Cats: Typically low Dogs: Mild hyperkalemia common

BKLK=big-kidney-little-kidney; PD=polydipsia; PU=polyuria

6

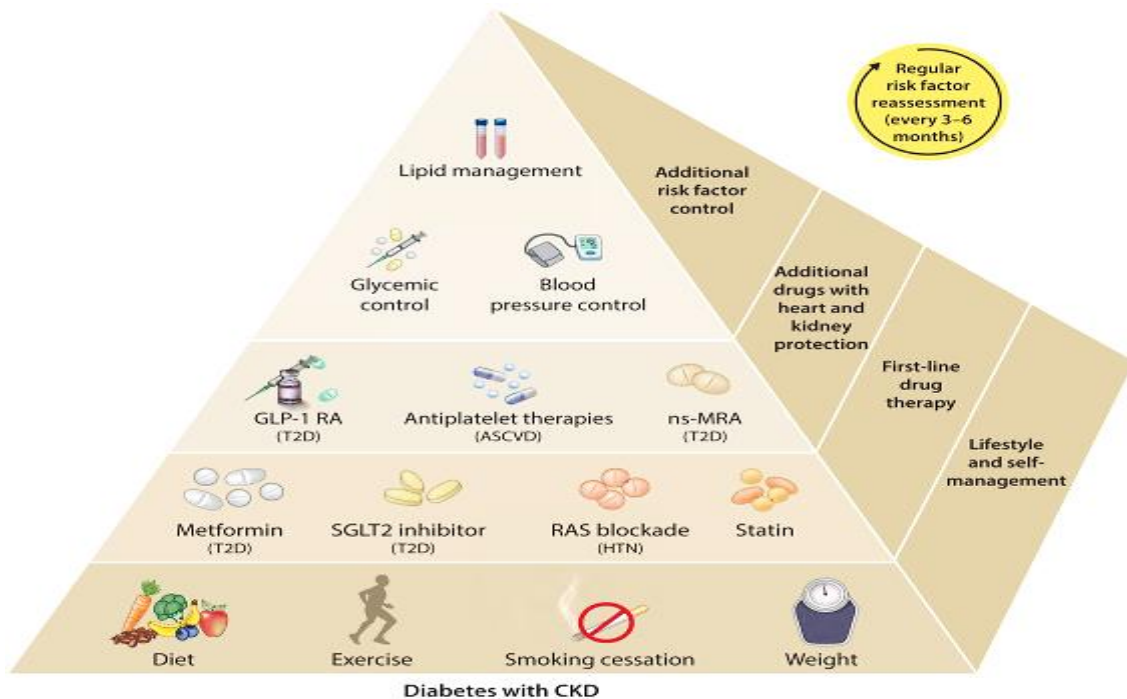
Treatment Chronic Kidney Disease

*

Treatment of CKD

- Treatment of the cause
- Treatment of associated conditions.
- Renal replacement therapy .

HOW TO TREAT THE PROGRESSION OF CKD



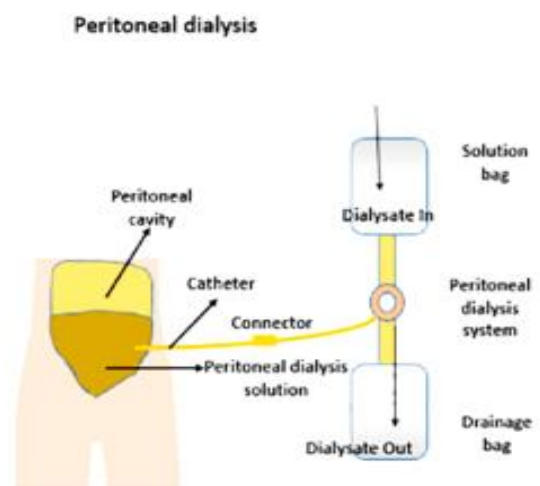
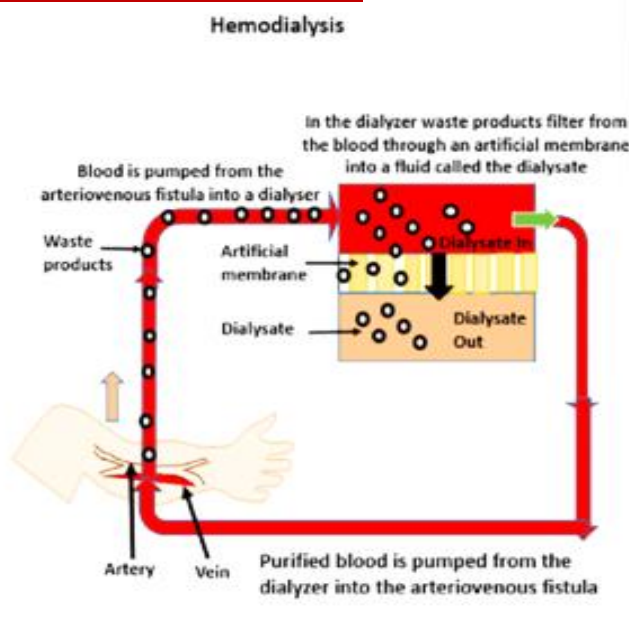
Renal replacement therapy (RTT)

- Hemodialysis.
- Peritoneal dialysis.
- Kidney transplantation

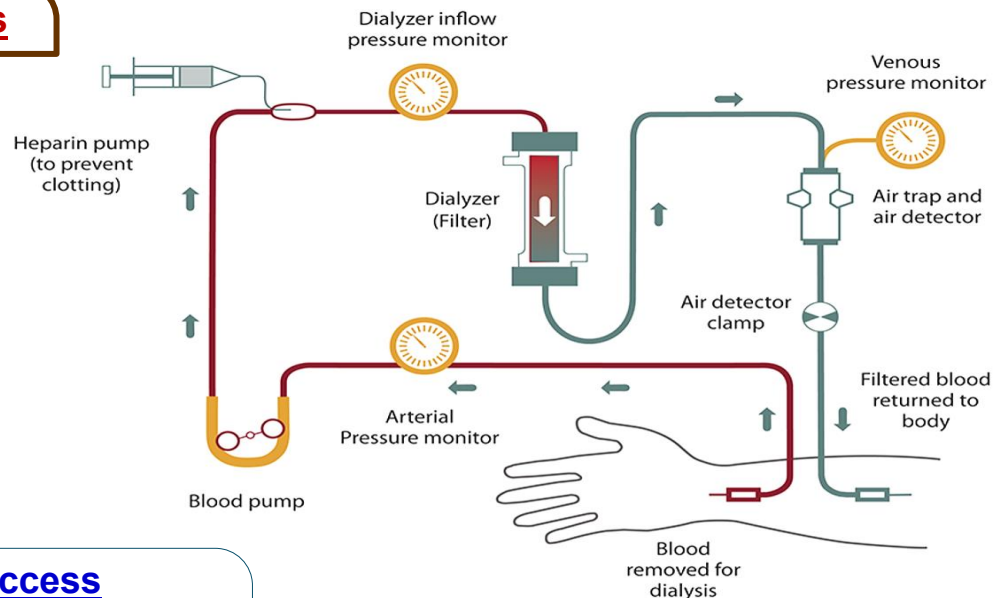
Table 1. Indications for initiation of renal replacement therapy

Renal indications	Rapidly evolving renal failure, or the development of uraemic complications (e.g. uraemic pericarditis or encephalopathy)
	Hyperkalaemia unresponsive to medical management
	Severe metabolic acidosis
	Oliguria or anuria resulting in fluid overload and diuretic-resistant pulmonary oedema
	Management of complex fluid balance, e.g. in cardiac failure
Non-renal indications	Drug toxicity, e.g. lithium
	Correction of electrolyte abnormalities, e.g. in hyponatraemia
	Temperature control (rarely performed nowadays because other methods work better)

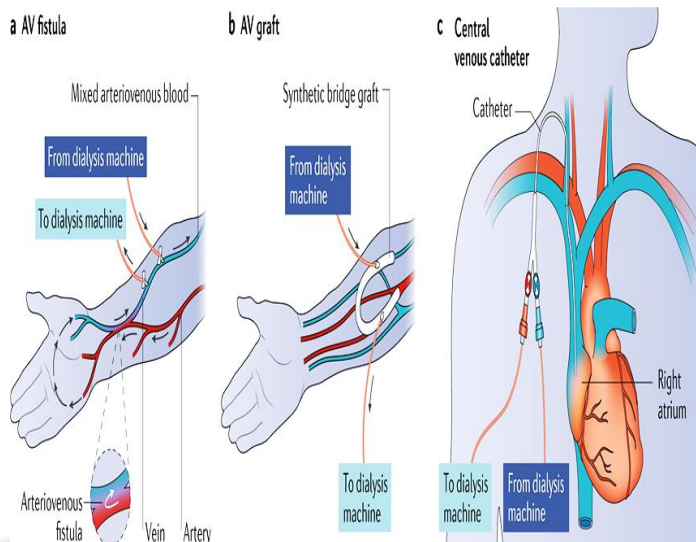
MECHNISM OF DIALYSIS



1-Hemodialysis

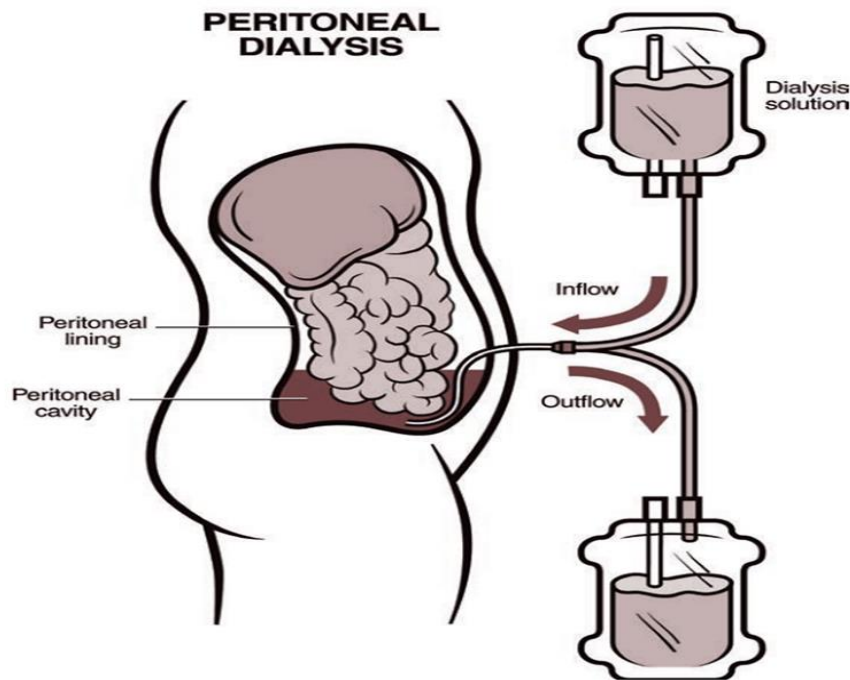


Hemodialysis access



	Fistula	Arteriovenous graft	Cuffed catheter
Advantages	Very low infection risk Long lasting once mature Association with lower mortality	Low infection risk Low primary failure rate Association with lower mortality	Ease of insertion Immediate use No cannulation
Problems	High primary failure rate Pain and bruising early on Risk of aneurysms and bleeding	High secondary failure rate Frequent need for interventions Pain and bruising early on Risk of bleeding	Higher infection risk High dysfunction rate and need for thrombolytic agents Damage to central veins Association with higher mortality

2- Peritoneal dialysis



Complications of dialysis

☐ Hemodialysis

- Blockage in vascular access site (entrance point)
- Muscle cramps
- Hypotension (low blood pressure)
- Weaknesses, dizziness, or nausea
- Blood loss

☐ Peritoneal dialysis:

- Hernia (weakness in abdomen muscle, often presenting as a lump or swollen area)
- Weight gain

☐ Both hemo and peritoneal dialysis:

- Infection of the skin, blood, and/or peritoneum (belly area) - if left untreated, these can cause sepsis (a life-threatening condition leading to multiple organ failure).
- Fatigue (feeling tired) - This can affect anyone but is usually more common for people who have been on dialysis for a long time.
It is often hard to tell for sure if this is a side effect of the dialysis or a symptom of long-term kidney disease.
- Pruritus- itchy skin that people with kidney disease may experience, especially in more advanced stages of CKD and people on dialysis. Like fatigue, it is often hard to tell for sure if this is a side effect of dialysis or a symptom of long-term kidney disease.

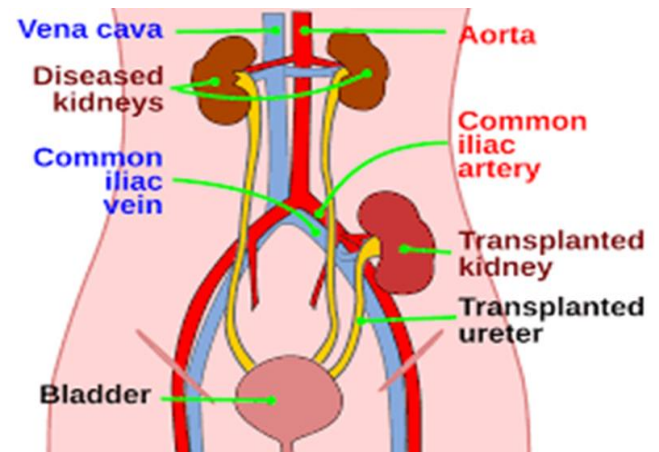
3- Kidney transplantation.

❑ Types of donors.

- Deceased-donor kidney transplant
- Living-donor kidney transplant

❑ Compared with dialysis, kidney transplants are associated with.

- Better quality of life
- Lower risk of death
- Fewer dietary restrictions
- Lower treatment cost



❑ Contraindications.

Absolute	Relative
• Uncontrolled malignancy • Active HIV infection • Life expectancy <2 years due to other illness	• Age > 65 years • Active infection • Chronic liver disease, cirrhosis, active hepatitis • Active substance abuse • Active TB, ATD therapy • COPD • Severe diffuse atherosclerotic disease • Morbid obesity • Psychosocial or behavioural abnormalities

❑ Complications of KT

- Immediate (within the first week)
 - Hyperacute rejection—accelerated rejection
 - Acute tubular necrosis
 - Acute tubular necrosis
 - Calcineurin inhibitors' toxicity
 - Infectious complications (acute graft pyelonephritis)
- Early (between the 1st and the 12th week)
 - Acute rejection
 - Calcineurin inhibitors' toxicity
 - Infectious complications (acute graft pyelonephritis)
- Late (after the twelfth week)
 - Chronic rejection
 - Calcineurin inhibitors toxicity
 - Infectious complications (acute graft pyelonephritis)
 - Nephropathy relapse

The ABCs of

Renal Transplant

UNDERSTANDING THE FACTS BEHIND

1. Definition

- Renal transplant is the curative treatment for End Stage Renal Disease (ESRD).
- Kidneys are usually recovered from deceased donors (70%) but can be obtained from living donors such as family members of the patients
- Deceased donors are either:
 - DBD → Donation after Brainstem Death
 - DCD → Donation after Circulatory Death
- The one year survival rate for living donor recipients is about 99% and for DBD donor recipients is about 97%.

2 Indications for renal transplant

- Patients with ESRD (stage 5) → GFR < 15ml/min
- Those with Chronic Kidney Disease (CKD) (stage 4) → GFR 15-29ml/min
 - Should be considered for transplant if they have rapidly progressing disease

STAGES OF CHRONIC KIDNEY DISEASE	GFR*	% OF KIDNEY FUNCTION
Stage 1	Kidney damage with normal kidney function	90 or higher
Stage 2	Kidney damage with mild loss of kidney function	60 to 89
Stage 3a	Mild to moderate loss of kidney function	45 to 59
Stage 3b	Moderate to severe loss of kidney function	30 to 44
Stage 4	Severe loss of kidney function	15 to 29
Stage 5	Kidney failure	Less than 15

* Your GFR number tells you how much kidney function you have. As kidney disease gets worse, the GFR number goes down.

3. Contraindications for renal transplant

Absolute	Relative
Untreated malignancy	Co-morbidities, e.g. diabetes mellitus
Active infection	Age >65 years
Untreated HIV infection or AIDS	Obesity
Any condition with a life expectancy <2 years	HBV or HCV infection
Malignant melanoma within the previous 5 years	Previous malignancy (depending on type)

4. Donor retrieval procedure (DBD/DCD)

- Full exposure of the abdomen to allow assessment of other organs whilst gaining access to the retroperitoneal space
- Patient is heparinised
- Renal vessels and ureters are identified and isolated
- Kidneys are then removed with the renal artery (containing a patch of the aorta), the renal vein (containing a patch of the IVC) and its ureter.

In both cases the kidney is flushed with preservation fluid as soon as it is removed

5. Living donor

- Procedure is commonly done laparoscopically
- Preference for left kidney → longer renal vein
- No patch of the IVC or aorta is taken

6. Recipient procedures

- In some cases the kidney will be arriving from another facility and must be stored in a specific way:
 - In perfusion fluid
 - Inside sterile bags
 - Surrounded by ice
- The kidney is then examined and flushed with preservation fluid to check for and repair any leaks in the vessels or ureter.
 - Iliac vessels are exposed and lymphatics identified and ligated
 - Anastomosis performed:
 - Donor's renal vein → recipient's external iliac vein
 - Donor's renal artery → recipient's internal/external iliac artery
 - Kidney is perfused
 - Ureter anastomosed to the bladder through a ureteroneocystostomy

7. Complications

- Delayed graft function → failure of the transplanted kidney to function immediately
 - Dialysis required within the first week
 - Most kidneys will function eventually but increased risk of rejection → recipient's immune system begins to destroy donor organ
- Vascular
 - Early
 - Renal artery/vein thrombosis
 - Late
 - Renal artery stenosis. Presents months post transplant with uncontrollable hypertension and worsening graft function
- Ureteral
 - Leaks
 - Decreased urine output & increasing abdominal pain
 - Urinary tract obstruction
 - Strictures in the distal ureter
- Long term
 - CVS complications
 - Immunosuppression → recurrent infection, malignancy

8. Warm vs Cold Ischemia time

- Warm → time between the cessation of organ perfusion by the donor's blood circulation until perfusion with preservation solution
- Cold → time from the perfusion of the organ with the preservation solution to re-perfusion of the organ with recipient blood after the implant's vascular anastomosis.

CREDITS

Content- Mufaro Mutoti; Design - Chia Yen Lek (SIGMUM 2021/2022)

SIGMUM

Definition:

Heterogeneous group of diseases characterized by structural changes and dysfunction of renal tubular structure & surrounding interstitium.

Often presented as:

- ✓ Acute or chronic renal failure.
- ✓ Electrolyte abnormalities.
- ✓ Acidosis.
- ✓ low-molecular weight proteinuria.
- ✓ Hematuria and pyuria are common.

Classification:

1-Tubular diseases:

- ✓ Acute Tubular Necrosis.
- ✓ Isolated defects of tubular function.

2-Interstitial nephritis.

- ✓ Acute interstitial nephritis.
- ✓ Chronic interstitial nephritis.

I

Tubular diseases

Tubular functions:

- ❑ **Selective reabsorption of water, essential electrolytes & other constituents.**
 - ❖ **Proximal tubules:**
60-80% of filtered water, Na^+ are reabsorbed together with K^+ , Hco_3^- , glucose and amino acids.
 - ❖ **Loop of Henle:**
urine conc. Via a countercurrent system.
 - ❖ **Distal tubules & collecting tubules:**
Reabsorption of salt & water under the influence of aldosterone and ADH.
- ❑ **Acid-base balance:**
 - ❖ Achieved mainly by reabsorption of Hco_3^- & H^+

1-Acute Tubular Necrosis:

Etiology:

- ✓ Renal ischemia; more common.
- ✓ Nephrotoxicity; chemical or bacterial toxins.

Pathogenesis:

1. Ischemic tubular necrosis:

- 1-Usually follows period of shock (marked ↓ of renal blood flow) → oliguria.
- 2-Blood flow is further reduced by release of vasoconstrictors as thromboxane, vasopressin, Angiotensin II & noradrenaline.
- 3-↓ oxygen delivery to tubular cells which are very metabolically active → impairment of cell membrane functions
- 4-→ Ca^{+2} influx and cell swelling, anaerobic glycolysis and intracellular acidosis → Denaturation of cellular proteins and apoptosis.
- 5-Finally, shedding of cells into tubular lumen → tubular obstruction & leak of tubular fluid into interstitial tissue.

2. Nephrotoxic acute tubular necrosis:

- Similar sequence occurs but initiated by direct toxicity of causative agent to tubular cells & binding of toxins to target intracellular proteins to interfere with cell functions.
- Examples of causative drugs includes aminoglycoside antibiotics, cytotoxic agents as cisplatin and antifungal drugs as amphotericin B.

Recovery phase:

- ✓ Tubular cells can regenerate and re-form basement membrane.
- ✓ Kidney functions may return to normal if the patient is supported during regeneration phase.
- ✓ Diuretic phase: often occurs during recovery, remains for several days. It occurs due to loss of medullary concentration gradient which depends on active tubular transport.

2- Isolated defects of tubular function:

□ Renal tubular acidosis:

- Failure of either reabsorption of bicarbonate in proximal tubule or acidification of urine in distal tubule with little or overall reduction in renal function.
- In both types the etiology can be:
 - ✓ Gene defect.
 - ✓ Causes of interstitial nephritis

➤ 1-Distal renal tubular acidosis:

- Ability to form a highly acidic urine is lost, urine PH can't be reduced below 5.3 even in the presence of systemic acidosis.

Presented with:

- ✓ Acidosis
- ✓ Hypokalemia
- ✓ Hypercalciuria, hyperphosphaturia, renal stones & osteomalacia.
- ✓ Children may present with polyuria and thirst.

➤ 2-Proximal renal tubular acidosis:

- ✓ proximal tubular Na^+/H^+ exchange is impaired, resulting in decreased bicarbonate reabsorption & large losses of bicarbonate in urine.

- ✓ May occur as isolated defect or as part of Fanconi's syndrome.

Fanconi's syndrome: generalized proximal tubular dysfunction with hypophosphatemia, glycosuria, aminoaciduria and proximal tubular acidosis

Causes

- ✓ Causes of interstitial nephritis.
- ✓ Some congenital metabolic disorders such as Wilson's disease, cystinosis, hereditary fructose intolerance.

II

Interstitial nephritis

1- Acute interstitial nephritis:

- Acute interstitial nephritis (AIN) is an immune-mediated disorder, characterized by acute inflammation affecting the tubulo-interstitium of the kidney.

❑ Causes:

1. Drugs: NSAIDs, penicillins, allopurinol, frusemide.
2. Systemic disease as sarcoidosis, Sjogren syndrome, myeloma.
3. Infections such as TB, leptospirosis, CMV, pyelonephritis.

❑ Clinical features:

- The clinical presentation is typically with renal impairment but, in some patients with drug-induced AIN, there may be signs of a generalized drug hypersensitivity reaction with fever, rash and eosinophilia.
- AIN should always be considered in patients with non-oliguric AKI. There may be a rapid deterioration of renal function in some cases of drug-induced AIN, causing the condition to be mistaken for RPGN.

❑ Investigations:

- The urine may contain white blood cells and white cell casts but is sterile on culture.
- Eosinophils are present in up to 70% of patients but this is a non-specific finding.
- Proteinuria is generally modest ($\text{PCR} < 100 \text{ mg/mmol}$) and tubular in type .

❑ Renal biopsy:

- Usually required to confirm the diagnosis.
- This typically shows evidence of intense inflammation, with infiltration of the tubules and interstitium by polymorphonuclear leucocytes and lymphocytes.
- Eosinophils may also be observed, especially in drug-induced AIN.
- Granulomas may be evident, especially in drug-induced AIN or sarcoidosis.
- The degree of chronic inflammation in a biopsy is a useful predictor of long-term renal function.

❑ Management:

- Some patients with drug-induced AIN recover following withdrawal of the drug alone.
- High-dose glucocorticoids (prednisolone 1 mg/kg/day) may accelerate recovery and prevent long-term scarring.
- Other specific causes should be treated, if possible.

2-Chronic interstitial nephritis

- Chronic interstitial nephritis (CIN) is characterized by renal dysfunction with fibrosis and infiltration of the renal parenchyma by lymphocytes, plasma cells and macrophages, in association with tubular damage.

❑ Causes:

1. Drugs: especially NSAIDS (analgesic nephropathy)
2. Chronic glomerular diseases.
3. Toxins as lead, cadmium.
4. Systemic diseases as diabetes mellitus, sarcoidosis, Sjogren syndrome, SLE.
5. Inherited diseases as Sick cell disease, Wilson's disease.
6. Reflux nephropathy.
7. Tumors as myeloma.

❑ Clinical features:

- Most patients with CIN present in adult life with CKD, hypertension and small kidneys.
- Urinalysis abnormalities are non-specific.
- A minority present with salt-losing nephropathy, characterized by hypotension, polyuria and features of sodium and water depletion.

❑ Management:

1. Management is supportive in nature.
2. Correction of acidosis and hyperkalemia.
3. Replacement of fluid and electrolytes.
4. Renal replacement therapy if irreversible renal damage has occurred.

Analgesic nephropathy:

Type of CIN & papillary necrosis occur as a result of chronic prolonged consumption of analgesics.

❑ Pathology:

- Diffuse interstitial fibrosis & tubular atrophy.
- Acute papillary necrosis is common.
- Predispose to uroepithelial carcinoma.

❑ Clinical features:

- History of drug intake.
- Asymptomatic hematuria.
- Hypertension.
- Salt & water wasting/acidosis (if damage is mainly tubular)
- Renal impairment / failure
- UT obstruction/R. colic (passage of fragments of necrotic papillae).

❑ Investigations:

- Biochemical evidence of tubular dysfunction.
- IVU “IntraVenous Urogram”; the appearance of the papillae is characteristic.

❑ Management:

- Discontinue drugs
- Maintain fluid intake,
- Correction of acidosis.
- Control of hypertension.
- When renal function is severely impaired → dialysis

8

Electrolyte Imbalance

1

HYPONATREMIA

- ▶ It is **defined** as a low serum sodium concentration **<135 mEq**.
- ▶ The physiologic effects of hyponatremia do not stem from the sodium per se, but rather from the **low serum osmolality**, which causes cerebral edema.
- ▶ Sodium is the main determinant of the serum osmolality

Classification of hyponatremia

- ▶ **Hypovolemic hyponatremia**: low extracellular volume (ECV), low total body sodium (dehydration, GI loss).
- ▶ **Euvolemic hyponatremia**: clinically normal ECV.
- ▶ **Hypervolemic hyponatremia**: high ECV (cirrhosis, CHF)

Clinical Presentation and Diagnosis.

- ▶ Symptoms of hyponatremia are predominantly neurologic, ranging from mild confusion and forgetfulness to disorientation and obtundation to seizure (or even coma).
- ▶ The history should focus on symptoms of malignancy, psychiatric, heart, renal, and liver disease.
- ▶ The drug history should ask about SSRI antidepressants, thiazides, and antipsychotics.
- ▶ The physical exam should determine the patient's extracellular volume status.

☐ **Hypovolemic hyponatremia**

- In hypovolemic hyponatremia, hyponatremia develops because of the loss of sodium and water through body fluids but sodium losses exceed water losses.
- Patients show signs of ECV depletion (orthostasis, hypotension, tachycardia, decreased skin turgor). Serum Na is usually >125 mEq/L, so hyponatremic symptoms are uncommon.
- Treatment is normal saline

☐ **Hypervolemic hyponatremia**

- In hypervolemic hyponatremia, hyponatremia is caused by high ADH levels
- Causes include CHF; nephrotic syndrome and low albumin states; and cirrhosis.
- Patients will show signs of ECV expansion: edema, ascites, pulmonary crackles, and specific signs of heart, liver, or renal disease. The serum Na is usually >125 mEq/L, so hyponatremic symptoms are uncommon.
- Treatment is of the underlying disorder.
- Furosemide may help

❑ Euvolemic (normal ECV) hyponatremia

- Patients will show normal clinical volume status.
- Serum Na varies, but may be very low (<110), so hyponatremic symptoms are common.
- For symptomatic hyponatremia (usually serum Na <120 mEq/L), administer hypertonic (3%) saline infusion.
- Monitor the rate of rise of sodium so as not to cause central demyelinating syndrome, which occurs if sodium is corrected too rapidly or is overcorrected.
- Central pontine myelinolysis is a neurological condition involving severe damage to the myelin sheath of nerve cells in the pons (an area of the brainstem).
- It is predominately iatrogenic (treatment-induced).
- characterized by acute paralysis, dysphagia (difficulty swallowing), dysarthria (difficulty speaking), and other neurological symptoms.
- If the serum sodium levels rise too rapidly, the increased extracellular tonicity will continue to drive water out of the brain's cells. This can lead to cellular dysfunction and central pontine myelinolysis.
- the hyponatremia should be corrected at a rate not exceeding 10 mmol/L/24 h or 0.5 mEq/L/h

2

HYPERNATREMIA

- ▶ Hyponatremia is generally defined as a sodium concentration more than 145 mmol/L.

Causes of hypernatremia

❑ Low volume

In those with low volume or hypovolemia:

- Inadequate intake of free water associated with total body sodium depletion. Typically in elderly who are unable to take in water. This is the most common cause of hypernatremia.
- Excessive losses of water from the urinary tract – which may be caused by [glycosuria](#), or other osmotic diuretics (e.g., [mannitol](#)) – leads to a combination of sodium and free water losses.
- Water losses associated with extreme sweating.
- Severe watery diarrhea

❑ Normal volume

- In those with normal volume or euvolemia:
- Excessive excretion of water from the kidneys caused by [diabetes insipidus](#).

❑ High volume

- In those with high volume or hypervolemia:
- Intake of a [hypertonic](#) fluid

Signs and symptoms:

- The major symptom is thirst.
- confusion, muscle twitching or spasms. With severe elevations, [seizures](#) and [comas](#) may occur.
- Severe symptoms are usually due to acute elevation of the plasma sodium concentration to above 157 mmol/L.
- Values above 180 mmol/L are associated with a high mortality rate

Treatment:

- ▶ All treatment for hypernatremia involves correcting the fluid and sodium balance in the body. Doing this usually means treating the underlying condition that is causing the increased blood sodium levels.
- ▶ Other treatment options for hypernatremia may include simply increasing fluid intake.
- ▶ In mild cases, increasing water consumption can help restore the proper balance of sodium in the blood.
- ▶ In more severe cases, a person may need IV fluids to help restore proper sodium levels.

HYPERKALEMIA

- ▶ Normal potassium levels are between 3.5 and 5.0 [mmol/L](#) .
- ▶ levels above 5.5 mmol/L defined as hyperkalemia.
- ▶ The symptoms of an elevated potassium level are generally few and nonspecific.
- ▶ Nonspecific symptoms may include feeling tired, numbness and weakness.
- ▶ [palpitations](#) and shortness of breath may occur.
- ▶ [cardiac arrhythmia](#) or [sudden cardiac death](#).

Causes:

- ▶ [Decreased kidney function](#) is a major cause of hyperkalemia.
- ▶ Drugs such as [ACE inhibitors](#), [angiotensin receptor blockers](#), [beta blockers](#), and [calcineurin inhibitor](#) immunosuppressants such as [ciclosporin](#) and [tacrolimus](#), potassium-sparing [diuretics](#).
- ▶ [Mineralocorticoid](#) (aldosterone) deficiency or resistance can also cause hyperkalemia. Primary adrenal insufficiency are: [Addison's disease](#)
- ▶ [Metabolic acidosis](#) can cause hyperkalemia
- ▶ Insulin deficiency can cause hyperkalemia as the [hormone insulin](#) increases the uptake of potassium into the cells.

Diagnosis:

- ▶ the measurement of potassium must be repeated.
- ▶ [electrocardiography](#) (ECG) may be performed to determine if there is a significant risk of abnormal heart rhythms.

Treatment of hyperkalemia:

- ▶ Calcium ([calcium chloride](#) or [calcium gluconate](#)) increases [threshold potential](#) .
- ▶ [Insulin](#) (e.g. intravenous injection of 10 units of regular insulin along with 50 ml of 50% dextrose to prevent [the blood sugar from dropping too low](#)) leads to a shift of potassium ions into cells.
- ▶ [Salbutamol](#) (albuterol), a β_2 -selective catecholamine, is administered by nebuliser (e.g. 10–20 mg). This medication also lowers blood levels of K^+ by promoting its movement into cells, and will work within 30 minutes.
- ▶ [Sodium bicarbonate](#) may be used with the above measures if it is believed the person has [metabolic acidosis](#).
- ▶ Potassium can bind to a number of agents in the gastrointestinal tract such as Sodium [polystyrene sulfonate](#).
- ▶ Severe cases require [hemodialysis](#), which are the most rapid methods of removing potassium from the body.

HYPOKALEMIA

- ▶ Hypokalemia is a low level of [potassium](#) (K⁺) in the [blood serum](#).
- ▶ Hypokalemia is one of the most common [water–electrolyte imbalances](#).

Signs and symptoms

- Mild hypokalemia is often without symptoms.
- Severe hypokalemia, with serum potassium concentrations of 2.5–3 meq/l, may cause [muscle weakness](#), [myalgia](#), tremor, and [muscle cramps](#), and [constipation](#).
- With more severe hypokalemia, [flaccid paralysis](#) and [hyporeflexia](#) may result.

Causes:

▶ **Inadequate potassium intake**

Not eating a diet with enough potassium-containing foods or fasting can cause the gradual onset of hypokalemia. This is a rare cause and may occur in those with anorexia nervosa.

▶ **Gastrointestinal or skin loss**

- ☐ Diarrhea
- ☐ Vomiting can also cause hypokalemia

▶ **Urinary loss**

- ☐ Certain medications can cause excess potassium loss in the urine. Ant hypertension drugs such as loop diuretic (e.g. furosemide) and thiazide diuretic (e.g. hydrochlorothiazide) commonly cause hypokalemia.
- ☐ Potassium loss occurs with alkalosis can cause temporary hypokalemia by causing a shift of potassium out of the plasma.
- ☐ High aldosterone levels can cause hypertension and excessive urinary losses of potassium. These include [renal artery stenosis](#) and [tumors](#) of the [adrenal](#) glands, e.g., [Conn's syndrome](#) (primary [hyperaldosteronism](#)). [Cushing's syndrome](#) can also lead to hypokalemia.

Diagnosis:

- ▶ Normal potassium levels are between 3.5 and 5.0 mmol/L with levels below 3.5 mmol/L (less than 3.5 mEq/L) defined as hypokalemia.
- ▶ Hypokalemia leads to characteristic ECG changes (PR prolongation, ST-segment and T-wave depression, U-wave formation).

Treatment of hypokalemia:

- ▶ **Treatment of the cause**, such as improving the diet, treating diarrhea, or stopping an offending medication.
- ▶ **Oral potassium supplementation:** Mild hypokalemia (>3.0 mEq/l) may be treated by eating potassium-containing foods or by taking potassium chloride supplements in a tablet or syrup form (by mouth supplements).
- ▶ **Intravenous potassium replacement:** Severe hypokalemia (<3.0 mEq/l) may require intravenous supplementation.
- ▶ **Potassium-sparing diuretics:** Hypokalemia which is recurrent or resistant to treatment may be amenable to a potassium-sparing diuretic, such as amiloride, spironolactone.

9

Renal cystic diseases

- Cyst are epithelium-lined cavities filled with fluid
- They occur single - or multiple
- Inherited or acquired
- Presented in infancy or older age
- Clinically silent or symptomatic

4

Simple cyst:

- 50% of the population has over 50 yrs:as asymptomatic and incidental finding, that should be differentiated from other solid lesions (US, rarely CT is needed).

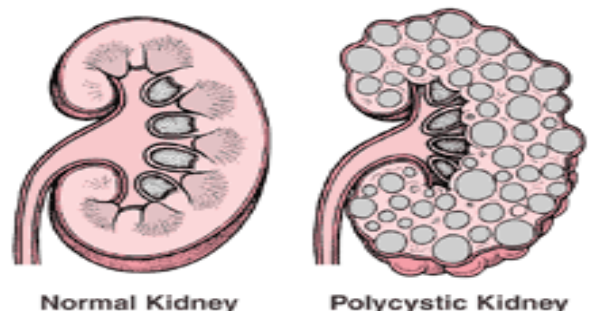
4

Polycystic kidney disease

- (PKD or PCKD, also known as **polycystic kidney syndrome**) is a [genetic disorder](#) in which the [renal tubules](#) become structurally abnormal, resulting in the development and growth of multiple [cysts](#) within the kidney. These cysts may begin to develop in utero, in infancy, in childhood, or in adulthood. Cysts are non-functioning tubules filled with fluid pumped into them, which range in size from microscopic to enormous, crushing adjacent normal tubules and eventually rendering them non-functional as well.
- PKD is caused by abnormal genes that produce a specific abnormal protein; this protein has an adverse effect on tubule development. PKD is a general term for two types, each having their own pathology and genetic cause: [autosomal dominant polycystic kidney disease](#) (ADPKD) and [autosomal recessive polycystic kidney disease](#) (ARPKD).

Types

- **1. Autosomal Dominant Polycystic Kidney Disease(ADPKD)**
 - Inherited as a Autosomal Dominant Disease...
most common in Adult
- **2. Autosomal Recessive Polycystic Kidney Disease (ARPKD)**
 - Inherited as a Autosomal Recessive Disease....most common in infants



Features

Feature	ADPKD	ARPKD
Severity	Less severe	Most severe
Occurrence	Most common	Less common
Onset	Onset age >30	Infants and children
Prognosis	Half get ESRD by 60	Infant renal failure
Appearance	Large nodules	Smaller nodules
Sequae	Cerebral aneurysm	Liver fibrosis, death

Clinical pictures

1. Abdominal Pain: flank or back (most common initial complaint).
2. Dull aching and an uncomfortable sensation of heaviness may result from a large polycystic liver or enlargement of one or more cysts.
3. Bleeding inside the cyst or lead to gross hematuria with passage of clots or a perinephric hematoma
4. Hypertension
5. UTI (e.g, acute pyelonephritis, infected cysts, perinephric abscess)
6. Nephrolithiasis and associated renal colic
7. Rarely, a coincidental hypernephroma
8. Palpable, bilateral flank masses: In advanced ADPKD
9. Nodular hepatomegaly: In severe polycystic liver disease
10. Rarely, symptoms related to renal failure (eg, pallor, dry skin, edema)
11. Decrease in urine-concentration
12. Increased plasma vasopressin etc

Diagnosis

1. Routine laboratory studies include the following

- ❖ Serum chemistry profile, including calcium and phosphorus
- ❖ CBC count from cysts
- ❖ Urinalysis
- ❖ Urine culture

2. Genetic testing

- ❖ This may be performed young adults potential kidney donors with negative ultrasonographic.

Imaging

- Ultrasonography: Technique of choice for case and routine screening patients.
- CT scanning: Not routine; useful in doubtful pediatric cases or in complicated cases (eg, kidney stone, suspected tumor)
- MRI: Not routine; helpful in distinguishing renal cell carcinoma from simple cysts and to monitor kidney size after treatment to assess progress
- MRA: Magnetic resonance angiogram is not routine but useful in viewing blood flow patterns

Diagnostic criteria

Age (years)	Number of cysts	
	Positive family history	Negative family history
<30	At least 2 in one or both kidneys	At least 5
30–59	At least 2 in each kidney	At least 5
>60	At least 3 in each kidney	At least 8

Treatment

- Control Blood Pressure using ACEIs or ARBs
- Treat Urinary Tract and Cyst Infections using Gyrase inhibitors (eg, ciprofloxacin, chloramphenicol, clindamycin, levofloxacin); dihydrofolic acid inhibitors (TMX/SMP_Septrim)
- Treat hematuria: Possibly analgesic plus copious oral hydration
- Reduce abdominal pain produced by enlarged kidneys
- Control abnormalities related to renal failure (Oedema, ascites, weakness, cephalic involvement etc)
- Prevent cardiac valve infection in patients with intrinsic valve disease
- In 2018, Jynarque (**Tolvaptan**) was introduced as the first FDA-approved treatment for PKD. In a recent long-term study, patients using Tolvaptan had a 6.4% higher kidney function after 5 years compared to standard of care.

Surgical treatment

- **Surgical drainage**: Ultrasonographically guided puncture
- **Open/fiberoptic-guided surgry**: for excision/drainage of the outer walls of cysts to ablate symptoms
- **Partial hepatectectomy** : To manage massive hepatomgaly
- **Liver tansplantaion**: incases of portal hypertension due to polycystic liver or hepatomgaly with nonresectable areas
- **Nephrectomy**: Last resort for pain control inpatient with inaccessible cycts inthe renal medulla; bilateral nephectomuy in patients with severe hepatic involmment

Medullary cystic disease:

- Uncommon
- Both autosomal dominant and recessive way of inheritance occur
- Pathogenesis and pathology: Small kidneys. Cysts at the corticomedullary junctions and in the medulla.
- Clinical manifestations: early adolescence polyuria, polydypsia, enuresis, defect in urinary concentrating ability, growth retardation, anaemia
- No specific therapy exists

Medullary sponge kidney

- Common: 1 in 5000 to 1 in 20000
- Pathogenesis: tubular dilatation within the medullary collecting ducts
- Signs: recurrent hematuria, UTI, renal calculi
- Diagnosis: US reveals normal size kidneys with medullary ductal ectasias
- Coincident hyperparathyreoidism is common, se Calcium should be tested
- Treatment: management of UTI and renal calculi

10

Infectious kidney disease

Definitions:

- **Pyuria:** White blood cells in urine.
- **Sterile pyuria:** White cells but sterile on standard culture (No bacteriuria).
- **Bacteriuria:** bacteria in urine.
- **Infection:** Bacteriuria+ pyuria.
- **Lower UTI:** Bladder(cystitis), Urethra(urethritis), Prostate (prostatitis)..
- **Upper UTI:** Infection of the kidney, renal pelvis (Pyelonephritis).

Incidence:

- Annual incidence of UTI 10–20%.
- 10% of men and 20% of women >65 years have asymptomatic bacteriuria.
- Pyelonephritis = 3 per 1000 patient years.

Classification

- **Uncomplicated:** normal renal tract structure and function.
- **Complicated:** structural/functional abnormality of genitourinary tract, eg. obstruction, catheter, stones, neurogenic bladder, renal transplant.

Risk factors

- Female.
- Low estrogen(menopause).
- Pregnancy.
- Urinary tract abnormalities.
- Urinary tract obstruction.
- A suppressed immune system (DM, Immunosuppression).
- Catheterization.

Symptoms:

- ❑ **Cystitis:** Frequency, dysuria, urgency, suprapubic pain, polyuria, hematuria.
- ❑ **Prostatitis:**
 - Pain: perineum, lower back.
 - Fever, malaise, nausea.
 - Urinary symptoms.
 - Swollen or tender prostate on PR.

❑ Acute pyelonephritis:

- Fever, rigor, vomiting, loin pain/tenderness, costovertebral pain.
- Associated cystitis symptoms.

Signs:

- Fever.
- loin tenderness.

Investigation

- ❑ **In non-pregnant women**, if ≥ 3 (or one severe) symptoms of cystitis and no vaginal discharge, treat empirically without further tests.

Dipstick : Use in non-pregnant women .

- **A negative dipstick** reduces probability of UTI to $<20\%$.
- Do not use in pregnant women.

• **MSU culture:** Conventional cut off $>100,000$ colonie-forming units (cfu)/mL .Use in :

- Pregnant women.
- Men.
- Children.
- If fail to respond to empirical antibiotics.

Organisms:

- **E. coli** is the main organism (75–95%).
- **Staphylococcus saprophyticus** (a skin commensal) in 5–10%.
- Other enterobacteriaceae such as **Proteus mirabilis** and **Klebsiella pneumonia**.

Causes of sterile pyuria:

1-Non infection related:

- Renal tract tumors.
- Calculi.
- Chemical cystitis.
- Pregnancy.
- Polycystic kidney.

2-Infection related:

- TB
- Inadequately treated UTI

Blood tests:

- CBC.
- Blood Urea& serum creatinine and Electrolytes.
- CRP.
- Fasting glucose.
- Blood culture (positive in only 10–25% of pyelonephritis).

• Imaging:

Consider Abdominal ultrasound, CT, cystoscopy, urodynamics) in:

- Failure to respond to treatment
- Recurrent UTI (>2/year).
- Pyelonephritis.
- Unusual organism.
- Persistent hematuria.

Managing

- Do not use antibiotics for the treatment of asymptomatic bacteriuria in non- pregnant women, men, and adults with catheters.

Non-pregnant women:

- ► If three or more symptoms (or one severe) of cystitis, and no vaginal discharge, treat empirically with 3-day course of trimethoprim, or nitrofurantoin (if eGFR>30).
- ► If first-line empirical treatment fails, culture urine and treat according to antibiotic sensitivity.
- ► In upper UTI, take a urine culture and treat initially with a broad-spectrum antibiotic eg co-amoxiclav.
- Hospitalization should be considered due to risk of antibiotic resistance.
- Avoid nitrofurantoin as it does not achieve effective concentrations in the blood.

Pregnant women:

- UTI in pregnancy is associated with preterm delivery and intrauterine growth restriction.
- Asymptomatic bacteriuria should be confirmed on a second sample.
- Treat with an antibiotic.
- Avoid ciprofloxacin, trimethoprim in 1st trimester, nitrofurantoin in 3rd trimester.
- Confirm eradication.

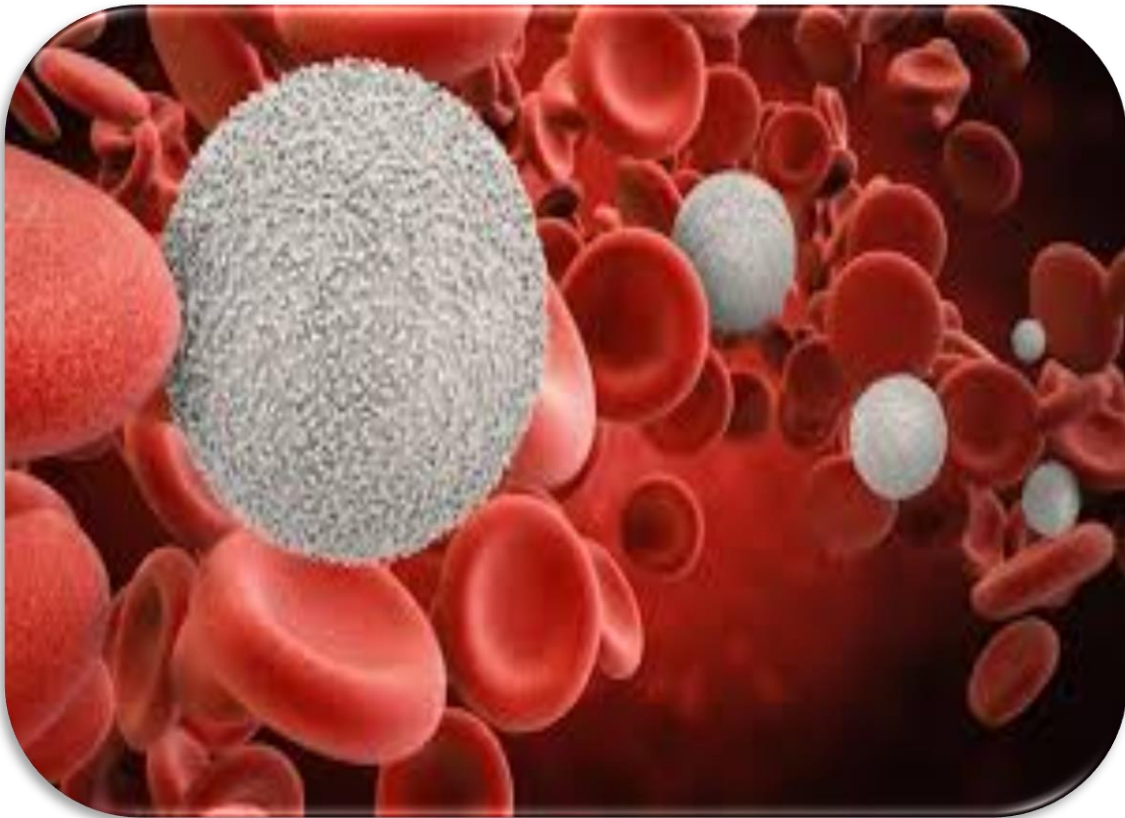
Men:

- Treat lower UTI with a 7-day course of trimethoprim or nitrofurantoin (if eGFR >30).
- If symptoms suggest prostatitis consider a longer (4-week) course of a fluoroquinolone (eg ciprofloxacin) due to ability to penetrate prostatic fluid.
- If upper or recurrent UTI, refer for urological investigation.

Catheterized patients:

- All catheterized patients are bacteriuric.
- Send MSU culture only if symptomatic.
- Change long-term catheter before starting an antibiotic.
- Use a narrow spectrum antibiotic according to culture sensitivity.

Hematology



Contents

A-Introduction	
B-Anemia	
1	Microcytic anemia- Iron deficiency anemia
2	Megaloblastic anemia - Macrocytic anemias
3	Aplastic anemia
4	Hemolytic Anemia (Inherited)
5	Hemolytic Anemia (Acquired)
C-Myeloproliferative disorders	
6	Polycythemia vera Essential thrombocythemia Myelofibrosis Chronic myeloid leukemia (discussed in chronic leukemia)
D-Lymphadenopathy and splenomegaly	
7	Hodgkin lymphoma
8	Non-Hodgkin lymphoma
E-Transfusion Medicine	
9	Blood Transfusion
F-Leukemia	
10	Acute leukemia
11	Chronic leukemia
G-Bleeding Tendency and thrombosis	
12	Platelet disorders
13	Coagulation disorders



A

Introduction

Blood consists of:

Cellular component

- Red blood cells (Erythrocytes).
- White blood cells (Leucocytes). Part of immune system
- Platelets.

Plasma About 55% of blood

- 90-92% water.
- 6-7% proteins
- 2-3%
 - Fats
 - Carbohydrates (glucose)
 - Electrolytes
 - Gases (O_2 , CO_2)

Normal values for peripheral blood

	Male	Female
Hb (g/dL)	13–18	11.5–15.5
PCV (haematocrit; L/L)	0.42–0.53	0.36–0.45
RCC ($10^{12}/L$)	4.5–6.0	3.9–5.
MCV (fl)		80–96
MCH (pg)		27–33
MCHC (g/dL)		32–35
WBC ($10^9/L$)		4.0–11.0
Platelets ($10^9/L$)		150–400
ESR (mm/h)		< 20
Reticulocytes (%)		0.5–2.5

ESR, erythrocyte sedimentation rate; Hb, haemoglobin; MCH, mean corpuscular haemoglobin; MCHC, mean corpuscular haemoglobin concentration; MCV, mean corpuscular volume of red cells; PCV, packed cell volume; RCC, red cell count; WBC, white blood count.

Hematopoiesis

Hemopoiesis is the formation of blood cells. The hemopoietic system includes the bone marrow, liver, spleen, lymph nodes and thymus.

There is a huge turnover of cells with the red cells surviving 120 days, platelets around 7 days but granulocytes only 7 hours. The bone marrow is the only source of blood cells during normal childhood and adult life.



Pluripotent stem cells give rise to:

1. lymphoid stem cells, which give rise to pre-t cells and pre-b cells.
2. Mixed myeloid progenitor cells, which give rise to colony-forming units committed to the production of red cells, platelets, monocytes, neutrophils, eosinophils and basophils.

Production is stimulated by growth factors such as erythropoietin (red cells), thrombopoietin (platelets), granulocyte colony stimulating factor[G-CSF] (neutrophils) and interleukin-5 (eosinophils). Other factors such as tumor necrosis factor (TNF) are inhibitory.

The Red cell erythropoiesis

Red cell precursors pass through several stages in the bone marrow; pronormoblasts, smaller normoblasts (less RNA and more Hb), and reticulocytes.

Reticulocytes contain residual ribosomal RNA and are still able to synthesize Hb. They are released into the circulation, where they lose their RNA and become mature red cells (erythrocytes) forming 0.5–2.5% of total circulating red blood cells. The reticulocyte count gives a guide to the erythroid activity in the bone marrow. Mature red cells are non-nucleated biconcave discs.

Erythropoietin is a hormone which controls erythropoiesis. It is produced in kidneys (90%) and in the liver (10%). Its secretion regulated by tissue oxygen tension.

Hemoglobin synthesis

Hemoglobin synthesis occurs in the mitochondria of the developing red cell. Normal adult Hb molecule (Hb A) consists of two α and two β globin polypeptide chains ($\alpha_2\beta_2$). A haem group, consisting of a single molecule of protoporphyrin IX bound to a single ferrous ion (Fe^{2+}) is linked covalently at a specific site to each globin chain.

Hb A comprises about 97% of the Hb in adults. Two other types, Hb A2 ($\alpha_2\delta_2$) and Hb F ($\alpha_2\gamma_2$), are found in adults in small amounts (1.5-3.2% and < 1%, respectively).

Hemoglobin function

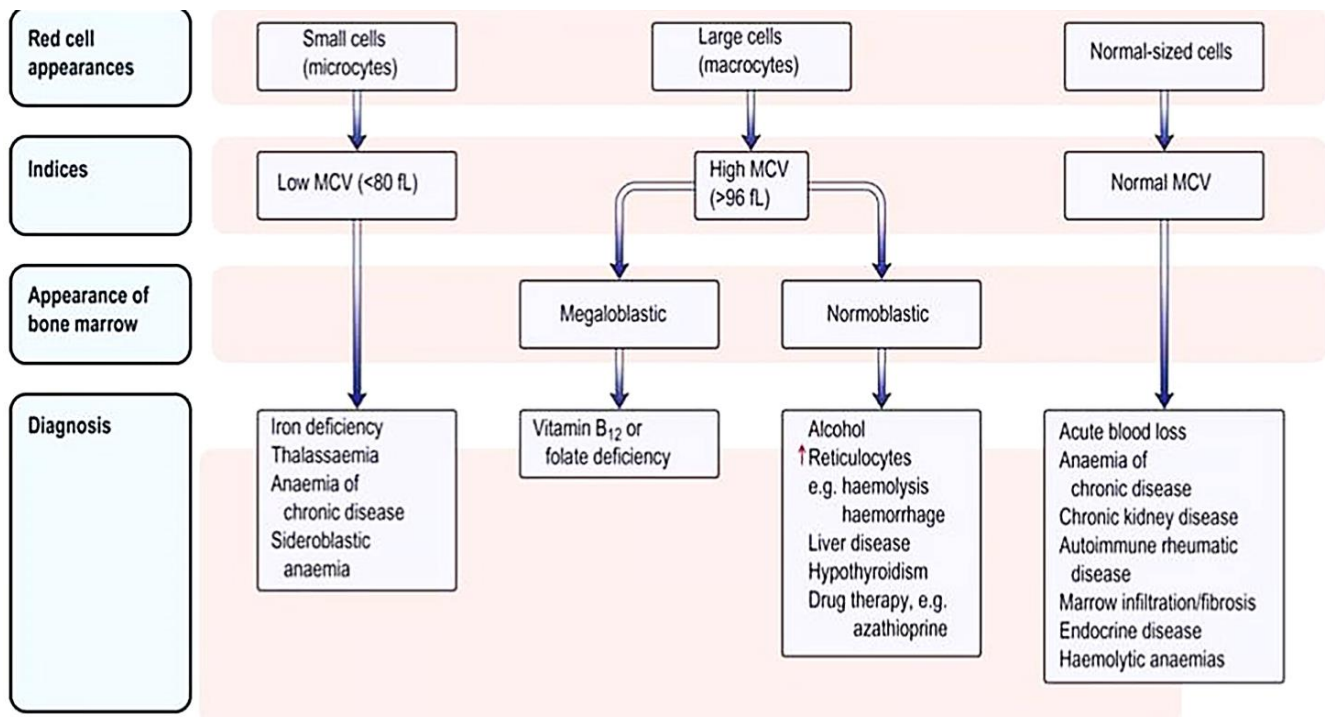
The biconcave shape of red cells provides a large surface area for the uptake and release of oxygen and carbon dioxide.

Hemoglobin becomes saturated with oxygen in the pulmonary capillaries where the partial pressure of oxygen is high and Hb has a high affinity for oxygen. Oxygen is released in the tissues where the partial pressure of oxygen is low and Hb has a low affinity for oxygen.

Anemia is present when there is a decrease in the level of hemoglobin in the blood below the reference level for the age and sex of the individual.

- Alterations in the level of Hb may occur as a result of changes in the plasma volume.
- Anemia is classified in terms of the red cell indices, particularly the MCV, into three major types:
 - Hypochromic microcytic with a low MCV
 - Normochromic normocytic with a normal MCV
 - Macrocytic with a high MCV.

Types of Anemias



Clinical features of anemia

Patients with anemia may be asymptomatic. A slowly falling level of Hb allows for hemodynamic compensation.

Symptoms (all non-specific)

- Decreased exercise tolerance
- Headache and dizziness
- Fainting
- Breathlessness
- Intermittent claudication
- Angina
- Palpitations.



Signs

- Pallor
- Tachycardia and wide pulse pressure
- Systolic flow murmur
- Cardiac failure

Specific signs of the different types of anemia

- koilonychia - spoon-shaped nails seen in iron deficiency anemia
- jaundice - found in hemolytic anemia
- bone deformities - found in thalassemia major
- leg ulcers - occur in association with sickle cell disease.

It must be emphasized that anemia is not a diagnosis, and a cause must be found

Investigations

Peripheral blood

A low hemoglobin should always be considered in relation to:

- White blood cell (WBC) count
- Platelet count
- Reticulocyte count (as this indicates marrow activity)
- Blood film, as abnormal red cell morphology may indicate the diagnosis e.g spherocyte, sickle cell, target cell, tear drop and schistocytes

Bone marrow

Obtained by aspiration or trephine biopsy. The following are assessed:

- Cellularity of the marrow
- Type of erythropoiesis (e.g Normoblastic or megaloblastic)
- Cellularity of the various cell lines
- Infiltration of the marrow
- Iron stores.
- Special tests may be performed: cytogenetic, immunological, cytochemical markers, biochemical analyses.

MICROCYTIC ANEMIA:

Causes

- 1) Iron deficiency anemia
- 2) Anemia of chronic disease
- 3) Sideroblastic anemia
- 4) Thalassemia



1

Iron deficiency anemia

- The most common cause of anemia worldwide.
- The prevalence of iron deficiency anemia is estimated to be 10% of the world population.
- The prevalence rates are especially high in developing countries where dietary insufficiency and intestinal parasites are prevalent.

Dietary iron

- In the average daily diet, whatever it's iron content, normally only 10-30% of iron is absorbed. Absorption may be increased to 20-35% in iron deficiency and pregnancy.
- Non-haem iron is mainly derived from cereals. Haem iron is derived from hemoglobin and myoglobin in red or organ meats. Haem iron is better absorbed than non-haem iron.

Iron absorption

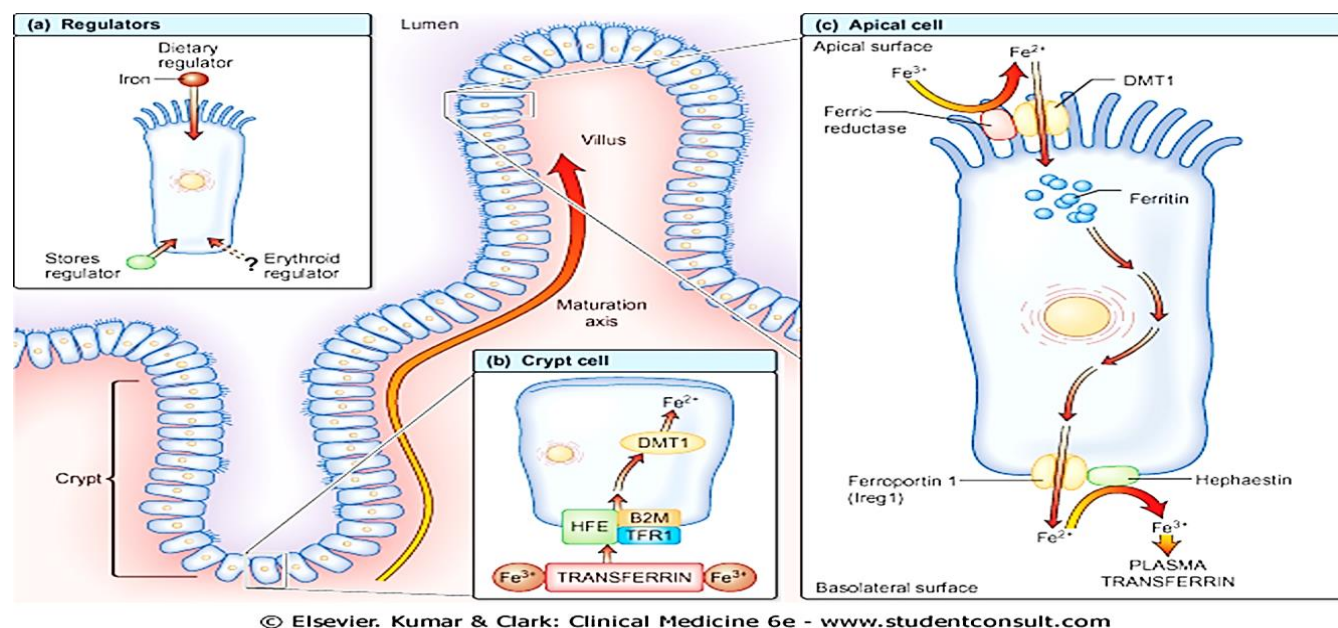
- Site of absorption is primarily in the duodenum.
- Non-haem iron is dissolved in the low pH of the stomach and reduced from the ferric to the ferrous form by a brush border ferrireductase. A protein, divalent metal transporter 1 (DMT1), transports iron (and other metals) across the apical (luminal) surface of the mucosal cells in the small intestine.
- Haem iron is absorbed transported across the apical membrane by Heme carrier protein 1 (HCP1)
- Once inside the mucosal cell, iron may be transferred across the cell to reach plasma or be stored as ferritin. Iron stored as ferritin may be lost into the gut lumen when the mucosal cells are shed; this regulates iron balance.

Regulation of iron absorption

The body is unable to excrete iron once it has been absorbed so the regulation of iron absorption is critical. Absorption of intestinal iron is regulated by:

- recent dietary iron intake (dietary regulator).
- in response to body iron stores (stores regulator).
- a signal communicates the state of bone marrow erythropoiesis to the intestine (erythroid regulator).

Hepcidin, a polypeptide, is the central regulator of iron homeostasis through its effects on intestinal absorption, macrophage recycling of iron from senescent RBCs, and iron mobilization from hepatic stores. Thus, hepcidin affects all major sites of iron uptake and storage. Hepcidin is produced in the liver and acts as a negative regulator of iron absorption by the intestine and of iron release from storage in the macrophages and hepatocytes.



Factors influencing Iron absorption

- Haem iron is absorbed better than non-haem iron
- Ferrous iron is absorbed better than ferric iron
- Gastric acidity helps to keep iron in the ferrous state and soluble in the upper gut
- Formation of insoluble complexes with phytate (in bran) or tannic acid (in tea) decreases iron absorption
- Iron absorption is increased with low iron stores and increased erythropoietic activity, e.g. bleeding, haemolysis, high altitude
- There is a decreased absorption in iron overload, except in hereditary haemochromatosis, where it is increased.

Transport in blood

- The normal serum iron level is about 13-32 $\mu\text{mol/L}$
- Iron is transported in the plasma bound to transferrin, a β -globulin that is synthesized in the liver.
- Transferrin-bound iron becomes attached by specific receptors to erythroblasts and reticulocytes in the marrow and the iron is removed.

Iron stores

- The majority of the approximately 4 g of iron in the adult human body is incorporated into hemoglobin (approximately 2100 mg) in erythrocytes or myoglobin (approximately 300 mg) in muscle.
- The remainder is chiefly present as storage iron in the liver (1000 mg) and in the reticuloendothelial macrophages of the bone marrow and spleen (600 mg)
- Only a small amount of iron (3 to 7 mg) is freely circulating in plasma bound to transferrin.

Iron requirement

- The majority of the approximately 4 g of iron in the adult human body is incorporated into hemoglobin (approximately 2100 mg) in erythrocytes or myoglobin (approximately 300 mg) in muscle.
- The remainder is chiefly present as storage iron in the liver (1000 mg) and in the reticuloendothelial macrophages of the bone marrow and spleen (600 mg)
- Only a small amount of iron (3 to 7 mg) is freely circulating in plasma bound to transferrin.

Causes of iron deficiency anemia

1. Chronic blood loss (ankylostoma & piles)
2. Increased demands such as growth and pregnancy
3. Decreased absorption (e.g. post gastrectomy and celiac disease)
4. Poor intake

Clinical features

Clinical features of anemia + Other clinical features occur as a result of tissue iron deficiency.

- Brittle nails & brittle hair
- Spoon-shaped nails (koilonychia)
- Atrophy of the papillae of the tongue
- Angular stomatitis
- A syndrome of dysphagia, esophageal web and glossitis (Plummer-Vinson syndrome)



Investigations

1. Blood count and film

- The red cells are microcytic (MCV < 80 fL) and hypochromic (MCH < 27 pg).
- There is poikilocytosis (variation in shape) and anisocytosis (variation in size). Target cells are seen.

2. Serum iron and total iron-binding capacity

- The serum iron decreases, and the total iron-binding capacity (TIBC) increase in iron deficiency compared with normal.

3. Transferrin saturation

- (i.e. serum iron divided by TIBC) decreases below 19%.

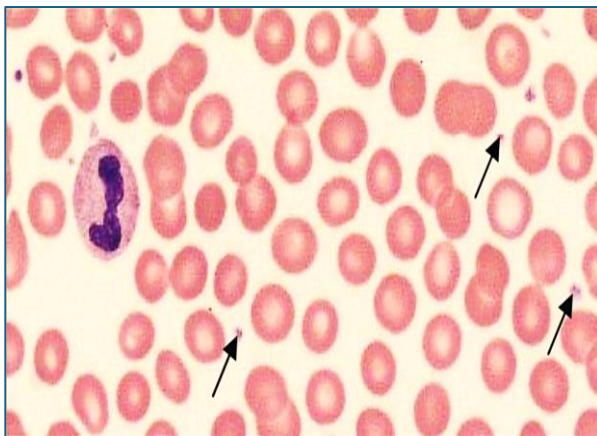
4. Serum ferritin

- In simple iron deficiency, a low serum ferritin confirms the diagnosis.

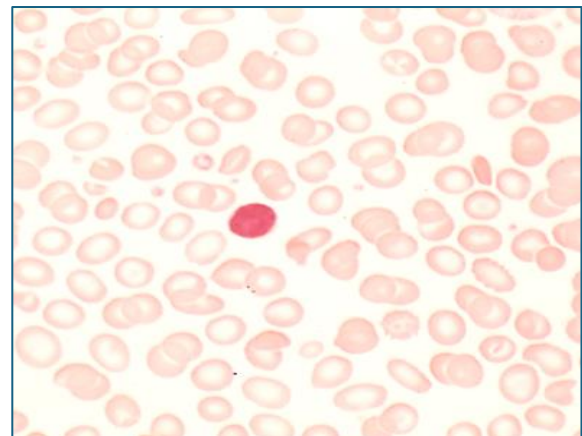
5. Serum soluble transferrin receptors

- The number of transferrin receptors increases in iron deficiency.

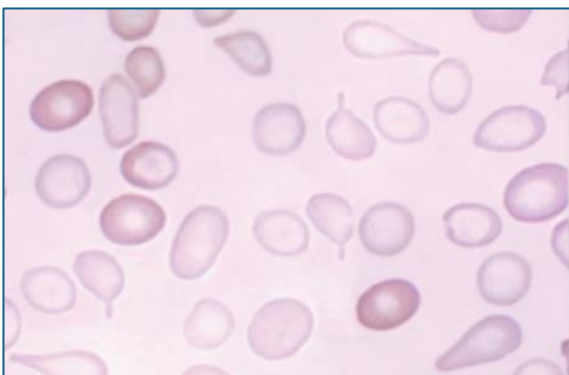
6. Bone marrow examination (is generally unnecessary) shows erythroid hyperplasia.



A normal blood film, with a neutrophil, red cells, and platelets (arrows).



Iron deficiency anemia. Many of these red blood cells are microcytic (smaller than the nucleus of the normal lymphocyte near the center of the field) and hypochromic (with central areas of pallor that exceed half the diameter of the cells).



Poikilocytosis and anisocytosis



Treatment

Iron deficiency is not a diagnosis per se. The correct management of iron deficiency is to:

1. find and treat the underlying cause,
2. give iron to correct the anemia and replenish iron stores.

Response to iron therapy can be monitored using the reticulocyte count and Hb level.

Iron preparation

The best preparation is oral ferrous sulphate (325 mg three times daily) which is absorbed best when the patient is fasting.

If the patient has side-effects such as nausea, diarrhea or constipation, taking the tablets with food or reducing the dose using a preparation with less iron such as ferrous gluconate (300 mg twice daily).

Oral iron should be given for a long time enough to correct the Hb level and to replenish the iron stores. This may take 4 - 6 months.

Causes of failure of response to oral iron are:

- 1- Lack of compliance
 - 2- Continuous bleeding
 - 3- Iron deficiency is not the primary cause of the anemia (e.g., coexisting anemia of chronic disease).
- These possibilities should be considered before parenteral iron is used.

Parenteral iron is required in special situations, including:-

- 1- Those who have general intolerance of oral preparations even at low dose.
 - 2- Those with severe malabsorption, chronic gastrointestinal diseases such as celiac disease or Crohn's disease.
- Parenteral iron can be given as slow intravenous infusion of sodium ferric gluconate.



Anemia of chronic illness

One of the most common types of anemia, particularly in hospital patients, occurring in patients with:

1. Chronic infections such as infective endocarditis or tuberculosis and osteomyelitis.
2. Chronic inflammatory diseases such as Crohn's disease, rheumatoid arthritis, systemic lupus erythematosus (SLE), polymyalgia rheumatica.
3. Malignant disease.

Pathophysiology

Dysregulated iron homeostasis, manifested by low serum iron in the presence of normal or elevated serum ferritin levels. Pro-inflammatory stimuli, including lipopolysaccharides, interferon- γ , and TNF- α , upregulate DMT-1 which increases iron uptake and downregulation of ferroportin expression; ferroportin is the protein required for the release of ferrous iron from storage cells and for the transport of dietary iron from duodenal enterocytes into the circulation.

Hepcidin is an iron-regulated, acute phase reactant peptide that blocks both iron uptake in the gut and iron release from hepatocytes and macrophage stores, its upregulation by lipopolysaccharides, interleukin (IL)-6. These mediators of inflammation act to increase erythroid progenitor **apoptosis**, downregulating erythropoietin receptors.

Investigations

1. The serum iron and the TIBC are low.
2. Serum ferritin is normal or raised because of the inflammatory process.
3. The serum soluble transferrin receptor level is normal
4. Stainable iron is present in the bone marrow, but iron is not seen in the developing erythroblasts.

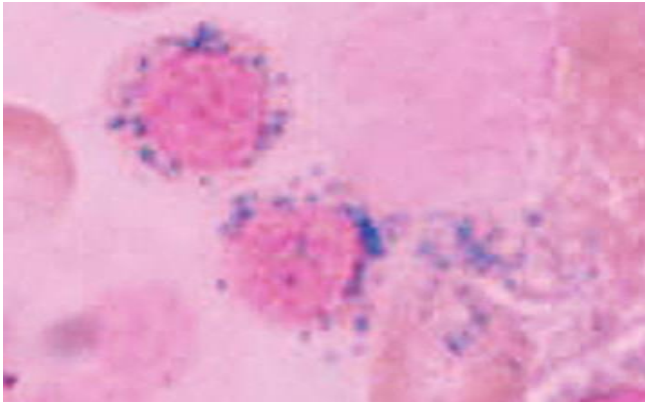
Treatment

Patients do not respond to iron therapy, and treatment is in general, that of the underlying disorder.

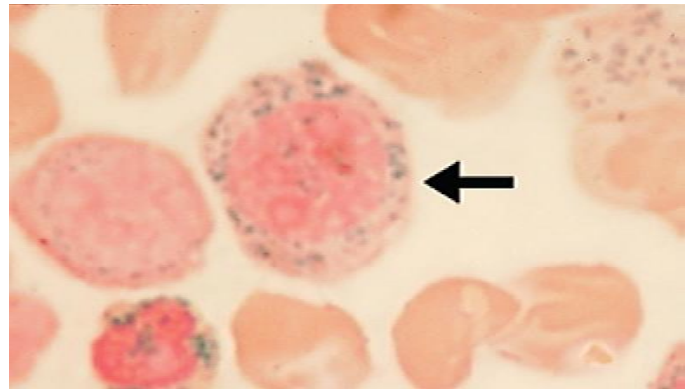
Sideroblastic Anemias

This heterogeneous group of anemias is distinguished by the characteristic finding of excessive mitochondrial iron in erythroblasts, as manifested by iron-laden, ringed sideroblasts in the bone marrow in the presence of moderate to severe anemia. These disorders result from mitochondrial defects either in the biosynthesis of the heme porphyrin ring or in the metabolism of iron.

Both hereditary and acquired types of sideroblastic anemia have been described. The presence of ring sideroblasts is the diagnostic feature of sideroblastic anemia



Two ringed sideroblasts showing how the distribution of peri nuclear mitochondrial ferritin can vary.



Ring sideroblasts in the marrow, with a perinuclear ring of iron granules.

Classification of sideroblastic anemia

Inherited

X-linked disease - transmitted by females.

Acquired

- 1- Myelodysplasia
- 2- Myeloproliferative disorders
- 3- Myeloid leukemia
- 4- Drugs e.g. isoniazid
- 5- Alcohol abuse
- 6- Lead toxicity
- 7- Other disorders, e.g. rheumatoid arthritis, carcinoma, megaloblastic and hemolytic anemias

Treatment

- Some patients respond when drugs or alcohol are withdrawn, if these are the causative agents.
- In some cases, there is a response to pyridoxine.



2

Macrocytic anemias

Macrocytosis (mean corpuscular volume (MCV) greater than 100 fL) in the setting of anemia. It is divided into two forms

- Megaloblastic (hyper segmented neutrophils)
- Non-megaloblastic.

Etiology:

A- Megaloblastic anemia occurs from deficiencies in folic acid and vitamin B12.

Folate deficiency is due to

- Diminished intake (alcohol abuse or malnutrition),
- Increased consumption (hemolysis or pregnancy),
- Malabsorption (gastric bypass, or medications like cholestyramine or metformin).

Vitamin B12 deficiency appears

- In diminished intake (malnutrition),
- Malabsorptive states (atrophic gastritis either autoimmune or non-autoimmune from *Helicobacter pylori* or Zollinger-Ellison syndrome, tapeworm infection, gastric bypass, ileal resection), or the presence of antagonists (nitrous oxide).
- Drugs that impair DNA synthesis are folic acid analogs (ex. methotrexate, trimethoprim-sulfamethoxazole), nucleic acid analogs (hydroxyurea)

B-Non-megaloblastic anemia, the absence of hyper segmented neutrophils, occurs in a variety of settings benign conditions

- Alcohol consumption (RBC toxicity),
- hereditary spherocytosis, hypothyroidism and liver disease (due to lipid deposition in the cell membrane), and marked reticulocytosis from states of excess RBC consumption such as hemolysis or turnover in pregnancy or primary bone marrow disease (reticulocytes are larger than the average RBCs).

Epidemiology

Macrocytosis affects 2% to 4% of the population, most commonly due to deficiencies in folate and vitamin B12 and medications.

- ✓ Autoimmune causes are more common in middle-aged women.
- ✓ Hypothyroidism and primary bone marrow disease account for more cases of macrocytic anemia in older patients.
- ✓ The prevalence of vitamin B12 deficiency increases in patients older than 60 years.

Pathophysiology

The equation for mean corpuscular volume [MCV (fL) = Hct (%) X 10 / RBC (10⁶/microgram)] explains how macrocytic anemia represents large red blood cells (RBCs) in comparison to total amount.

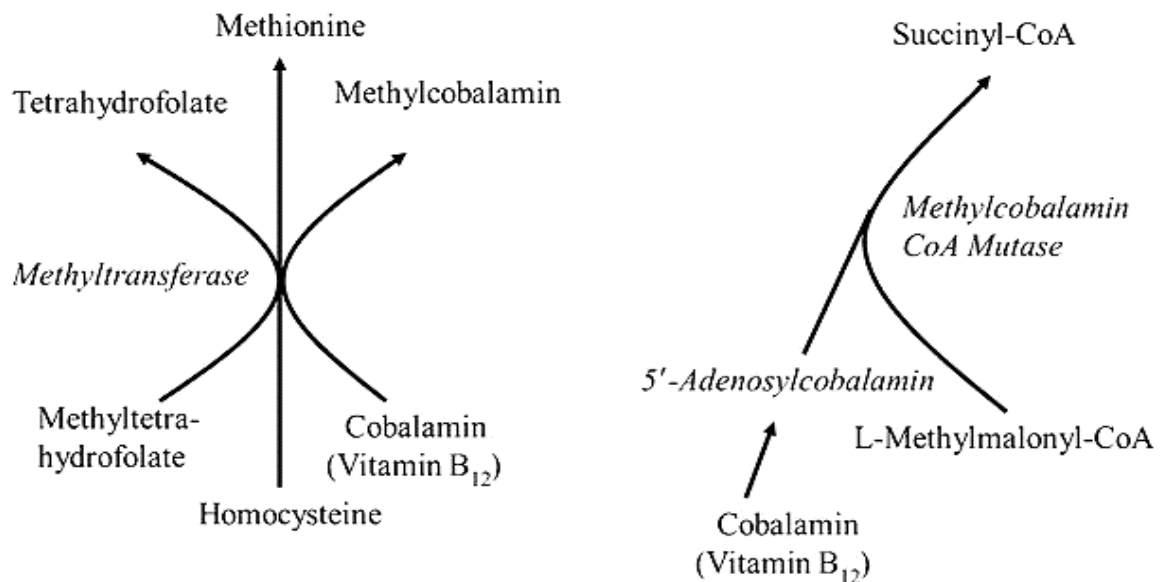
Folate and vitamin B12 are necessary for RBC nucleic acid synthesis. Without DNA or RNA, erythropoiesis is ineffective with nuclear/cytoplasmic asynchrony, resulting in larger erythrogenic precursors with abnormal nuclei (ex. hypersegmentation) but normal cytoplasms.

Megaloblastic anemias

Megaloblastic anemia is a macrocytic anemia that is characterized by large RBC precursors (megaloblasts) in the bone marrow and that is usually caused by vit b12 and folic acid deficiency.

Vitamin B12: Cellular Processing

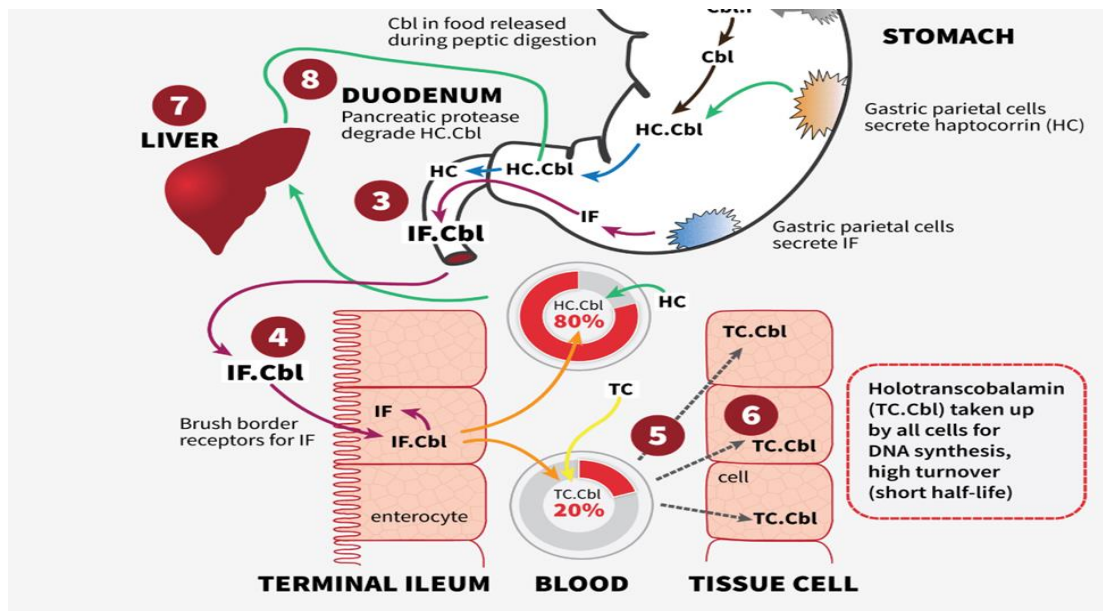
- The main function of vitamin B12 is serving as a cofactor for two major metabolic reactions.
 - In the cytoplasm, the methylation of homocysteine to methionine with the demethylation of methyl THF polyglutamate to THF. It plays an important role in DNA synthesis.
 - In mitochondria, conversion of methylmalonyl-CoA into succinyl-CoA which is necessary for proper myelin synthesis.



Daily requirement of vit B12

- Daily vitamin B12 requirements are one microgram, and the body can absorb 2 to 3 micrograms/day. Most patients have several years of B12 stores.
- Vitamin B12 is absorbed in the ileum when bound by an intrinsic factor (IF), a protein produced by the gastric parietal cells.
- Abnormalities in these cascades cause deficiencies in folate or vitamin B12, respectively.

Vit B12 absorption, transport, excretion



- **Intrinsic factor :** is a glycoprotein secreted by gastric parietal cells. It combines with vitamin B12 and carries it to specific receptors on the surface of the mucosa of the ileum.
- **R factor or R protein.** Glycoprotein is produced by the salivary glands of the mouth. It primarily serves to protect cobalamin (Vitamin B12) from acid degradation in the stomach.
- **Transcobalamin II (TC II).** TC II is synthesized by epithelial and endothelial cells, monocytes and fibroblasts. It binds circulating cobalamin and is its only real carrier, transporting it to target cells where it will be used.

Folic acid absorption

- Normally absorbed from duodenum and upper jejunum and to a lesser extent from lower jejunum and ileum.
- It exists in nature as folates in polyglutamate form. (conjugated folates)
- Polyglutamate form is first cleaved to mono and di glutamates, which further undergo reduction in the mucosal cells to form tetrahydrofolate, a monoglutamate.



Daily requirements of folic acid:

Daily folate needs are 100 to 200 micrograms, and the body can absorb 400 micrograms/day. Healthy patients have stores for 4 months. Folate is largely absorbed in the small bowel.

Pernicious anemia

Pernicious anemia (PA) is an autoimmune disorder in which there is atrophic gastritis with loss of parietal cells in the gastric mucosa with consequent failure of intrinsic factor production and vitamin B12 malabsorption.

Pathogenesis of Pernicious anemia

- This disease usually occurred at age of 40-70 yrs., more common in females.
- About 30% of patients have a positive family history.
- There is an association with other autoimmune diseases, particularly Graves' disease, Hashimoto thyroiditis, Addison's disease, vitiligo and type 1 diabetes.
- There is a higher incidence of gastric carcinoma with PA than in general population.
- Parietal cell antibodies are present in the serum in 90% of patients with PA. while intrinsic factor antibodies, found in only 50% of patients.

Pathology of Pernicious anemia

- Autoimmune gastritis affecting the fundus is present.
- The parietal and chief cells are replaced by mucin-secreting cells
- There is achlorhydria and absent secretion of intrinsic factor.

Clinical presentation of Megaloblastic anemia

- Mood disturbances and neurologic symptoms like loss of balance, memory loss, paresthesias, peripheral neuropathy.
- Symptoms specific to the underlying cause (constitutional symptoms from primary bone marrow disease, gastrointestinal upset from enteral malabsorption, or hypoactive metabolism from hypothyroidism).
- Common features of anemia dizziness, easy fatigability, palpitation.
- Physical examination may reveal
 - Nonspecific anemia findings (conjunctival pallor),
 - Neurologic deficits if vitamin B12 deficient and stigmata of underlying diseases (glossitis from autoimmune atrophic gastritis,
 - Hepatosplenomegaly from familial hemolytic anemias,

Investigations:

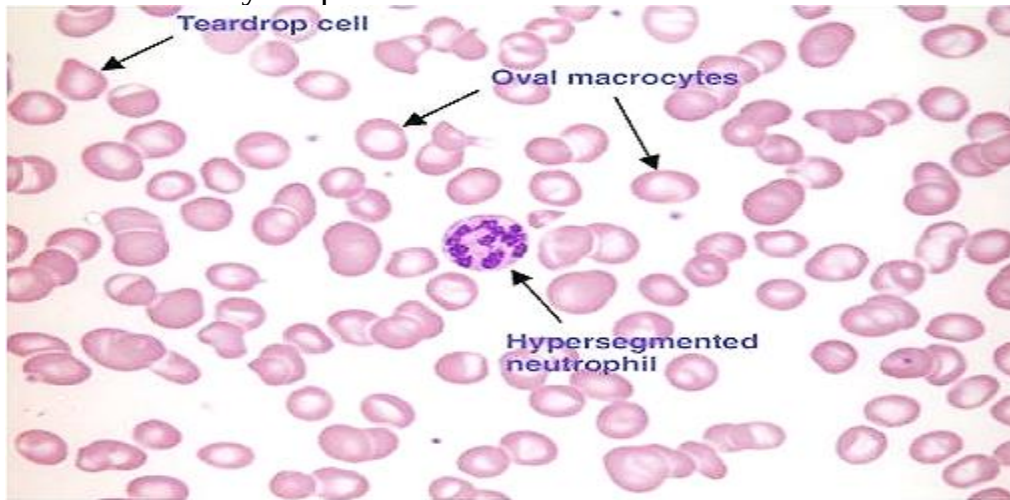
A- CBC

- Hemoglobin: Decreased
- Retic. count: Low to normal

- Leucocytes: May be reduced. Presence of Hypersegmented Neutrophil is characteristic.
- Platelets: Moderately reduced
- The MCV is characteristically > 96 fL unless there is a coexisting cause of microcytosis when there may be a dimorphic picture.

B- Peripheral blood smear (PBS)

- Megaloblastic contains hypersegmented neutrophils and macro-ovalocytes on peripheral blood smear, anisocytosis and poikilocytosis are not uncommon due to ineffective erythropoiesis.



- Non megaloblastic macrocytic anemia will not have hypersegmented neutrophils; instead, its PBS will show round macrocytes or macro reticulocytes, in addition to other cells representative of the underlying etiology (schistocytes from hemolysis, or spherocytes in inherited hemolytic anemias such as hereditary spherocytosis).

C- **Serum bilirubin** may be raised as a result of ineffective erythropoiesis where the destruction of developing red cells is much increased.

D- **Serum vitamin B12** is usually low .

E- **Serum folate and red cell folate** are low.

F- **LDH**: may be raised due to hemolysis.

G- **Gastrointestinal investigations**: In Pernicious anemia there is achlorhydria.

H- **Endoscopy or barium meal** examination of the stomach is performed only if **gastric symptoms are present**.

I- **Bone marrow**:

- Bone marrow evaluation will show hypercellularity with abnormal maturation and proliferation particularly the erythroid lineage.



Treatment and prevention of megaloblastic anemia

- Treatment depends on the type of deficiency.
- Dietary education
- Treatment of the cause
- Blood transfusion may be indicated in severe cases.
- In severely ill patients, it may be necessary to treat with both folic acid and vitamin B12 while awaiting serum levels.
- Folic acid may produce a hematological response in vitamin B12 deficiency but did not prevent the neuropathy of vitamin B12 deficiency.
- For patients with severe cobalamin deficiency from any cause, an aggressive scheme to replace cobalamin rapidly is 1 mg/day of intramuscular cyanocobalamin (week 1), 1 mg twice weekly (week 2), 1 mg/week for 4 weeks, and then 1 mg/month for life
- Alternatively, it is recommended that oral B12 2mg per day is given, as 1-2% of an oral dose is absorbed by diffusion and therefore does not require intrinsic factor.
- The use of sublingual nuggets of B12 ($2 \times 1000 \mu\text{g}$ daily) has been suggested to be an effective and more convenient option.

Monitoring response to therapy

- Clinical improvement within 48 hrs
- Reticulocytosis may occur within 2-3 days and peaks in 4-10 days.
- LDH and bilirubin fall rapidly, leucocyte and plt counts are restored within days, hypersegmented neutrophils may persist for 1-2 wks and Hb level rise approximately 1 g/dl each week
- Improvement of the polyneuropathy may occur over 6-12 months, but long-standing spinal cord damage is irreversible.

Treatment of folate deficiency

- Folate deficiency can be corrected by giving 5 mg of folic acid daily;
- The same hematological response occurs as seen after treatment of vitamin B12 deficiency.
- Treatment should be given for about 4 months to replace body stores.
- Any underlying cause, e.g. coeliac disease, should be treated.
- Prophylactic folic acid is given in chronic hematological disorders where there is rapid cell turnover. A dose of 5 mg each week is probably sufficient.

Sources of vit B12 and folic acid



Macrocytosis without megaloblastic changes

A raised MCV with macrocytosis on the peripheral blood film can occur with a normoblastic rather than a megaloblastic bone marrow.

- A common physiological cause of macrocytosis is pregnancy and newborn.
- Common pathological causes are:
 - Alcohol excess
 - Liver disease
 - Reticulocytosis
 - Hypothyroidism
 - Some hematological disorders (e.g. myelodysplasia, sideroblastic anaemia, pure red cell aplasia)
 - Drugs (e.g. Cytotoxics -azathioprine)
 - Spurious (agglutinated red cells measured on red cell counters)
 - Cold agglutinins due to autoagglutination of red cells (the MCV decreases to normal with warming of the sample to 37°C).

In all these conditions, normal serum levels of vitamin B12 and folate will be found. The exact mechanisms in each case are uncertain, but in some there is increased lipid deposition in the red cell membrane.



Items of aplastic anemia (AA)

- Definition
- Epidemiology (statistics, race, age, sex, mortality/morbidity)
- Etiology
- Pathophysiology of AA
- Types.
- Clinical Manifestations
- Diagnostic Evaluation & severity .
- Differential Diagnosis of Pancytopenia
- Treatment
 - Supportive treatment of patients with AA
 - Immunosuppressive treatment (IST) of Acquired AA
 - Haemopoietic stem cell transplant in AA

1-Definition of Aplastic Anemia (AA)

AA is defined as pancytopenia with a hypocellular bone marrow in the absence of an abnormal infiltrate or marrow fibrosis.

To diagnose AA there must be at least **two** of the following:-

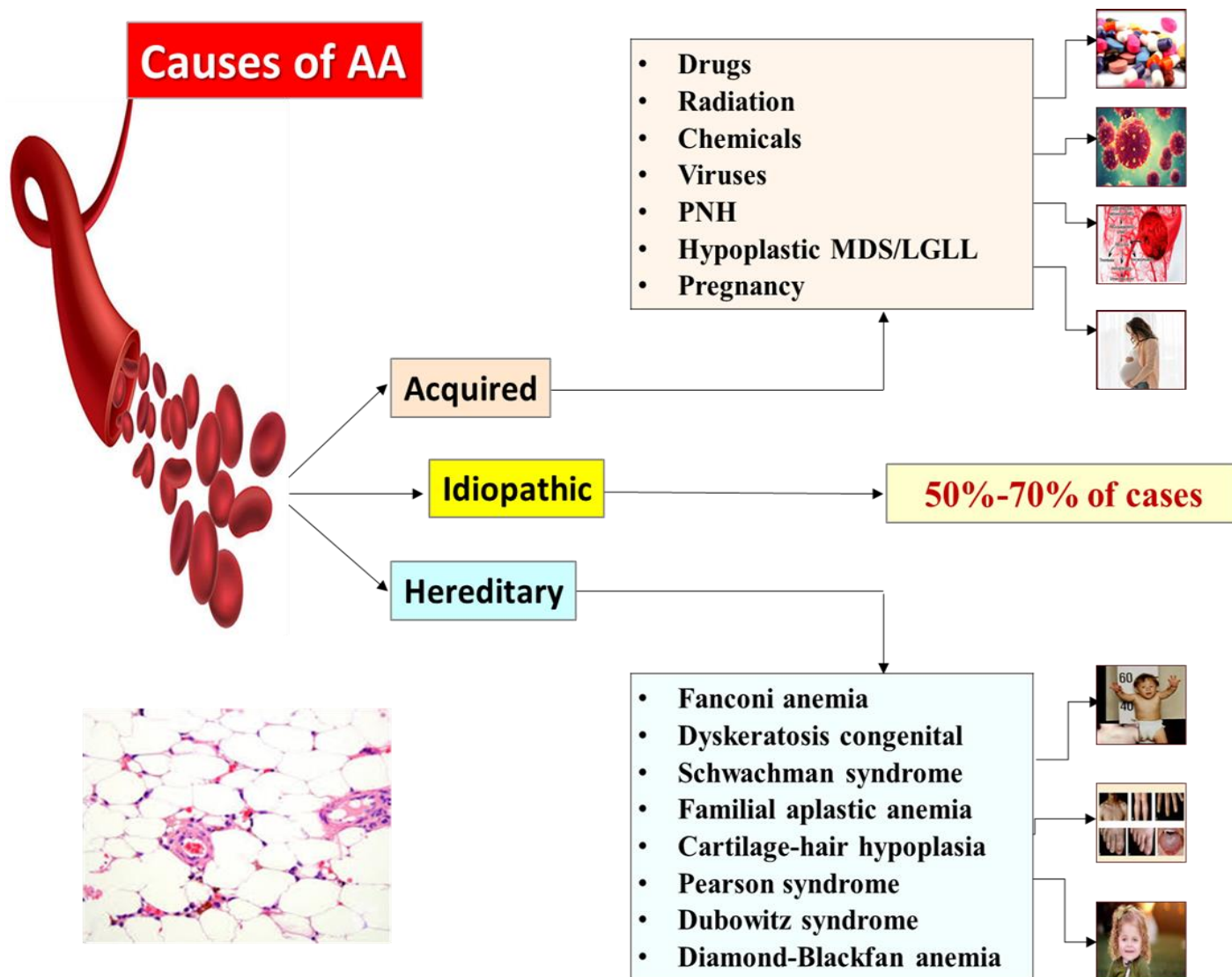
- Hemoglobin (Hgb) $<10^9$ g/dl,
- Platelet count $<50 \times 10^3/L$,
- Neutrophil count $<1.5 \times 10^9/L$

2-Epidemiology

- **International statistics**
 - AA is more common in Asia than in the West
 - No accurate data of AA in Egypt but increasing of the incidence is observed which may be related to environmental factors, exposure to toxic chemicals, rather than to genetic factors
- **Race and sex**
 - The prevalence is increased in the Far East
 - The male-to-female ratio for acquired AA is approximately 1:1
 - A male preponderance may be observed in the Far East

- **Age-related demographics**
 - AA occurs in all age groups
 - A small peak in childhood because of inherited AA
 - There is a biphasic distribution, with peaks at 10-25 years and over 60 years in non-Hereditary types.
 - A common peak is observed in people aged 20-25 years
- **Mortality/morbidity**
 - The causes are infection and bleeding
 - Infections remain the major cause of death in AA

3-Causes





Drugs associated aplastic anemia

Antineoplastic drugs

Antimetabolites: Methotrexate

Alkylating agents: Cyclophosphamide, nitrogen mustard, melphalan

Sulfonamides and derivatives

Antibacterials: sulfonamides

Diuretics: Furosemid

Other antimicrobial drugs

Antibacterials: chloramphenicol,

Antifungals: amphotericin, flucytosine

Anticonvulsant drugs: Carbamazepine

Antithyroid drugs: Carbimazole

Antihypertensive drugs: Captopril,

Viruses associated aplastic anemia

Hepatitis-associated aplastic anemia:

Hepatitis viruses (**A, B, and C**) and for putative agents, such as Hepatitis D, E, G and transfusion-transmitted viruses

Parvovirus B19:

This virus directly affects blood cell production but almost always only as a result of highly specific infection of **erythroid progenitor cells**, mostly manifested in **hereditary hemolytic anemia**

Other viruses:

EBV, HHV6, and HIV

4-Clinical Features

- **Can present acutely or insidiously.**
- **Symptoms of**
 - A-Anemia (progressive weakness, fatigue, headaches, dyspnea),
 - B-Thrombocytopenia (petechia, ecchymoses, epistaxis,),
 - C-Infection

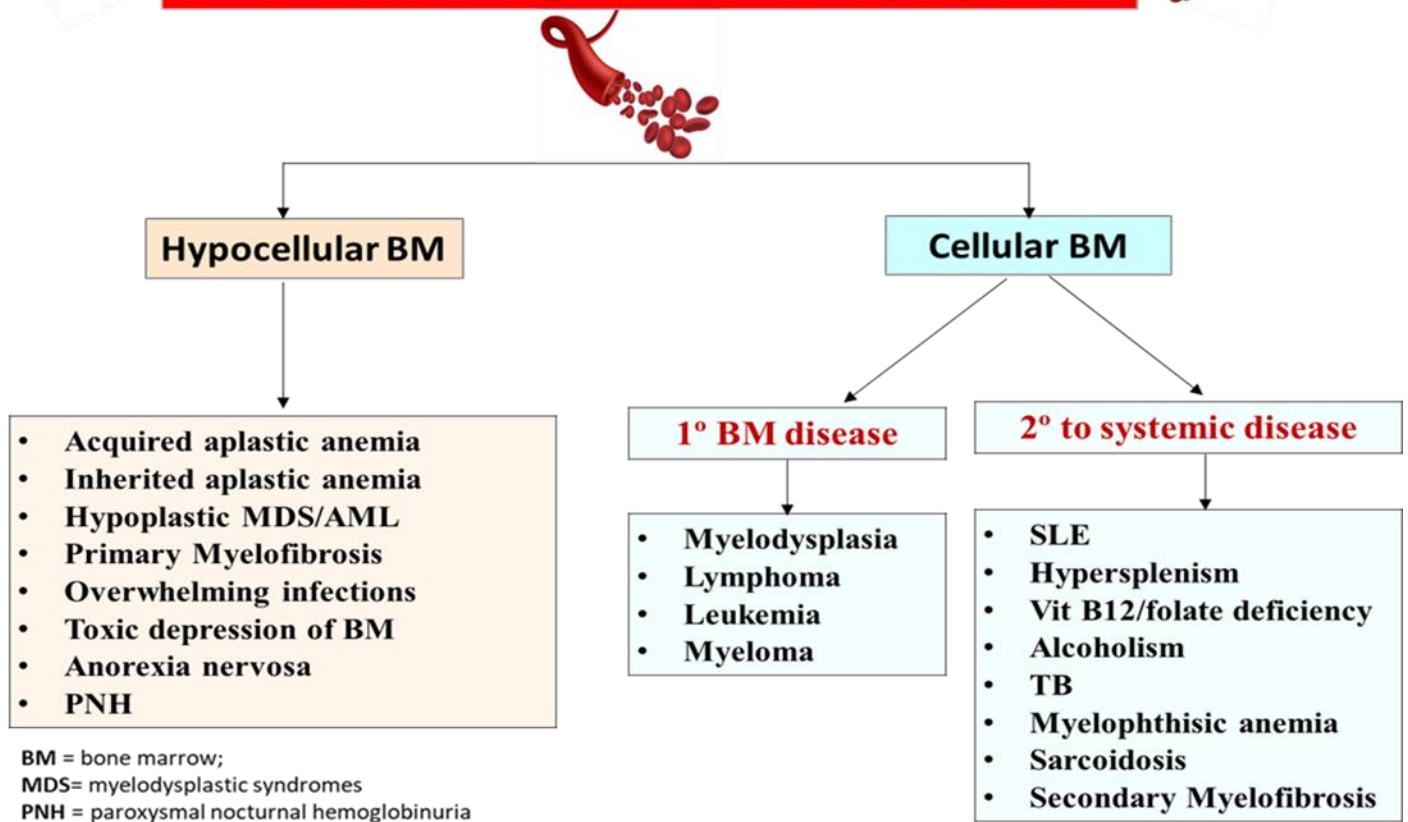
The most frequent physical findings are:

A-Cutaneous and conjunctival **pallor and hemorrhages**

B-If the anemia is severe, the patient may be tachycardic

C-Hepatosplenomegaly and lymphadenopathy are **notably absent**.

Differential Diagnosis of Pancytopenia



5-Investigations

A-To confirm diagnosis and severity

- **Full blood count** (Anemia or neutropenia or thrombocytopenia ± All)
- **Reticulocyte count** (Decreased reticulocytes (<1% of the total RBC count))
- **Blood film examination** (No blasts nor other abnormal cells)
- **Bone Marrow Aspiration and Biopsy**
 - Aplasia or hypoplasia of marrow cells with fat replacement
 - Decreased cellularity ■No fibrosis

To exclude other causes of pancytopenia

- Liver function tests
- Viral studies: hepatitis A/B/C, EBV, CMV, HIV and Parvovirus B19
- Anti-nuclear antibody and anti-double stranded DNA
- Chest X-ray, Abdominal ultrasound and other radiology
- Vitamin B12 and folate level
- Peripheral blood chromosomal breakage analysis: to exclude Fanconi anemia
- Flow cytometry for glycosylphosphatidyl-inositol anchored proteins to detect PNH
- Cytogenetic and FISH analysis to exclude MDS/AML or other malignancies



6-Disease severity

AA is classified according to disease severity based on bone marrow cellularity and peripheral blood cytopenias:

1-Severe AA (SAA)

- Bone Marrow cellularity <25% or
- BM 25–50% with <30% residual hematopoietic cells), plus at least 2 of:
 - Neutrophils <0.5 x 10⁹/L,
 - Platelets <20 x 10⁹/L
 - Reticulocyte count <20 x 10⁹/L

2-Very Severe AA (VSAA)

- As for SAA but neutrophils <0.2 x 10⁹

3-Non-Severe AA

- Not meet criteria of VSAA and SAA

7-Treatments

A- Supportive treatment of patients with aplastic anemia

1. Transfusions

- Red blood cells transfusion
- Platelets' transfusion
- Granulocyte transfusions
 - Judicious use to not increase the risk of immune sensitization to blood products
 - Use of irradiated cellular blood components for AA patients
 - Avoid blood products from sibling donors

2. Growth factors

- Eltrombopag (Thrombopoietin receptor agonists): **good results**
- Granulocyte stimulating factor (G-CSF) and erythropoietin are **not effective**
- Androgens especially for patients with inherited bone marrow failure

3. Remove offending agents

4. Prevention and treatment of infection

5. Vaccinations in non-transplanted patients

- Vaccinations, including influenza vaccination, should be avoided if possible.

B-Immunosuppressive treatment (IST) of Acquired aplastic anemia

Antithymocyte globulin (ATG) and Cyclosporine (CsA)

ATG + CsA is indicated as first line therapy for:

1. NSAA patients who are transfusion dependent, bleeding, encountering infections or for lifestyle (activities).
2. In the absence of an HLA-matched sibling.
3. SAA/VSAA patients >35-50 years of age

C-Allogenic Haemopoietic stem cell transplant (HSCT) in aplastic anemia for

1. SAA after failure to respond to one course of IST
2. SAA with MDS/ VSAA
3. Hereditary BMF

Types: of HSCT that can used in AA

- HLA identical sibling donor
- Unrelated donor
- Alternative donor: cord blood and haploidentical
- Syngeneic donor

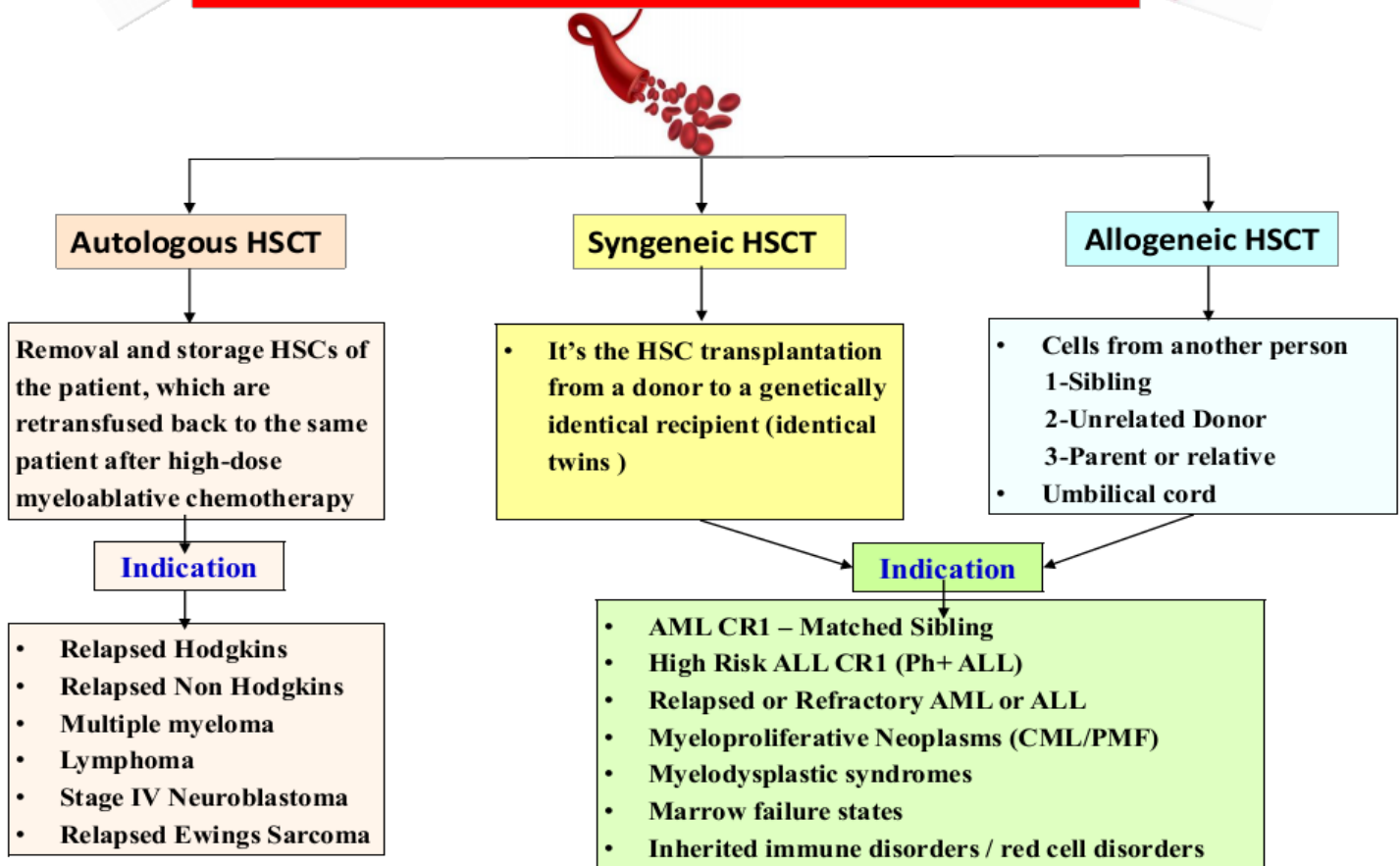
Hematopoietic stem cell transplantation (HSCT)

Is a procedure by which hematopoietic stem and progenitor cells (HSCs) are infused intravenously to restore hematopoiesis and immune function following a “conditioning” or “preparative” regimen, consisting of high-dose chemotherapy with or without radiation therapy.

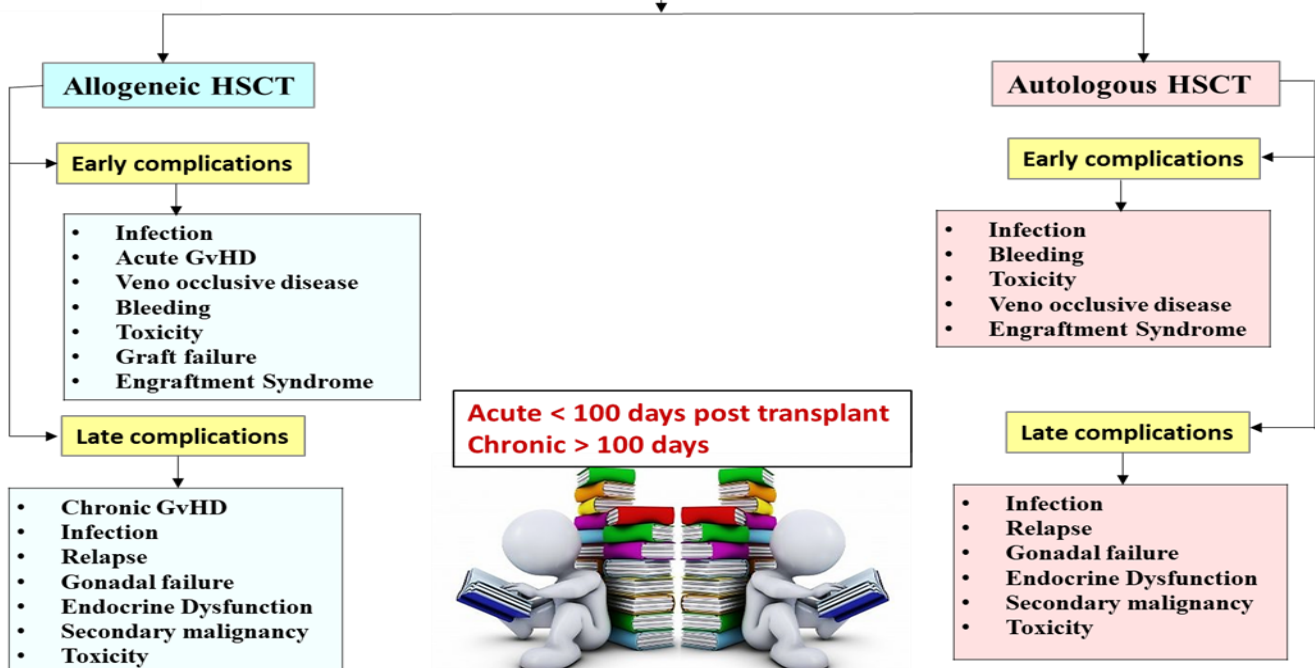
Sources of HSCT

	Bone marrow transplant	Peripheral blood stem cell transplant	Umbilical cord blood transplant
Source	Bone marrow aspiration from the posterior superior iliac spine	Pheresis of peripheral blood after administering a myeloid growth factor for 4–5 days	Umbilical cord blood that is collected at the time of delivery and stored in stem cell banks
Risk of GvHD	Lowest	Highest	Low
Engraftment	By 3 weeks	By 2 weeks	By 4 weeks

Types of stem cell transplantation



Complication of HSCT



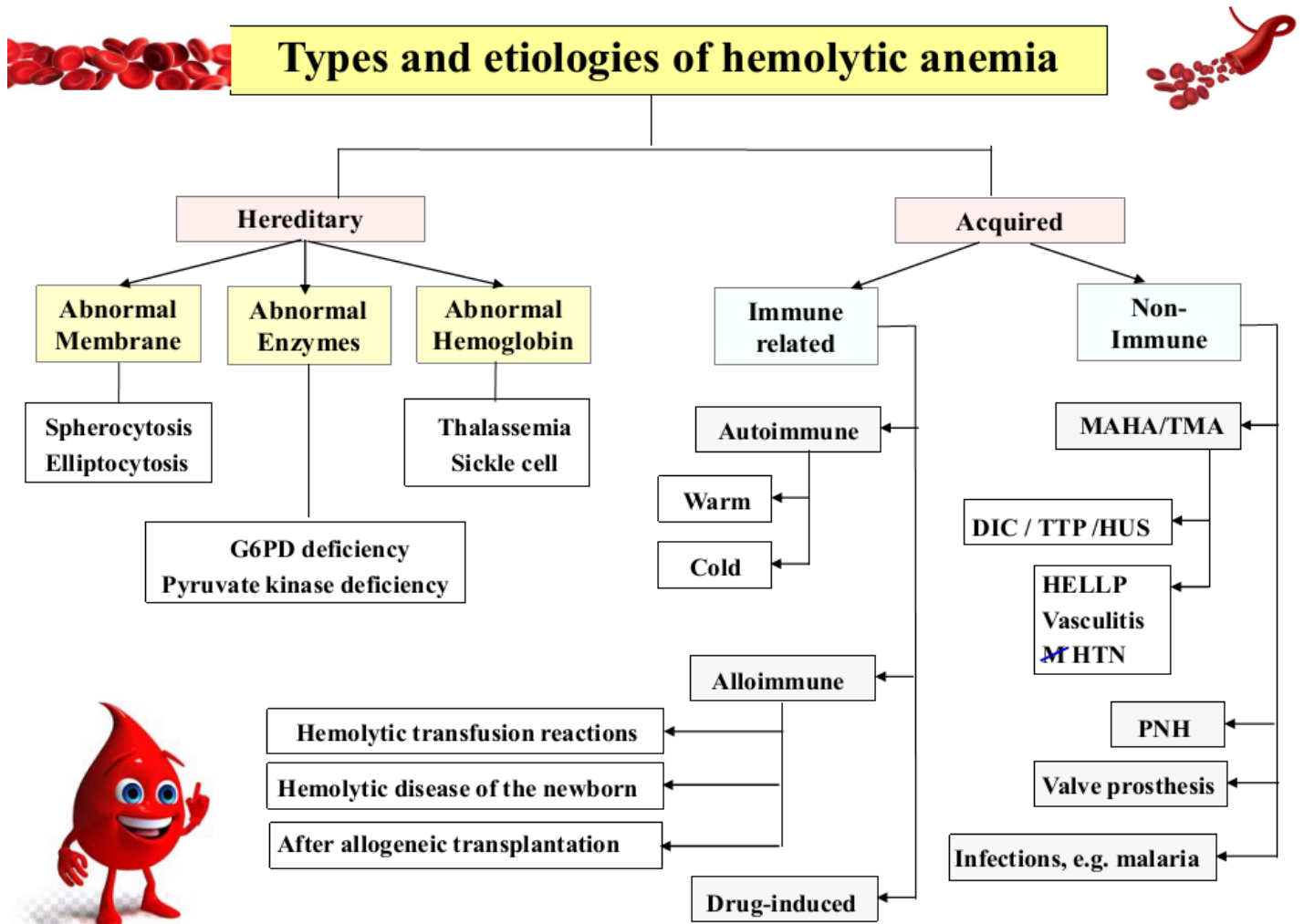
GvHD= graft-vs-host disease

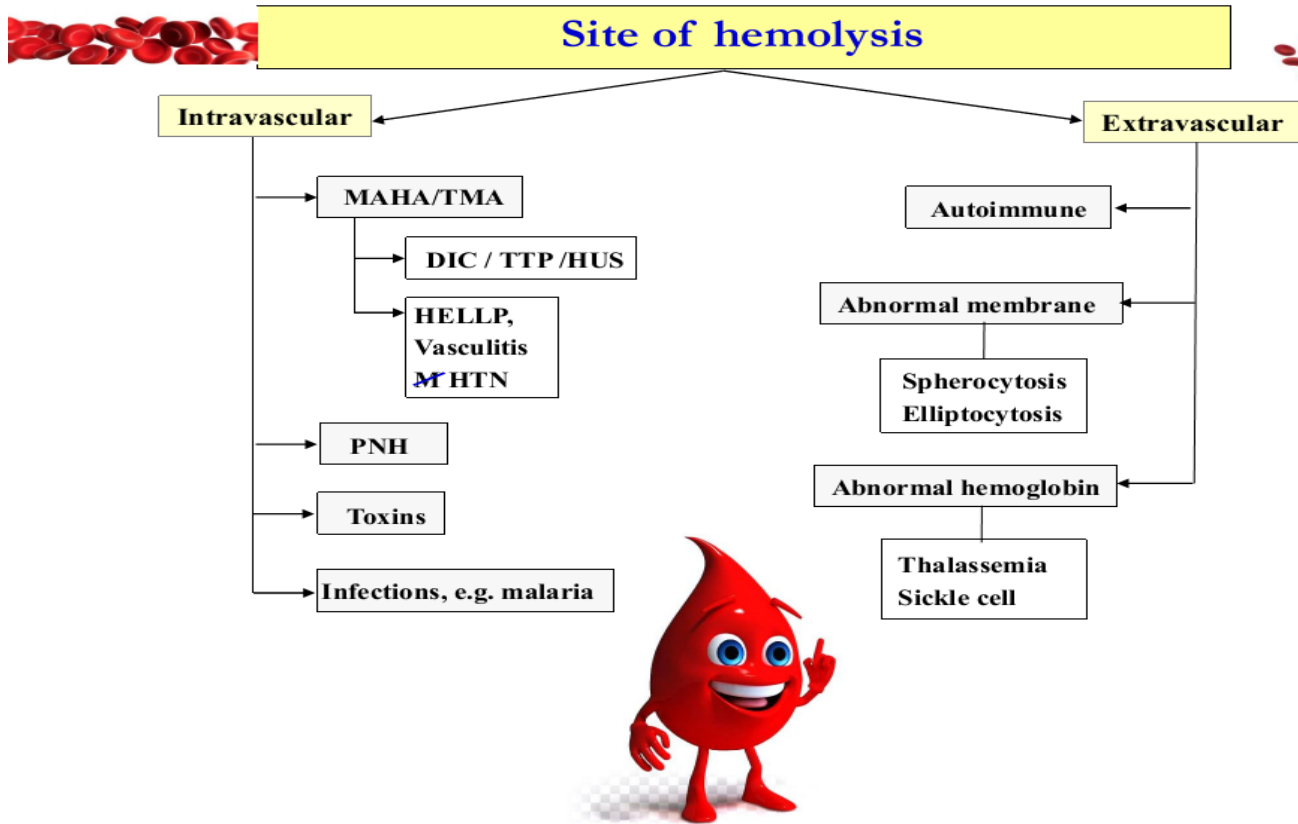
Hemolytic anemia

Hemolytic anemia is characterized by the breakdown of red blood cells (RBCs).

Defining features

- **Intrinsic hemolytic anemia:** Hemolysis can be caused by abnormalities in RBCs (hemoglobin, the RBC membrane, or intracellular enzymes),
- **Extrinsic hemolytic anemia:** Hemolysis by external causes (immune-mediated or mechanical damage)
- **Intravascular hemolysis:** RBCs destroyed within the circulation and hemoglobin is liberated in plasma.
- **Extravascular hemolysis:** RBCs are removed from the circulation by macrophages in the reticuloendothelial system, particularly the spleen.





Abbreviations: MAHA: Microangiopathic hemolytic anemia, TMA: thrombotic microangiopathy, DIC: disseminated intravascular coagulopathy, TTP: thrombotic thrombocytopenic purpura, HUS: hemolytic uremic syndrome

General Clinical features

Signs of anemia

- Pallor
- Fatigue
- Exertional dyspnea
- In severe cases: tachycardia, angina pectoris, leg ulcers

Signs of hemolysis

- Jaundice
- Pigmented gallstones
- Splenomegaly
- Back pain and dark urine in severe hemolysis with hemoglobinuria

Signs of increased hematopoiesis (mostly in severe chronic anemias)

- **Bone marrow expansion:** widening of the diploic space of the skull, biconcave deformity of the vertebral bodies
- **Cortical thinning and softening of bone** → ↑ risk of pathologic fractures
- **Extramedullary hematopoiesis:** hepatosplenomegaly

Approach

- ❑ Detailed history, drug history, family members with anemia, and full examination
 - ❑ Perform initial **laboratory studies** to confirm anemia and hemolysis and **classify** anemia by **morphology**.
 - Anemia workup: CBC with MCV, reticulocyte count
 - Hemolysis workup: Add LDH, haptoglobin, bilirubin, urinalysis, and peripheral blood smear (PBS)
 - ❑ Obtain a direct **Coombs test** (i.e., DAT) to narrow the differential diagnosis
 - ❑ Consider further investigations of the **underlying etiology** based on clinical suspicion and DAT results
 - ❑ If the patient has **severe** symptoms of anemia or a **life-threatening** cause is suspected (i.e., **TTP/HUS, DIC, HELLP syndrome, acute hemolytic transfusion reaction**), proceed directly to treatment in parallel with diagnostic evaluation.

Investigation

1-Routine laboratory studies

❑ CBC

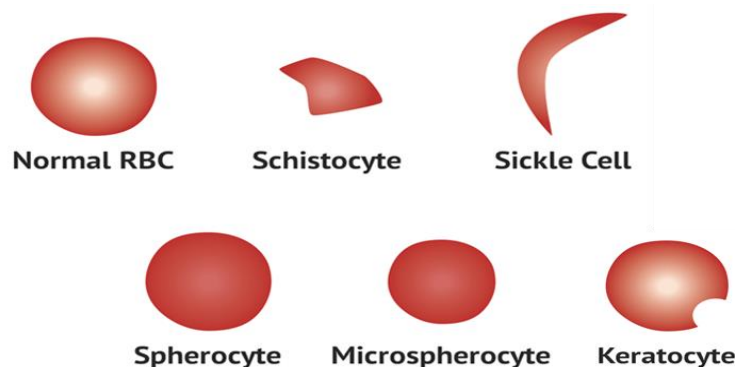
- ↓ **Hb, Hct, and RBC count**
- **MCV**

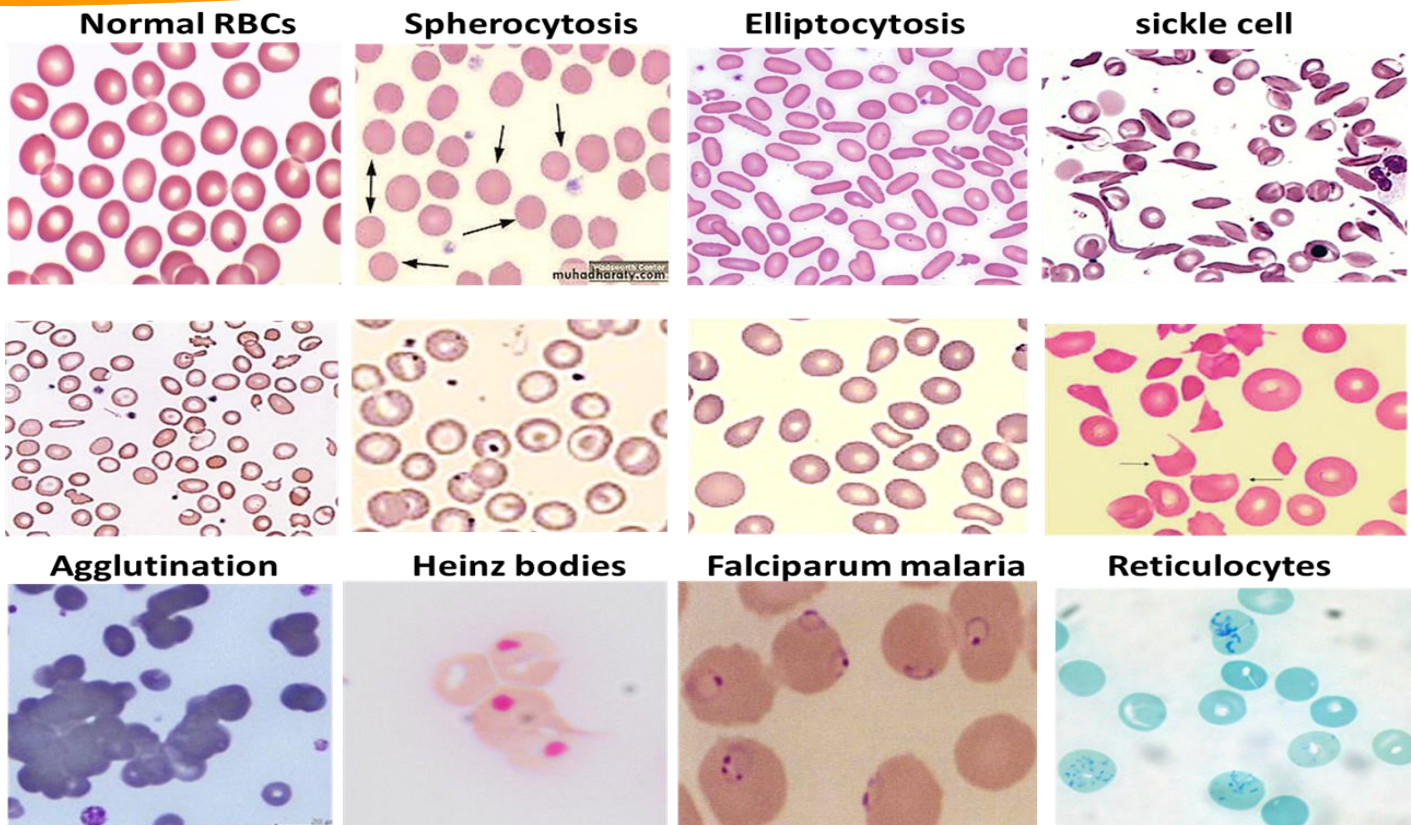
Normocytosis: typical finding for **most hemolytic anemias**; consider other anemia of chronic disease.

↑ **MCV:** Can be due to severe **reticulocytosis**; consider other causes of macrocytosis (e.g., **megaloblastic anemia**).

Microcytosis: consider **thalassemia** and/or **iron** deficiency

- **WBC count:** can be elevated due to inflammation or malignancy
- **Platelets:** **decreased** in **MAHA** and in Evan syndrome
- **Reticulocyte count:** **Reticulocytosis:** most common. Can be normal in patients with concomitant BM suppression
- **Peripheral blood smear (PBS) :** Spherocytes, sickle cell, Schistocytes





Laboratory Findings in Hemolytic anemia

- Increased LDH
- Decreased haptoglobin
- Increased unconjugated bilirubin
- Increased urobilinogen
- Reticulocytosis

Tests Specific For Intravascular Hemolysis

- Schistocytes on blood film
- Free hemoglobin in serum
- Methemalbuminemia (heme + albumin)
- Hemoglobinuria (immediate)
- Hemosiderinuria (delayed) – most sensitive

Tests Specific For Autoimmune hemolytic A

- **Direct Antiglobulin Test (direct Coombs)**
Detects IgG or complement on the RBCs surface
- **Indirect Antiglobulin Test (indirect Coombs)**
Detects antibodies in serum that can recognize antigens on RBCs



- ☐ **Iron studies** are usually normal in hemolytic anemia, however, **iron deficiency** can be seen in chronic **intravascular hemolysis** **iron overload with extravascular hemolysis**.
- ☐ **Urinalysis,**
- ☐ **Direct and indirect Coombs tests**
- ☐ **In patients with AIHA:** Autoimmune diseases screening as ANA, dsDNA, lupus anticoagulant

☐ Investigation of underlying causes

- **Hb electrophoresis (HE):** abnormal Hb patterns, e.g., in thalassemia
- **High performance liquid chromatography (HPLC):** more accurate than HE

Normal	A	$\alpha_2\beta_2$	Comprises 97% of adult Hb
	A2	$\alpha_2\delta_2$	Comprises 2-3.4% of adult Hb Elevated in β -thalassemia
	F	$\alpha_2\gamma_2$	Normal Hb in fetus from 3rd to 9th month Elevated in β -thalassemia
Abnormal chain production	H	β_4	Found in α -thalassemia, biologically useless
	Barts	γ_4	Comprises 100% of Hb in homozygous α -thalassemia, biologically useless
Abnormal chain structure	S	$\alpha_2\beta_2$	Substitution of valine for glutamine acid in position 6 of β chain
	C	$\alpha_2\beta_2$	Substitution of lysine for glutamic acid in position 6 of β chain

- **Flow cytometry:**
 - Detection of **EMA negative** RBCs, in hereditary **spherocytosis**
 - Absence of **CD55** and **CD59** on RBC surface in **PNH**
- **Genetic analysis:** mutations in congenital hemolytic anemia
- **Bone marrow biopsy**

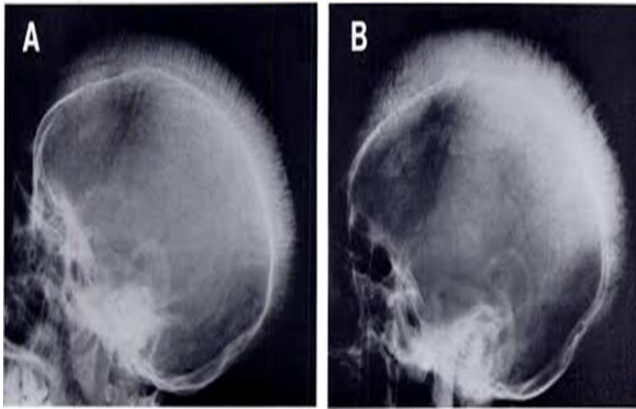
Indication:

- ☐ **Pancytopenia,**
- ☐ Presence of **abnormal** cells(e.g., blasts) in CBC/peripheral blood smear.

□ 2-Radiological investigation

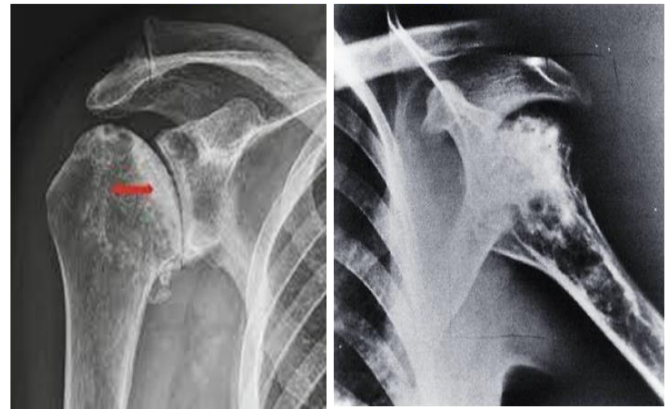
- **X rays:** Skull, hands, Joints and chest

Thalassemia



The hair on end sign of the diploic space expansion by a thickening of trabeculae

Sickle cell anemia



osteonecrosis of the humeral head and bone infarction in patient with sickle cell disease

- **Abdominal U/S:** detection of hepatosplenomegaly and gall stones, lymph nodes
- **Echocardiography:** for detection of cardiac complication
- **CT and PET scan :** to exclude lymphoproliferative disorders with autoimmune hemolytic anemia (AIHA)

4

Inherited Hemolytic anemia

Thalassemia

Definition

- Defects in production of (globin genes) the α or β chains of hemoglobin resulting imbalance in **ineffective erythropoiesis** and hemolysis in the spleen or BM
- Autosomal recessive disorder: **Hypochromic microcytic anemia**
- The most common cause of hemolytic anemia in Egypt

β-thalassemia

Pathophysiology and types

- The most common cause of hemolytic anemia in Egypt
- **1 gene deletion** ⇨ β thalassemia **minor** ----slightly increased Hb A2
- **2 genes deletions** ⇨ β thalassemia **major & intermedia** ⇨ markedly reduced HbA (10-30%) and markedly increased Hb F (70-90%).

Clinical classification

- Thalassemia minor (or trait), the symptomless heterozygous carrier state
- Thalassemia intermedia, with moderate anemia, rarely requiring transfusions
- Thalassemia major, with severe anemia requiring regular transfusions.

β-thalassemia Trait

Clinical Features

- Usually asymptomatic; a palpable spleen is very rare
- It may be confused with iron deficiency.

Investigations

- Hb (**10-14** g/dL), MCV(<70) – microcytosis , Fe (normal), RBC count
- Hb electrophoresis: specific: **HbA2 increased** to 3.5-5% (normal 1.5-3.5%)
non-specific: 50% have slight increase in HbF

Treatment

- No treatment required
- Genetic counselling for patient and family

β-thalassemia Intermedia

Clinical Features

- Includes patients who are symptomatic with moderate anemia (Hb 7-10 g/dL) and not require regular transfusions.

Patients may have:

1) Splenomegaly	3) Leg ulcers
2) Bone deformities.	4) Gallstones

Investigations

- Hb (Hb **7-10** g/dL), MCV(<70) – microcytosis ,
- Hb electrophoresis : HbA: 0-10% (normal >95%)
HbA2 >2.5%
HbF: 70-100%

β -thalassemia Major (Cooley's anemia)

Clinical Features

- Initial presentation at **age 6-12 mo** with severe anemia, jaundice
- **Iron overload** progressing to hemochromatosis
 - Secondary to repeated transfusions and ineffective erythropoiesis
 - leads to iron-induced organ damage
- **Stunted growth** and development (hypogonadal dwarf)
- **Gross hepatosplenomegaly** (due to extramedullary hematopoiesis)

Classical thalassemic facies

- Enlarged head,
- Frontal and parietal bossing,
- Prominent malar eminence,
- Protrusion of maxilla,
- Prominent upper teeth



Manifestations of complications

- 1- Complication of blood transfusion (volume overload, infections).
- 2- Iron overload ➤ hemosiderosis.
- 3- 2ry hypersplenism ➤ increased transfusions requirements and pancytopenia.
- 4- Pathological fractures, gall stones and leg ulcers.
- 5- Crises ➤
 - **Acute hemolytic crises.**
 - **Aplastic crises;** caused by infections (Human parvovirus B19).
 - **Megaloblastic crises** due to increased folic acid requirements.

Investigations

- Hb (Hb <6 g/dL), MCV(<70) – microcytosis , Fe (normal)
- Hb electrophoresis
HbA: 0-10% (normal >95%)/ HbA2 >2.5%/ HbF: 70-100%
- **Radiologic changes** (due to expanded marrow cavity) and extramedullary hematopoietic masses
 - Skull x-ray has “hair-on-end” appearance
 - Pathologic fractures common



Treatment

- lifelong regular transfusions to suppress endogenous erythropoiesis
- HBV vaccination
- Iron chelation (e.g. deferoxamine, deferasirox, deferiprone) to prevent iron overload in organs and the formation of free radicals
- Folic acid supplementation if not transfused
- Allogenic BM transplantation (potentially curative) or cord blood transplant
- Splenectomy (now performed less frequently)
- Luspatercept is an erythroid maturation agent approved by the FDA for treating anemia in patients with β -thalassemia who need regular red blood cell transfusions.
- Recently the U.S. FDA approved Zynteglo (betibeglogene autotemcel), the first cell-based gene therapy for the treatment of adult and pediatric patients with β -thalassemia who require regular RBC transfusions.

Prevention

- Prenatal diagnosis and gene therapy by insertion of normal globin gene in stem cells
- Carrier screening and genetic counselling

α -thalassemia

Clinical Features

- Hemolytic anemia due to defect(s) in **α genes**.
- 1 or 2 defective α genes no abnormalities
- **3 defective α genes (α -/--): HbH (β 4) disease; presents in adults, decreased MCV, decreased Hb, splenomegaly**
- **4 defective α genes (--/--): Hb Barts (γ 4) disease (hydrops fetalis); incompatible with life**

Investigations

- Peripheral blood film – screen for HbH inclusion bodies with supravital stain
- Electrophoresis – Hb H disease. Definitive diagnosis with DNA genotyping

Treatment

- Depends on degree of anemia,
 - 1 or 2 defective α genes: no treatment required
 - HbH disease: similar to β -thalassemia intermedia
 - HbBarts: no definitive treatment, intrauterine HSCT.



Sickle Cell Disease

Definition

- Autosomal recessive sickling
- A mutant β -globin chain, most commonly caused by a Glutamine $\rightarrow$ Valine substitution at position 6 (chromosome 11) resulting in HbS variant, rather than HbA
- In the homozygous (sickle cell anaemia) both genes are abnormal (Hb SS),
- The heterozygous state (sickle cell trait, Hb AS) only one chromosome carries the gene

Pathophysiology

- At low pO₂, deoxy HbS polymerizes leading to rigid crystal-like rods that distort membranes $\rightarrow$ 'sickles'
- Sickling aggravated by acidemia, increased CO₂, fever, dehydration and infections
- Fragile sickle cells then cause injury in two main ways
 1. fragile sickle cells hemolyze
 2. Occlusion of small vessels (hypoxia, ischemia-reperfusion injury)

Manifestations

- **HbAS (sickle cell trait):** patient will be asymptomatic except during extreme hypoxia or infection increased risk of renal medullary carcinoma
- **SCD-SS (HbSS)**
 - Chronic hemolytic anemia
 - Jaundice in the first yr of life
 - Retarded growth and development $\pm$ skeletal changes
 - Splenomegaly in childhood; splenic atrophy in adulthood.
 - **Crises**
 - Vaso-occlusive crises
 - Splenic sequestration crises
 - Aplastic crises
 - Hemolytic crises
 - Megaloblastic crises
 - Acute sickle chest syndrome



❑ Vaso-occlusive crises

- Acute pain in the hands and feet (dactylitis)-Vaso occlusion of the small vessels.
- Severe pain in other bones as femur, humerus, vertebrae, ribs, pelvis
- Mesenteric vascular occlusion.
- Autosplenectomy (hyposplenism) due to multiple painful splenic infarctions.
- Renal, pulmonary and cerebral infarctions.

❑ Splenic sequestration crises

- Usually in children, significant pooling of blood in spleen resulting in acute Hb drop and shock, uncommon in adults due to asplenia from repeated infarction.

❑ Aplastic crises

- Toxins and infections (especially parvovirus B19) transiently suppress BM

❑ Megaloblastic crises

- Anemia can also result from folate deficiency

❑ Hemolytic crises

- Severe anemia, deep jaundice, fever and abdominal or bone pain

❑ Acute sickle chest syndrome

- Affects 30% of patients with sickle cell disease and may be life threatening.
- Presentation includes dyspnea, chest pain, fever, tachypnea, leukocytosis, and pulmonary infiltrate on CXR.
- Caused by vaso-occlusion, infection, or pulmonary fat embolus from infarcted BM
- Initial management is with oxygen, antibiotics and exchange transfusion to reduce the amount of Hb S to < 20%

Long-term problems

- **Delayed growth** and development
- **Respiratory:** pulmonary hypertension
- **Cardiac problems;** cardiomegaly, arrhythmias, iron overload cardiomyopathy and myocardial infarctions
- **Bones:** Avascular necrosis of hips, shoulders. Osteomyelitis
- **Neurological complications;** TIA, fits, cerebral infarction, cerebral hemorrhage
- **Cholelithiasis:** Pigment stones occur.
- **Liver;** hepatomegaly and liver dysfunction
- **Renal;** Chronic interstitial nephritis
- **Eye;** retinopathy, vitreous hemorrhages and retinal detachments all occur.
- **Pregnancy;** spontaneous abortion, intrauterine growth retardation, pre-eclampsia



Investigations

- Hb (Hb **6 -8**g/dL), MCV(normocytic) , high reticulocyte count
 - Sickling of red cells on a blood film
 - Sickle solubility test.
- **Hb electrophoresis**

Hb electrophoresis distinguishes HbAS, HbSS, HbSC eg sickle cell disease 80-95% Hb SS, and 2-20% Hb F

Treatment

- Genetic counselling
- Precipitating factors should be avoided or treated quickly.
- **HbAS**: no treatment required
- **HbSC**: treatment as per HbSS, but is dictated by symptom severity
- **HbSS**
 1. Folic acid to prevent folate deficiency
 2. Hydroxyurea to enhance production of HbF.

Treatment of vaso-occlusive crisis

- **Supportive care**: oxygen, hydration/ intravenous fluids (reduces viscosity), correct acidosis, analgesics/opiates.
- **Indication for exchange transfusion**: Hb <5-6 g/dL, SCD complications (acute chest syndrome, aplastic crisis, hepatic or splenic sequestration, stroke), prevention of complications, pre-operative
- **Antimicrobials** for suspected infection
- **Prevention of crises**
 - a. Avoid conditions that promote sickling (hypoxia, acidosis, dehydration, fever)
 - b. Vaccination in childhood (pneumococcus, meningococcus, H. influenzae B)
 - c. Good hygiene, nutrition, and social support
- **Screen for complications**
- **Bone marrow transplantation**:
- **Gene therapy**



Glucose-6-Phosphate Dehydrogenase Deficiency

Definition

Deficiency in G6PD, corresponding to a lack of reduced glutathione (GSH) and leading to RBC sensitivity due to oxidative stress

Pathophysiology

X-linked recessive, prevalent in individuals of African, Asian, and Mediterranean descent

Manifestations

Frequently presents as episodic hemolysis precipitated by: oxidative stress

- Drugs (e.g. Analgesics, sulfonamide, antimalarials, nitrofurantoin, Vitamin K)
- Infection
- Favism: Food (fava beans)
- In neonates: can present as prolonged, pathologic neonatal jaundice.

Investigations

- Neonatal screening
- G6PD assay : should not be done in acute crisis when reticulocyte count is high (reticulocytes have high G6PD levels)
- Blood film
 - Heinz bodies. Bite cells
 - May have features of intravascular hemolysis (e.g. RBC fragment)

Treatment

- Folic acid
- Stop offending drugs and avoid triggers
- Underlying infection should be treated
- Transfusion in severe cases
- Splenectomy is not usually helpful.



Hereditary Spherocytosis

Definition

- Most common type of hereditary hemolytic anemia

Pathophysiology

- Abnormality in RBC membrane proteins (e.g. spectrin)-increasing permeability
- Spleen makes defective RBCs more spherocytotic (and more fragile) by membrane removal; also acts as site of RBC destruction
- Autosomal dominant with variable penetrance

Manifestations

- The patient could be completely asymptomatic
- Jaundice at birth. can be delayed for many years.
- Anemia, splenomegaly, leg ulcers, and pigment gallstones.
- The course of the disease may be interrupted by (Aplastic or Megaloblastic anemia)

Investigations

- **Raised MCHC**, Blood film (shows **spherocytes**),
- Osmotic fragility (increased),
- Flow cytometry: **Detection of EMA negative RBCs**, in hereditary spherocytosis
- Molecular analysis for spectrin gene

Treatment

- Folic acid
- In severe cases, splenectomy and vaccination against pneumococcus, meningococcus, and H. influenza B (avoid in early childhood)

Hereditary Elliptocytosis

Autosomal dominant abnormality in spectrin interaction with other membrane proteins

- 25-75% elliptocytes and hemolysis is usually mild
- **Treatment** : Immunizations; splenectomy for severe hemolysis

5

Acquired Hemolytic anemia

Autoimmune Hemolytic Anemia (AIHA)

	Warm (75-90% cases)	Cold
Antibody Allotype	• IgG	• IgM
Agglutination Temp.	37°C	4-37°C
Direct Coombs Test	Positive for IgG ± complement	Positive for complement
Etiology	• Idiopathic • Secondary to lymphoproliferative disorder (e.g. CLL, Hodgkin lymphoma) • Secondary to autoimmune disease (e.g. SLE) • Drug-induced (e.g. penicillin, quinine, methyldopa)	• Idiopathic • Secondary to infection (e.g. mycoplasma pneumonia, EBV) • Secondary to lymphoproliferative disorder (e.g. macroglobulinemia, CLL)



	Warm (75-90% cases)	Cold
Blood Film	• Spherocytes	• Agglutination
Treatment	• Treat underlying cause • Corticosteroids • Immunosuppression as azathioprine and cyclophosphamide, may be effective in patients who fail to respond to steroids and splenectomy • Splenectomy • Folic acid • Rituximab (2nd line to steroids)	• Treat underlying cause • Warm patient/avoid cold • Rituximab regimens (1st line) • Plasma exchange (2nd line for high IgM levels) • Folic acid • Low dose alkylating agents (chlorambucil, cyclophosphamide) or interferon may be useful but less effective





Drug-induced immune hemolytic Anemia

Definition

- The interaction between a drug and red cell membrane produces a composite antigenic structure, provoking two types of antibodies:

Drug-dependent antibodies

- Bind to both the drug and the cell membrane but not to either separately.
- Clinically there is usually severe complement-mediated intravascular haemolysis.
- - Resolves quickly after withdrawal of the drug

Drug-independent antibodies

- Induced by change of the red cell membrane.
- There is extravascular hemolysis and the clinical course tends to be more prolonged.

This concept also applies to drug-induced thrombocytopenia and neutropenia.

Acquired membrane Defects

Paroxysmal nocturnal hemoglobinuria (PNH)

Definition

- PNH caused by **somatic mutations** in HSC.
- RBCs lacking **CD55 & CD59** allows increased binding of complement to the red cell, leading to **increased intravascular hemolysis** and so can develop into aplastic anemia and leukemia

Clinical features

- Anemia, jaundice
- Dark urine from intravascular hemolysis (hemoglobinuria).
- Thrombosis
 - Venous structures
 - Most common is Hepatic vein (Budd-Chiari syndrome)
- Renal damage from hemoglobin by **acute tubular necrosis**

Diagnosis

Flow cytometry

↓ **CD55 and CD59 on white and red cells levels are low or absent**

Treatment

- Prednisone is often given to slow the rate of RBCs destruction
- Acute thrombosis, thrombolytic therapy (streptokinase, TPAs)
- Long-term anticoagulation drugs to help prevent further blood clots
- PNH with bone marrow failure (antithymocyte globulin)
- Red cell and/or platelet transfusions
- Eculizumab

Allogeneic bone marrow transplantation has been the mainstay of curative therapy for PNH

Mechanical destruction of RBCs

- ❑ **Microangiopathic hemolytic anemia (MAHA)**
- ❑ **Macroangiopathic hemolytic anemia**
- ❑ **Other:**
 - **Toxin-mediated RBC destruction**, e.g., from a snake bite, oxidizing agents
 - RBC destruction by intracellular **pathogens**, e.g. in malaria, rickettsiosis
 - RBC destruction due to **hypersplenism**

Microangiopathic hemolytic anemia	Macroangiopathic hemolytic anemia
Primary MAHA • Thrombotic thrombocytopenic purpura (TTP) • Hemolytic uremic syndrome (HUS) Secondary MAHA ; causes include: • Autoimmune disease (e.g., SLE) • HELLP syndrome • Hypertensive emergency • Disseminated intravascular coagulation (DIC) • Drug induced (e.g. cyclosporine)	Congenital cardiovascular anomalies (e.g., bicuspid aortic valve, coarctation of the aorta) Moderate and severe aortic stenosis: Prosthetic heart valves can cause anemia. Extracorporeal circulation , dialysis Exertional hemoglobinuria ("march hemoglobinuria") RBC destruction in the feet during strenuous exercise (e.g., running on hard surfaces)

Thrombotic thrombocytopenic purpura (TTP) / Hemolytic uremic syndrome (HUS)

	TTP	HUS
Epidemiology	• Adult individuals with mortality 10–20% • Common in women	• Predominantly affects children • 10% mortality .
Etiology	1-ADAMTS 13 abnormalities 2-Secondary TTP • Cancer , HIV • Drugs	1-Shiga-liketoxin HUS <i>E.coli O157:H7</i> 2-Strep Pneumonia HUS 3-Shigella HUS
Clinical features	1- Fever 2- Neurological ▪ Delirium, Seizure, focal defects, stroke 3-Thrombocytopenia) 4- Microangiopathic hemolytic anemia 5- Impaired renal function	• History of diarrhea (usually bloody) followed by 1. Microangiopathic hemolytic anemia 2. Thrombocytopenia 3. Impaired renal function
Diagnostics test	Plasma ↓ ADAMTS13 activity ± inhibitors	Rule out infection causes e.g. Culture, ELISA or PCR for shiga toxin
Main Treatment	Plasma exchange therapy Glucocorticoids: Rituximab	Supportive treatment. Mild disease resolves spontaneously Do not give antibiotics

Do not transfuse platelets

Disorders that cause hemolytic anemia, thrombocytopenia and thrombosis

Differential diagnosis

- Thrombotic thrombocytopenic purpura (TTP)
- Hemolytic-uremic syndrome (HUS)
- Disseminated intravascular coagulation (DIC)
- Paroxysmal nocturnal hemoglobinuria (PNH)
- Vasculitis (such as systemic lupus erythematosus)
- Antiphospholipid antibody syndrome (APS)



6

Myeloproliferative disorders

Myeloproliferative diseases (MPDs) being clonal stem cell disorders characterized by leukocytosis, thrombocytosis, erythrocytosis, splenomegaly, and bone marrow hypercellularity.

Old classification: a group of disease characterized by overgrowth of one or more hematologic cell lines in BM.

- 1) Chronic myelogenous leukemia (CML).
- 2) Polycythemia vera (PV).
- 3) Essential thrombocythemia.
- 4) Agnogenic myeloid metaplasia/myelofibrosis.

2008 WHO classification

- Polycythemia vera
- Essential thrombocytosis
- Chronic neutrophilic leukemia
- Hyper eosinophilic syndrome
- Myelodysplastic/myeloproliferative diseases
- Chronic idiopathic myelofibrosis.
- Chronic myeloid leukemia (CML).
- Chronic eosinophilic leukemia
- Mast cell disease
- Chronic myelomocytic leukemia.

Myeloproliferative diseases are grouped together as there can be transition from one disease to another, e.g. PV can lead to myelofibrosis. They may also transform to acute myeloblastic leukemia.

Polycythemia rubra vera (PV)

Polycythemia rubra vera is one of the chronic myeloproliferative neoplasms (MPNs), which are collectively characterized by clonal proliferation of myeloid cells with variable morphologic maturity, distinguished clinically from the other MPNs by the presence of an elevated red blood cell mass. However, an increased red blood cell mass alone is insufficient to establish the diagnosis, since this is also observed in conditions associated with chronic hypoxia and with erythropoietin-secreting tumors.

PV: is defined as an increase in hemoglobin, packed cell volume (PCV) and red cell count.

PCV is a more reliable indicator of polycythemia than Hb, which may be disproportionately low in iron deficiency.



Polycythemia can be divided into:

- **Absolute erythrocytosis** where there is a true increase in red cell volume.
- **Relative erythrocytosis** where the red cell volume is normal but there is a decrease in the plasma volume.

Absolute erythrocytosis

- ❑ **Primary; Polycythemia vera.**
- ❑ **Secondary.**

A) Due to an appropriate increase in erythropoietin in response to hypoxia as

- High altitude.
- Lung disease.
- Cardiovascular disease (right-to-left shunt).
- Heavy smoking.
- Mutant high oxygen affinity haemoglobin, e.g. congenital polycythaemia.

B) Due to an inappropriate increase in erythropoietin

- Congenital polycythemia.
- Renal disease: tumors, cysts.
- Liver disease: hepatocellular carcinoma, cirrhosis.
- Endocrine: adrenal tumors.
- Tumors: cerebellar hemangioblastoma, massive uterine fibroma, bronchial carcinoma.
- Drugs: erythropoietin, androgens.

Polycythemia Vera (PV)

Definition: PV is a clonal stem cell disorder in which there is an alteration in the pluripotent progenitor cells leading to excessive proliferation of erythroid, myeloid and megakaryocytic progenitor cells.

- Over 95% of patients with PV have acquired mutations of the gene Janus Kinase 2 (JAK2).
- JAK2 is a cytoplasmic tyrosine kinase that transduces signals, especially those triggered by hematopoietic growth factors such as erythropoietin, in normal and neoplastic cells.



CLINICAL PICTURE:

- The onset is insidious.
- It usually presents in patients aged over 60 years
- Symptoms of hyperviscosity like tiredness, depression, vertigo, tinnitus and visual disturbance.
- Together with hypertension, angina, intermittent claudication and a tendency to bleed, are suggestive of PV.
- Severe itching after a hot bath or when the patient is warm is common due to release of histamine.
- Gout due to increased cell turnover may be a feature, and peptic ulceration occurs in a minority of patients.
- Thrombosis and hemorrhage are the major complications of PV.
- The patient is usually plethoric and has deep dusky cyanosis.
- Injection of the conjunctivae
- The spleen is palpable in 70% and is useful in distinguishing PV from secondary polycythemia.
- The liver is enlarged in 50% of patients.

INVESTIGATIONS:

- CBC; Hb and PCV are increased.
 - The WBC is raised in about 70% of cases of PV
 - The platelet count is elevated by about 50%.
- Bone marrow shows erythroid hyperplasia and increased numbers of megakaryocytes.
- JAK 2 kinase in > 95%.
- Plasma volume shows normal or increased values.
- Serum uric acid levels may be raised.

DIFFERENTIAL DIAGNOSIS:

When increased red cell volume is established, it should be differentiated mainly from secondary polycythemia.

- Raised WBC and platelet counts with splenomegaly makes a diagnosis of PV very likely.
- The principal secondary causes can often be excluded by history and examination, but a renal ultrasound, an arterial Po₂.



COURSE AND MANAGEMENT:

- Treatment is designed to maintain HCT (PCV) below 0.45 to prevent the complications of the disease.
- **There are three types of specific treatment:**
 - ❑ **Venesection.** This will successfully relieve many of the symptoms of PV.
 - ❑ **Iron deficiency** limits erythropoiesis.
 - ❑ **Chemotherapy:**
 - Hydroxyurea.
 - Interferon alfa.
- **General treatment:**
 - ❑ Allopurinol is given to block uric acid production.
 - ❑ Low-dose aspirin (100 mg daily) prevent thrombotic complications
 - ❑ H1-receptor antagonists for itching.
 - ❑ Patients with uncontrolled PV have a high operative risk; Polycythemia should be controlled before surgery.
 - ❑ In an emergency, reduction of the hematocrit by venesection and appropriate fluid replacement must be carried out.

Thrombocythaemia (ET)

Thrombocythemia (ET) is a myeloproliferative disorder closely related to PV, Patients have normal Hb levels and WBC but elevated platelet counts.

- At diagnosis the platelet count will usually be $>600 \times 10^9/L$ and may be as high as $2000 \times 10^9/L$ or rarely even higher.
- ET presents either symptomatically with thromboembolic or less commonly bleeding problems or incidentally (e.g. at a routine medical check).
- The diagnosis of ET is not straightforward as there is no global gold standard test. • The JAK2 mutation tests are useful in that the gene is mutated in about half of all cases of ET, confirming a myeloproliferative disorder.

CLINICAL MANIFESTATIONS:

- Headache, dizziness, visual changes or complications (eg, thrombosis, bleeding, first trimester fetal loss).
- Unlike in polycythemia vera, pruritus is very uncommon in ET.

Reactive high platelet counts

- Include autoimmune rheumatic disorders and malignancy.
- Individuals who have been splenectomized (for any reason, including trauma) sometimes have high platelet counts.



Major criteria of ET diagnosis

- Platelet count $\geq 450 \times 10^9 /L$
- Bone marrow biopsy showing proliferation mainly of the megakaryocyte lineage.
- Not meeting WHO criteria for BCR-ABL1 + CML, PV, PMF, myelodysplastic syndromes, or other myeloid neoplasms
- Presence of JAK2, CALR, or MPL mutation.

Minor criterion

- Presence of a clonal marker or absence of evidence for reactive thrombocytosis.

Diagnosis of ET requires meeting all 4 major criteria or the first 3 major criteria and the minor criterion WHO diagnostic criteria essential thrombocythemia.

TREATMENT

- Aspirin in low-risk patients
- Hydroxycarbamide (hydroxyurea), and α -Interferon is in high risk patients.
- Plateletpheresis in emergencies.

PROGNOSIS:

- ET may eventually transform into PV, myelofibrosis or but the disease may not
- progress for many years.

Myelofibrosis

DEFINITION:

- Increased fibrosis in the bone marrow is caused by hyperplasia of abnormal Megakaryocytes which release fibroblast-stimulating factors.
- There is clonal proliferation of stem cells and myeloid metaplasia in the liver, spleen and other organs.
- In about 25% of cases there is a preceding history of PV.

CLINICAL PICTURE

- The disease presents insidiously with lethargy, weakness and weight loss.
- Patients often complain of a 'fullness' in the upper abdomen due to splenomegaly.
- Severe pain related to respiration may indicate peri splenitis secondary to splenic infarction.
- Bone pain and attacks of gout can complicate the illness.
- Bruising and bleeding occur because of thrombocytopenia or abnormal platelet function.
- Other physical signs include anemia, opportunistic infections.



INVESTIGATIONS

- CBC: Anemia with leucoerythroblastic features is present. The WBC count may be over $100 \times 10^9 /L$, and its differential count may be very similar to chronic myeloid leukemia (CML); later leucopenia may develop.
- The platelet count may be very high, but in later stages, thrombocytopenia occurs.
- Blood film: tear drop shaped RBC (pass in fibrotic BM).
- Bone marrow aspiration is often unsuccessful, and this gives a clue to the presence of the condition.
- A bone marrow trephine is necessary to show the markedly increased fibrosis.
- Presence of JAK2, CALR, or MPL mutation.

DIFFERENTIAL DIAGNOSIS

1. CML:

- In both conditions there may be marked splenomegaly and a raised
- WBC count with many granulocyte precursors seen in the peripheral blood.
- The main distinguishing features are:
 - a. the appearance of the bone marrow
 - b. the absence of the Philadelphia chromosome in myelofibrosis.

B) Fibrosis of the marrow:

often with a leucoerythroblastic anaemia, can also occur secondarily to leukaemia or lymphoma, tuberculosis or malignant infiltration with metastatic carcinoma, or to irradiation

TREATMENT:

- The only cure is stem cell transplantation.
- General supportive measures such as blood transfusion, PLT transfusion, folic acid, analgesics and allopurinol.
- Drugs such as tyrosine kinase inhibitors
- Hydroxyurea is used in prefibrotic stage to reduce metabolic activity and high WBC count and platelet levels.
- If the spleen becomes very large and painful, it may be advisable to perform splenectomy.

PROGNOSIS

- Patients may survive for 10 years or more; median survival is 3 years.
- Death may occur in 10-20% of cases from transformation to AML
- The most common causes of death are cardiovascular disease, infection and gastrointestinal bleeding.

D

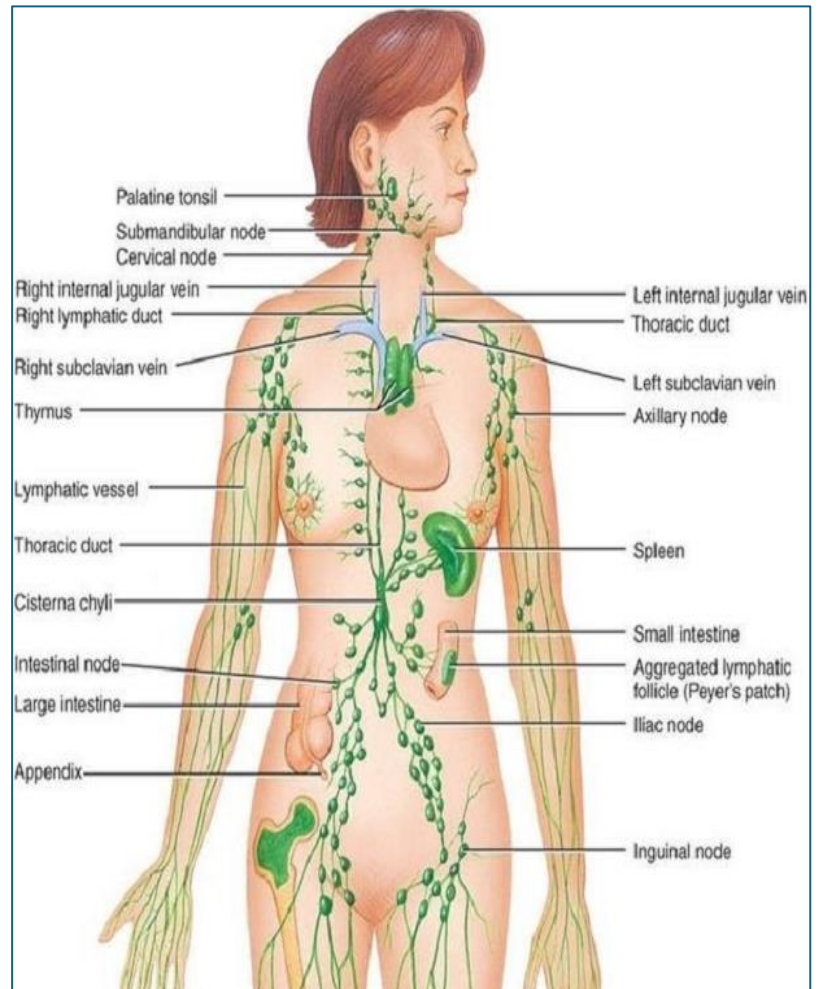
Lymphadenopathy and splenomegaly

Introduction

- The lymphatic system is the part of the immune system comprising lymphatic vessels that carry a clear fluid called lymph (from Latin lymph a "water") in a unidirectional pathway.
- The components of the lymphatic system are: -
 - **Lymph**, the recovered fluid
 - **Lymphatic vessels**, which transport the lymph
 - **Lymphatic tissue**, composed of aggregates of lymphocytes and macrophages that populate many organs of the body
 - **Lymphatic organs**, in which these cells are especially concentrated and which are set off from surrounding organs by connective tissue capsules.

Definitions

- **Lymphadenopathy:** refers to lymph nodes that are abnormal in size and consistency, Whether because of normal reactive process or pathology.
(Abnormalities may be localized or generalized)
- **Generalized lymphadenopathy** is defined as: →Enlargement of ≥ 2 non-contiguous lymph node groups
- **Localized lymphadenopathy** If: →Enlargement of a single node or multiple contiguous nodal regions.
- **Lymphoma** is a general term used for proliferations that arise as discrete tissue masses.
- **Leukemia** is used for neoplasms that present with widespread involvement of the bone marrow and the peripheral blood(usually) .





7

Hodgkin lymphoma

Lymphoma

Lymphoma is a term given to tumors of the lymphoid system and specifically of lymphocytes and their precursor cells.

Hodgkin lymphoma:

Cancers that originate from lymphatic systems. • It was named after Thomas Hodgkin who first described it in 1832. • Dorothy Reed and Carl Sternberg first described the malignant cells of Hodgkin lymphoma call Reed Sternberg cells.

Epidemiology:

Hodgkin disease has bimodal age distribution--one peak in the 20s and 60s.

- Early peak middle to late 20s
- Second peak after 50 yr
- men > women
- whites > blacks > Asians.

Risk factors, incidence:

No clear risk factors, several implicated

- EBV (pathogen or passenger)
- HIV
- Rare familial aggregations. First degree relatives have five fold increase in risk for Hodgkin lymphoma.
- Associated with EBV infection mainly with mixed cellularity type
- High socio-economic status.
- Prolonged use of human growth hormone.

Clinical presentations:

The most common presentation is asymptomatic lymph node enlargement, typically in the neck.

- Cervical lymph nodes are involved in 80% cases, mediastinal involvement is seen in about 50% cases, they produce symptoms like chest pain, cough and dyspnoea.
- Infradiaphragmatic involvement is seen in 5% cases and usually seen with older patients.

- Other less common manifestations are: Pruritis, nephrotic syndrome, erythema nodosum, cerebellar degeneration, immune hemolytic anaemia, thrombocytopenia, hypercalcemia.
- Hepatosplenomegaly

B symptoms

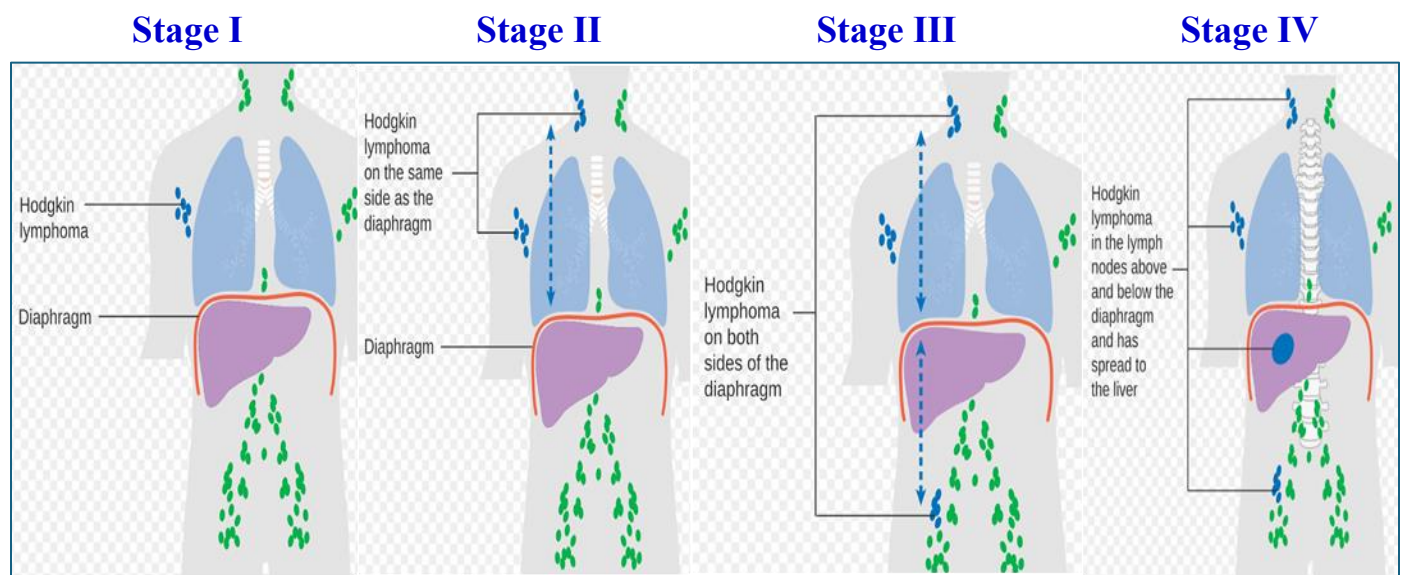
Only 15-20% of stage I-III have B symptoms like

- Fever($>38^{\circ}\text{C}$) may first present as fever of unknown origin, fever persists for days to weeks followed by afebrile intervals and then recurrence this pattern is called Pel Ebstein fever.
- Drenching night sweats.
- Weight loss ($>10\%$ in 6 months).

Classification for Hodgkin Disease

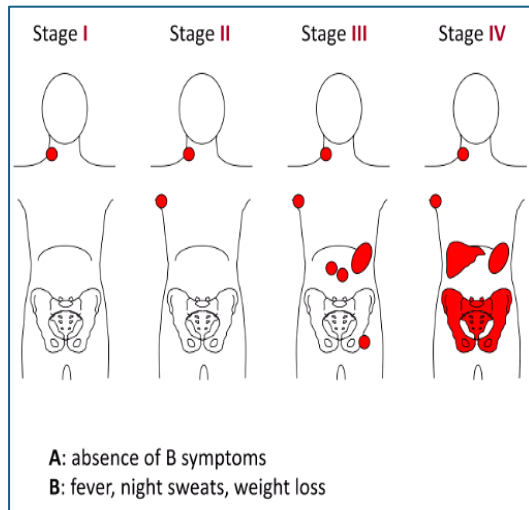
Ann Arbor staging system

- **Stage I** is involvement of a single lymph node region (I) (mostly the cervical region) or single extralymphatic site (Ie);
- **Stage II** is involvement of two or more lymph node regions on the same side of the diaphragm (II) or of one lymph node region and a contiguous extralymphatic site (IIe);
- **Stage III** is involvement of lymph node regions on both sides of the diaphragm, which may include the spleen (III_s) or limited contiguous extra lymphatic organ or site (III_e, III_{es});
- **Stage IV** is disseminated involvement of one or more extra lymphatic organs.



The absence of systemic symptoms is signified by adding "A" to the stage; the presence of systemic symptoms is signified by adding "B" to the stage. For localised extranodal extension from mass of nodes that does not advance the stage, subscript "E" is added. Splenic involvement is signified by adding "S" to the stage. The inclusion of "bulky disease" is signified by "X".

Ann Arbor staging system



WHO Histopathological Classification Of HL

	Histologic Subtypes
	Nodular lymphocyte predominant Hodgkins lymphoma (NLPHL)
	Classical Hodgkins lymphoma(CHL)
1	Nodular sclerosis Hodgkins lymphoma
2	Lymphocyte rich classical Hodgkins lymphoma
3	Mixed cellularity Hodgkins lymphoma
4	Lymphocyte depletion Hodgkin lymphoma

Diagnosis

History Taking:

- Swellings.
- B symptoms.
- Previous infection.
- Immune deficiencies.
- Exposure to toxins.
- Familial.

Examination:

- Lymph node (size, number, location, consistency, fixed or not).
- Liver and spleen (size)
- Skin involvement
- Pulmonary findings
- Neurological signs

Laboratory:

- CBC with differential count and blood film - Normal/Lymphopenia/lymphocytosis.
- ESR Usually raised/ Indicator of disease activity.
- Liver Biochemistry is often abnormal with or without liver involvement. Cholestatic Jaundice in LN of porta hepatis involvement.
- Serum LDH -Raised level is adverse prognostic factor.
- Serum uric acid -Normal or raised (Tumor lysis syndrome).
- Renal function and Electrolytes.
- Virology: HIV, hepatitis B and C -when potentially immunosuppressive therapy is to be recommended.
- Fertility (Pregnancy test if married female).

Imaging:

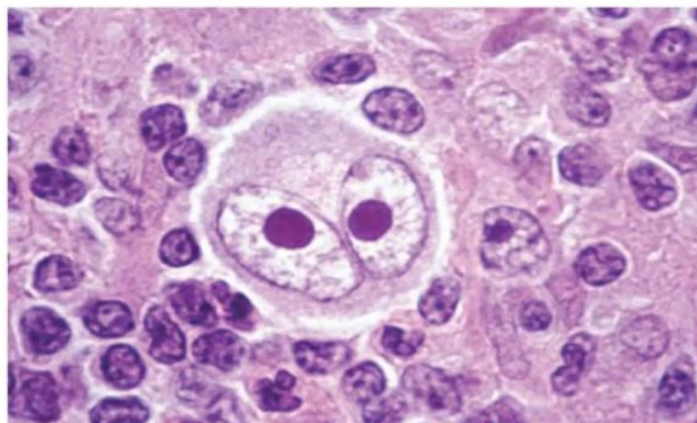
- Chest Xray for mediastinal widening.
- CT scan of chest, abdomen, pelvis $\pm$ neck It is the investigation of choice for staging.
- PET scan (It is the examination of choice for detecting disease activity during and after treatment, to stage a patient accurately and to establish sites of extranodal disease.)
- Cardiac evaluation (ECG, Echo).

Histopathological diagnosis:

Lymph node biopsy is required for a definitive diagnosis.

- The hallmark of HL is the Reed–Sternberg cell. (Large malignant B cells have two mirror-image nuclei “owleye” appearance and surrounded by large numbers of reactive nonmalignant T cells, plasma cells and eosinophils).
- CD30 and CD15 are almost always expressed in the majority of cases of classical HL.

Reed–Sternberg cell





BM examination:

-BMA & BMB are less commonly needed as overall BM involvement in HL is less than 5% so it is recommended only for patients with:

- Advanced stage disease (stage III, IV),
- B symptoms, or • Any abnormality on CBC that is suspicious for bone marrow involvement (anemia).
- Clinical evidence of subdiaphragmatic disease.
- Recurrence.

Treatment

- Specific treatment is based on the anatomical distribution of disease, its 'bulk' and 'B' symptoms.
- 90% of patients with early stage achieve complete remission, while 50-70% of those with advanced disease can be cured.

Treatment methods

- Radiotherapy.
- Chemotherapy.
- combined treatment modality
- Hematopoietic stem cell transplantation.

Early stage (IA, IIA + no bulk)

Chemotherapy ABVD ➤ (doxorubicin, bleomycin, vinblastine, and dacarbazine) ➤ 2-4 cycles followed by involved field irradiation.

Advanced disease (stages IB, IIB, or any stage III or IV or bulky disease)

- This is also curable for a significant proportion of patients, the median survival exceeding 5 years for 50-60 %.
- Cyclical combination chemotherapy with or without irradiation to sites of 'bulk' disease is the treatment of choice for all these patients.
- The 'gold standard' combination is ABVD given to a total of 6-8 cycles.
- All patients with mediastinal bulk receive irradiation.
 - the major potential toxicity in the short term is myelosuppression and mucositis with a mortality of ~ 1%.
 - long term risks → heart and lungs affection, infertility and second malignancy.
- Autologous bone marrow transplant may be considered in resistant and relapsing disease.



8

Non Hodgkin lymphoma

Non-Hodgkin's Lymphoma (NHL) represents a monoclonal proliferation of lymphoid cells of B cell origin (80%) or T cell origin (20%).

Incidence

- The incidence of NHL is 5- 15/100000 per year in developed countries, the incidence has increased over the last 20–30 years.
- 3rd most frequently diagnosed cancer in males and 4th in females.
- There is a slight male predominance.
- The median age of presentation is 55–75 years.

Etiology

NHLs may result from chromosomal translocations, infections, environmental factors, immunodeficiency states, and chronic inflammation.

a) Chromosomal translocations

- Chromosomal translocations and molecular rearrangements play an important role in the pathogenesis of many lymphomas.
- Example: The t (14;18) translocation is the most common chromosomal abnormality associated with NHL. This translocation occurs in 85% of follicular lymphomas and 28% of higher-grade NHLs.

b) Infections

- Some viruses are implicated in the pathogenesis of NHL as Epstein-Barr virus (EBV) which is associated with Burkitt lymphoma and lymphomas in immunocompromised patients.
- Also, Herpesvirus, Hepatitis C virus (HCV) and the human T cell lymphotropic virus are associated with NHL.
- Helicobacter pylori is an etiological factor in gastric MALT lymphoma.

c) Environmental factors

- Chemicals (eg, pesticides, herbicides, solvents, organic chemicals, hair dye),
- Chemotherapy
- Irradiation exposure

d) Immunodeficiency states

- Congenital and acquired immunodeficiency states
- Induced immunodeficiency states (e.g. immunosuppression)
- Celiac disease has been associated with an increased risk of malignant lymphomas.

e) Chronic inflammation

- Autoimmune disorders, such as Sjogren syndrome and Hashimoto thyroiditis, promotes the development of MALT.

Classification

- The NHLs are subclassified according to the cell of origin (T or B lineage) and the stage of lymphocytic maturation at which they develop (precursor or mature).
- NHL includes many subtypes, each with distinct epidemiology; etiology; morphology, immunophenotypic, genetic, and clinical features; and responses to therapy.

**TABLE
73.1**

World Health Organization Classification of Lymphomas^a

Mature B-Cell Neoplasms <i>Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma</i> B-cell prolymphocytic leukemia Splenic B-cell marginal zone lymphoma Hairy cell leukemia <i>Splenic B-cell lymphoma/leukemia, unclassifiable</i> <i>Splenic diffuse red pulp small B-cell lymphoma</i> <i>Hairy cell leukemia-variant</i> Lymphoplasmacytic lymphoma Waldenström macroglobulinemia Heavy Chain Diseases Alpha heavy chain disease Gamma heavy chain disease Mu heavy chain disease	High grade B-cell lymphomas, with <i>MYC</i> and <i>BCL2</i> and/or <i>BCL6</i> rearrangements B-cell lymphoma unclassifiable, with features intermediate between diffuse large B-cell lymphoma and classic Hodgkin lymphoma
<i>Plasma Cell Myeloma</i> Solitary plasmacytoma of bone Extramedullary plasmacytoma	Mature T-Cell and NK-Cell Neoplasms T-cell prolymphocytic leukemia T-cell large granular lymphocytic leukemia <i>Chronic lymphoproliferative disorder of NK-cells</i> Aggressive NK leukemia Systemic EBV-positive T-cell lymphoma of childhood Hydroa vacciniforme-like lymphoma Adult T-cell leukemia/lymphoma Extranodal NK/T-cell lymphoma, nasal type Enteropathy-associated T-cell lymphoma Monomorphic epitheliotropic intestinal T-cell lymphoma Hepatosplenic T-cell lymphoma Subcutaneous panniculitis-like T-cell lymphoma
<i>Extranodal Marginal Zone Lymphoma of Mucosa-Associated Lymphoid Tissue (MALT Lymphoma)</i> Nodal Marginal zone lymphoma <i>Pediatric nodal marginal zone lymphoma</i>	<i>Mycosis Fungoides</i> Sézary syndrome Primary cutaneous CD30-positive T-cell lymphoproliferative disorders Lymphoid papulosis Primary cutaneous anaplastic large cell lymphoma Primary cutaneous gamma-delta T-cell lymphoma Primary cutaneous CD8-positive aggressive epidermotropic cytotoxic T-cell lymphoma Primary cutaneous CD4-positive small/medium T-cell lymphoproliferative disease
<i>Follicular Lymphoma</i> <i>Pediatric-type follicular lymphoma</i> Primary cutaneous follicle center lymphoma	<i>Peripheral T-Cell Lymphoma, NOS</i> <i>Angioimmunoblastic T-Cell Lymphoma</i> <i>Anaplastic Large Cell Lymphoma, ALK-Positive</i> Anaplastic large cell lymphoma, ALK-negative
<i>Mantle Cell Lymphoma</i> <i>Diffuse Large B-Cell Lymphoma (DLBCL), NOS</i> T-cell/histiocyte-rich large B-cell lymphoma Primary DLBCL of the CNS Primary cutaneous DLBCL, leg type EBV-positive DLBCL DLBCL associated with chronic inflammation Lymphomatoid granulomatosis Primary mediastinal (thymic) large B-cell lymphoma Intravascular large B-cell lymphoma ALK-positive large B-cell lymphoma Plasmablastic lymphoma HHV8-positive diffuse large B-cell lymphoma & primary effusion lymphoma Burkitt lymphoma	Hodgkin Lymphoma <i>Nodular Lymphocyte Predominant Hodgkin Lymphoma</i> <i>Classic Hodgkin Lymphoma</i> <i>Nodular Sclerosis Hodgkin Lymphoma</i> Lymphocyte-rich classic Hodgkin lymphoma Mixed cellularity classic Hodgkin lymphoma Lymphocyte-depleted classic Hodgkin lymphoma

^aMost common entities are underlined. Provisional entities are in italics. Some rare entities or variants are omitted. (See reference 2 for complete list.)

ALK, Anaplastic lymphoma kinase; CNS, central nervous system; EBV, Epstein-Barr virus; HHV-8, human herpesvirus-8; NK, natural killer; NOS, not otherwise specified.

Classification

Mature B cell lymphomas	Mature T-cell and natural killer (NK) cell lymphomas
• Diffuse large B cell (37%) • Follicular (29%) • MALT lymphoma (9%) • CLL/SLL (12%) • Mantle cell lymphoma (7%) • Burkitt lymphoma (0.8%)	• Primary T cell lymphoma NOS (29%) • Angioimmunoplastic lymphoma (18%) • Extranodal T cell or NK lymphoma (10%) • Adult T-cell lymphoma or leukemia • Anaplastic large cell lymphoma • Primary cutaneous lymphoma

Clinical features

Peripheral lymphadenopathy

- H/O of Multiple body swellings at anatomical sites of L.Ns.
- Criteria → painless, progressive lymph node enlargement in most of cases
- Pain at sites of nodal disease, precipitated by drinking alcohol, occurs in fewer than 10% of patients
- Unlike HL, NHL is often widely disseminated or multicentric at presentation
- NHL is more likely to be advanced stage at presentation

B symptoms

- More common than in Hodgkin lymphoma
- Their presence indicates advanced stage and poor prognosis.
 - Fever(>38°C)
 - May first present as fever of unknown origin.
 - Fever persists for days to weeks followed by afebrile intervals and then recurrence. This pattern is called Pel-Ebstein fever.
 - Recurrent drenching night sweats
 - Unexplained significant Weight loss (>10% in 6 months or less)

Extra nodal involvement

- More common than in Hodgkin lymphoma
- In more than one third of patients especially in high grade NHL
- May be present at the initial presentation
- The most common extra nodal sites → GIT, GU, CNS, lung, bone marrow, thyroid, and skin

Compression symptoms may occur (Gut obstruction, ascites, SVC obstruction and spinal cord compression).



Diagnosis

A. History Taking

- Swellings (peripheral lymphadenopathy)
- B symptoms
- Previous infection
- Immune deficiencies
- Exposure to toxins
- Familial

B. Examination

- Lymph node (size, number, location, consistency, fixed or not, tender or not, skin overlying) → large, hard, fixed, amalgamated
- Hepatosplenomegaly
- Skin involvement → erythema nodosum
- Pulmonary findings
- Neurological signs

C. Laboratory investigations

- CBC with differential count and blood film
 - Normal/ NNA/Lymphopenia/Lymphocytosis/Eosinophilia.
- ESR: Usually raised/ Indicator of disease activity.
- Liver Biochemistry
 - Often abnormal with or without liver involvement.
 - Cholestatic Jaundice in LN of porta hepatis involvement.
- Serum LDH
 - Raised level is adverse prognostic factor.
- Serum uric acid
 - Normal or raised (Tumor lysis syndrome).
- Renal function and Electrolytes
- Virology: HIV, hepatitis B and C.
 - HIV predisposes to NHL.
 - when potentially immunosuppressive therapy is to be recommended.

D. Imaging

- Chest X-ray for mediastinal widening
- CT scan of chest, abdomen, pelvis ± neck
It is the investigation of choice for staging

- PET scan a functional imaging technique
 - It is the investigation of choice for →
 - 1- detecting disease activity during and after treatment,
 - 2- detect relapse
 - 3- accurate staging of the patient and
 - 4- establish sites of extra nodal disease
- Cardiac evaluation (ECG, Echo.)

E. Histopathological diagnosis:

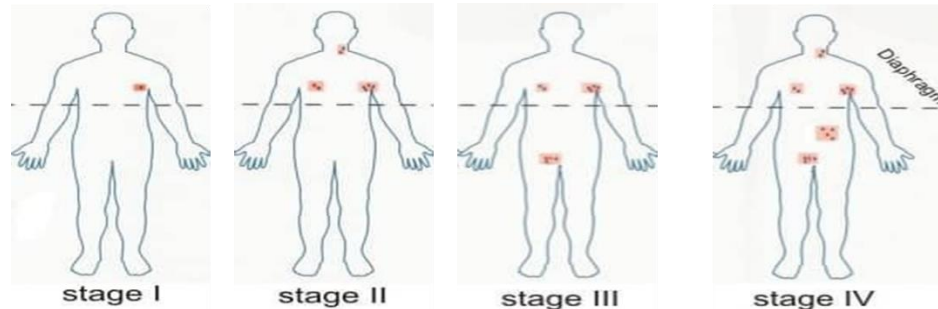
- Lymph node or extra nodal disease biopsy, is required for a definitive diagnosis → the gold standard investigation
- Immunophenotyping: CD30 → in anaplastic large cell lymphoma

F. BM examination:

- BM aspiration & biopsy are routinely recommended mainly for staging rather than for diagnosis.

G. Staging (Ann Arbor staging system)

- The same Ann Arbor staging system of Hodgkin Disease
- A and B for each stage according to absence or presence of B symptoms



H. Grading

	High grade NHL (85%)	Low grade NHL (15%)
Onset	Rapidly growing	Slowly growing
B symptoms	Common	Uncommon
Extra-nodal involvement	Common Affecting GIT, GU, CNS, lung, skin, thyroid	Uncommon
BM Infiltration	Uncommon	Common



I. Treatment

□ Low-grade NHL:

- Asymptomatic patients may not require treatment as there is no survival advantage with immediate treatment versus a watch-and-wait approach.
- Indications of treatment:
 - 1- Marked systemic symptoms.
 - 2- Compression symptoms.
 - 3- Bone marrow infiltration.
 - 4- Lymphadenopathy causing discomfort.
- Options of therapy:
 - Radiotherapy for stage I
 - Chemotherapy → Oral chlorambucil, cyclophosphamide
 - Chemotherapy in combination with Rituximab (monoclonal antibody therapy) to target surface antigens on tumor cells → for stage III & IV
 - BM transplantation in patients with relapsed disease.

□ High-grade NHL:

- Chemotherapy, commonly CHOP regimen in combination with Rituximab (**R-CHOP**) → first line treatment
 - CHOP → Cyclophosphamide,
Doxorubicin (Hydroxydaunorubicin),
Vincristine (Oncovin),
Prednisolone
- Radiotherapy: for a residual localized bulky disease after chemotherapy and for spinal cord compression.
- BM transplantation: in patients with relapsed or refractory disease

SPLENOMEGALY

- ✚ **Massive splenomegaly:** Beyond umbilicus, crosses midline, >8 cm below left costal margin and or >1000gm weight
- ✚ **Moderate splenomegaly:** Between costal margin & umbilicus 4-8 cm below left costal margin
- ✚ **Mild splenomegaly:** Just palpable 1-3 cm below left costal margin

Causes of splenomegaly

1) Increased function

a) Removal of defective RBCs

- Spherocytosis
- Thalassemia
- Early sickle cell anemia

b) Extramedullary hematopoiesis

- Myelofibrosis
- Marrow infiltration by leukemias, lymphomas, metastatic tumors
- Marrow damage by radiation and toxins

c) Immune hyperplasia

1. Response to infection (viral, bacterial, fungal, parasitic)

- Viral → infectious mononucleosis, AIDS, viral hepatitis
- Bacterial → TB, typhoid fever, brucellosis, leptospirosis
subacute bacterial endocarditis, bacterial sepsis
splenic abscess
- Parasitic → malaria, bilharziasis, leishmaniasis

2. Disordered immunoregulation

- Rheumatoid arthritis, including cases of Felty's syndrome
- Systemic lupus erythematosus
- Serum sickness
- Autoimmune hemolytic anemia
- Sarcoidosis



2) Abnormal blood flow

- a) Organ Failure → liver cirrhosis & portal hypertension
- b) Vascular → hepatic vein obstruction, portal vein obstruction,
Budd–Chiari syndrome & splenic vein obstruction
- c) Infections → hepatic schistosomiasis

3) iii) Infiltration

a) Metabolic diseases

- Gaucher disease & Niemann–Pick disease
- Amyloidosis

b) Benign and malignant “infiltrations”

- Leukemias (acute, chronic, lymphoid, and myeloid)
- Lymphomas (Hodgkin's and non-Hodgkin's)
- Myeloproliferative disease
- Metastatic tumors (commonly melanoma)
- Hemangioma, lymphangioma
- Splenic cysts
- Hamartomas
- Eosinophilic granuloma

Causes of massive splenomegaly

- Bilharzial splenomegaly
- Thalassemia major
- Myeloproliferative disorders
 - Polycythemia vera
 - Essential thrombocytosis
 - Primary myelofibrosis
 - Chronic myeloid leukemia
- Non-Hodgkin lymphoma
- Hairy cell leukemia
- Chronic malaria (tropical splenomegaly)
- kala-azar (visceral leishmaniasis)
- Sarcoidosis & storage diseases



Transfusion medicine is a specialized field of healthcare that focuses on the safe and effective use of blood and blood products in medical treatments. It involves the collection, testing, processing, storage, and transfusion of blood components to patients in need. Transfusion medicine plays a critical role in various medical conditions, including trauma, surgery, anemia, and hematologic disorders, ensuring that blood products are matched and administered to enhance patient outcomes while minimizing risks of complications.

9**Blood Transfusion****Definition**

Safe transfusion of the whole blood or its components such as blood cells or plasma from a donor to a recipient.

- ☐ Blood should only be given if strictly necessary and there is no alternative.

Purpose

- ☐ Restore blood volume.
- ☐ Replace clotting factors.
- ☐ Improve oxygen carrying capacity.
- ☐ Restore blood elements that are depleted.
- ☐ Prevent complications.
- ☐ To raise the hemoglobin level.

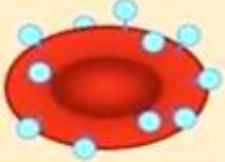
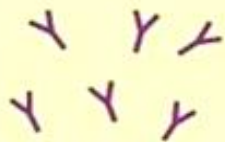
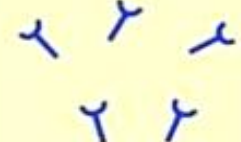
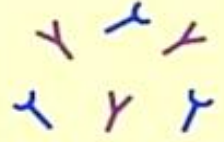
Blood grouping and systems:

- The clinical significance of blood groups in blood transfusion is that individuals who lack a particular blood group antigen may produce antibodies reacting with that antigen which may lead to a transfusion reaction.
- The blood groups, and systems determined by antigens on the surface of red cells.
- Approximately 400 red blood cell group antigens have been described.
- The ABO and Rh systems are the two major blood grouping systems.
- There are some other systems other ABO. P, Lewis and MN system are not uncommon, and they are of no clinical consequence.
- This blood group system (ABO) involves naturally occurring anti-A and anti-B antibodies (IgM), which can produce rapid and severe intravascular hemolysis of incompatible red cells. They react optimally at cold temperatures (4°C) so, are called cold antibodies.



- IgG antibodies are called warm antibodies, as they react optimally at 37°C.
- Only IgG antibodies are capable of trans placental passage from mother to fetus and the most important immune antibody is the Rh antibody, (anti-D).
- Rh antibodies rarely occur naturally and are therefore immune antibodies that result from previous transfusion or pregnancy.
- Anti-D is responsible for most of the clinical problems associated with the system and a simple subdivision of subjects into Rh D+ and
- Rh D– using anti-D antibody is sufficient for routine clinical purposes.

BLOOD GROUPS

ABO Blood Groups				
Antigen (on RBC)				
Antibody (in plasma)				
Blood Type	Type A Cannot have B or AB blood Can have A or O blood	Type B Cannot have A or AB blood Can have B or O blood	Type AB Can have any type of blood Is the universal recipient	Type O Can only have O blood Is the universal donor

The Rhesus system

- The red blood cell membrane antigen important here is the Rhesus (Rh) antigen, or Rhesus factor.
- About 85% of people have this antigen; they are Rhesus positive (Rh+) and therefore not make anti-Rhesus antibodies. The remaining 15% have no Rhesus antigen (they are Rhesus negative, or Rh-) Rh- individuals can make anti Rhesus antibodies but are stimulated to do so only in certain circumstances, e.g. in pregnancy, or as the result of an incompatible blood transfusion.



Blood donation:

- In most countries blood donors contribute on a voluntary basis and this is generally preferable in terms of product safety.
- The measures to protect donors and for donor selection must be considered.
- Cross matching and pre-transfusions tests:
 - Several steps are taken to ensure that patients receive compatible blood at the time of transfusion.

From the donor

- An appropriate ABO and Rh unit are selected.
- Routine screening of all blood donations (HCV, HBV, HIV).

From the patient

- The ABO and Rh blood group is determined.
- Serum is screened for important antibodies by an indirect antiglobulin test on a large panel of anti-genically typed group O red cells.
- If a red cell alloantibody is discovered in the recipient, donor blood is selected lacking the relative antigen.
- N.B: In acute hemorrhage, use cross matched blood, if possible, but if not, use universal donor' group O Rh-ve blood, changing to cross matched blood as soon as possible.

Blood products and indications of transfusion:

- ☐ Whole blood: The only option for the beginning years of transfusion history, but now rarely used.
 - ☐ Packed RBCs: Use to correct anemia or blood loss and in patients with heart failure (Give each unit over 4hours with furosemide).
 - ☐ Platelets Concentrates: Usually only needed if bleeding or count is <20,000 or platelet function defects.
 - ☐ Fresh frozen plasma (FFP): Use to correct clotting defects:
 - DIC; warfarin over dosage where vitamin K would be too slow; liver disease; thrombotic thrombocytopenic purpura. It is expensive and carries all the risks of blood transfusion. Do not use as a simple volume expander.
- ☐ Human albumin solution is produced as 4.5% or 20% protein solution.
4.5% is used to replace protein and this is a useful as a plasma volume expander. It is also used for fluid replacement in patients undergoing plasma pheresis.
20% albumin can be used temporarily in the hypoproteinemia patient (e.g. liver disease; nephrosis) who is fluid overloaded, without giving an excessive salt load.



Cryoprecipitate (a source of fibrinogen):

This is obtained by thawing FFP at 4°C and contains concentrated factor VIII, Von Willebrand factor (vWF), and fibrinogen. Its main use is in fibrinogen replacement in patients with massive hemorrhage or consumptive coagulopathy (DIC).

Granulocyte concentrate:

These are prepared from normal healthy donors or from patients with chronic myeloid leukemia. They have been used in patients with severe neutropenia ($<0.5 \times 10^9 /L$) who are not responding to antibiotic therapy, but it is not usually possible to give sufficient amounts.

Blood donor

- ☐ Shall be free of diseases
- ☐ No history of any diseases.
- ☐ They have not donated blood within the previous 90 days. (3 months).
- ☐ They should be physically.
- ☐ Donor must have a normal temperature, pulse and blood pressure.
- ☐ Their hemoglobin level must be above 12 grams per 100ml.

Hazards or complications of blood transfusion:

- Time related; Early or Late.
- Immunological. OR non immunological

Early (within 24h):

1. Acute hemolytic reactions (e.g. ABO or Rh incompatibility).
2. Anaphylaxis.
3. Allergic reactions (itch, urticaria, mild fever).
4. Bacterial contamination.
5. Febrile reactions (e.g. from HLA antibodies).
6. Fluid overload.
7. Transfusion-related acute lung injury (TRALI, i.e.; ARDS due to anti leucocyte antibodies).

Delayed (After 24h):

1. Infections (e.g. viruses: hepatitis B/C, HIV; bacteria; protozoa.prions).
2. Iron overload.
3. Graft-versus-host disease (GVHD).
4. Post-transfusion purpura.



Immunological (Alloimmunization and incompatibility)

1- Red cells

- ☐ Immediate hemolytic transfusion reactions.

Delayed hemolytic transfusion reactions.

2- Leucocytes and platelets

- ☐ Non-hemolytic (febrile) transfusion reactions
- ☐ Post-transfusion purpura
- ☐ Poor survival of transfused platelets and granulocytes Lung injury.

3- Plasma proteins

- ☐ Urticarial and anaphylactic reactions

Non-immunological complications

- ☐ Transmission of infection.
- ☐ Circulatory failure due to volume overload.
- ☐ Iron overload due to multiple transfusions.
- ☐ Massive transfusion of stored blood may cause bleeding reactions and electrolyte changes.
- ☐ Thrombophlebitis and Air embolism.

Acute hemolytic reactions:

- ☐ Immediate life-threatening or acute reactions are the result of complement activating antibodies of IgM or IgG classes, usually with ABO specificity.
- ☐ Clinical features include urticaria, pain in the lumbar region, flushing, headache, precordial pain, shortness of breath, vomiting, rigors, pyrexia, hypotension, jaundice, hemoglobinuria, and disseminated intravascular coagulation (DIC) may all become apparent.
- ☐ At the first suspicion of any serious transfusion reaction, the transfusion should always be stopped, and the donor units returned to the blood transfusion laboratory with a new blood sample from the patient to exclude a hemolytic transfusion reaction.
- ☐ The principal object of initial therapy is to maintain the blood pressure and renal perfusion.
- ☐ Hydrocortisone 100 mg intravenously and an antihistamine may help to alleviate shock.
- ☐ Intravenous dextran, plasma or saline and furosemide are sometimes needed.



Delayed hemolytic reactions:

- ☐ The antibody level is too low to be detected by pre transfusion compatibility testing, but a secondary immune response occurs after transfusion, resulting in destruction of the transfused cells, usually by IgG antibodies.
- ☐ Haemolysis is usually extravascular, and the patient may develop anemia and jaundice about a week after the transfusion.
- ☐ The blood film shows spherocytosis and reticulocytosis.
- ☐ The direct anti globulin (comb) test is positive.

Non-haemolytic febrile transfusion reactions:

- ☐ The most frequent reaction associated with transfusion of cellular blood components and are common complication in patients who have previously been transfused or pregnant women.
- ☐ The usual causes are the presence of leucocyte antibodies acting against donor leucocytes leading to release of pyrogens, or cytokines from donor leucocytes.
- ☐ They produce fever ($> 38^{\circ}\text{C}$), chills and rigors usually $\frac{1}{2}$ –1h after starting transfusion.
- ☐ Treatment by SLOW or STOP the transfusion.
- ☐ Monitor closely, and give an anti-pyretic, e.g. Paracetamol 1g.
- ☐ If recurrent, use leucocyte depleted (i.e. WBC filter) cells.

Anaphylactic shock:

- ☐ Is usually caused by hypersensitivity to donor plasma proteins.
- ☐ The clinical features are urticaria, pyrexia and, in severe cases, dyspnea brochospasm, cyanosis, facial edema and rigors.
- ☐ Immediate treatment is STOP the transfusion.
- ☐ Epinephrine (adrenaline) 0.5 mg i.m. and chlorphenamine 10 mg i.v. should be given immediately; endotracheal intubation may be required.

Urticarial reactions:

- ☐ are often attributed to plasma protein incompatibility, but in many cases, they are unexplained. They are common but rarely severe and treatment by Stopping or slowing the transfusion and administration of chlorphenamine 10 mg i.v.
- ☐ Patients who have severe urticarial or anaphylactic reactions should receive either washed red cells or autologous blood.



Post transfusion transmission of infections:

- ☐ Hepatitis may be caused by one of the hepatitis viruses, although CMV and Epstein–Barr virus (EBV) have also been implicated.
 - ☐ Post transfusion viral hepatitis, or HIV infection are very rarely seen because of routine screening of all blood donations.
- Other infections: Toxoplasmosis, malaria and syphilis may be transmitted by blood transfusion.

Massive blood transfusion:

- ☐ This is defined as replacement of an individual's entire blood volume ($>10U$) within 24h.
- ☐ Complications: platelets; Ca^{2+} ; clotting factors; increase K^{+} ; hypothermia.

Post-transfusion circulatory fluid overload:

- ☐ Dyspnea, hypoxia, tachycardia, raised JVP and basal crepitations.
- ☐ These reactions are prevented by a slow transfusion of packed red cells or of the blood component required, accompanied by diuretic therapy.

Treatment:

- ☐ LOW or STOP the transfusion.
- ☐ Give oxygen and a diuretic, e.g. furosemide 40mg IV initially.
- ☐ Consider CVP line.

Post-transfusion purpura:

- ☐ This is a rare problem of severe thrombocytopenia 7–10 days after transfusion of a platelet containing product, usually red cells.
- ☐ It is caused by an antibody in the recipient (resulting from previous transfusion or pregnancy) which is usually directed against a platelet-specific antigen HPA1a (PIA1).
- ☐ Both the transfused and recipient platelets are destroyed by the immune complexes.



Leukaemia divides into four main types depending on the cell line involved.

- 1. Acute lymphoblastic leukaemia (ALL)
- 2. Acute myeloid leukaemia (AML)
- 3. Chronic lymphocytic leukaemia (CLL)
- 4. Chronic myeloid leukaemia (CML)

10

Acute leukemia

Acute Myeloid Leukemia (AML)

Definition

Rapidly progressive malignancy characterized by failure of myeloid cells to differentiate beyond blast stage.

- **It progresses rapidly (100% death rate in 2 months)**
- **If untreated; ~20% 3yr survival after treatment).**

Epidemiology & Risk Factors

International statistics

- **The commonest acute leukemia of adults**
- incidence increases with age; median age of onset is 65 yr old; 80% of acute adult leukemias

Risk Factors

- Myelodysplastic syndromes (MDS),
- Benzene,
- Smoking
- Radiation,
- Down Syndrome,
- Alkylating agents as treatment for previous malignancy

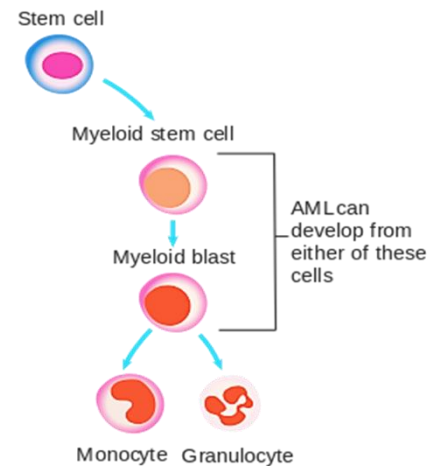
Pathophysiology

Uncontrolled growth of blasts in marrow leads to suppression of normal hematopoietic cells

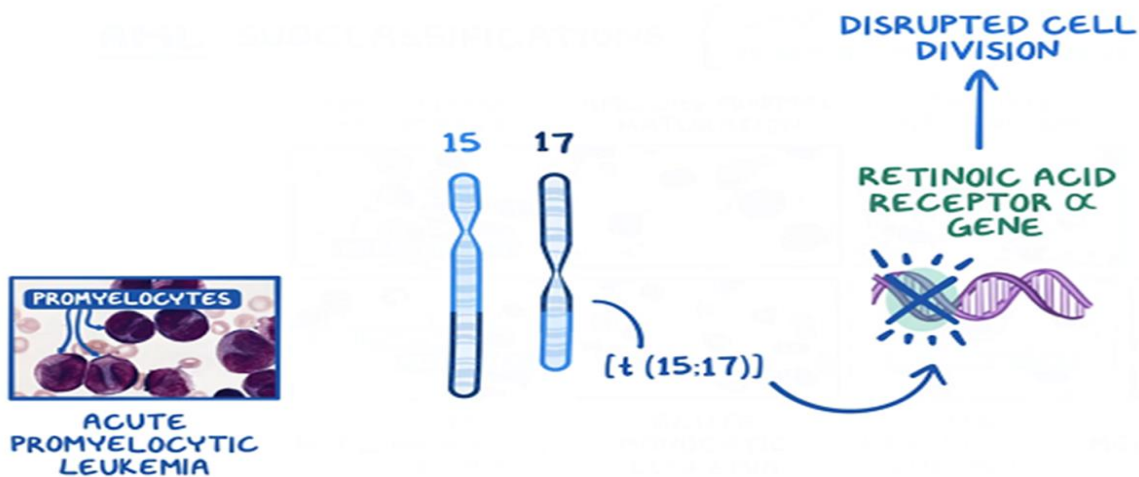
- Appearance of blasts in peripheral blood – risk of leukostasis
- Accumulation of blasts in other sites (e.g. skin, gums)
- Metabolic consequences; tumour lysis syndrome

Classification - French-American-British (FAB) (AML)

- M0 - undifferentiated
- M1 - without maturation
- M2 - with granulocytic maturation
 - the **most common** (25% of adult AML)
 - associated with a t(8;21) translocation.
- **M3 - acute promyelocytic (APL)**
 - has the **best prognosis** of all the subtypes of AML.
 - Unlike the other AML subtypes, APL is treated with all-trans retinoic acid (ATRA).
 - t(15;17)
- M4 - granulocytic and monocytic maturation
 - associated with a t(16;16) translocation
- M5 - monocytic
- M6 - erythroleukaemia
- M7 – megakaryoblastic



Acute Promyelocytic leukemia (M3)



Disseminated intravascular coagulation (DIC) is **common complication in AML-M3**



WHO classification of acute myeloid leukemia (FOR VIEWING ONLY)

Acute myeloid leukemia with recurrent genetic abnormalities
Acute myeloid leukemia with t(8;21)(q22;q22), (<i>AML1/ETO</i>)
Acute myeloid leukemia with abnormal bone marrow eosinophils and inv(16)(p13q22) or t(16;16)(p13;q22), (<i>CBFβ/MYH11</i>)
Acute promyelocytic leukemia with t(15;17)(q22;q12), (<i>PML/RARα</i>) and variants
Acute myeloid leukemia with 11q23 (<i>MLL</i>) abnormalities
Acute myeloid leukemia with multilineage dysplasia
Following MDS or MDS/MPD
Without antecedent MDS or MDS/MPD, but with dysplasia in at least 50% of cells in 2 or more myeloid lineages
Acute myeloid leukemia and myelodysplastic syndromes, therapy related
Alkylating agent/radiation–related type
Topoisomerase II inhibitor–related type (some may be lymphoid)
Others
Acute myeloid leukemia, not otherwise categorized
Classify as:
Acute myeloid leukemia, minimally differentiated
Acute myeloid leukemia without maturation
Acute myeloid leukemia with maturation
Acute myelomonocytic leukemia
Acute monoblastic/acute monocytic leukemia
Acute erythroid leukemia (erythroid/myeloid and pure erythroleukemia)
Acute megakaryoblastic leukemia
Acute basophilic leukemia
Acute panmyelosis with myelofibrosis
Myeloid sarcoma

Clinical Features

1. Bone Marrow failure:

- **Symptoms of**
 - A-Anemia (progressive weakness, fatigue, headaches, dyspnea),
 - B-Thrombocytopenia (petechia, ecchymoses, epistaxis,),
 - C-Infection, fever
- **DIC** occurs in acute promyelocytic leukemia, where there is release of thromboplastin.

2- Accumulation of blast cells in marrow:

- Skeletal pain, bony tenderness (especially sternum)

3. organ infiltration:

- Gingival hypertrophy (particularly myelomonocytic leukemia)
- Hepatosplenomegaly
- Lymphadenopathy
- Gonads (uncommon)
- Skin: leukemia cutis or myeloid sarcoma
- Eyes: hemorrhages and/or whitish plaques, Roth spots, cotton wool spots, vision changes

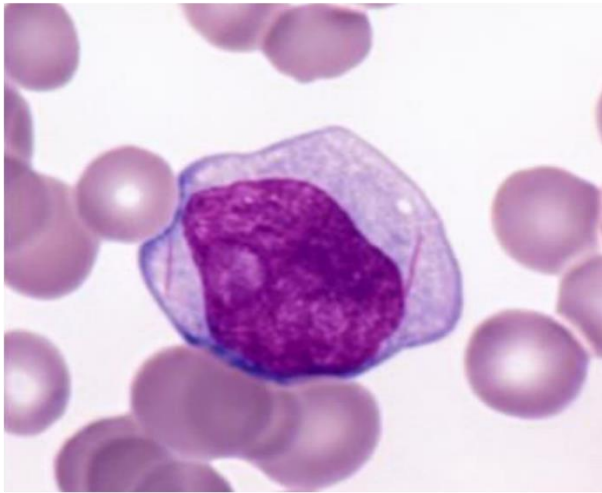
4- Metabolic effects;

- Increased uric acid → nephropathy, gout
- Release of phosphate → decreased Ca^{2+} , decreased Mg
- Release of procoagulants → DIC (higher risk in M3)
- Decreased or normal K before treatment, increased K after treatment (from lysed cells)

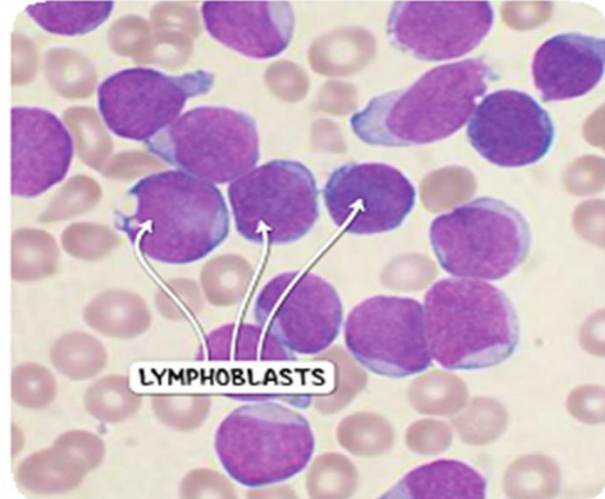
Investigations

A- To confirm diagnosis

1. **Full blood count:** Anemia, thrombocytopenia, variable WBC
2. **Blood film examination**
circulating blasts with Auer rods (azurophilic granules) are pathognomonic for AML
3. **Bone Marrow Aspiration and Biopsy for definitive diagnosis**
Hypercellular,-----Blast count: AML >20% (normal is <5%)



Auer rods (azurophilic granules) are pathognomonic for AML



Acute Lymphoblastic Leukemia

4. **Cytochemistry:** Special stains to differentiate AML from ALL; Positivity with Sudan black & Myeloperoxidase (MPO) in AML
5. **Immunophenotyping**
6. **Molecular genetics / Fluorescence in situ hybridization (FISH)**
7. **Cytogenetics:** Chromosomal abnormalities guide treatment and prognosis

B- To diagnosis the complications

1. INR, aPTT, fibrin degradation products (FDP), fibrinogen (in case of DIC)
2. Increased LDH, increased uric acid, increased PO₄ (released by leukemic blasts), decreased Ca, decreased K
3. Baseline renal and liver function tests
4. Chest X ray to rule out pneumonia, ECG, ECHO scan prior to chemotherapy (cardiotoxic)

Treatments

A- Supportive treatment of patients with AML

- **Transfusions:** Red blood cells transfusion - Platelets' transfusion
 - Use of irradiated cellular blood components for AML patients
 - Avoid blood products from sibling donors,



- **Infection** –
 - Broad spectrum IV antibiotics
- **Indwelling central venous catheter**
- **Screening for infection** via regular Culture and Sensitivity of urine, stool, sputum, oropharynx, catheter sites
- **IV fluids, and allopurinol** (prevents tumour lysis syndrome)

B-Chemotherapy

Induction chemotherapy to induce complete remission of AML

1. Very intensive, resulting in long periods of marrow suppression with neutropenia + decreased platelets.
2. The main drugs used include daunorubicin and cytarabine, with ~5 cycles given in 1-week blocks to get a remission

Consolidation: to prevent recurrence: intensive consolidation chemotherapy

C-Haemopoietic stem cell transplant in AML

1. Indicated in refractory, poor prognosis or relapsing disease.
2. The idea is to destroy leukemic cells and the immune system by, eg cyclophosphamide + total body irradiation, then repopulate the marrow with donor cells infused IV.

Haemopoietic stem cell transplant Types

- HLA identical sibling donor
- Unrelated donor
- Alternative donor: cord blood and haploidentical
- Syngeneic donor

Acute Lymphoblastic Leukemia (ALL)

Definition

- Malignant disease of the bone marrow in which early lymphoid precursors proliferate and replace the normal hematopoietic cells of the marrow

Epidemiology & Risk Factors

International statistics

- **It is the commonest cancer of childhood**
- Usually occurs before 14 years of age peak incidence is between 2-9 years of age.

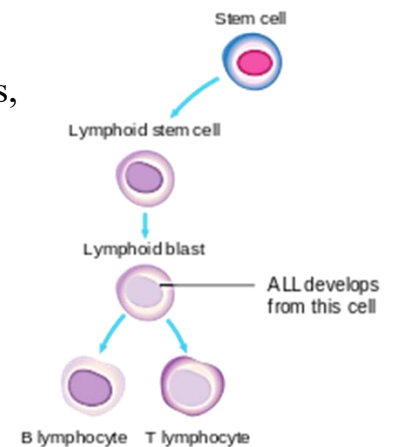
Risk Factors

- Ionizing radiation (eg X-rays) during pregnancy,
- Down's syndrome

Pathophysiology

Uncontrolled growth of blasts in marrow . A malignancy of lymphoid cells, affecting B- or T lymphocyte cell lineages, lead to

- Arresting maturation
- Promoting uncontrolled proliferation of immature blast cells,
- Marrow failure
- Tissue infiltration.



Classification - French-American-British (FAB) (ALL)

(L1, L2, L3), as morphology is not prognostic

WHO subdivides ALL into two types depending on cell of origin

1. B-cell: precursor B lymphoblastic leukemia
2. T-cell: precursor T lymphoblastic leukemia



Clinical Features

1. Bone Marrow failure:

- **Symptoms of**
 - A-Anemia (progressive weakness, fatigue, headaches, dyspnea),
 - B-Thrombocytopenia (petechia, ecchymoses, epistaxis,),
 - C-Infection, fever
- (opportunistic micro-organisms) Especially chest, mouth, perianal, and skin. Bacterial septicaemia, zoster, CMV, measles, candidiasis, Pneumocystis pneumonia

2. Organ infiltration:

Tender bones, lymphadenopathy, hepatosplenomegaly, orchidomegaly
Meningeal signs (headache, visual symptoms; especially in ALL relapses)

3. Leukostasis:

If WBC Count is $>100 \approx 10^9/L$, WBC thrombi may form in brain, lung, and heart. Avoid transfusion before lowering WBCs, eg with hydroxycarbamide or leukapheresis, as viscosity rises.

4- Metabolic effects.

- Increased uric acid $\rightarrow$ nephropathy, gout
- Release of phosphate $\rightarrow$ decreased Ca^{2+} , decreased Mg
- Release of procoagulants $\rightarrow$ DIC
- Increased K after treatment (from lysed cells)

Investigations

A- To confirm diagnosis

1. **Full blood count:** Anemia, thrombocytopenia,
WBC increased leukocytes $>10 \times 10^9/L$
2. **Blood film examination**
circulating lymphoblasts lack specific morphological (no granules) or cytochemical features
3. **Bone Marrow Aspiration and Biopsy for definitive diagnosis**
Hypercellular,-----Blast count: ALL $>20\%$ (normal is $<5\%$)



4. **Cytochemistry:**
Special stains to differentiate ALL from AML; Positivity with PAS (periodic acid-Schiff) in ALL
5. **Immunophenotyping**
6. **Molecular genetics / Fluorescence in situ hybridization (FISH)**
7. **Cytogenetics:** Chromosomal abnormalities guide treatment and prognosis
Philadelphia (Ph) chromosome in ~25% of adult ALL cases

B- To diagnosis the complications

- INR, aPTT, fibrin degradation products (FDP), fibrinogen (in case of DIC)
- Increased LDH, increased uric acid, increased PO₄ (released by leukemic blasts), Ca, K
- Baseline renal and liver function tests
- Chest X ray to rule out mediastinal mass, ECG, ECHO scan prior to chemotherapy (cardiotoxic)
- Lumbar puncture prior to systemic chemotherapy to assess for CNS involvement (ensure adequate platelet count and PT/PTT)

Treatments

A- Supportive treatment of patients with ALL

- **Transfusions:** Red blood cells transfusion - Platelets' transfusion
 - Use of irradiated cellular blood components for ALL patients
 - Avoid blood products from sibling donors,
- **Infection –**
 - Broad spectrum IV antibiotics
- **Indwelling central venous catheter**
- **Screening for infection via regular Culture and Sensitivity of urine, stool, sputum, oropharynx, catheter sites**
- **IV fluids, and allopurinol (prevents tumour lysis syndrome)**

B-Chemotherapy

Remission induction (4-6 wks): chemotherapy to induce complete remission of ALL

Vincristine, prednisolone, L-asparaginase + daunorubicin.

Hematological remission: Means no evidence of leukemia in the blood, a normal or recovering blood count, and <5% blasts in a normal regenerating marrow.

Consolidation (6-9 month): high-medium-dose therapy in 'blocks' over several weeks.

CNS prophylaxis: intrathecal (or high-dose IV) methotrexate ± CNS irradiation



Maintenance (2 years): Prolonged chemotherapy, eg mercaptopurine (daily), methotrexate (weekly), and vincristine + prednisolone (monthly) for 2yrs.

Relapses are common in blood, CNS, or testis (examine these sites at follow-up)

C-Haemopoietic stem cell transplant in ALL

- Once in 1st remission is the best option in standard-risk younger adults.
- Can be curative in younger patient (< 40-50 yrs)

Haemopoietic stem cell transplant Types

- HLA identical sibling donor
- Unrelated donor
- Alternative donor: cord blood and haploidentical
- Syngeneic donor

D-Type of immunotherapy

- 1-Immune checkpoint inhibitors
- 2-Chimeric antigen receptor (CAR) T-cell therapy

Prognosis

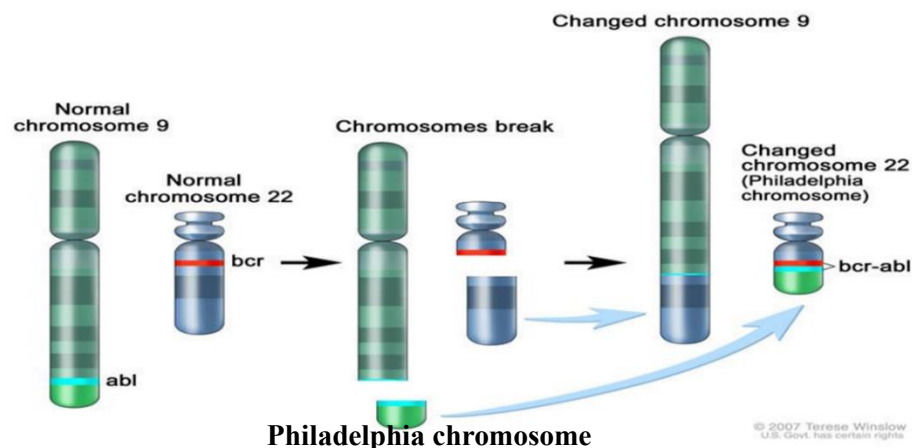
- Cure rates for children are 70–90%; for adults only 40%.
- Poor prognosis if: adult, male, Philadelphia chromosome, presentation with CNS signs, anemia, WCC >100≈109/L , or B-cell ALL.

11

Chronic leukemia

Chronic myeloid leukemia (CML)

- **Definition** → a myeloproliferative disorder of hematopoietic myeloid stem cells in the bone marrow, leading to clonal expansion & uncontrolled proliferation of the myeloid lineage cells.
- **Cells of origin** → mid-stage progenitor cells in the myeloid lineage (promyelocytes, myelocytes, and metamyelocytes)
- **Epidemiology**
 - The incidence of CML is 1/100000 per year
 - It accounts for 20% of all leukemias
 - It is a disease of middle and old aged population with a median age of presentation 40 –60 years
 - There is a slight male predominance.
- **Pathophysiology**
 - CML is characterized by a specific chromosomal abnormality in myeloid stem cells in BM resulting from a reciprocal translocation between long arm of chromosomes 9 & long arm of chromosome 22 t (9;22) leading to *shortening of chromosome 22* and forming abnormal hybrid chromosome called Philadelphia (Ph1) chromosome.
 - This forms BCR-ABL fusion oncogene which has a strong tyrosine kinase activity → inhibition of apoptosis & uncontrolled proliferation of myeloid progenitor cells.
 - Detected in 95% of patients with CML.





- The initiating factor of CML is still unknown but there are possible predisposing factors →
 - Ionizing radiation: Increased prevalence among survivors of Hiroshima & Nagasaki atomic bombing
 - Chemicals e.g. benzene

- **Diagnosis**

- **History:**

- Insidious onset & discovered incidentally in most cases
- 85% of cases presented at chronic phase
- Non-specific symptoms → chronic fatigue, weight loss, fever, sweats due to hypermetabolic state
- Symptoms due to splenomegaly → Left upper quadrant abdominal heaviness & discomfort

- **Examination**

- Massive (huge) splenomegaly →
 - The most common finding (75% of cases)
 - Size of spleen correlates with WBCs count
- Hepatomegaly → less common
- Lymphadenopathy → uncommon in CML
- Petechiae & ecchymosis
- Skin or soft tissue leukemic infiltrates in blast phase

- **Investigations**

- **CBC with differential count and blood film**

WBCs → markedly elevated (>20,000 up to 1 million/ml) with:

- predominance of different mid-stage myeloid progenitor cells (promyelocytes, myelocytes & metamyelocytes) & neutrophils which differentiates CML from AML
- mild increase in basophils & basophils which becomes more prominent in accelerated phase
- Typical leucoerythroblastic picture in blood film
- Mild to moderate normocytic normochromic anemia
- Platelets → variable at presentation & low in accelerated & blast phases.



- **Bone marrow aspiration & biopsy**
 - Hypercellular
 - Myeloid hyperplasia with shift to the left
 - Increased megakaryocytes
 - Mild fibrosis in reticulin stain
- **Cytogenetic analysis (Karyotyping) of bone marrow or peripheral blood →**
Philadelphia chromosome t (9;22) →
The gold standard investigation. It is the hallmark of CML & present in > 90% of those with CML, its presence means good prognosis.
- **BCR-ABL fusion oncogene detection by FISH**
- **Leucocyte alkaline phosphatase** → low or absent
- **Serum uric acid** → elevated (high cell turnover)
- **Abdominal ultrasound** → for spleen size

▪ **Natural history**

Variable, median survival 5–6 years.

There are three phases:

1. **Chronic phase:** lasting months or years of few, if any, symptoms.
2. **Accelerated phase:** with increasing symptoms, spleen size, and difficulty in controlling counts.
3. **Blast transformation (blastic crisis):** with features of acute leukaemia ± death.

▪ **Treatment**

□ **Goals of treatment:**

- **Hematological remission**
 - Normal CBC
 - No splenomegaly
- **Cytogenetic remission**
 - Undetectable Ph chromosome in the bone marrow
- **Molecular remission**
 - Negative PCR result for BCR-ABL mRNA

□ **Lines of treatment:**

➤ **Chronic phase**

A. **First-line treatment** → *Tyrosine Kinase Inhibitors (TKIs)*

Induce apoptosis & inhibit proliferation of myeloid Ph +ve cells



✓ **Imatinib:** -

- 1st generation TKI
- 1ST line standard TKI therapy for CML. This has transformed therapy over the last 10yrs. Side effects are usually mild: nausea, cramps, oedema, rash, headache, arthralgia. May cause myelosuppression.

✓ **Dasatinib, nilotinib, bosutinib :-**

- 2nd generation TKIs more potent than imatinib.
- So, they may be used as 1st line TKI therapy instead of imatinib especially in high-risk patients.

B. Second-line treatment

- ✓ Switch to 2nd generation TKI (dasatinib, nilotinib, bosutinib) in patients resistant or intolerant to imatinib
- ✓ Switch to 3rd generation TKI (Ponatinib, asciminib) → in patients resistant or intolerant to 2 or more TKIs (imatinib & one or more 2nd generation TKI) and in patients with T315I mutation (agents of choice in patients with this mutation)

C. Allogenic hematopoietic stem cell transplant (AHSCT)

- ✓ the only curative.
- ✓ *Indicated in chronic phase CML in the following situations*
 - Progression to accelerated phase or blast phase while on TKI ttt.
 - Patients resistant or intolerant to 2 or more TKIs or in patients with T315I mutation (after a trial of ponatinib therapy)

D. Other medications

- ✓ Interferon-alpha → combined with 2nd generation TKI to improve their response, typically now used for pregnant patients
- ✓ Hydroxyurea → myelosuppressive agent for hematological remission (↓↓ WBCs count & ↓↓ CML burden)
- ✓ Omacetaxine → protein translation inhibitor for patients resistant or intolerant to 2 or more TKIs.

➤ **Accelerated or blast phase**

a) **Allogenic hematopoietic stem cell transplant (AHSCT)**

- ✓ the only curative but carries significant morbidity and mortality

b) **TKIs combinations** (2nd generation e.g. dasatinib or 3rd generation e.g. ponatinib) **with chemotherapy** (like that of AML) → used as a bridge to AHSCT to reduce CML burden



Chronic lymphocytic leukemia (CLL)

- **Definition** → uncontrolled clonal proliferation of mature lymphocytes, mostly B-lymphocytes (99%) in blood, bone marrow, lymph nodes, and spleen.
- **Epidemiology** →
 - The most common form of leukemia in adults in western world (Incidence: 5/100000/year)
 - Mainly affects elderly patients with a median age of presentation 70 years
 - More common in males (male: female → 2:1)
- **Diagnosis**
- **History**
 - Often asymptomatic & discovered incidentally
 - Constitutional symptoms (5-10%) → extreme fatigue, anorexia, fever, sweats, and unexplained significant weight loss
 - Lymphadenopathy (50-90%) → multiple painless & progressive swellings in anatomical sites of L.Ns.
 - Bleeding & infections → if BM failure occurs (late)
- **Examination**
 - Lymphadenopathy → more common than in CML
 - Splenomegaly → 25-55% of cases
 - Hepatomegaly → less common (15-25%)
 - Fever, pallor, petechiae & ecchymosis
- **Investigations**
 - **CBC with differential count and blood film**
 - WBCs → elevated (>100000/L) mainly lymphocytes
 - Anemia & thrombocytopenia
 - Blood film →
 - A - Smudge cells (smear cells) → artifacts produced by the lymphocytes damaged during the slide preparation
 - B - Small mature lymphocytes
 - **Bone marrow aspiration & biopsy**: show extensive infiltration by large number of lymphocytes (30%)
 - **Immunophenotyping** → the gold standard investigation
 - Show the cells to be B- cells (CD19 positive)
 - CD5 & CD23 → characteristically positive in CLL



- **Cytogenetics** → can help assess prognosis e.g. del 17p (poor prognostic factor) & del 13q (good prognosis)
- **Direct Coombs' test** → may be positive if there is autoimmune hemolysis

➤ **Complications**

1. **Bone marrow failure** → pancytopenia → anemia, bleeding, fever & recurrent infections
2. **Immune complications (immune dysregulation)**
 - Autoimmune hemolytic anemia (AIHA)
 - Hypogammaglobulinemia → recurrent infections (bacterial & viral especially herpes zoster)
3. **Richter's Transformation (5%)** to aggressive high-grade lymphoma (diffuse large B-cell lymphoma).

➤ **Natural history**

- 1/3 never progress (or even regress).
- 1/3 progress slowly.
- 1/3 progress actively.
- Death is often due to recurrent infections or transformation to aggressive lymphoma (Richter's transformation).

➤ **Treatment**

☐ **Lines of treatment:**

- a) **Observation** → in asymptomatic patients in early stage
- b) **Chemotherapy** →
 - **Indications:** if patients are symptomatic (systemic constitutional symptoms, progressive lymphadenopathy, progressive splenomegaly) or complicated by progressive BM failure or autoimmune cytopenia e.g. AIHA
 - **Fludarabine + Cyclophosphamide + Rituximab (FCR)** → 1st line treatment of choice
 - **Ibrutinib (BTK inhibitor)** → if p53 positive
- c) **Radiotherapy** →
 - for isolated bulky lymph nodes & splenomegaly
- d) **Stem cell transplantation** → in carefully selected patients
- e) **Supportive care** →
 - Packed RBCs transfusions in severe anemia
 - Platelets transfusion in severe thrombocytopenia or in significant life-threatening bleeding
 - IV human immunoglobulin → for recurrent infections



Haemostasis

- Is a series of processes which stops bleeding, when the blood vessel is broken or damaged.
- It is a complex process depending on interactions between the vessel wall, platelets, coagulation and fibrinolytic mechanisms.
- This occurs through three processes:
 - Vasoconstriction-----vessel wall.
 - Gap-plugging-----Platelets.
 - Coagulation cascade-----Coagulation and fibrinolytic mechanisms.

I- Vessel wall:

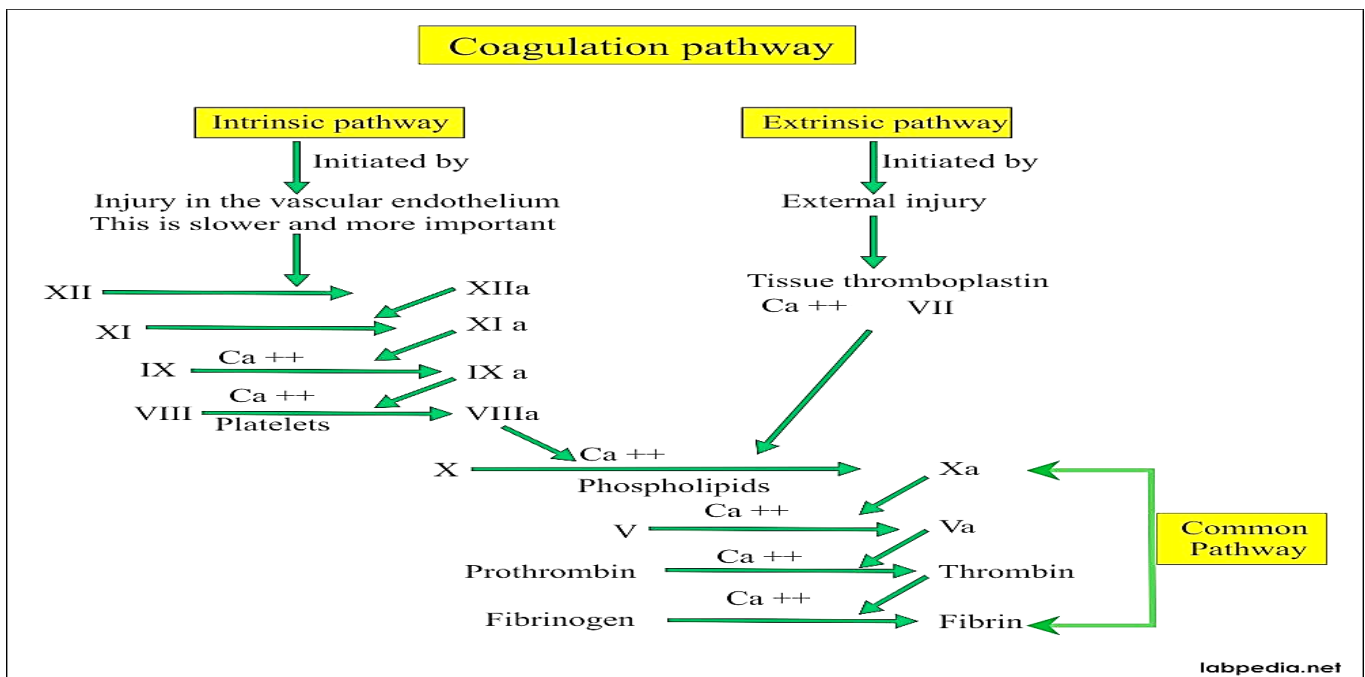
- The vessel wall is lined by endothelium which, in normal conditions, prevents platelet adhesion and thrombus formation. This property is due to its negative charge
- Thrombomodulin expression (protein expressed on the surface of endothelial cells. It reduces blood coagulation)
- Synthesis of prostacyclin and nitric oxide, which cause vasodilation and inhibit platelet aggregation.
- Production of plasminogen activator.
- Injury to vessels $\rightarrow$ reflex vasoconstriction, while endothelial damage $\rightarrow$ loss of antithrombotic properties, activation of platelets, coagulation and inhibition of fibrinolysis.

II- Platelets:

- Platelet **adhesion** to vascular wall structures.
- Platelet release **ADP & serotonin**.
- Platelet **aggregation** to one another or to other blood cells.
- **Coagulation:** aggregation $\rightarrow$ further platelet membrane is exposed $\rightarrow$ interaction of coagulation factors.

III- Coagulation and fibrinolysis:

- Coagulation involves a series of enzymatic reactions leading to → the conversion of *soluble plasma fibrinogen to fibrin clot*.
- The coagulation factors are primarily synthesized in the *liver* and are either enzyme precursors (factors *XII, XI, X, IX and thrombin*) or co-factors (*V and VIII*).
- Factors I and II are referred to as fibrinogen and prothrombin respectively.
- The active forms are denoted by '**a**'.
- This system is divided into 'extrinsic' and 'intrinsic' pathways.
- Coagulation is initiated by binding of activated factor VII (VIIa) in plasma to tissue factor (TF which is a glycoprotein which is expressed after tissue injury).
- The complex of VIIa and tissue factor (TF:VIIa) → → activates factor X and factor IX.
- Activated factor IX then works with factor VIII → initiate the crucial activation of factor X → (The reaction which increases the generation of thrombin, and which fails in hemophilia A and B).
- Prothrombin to thrombin. Thrombin hydrolyses the peptide bonds of fibrinogen to form fibrin.
- At the same time, thrombin, in the presence of calcium ions, activates factor XIII, which stabilizes the fibrin clot by cross-linking adjacent fibrin molecules.



Limitations of coagulation

1- Antithrombin (AT)

- A potent inhibitor of coagulation. Its action is greatly potentiated by heparin.

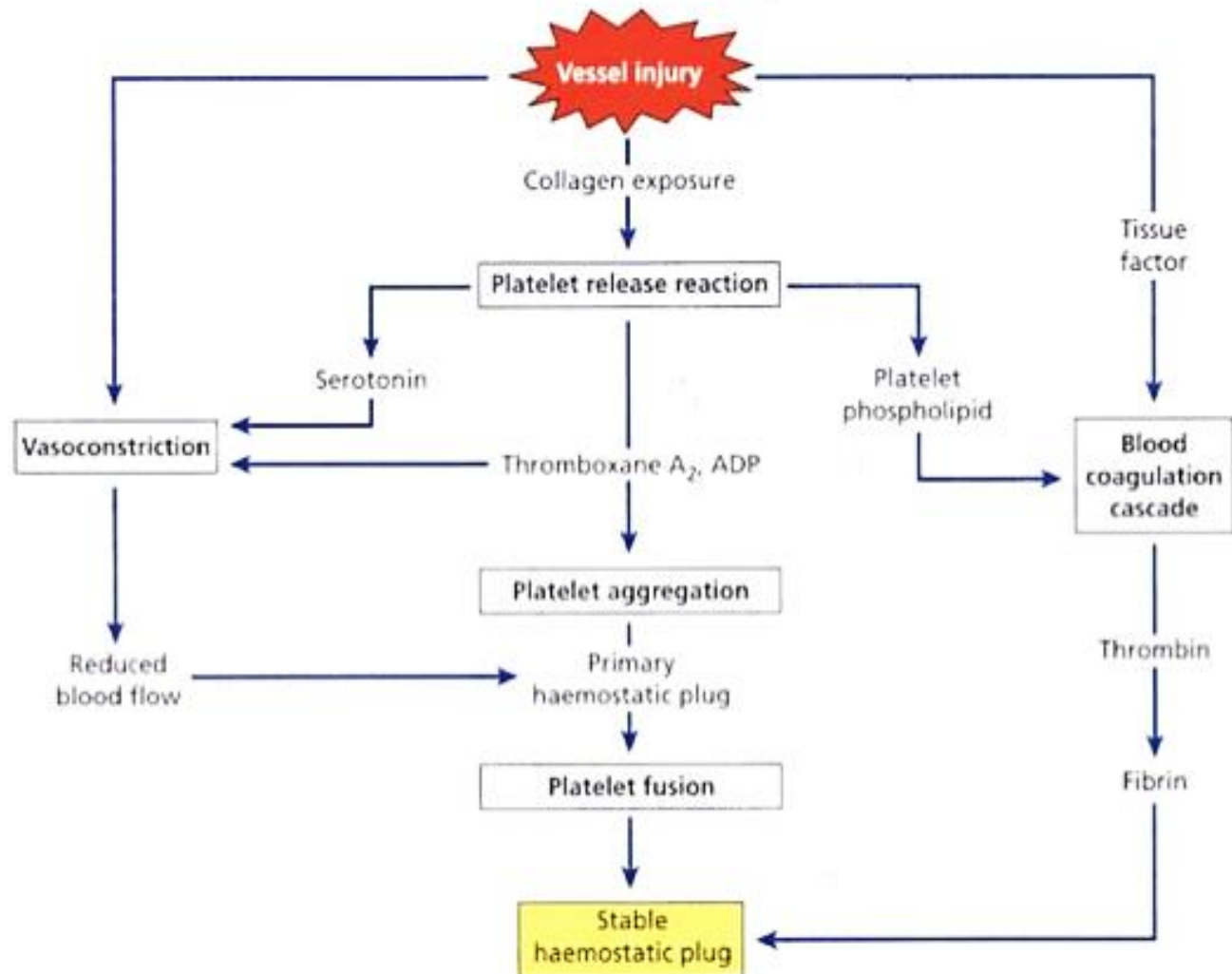
2- Activated protein C

- Generated from its precursor by the action of thrombin; Activated protein C destroys factor V and factor VIII, reducing further thrombin generation.

3- Protein S: This is a cofactor for protein C

4- Other inhibitors: Other natural inhibitors of coagulation include α_2 macroglobulin, α_1 -antitrypsin and α_2 -antiplasmin.

Hemostatic response





Fibrinolysis

- It occurs also in response to vascular damage and helps to restore vessel patency. The fibrinolytic system is activated by the presence of fibrin.
- The fibrinolytic system works by generating plasmin, which causes fibrin dissolution.
- An inactive plasma protein - plasminogen - is converted to plasmin by plasminogen activators derived from the plasma or blood cells (intrinsic activation) or the tissues (extrinsic activation).

Fibrinolytic agents

- The major plasminogen activator is **tissue-type plasminogen activator (t-PA)**; which is released by the vascular endothelium (Alteplase is recombinant t-PA, tenecteplase and reteplase)
- **Streptokinase**, a streptococcal exotoxin that binds and activates plasminogen, was the first licensed agent but risks anaphylaxis on repeat dosing.
- Another plasminogen activator is **urokinase** which is synthesized in the kidney.

Types of bleeding disorders:

- Therefore, the disorders of hemostasis fall into these three groups:
 - 1) **Vascular disorders.**
 - 2) **Platelet disorders.**
 - 3) **Coagulation disorders.**
- **Vascular/platelet bleeding** is characterized by easy bruising and spontaneous bleeding from small vessels. There is often bleeding into the skin.
- **Coagulation disorders** are typically associated with hemarthrosis and muscle hematomas and bleeding after injury or surgery.



12

Platelet -vascular disorders

Vascular disorders

1- Congenital:

- **Hereditary hemorrhagic telangiectasia** (Osler–Weber–Rendu syndrome)
- **Connective tissue disease** (e.g. Ehlers–Danlos syndrome, osteogenesis imperfecta, and Marfan's syndrome)

2- Acquired:

- **Senile purpura.**
- **Infections** (e.g. meningococcal, measles, dengue fever).
- **Henoch Schönlein purpura.**
- **Drugs** (Steroids).

Platelets disorders

Etiology

1- Decreased marrow production:

- Aplastic anemia
- Megaloblastic anemia
- Marrow infiltration (e.g. leukemia, myeloma)
- Marrow suppression (cytotoxic drugs, radiotherapy)
- Infections as HIV
- Rare congenital defect as Selective megakaryocyte depression

2- Excess destruction:

A- Immune:

- Immune thrombocytopenia purpura (ITP)
- Associated with autoimmune causes: e.g. SLE
- Associated with CLL or lymphoma
- Associated with infections: HIV, malaria
- Post-transfusion purpura
- Drugs: e.g. heparin



B- Non-immune:

- Disseminated intra vascular coagulation (DIC)
- Thrombotic thrombocytopenic purpura (TTP)
- Hemolytic uremic syndrome (HUS)
- Abnormal distribution of platelets (sequestration in hypersplenism)

3. Poorly functioning platelets (Thrombasthenia):

A- Hereditary:

- Glanzmann's disease
- Bernard–Soulier syndrome
- Storage pool disease

B- Acquired:

- Drugs e.g. NSAIDs & anti-platelets
- Renal Failure
- Myeloproliferative disorders

Immune thrombocytopenia purpura (ITP)

- Autoimmune (idiopathic) thrombocytopenic purpura (ITP) may be divided into **acute and chronic** forms

Acute ITP

- Usually in children, **2 weeks** follows *vaccination or an infection* such as chickenpox or infectious mononucleosis, with *sudden self-limiting purpura*.
- Spontaneous remissions are usual but in 5–10% of cases, the disease becomes chronic lasting more than 6 months.
- Most children *do not have any bleeding even with platelet counts <10,000 cell* but need to avoid trauma such as contact sports.
- *If the platelet count is over 30,000 → no treatment is necessary unless the bleeding is severe.*
- Treatment is with *steroids and/or intravenous immunoglobulin*, especially if there is *significant bleeding or for surgery or pregnancy*.



Chronic ITP

- This is a relatively **common disorder**.
- ITP is caused by **antiplatelet auto anti bodies (IgG)**, result in the premature removal of platelets from the circulation by macrophages of the reticuloendothelial system, *especially the spleen*.
- It is usually **idiopathic** but may be seen in association with other diseases such as ***SLE, HIV, viral hepatitis, or autoimmune hemolytic anemia***.
- The onset is often ***insidious with petechial hemorrhage, purpura*** (esp. dependent pressure areas), epistaxis, and menorrhagia.
- Chronic ITP runs ***a fluctuating course*** as it tends to relapse and remit spontaneously.
- Many asymptomatic cases are discovered by a routine blood count.
- ***The is no splenomegaly.***

Investigations

- **The blood film** shows reduced numbers of platelets, with no morphological abnormalities in the other cell lines.
- **The bone marrow** shows increase numbers of megakaryocytes.
- **Antiplatelet autoantibodies** often present (Sensitive tests are able to demonstrate specific anti-glycoprotein GPIIb/IIIa or GPIb antibodies on the platelet surface or in the serum in most patients (These tests are not usually used in clinical practice)).

Treatment

- The aim of treatment should be to ***maintain a platelet count above >30,000***, the level at which spontaneous bruising or bleeding occurs with the minimum of intervention.
- ***None if platelets >20,000 without symptoms.***
- ***If symptomatic or platelets <20,000 cell:*** prednisolone 1mg/kg/day. The dosage is gradually reduced after 10–14 days after remission.
- **In poor responders** the dosage is reduced more slowly but alternative ***immunosuppression or splenectomy*** is considered.
- **If no response** to steroid use or relapse occurs, ***monoclonal antibody as Rituximab and Immunosuppressive drugs*** (e.g. cyclophosphamide, and azathioprine) are used.



- **In refractory cases:** **Eltrombopag** (an oral thrombopoietin-receptor agonist) and **romiplostim** (an injectable thrombopoietin analogue), can be used as they *stimulate thrombopoiesis*.
- **Splenectomy:** is now performed less frequently for ITP than previously.
- **Platelet transfusions are not used** (*except during splenectomy or life-threatening hemorrhage*) as these are quickly destroyed by the *autoantibodies*.

Drug-induced immune thrombocytopenia

- It occurs usually due to an *immunological* mechanism.
- **Quinidine and heparin** are particularly common causes.
- The bone marrow shows *normal or increased numbers of megakaryocytes*.
- **Drug-dependent antibodies** against platelets may be demonstrated in the sera of some patients.
- The treatment is to **stop all suspected drugs**, but platelet concentrates should be given to patients with dangerous bleeding.

Disorders of platelet function (Thrombasthenia)

- May be hereditary or acquired

a- Hereditary

- Glanzmann's disease

- This autosomal recessive disorder leads to *failure of platelet aggregation*.

- Bernard–Soulier syndrome

- Autosomal recessive disease.
 - There is *defective binding to VWF, defective adherence* to exposed sub-endothelial connective tissues and *platelets do not aggregate* with ristocetin.

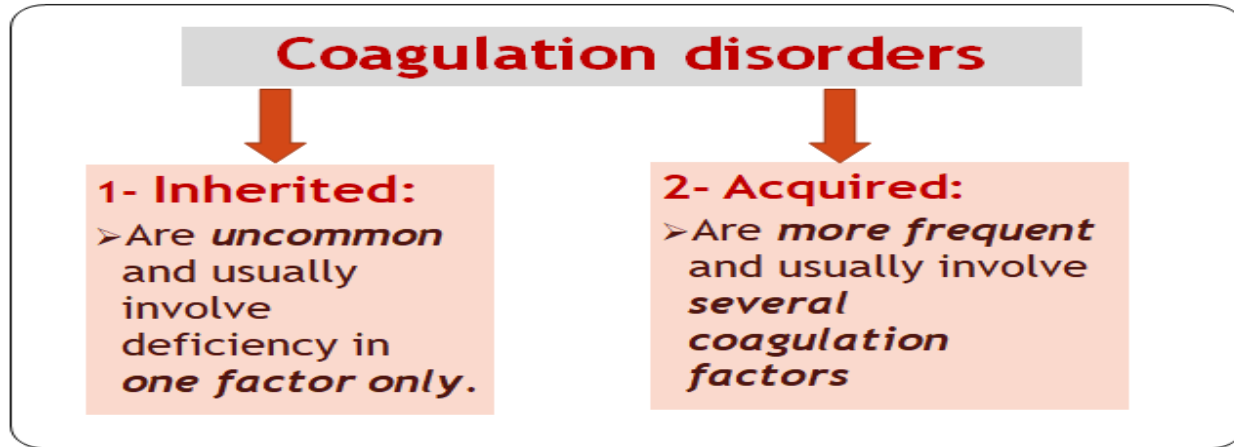
b- Acquired

- Drugs e.g. NSAIDs & antiplatelets
- Renal failure
- Myeloproliferative disorders



13

Coagulation disorders



1- Inherited

- Hemophilia A (factor VIII deficiency)
- Hemophilia B (Christmas disease, factor IX deficiency)
- Von Willebrand's disease (VWD)

2- Acquired

- Vitamin K deficiency
- Liver disease
- Disseminated intravascular coagulation (DIC)

Hemophilia A

- Factor VIII deficiency.
- Inherited in an X-linked recessive pattern in 1:10,000 male births, usually due to a 'flip tip' inversion in the factor VIII gene in the X chromosome.
- Hemophilia is typically expressed in *males and carried by females*.
- *30 % of cases of hemophilia are new mutations (Has no family H/O).*
- Clinical presentation depends on severity.
- Is often *early in life or after surgery/ trauma*.
- Infants may develop profuse post-circumcision hemorrhage or joint and soft tissue bleeding leading to crippling arthropathy, and into muscles causing hematoma (Increase pressure can lead to nerve palsies and compartment syndrome).



Hemophilia A: acute hemarthrosis of the right knee joint

Degrees of Severity of Hemophilia

- **Normal factor** VIII or IX level = 50-150%
- **Mild hemophilia:** factor $>5\%$ (factor VIII or IX level = 6-50%): *bleeding only after surgery or injury & diagnosis is often delayed*
- **Moderate hemophilia:** 1-5% (factor VIII or IX level = 1-5%): *Sever bleeding after injury & occasional spontaneous bleeding*
- **Severe hemophilia:** $<1\%$ (factor VIII or IX level = $<1\%$): *Spontaneous bleeding from early life, hemarthrosis & muscle hemorrhage.*

Diagnosis

- Family history (in 70%)
- Symptoms (depend on level of factor deficiency): *Bruising, bleeding with circumcision, and/or Muscle, joint or soft tissue bleeding*
- Increased activated partial thromboplastin time (APTT)
- Decrease factor VIII assay
- Normal bleeding time
- Genetic testing

Treatment

- Replacement of missing coagulation factor (factor VIII)
 - ✓ On demand (after bleeding, before surgery)
 - ✓ Prophylaxis



- Desmopressin (Synthetic vasopressin) used in *mild cases*, it releases of stores from endothelial cells raising factor VIII and vWD serum levels and it is given IV, SC, and nasally.
- Antifibrinolytic Agents.
- Supportive measures:
 - ✓ Immobilization
 - ✓ Rest

Factor VIII Concentrate:

- Intravenous infusion
- Dose varies depending on the type of bleeding.
- Ranges from 20-50 units/kg. body weight.
- Half-life: 8-12 hours.
- Each unit infused *raises serum factor VIII level by 2 %*.

Prophylaxis

- In severely affected patients
- Scheduled infusions of factor concentrate to prevent most bleeding and joint deformity
- Frequency: *2 to 3 times weekly*.

Genetic counselling

- Carriers are detected with DNA probes. A known specific mutation can be identified.
- Chorionic biopsies at 8–10 weeks' gestation provide sufficient fetal DNA for analysis.

Hemophilia B (Christmas disease)

- Factor IX deficiency (inherited, X-linked recessive).
- Behaves clinically like hemophilia A.
- The incidence is one-fifth that of hemophilia A.

Treatment

- with recombinant factor IX.
- Intravenous infusion
- Half-life: 12-24 hours
- Each unit infused *raises serum factor IX level by 1%*.



Von Willebrand disease (vWD)

Von Willebrand Factor (vWF)

- Large multimeric glycoprotein
- Primary function (platelet adhesion and aggregation)
- Secondary (FVIII carrier): *protects FVIII from degradation and delivers it to the site of injury*

Von Willebrand disease (vWD)

- Family of bleeding disorders caused by an abnormality of the vWF.
- *The most common inherited bleeding disorder.*
- Prevalence of 1%.

Causes

1. **Inherited**: Defect of short arm of chromosome 12
2. **Acquired**:
 - ✓ Systemic lupus erythematosus
 - ✓ Hematological malignancies e.g. CLL
 - ✓ Plasma cell disorders
 - ✓ Hypothyroidism
 - ✓ Uremia
 - ✓ Angiodysplasia

Treatment

- Type 1: Desmopressin
- Type 2 & 3: Give factor VIII concentrates with preserved vWF.
FFP (contains factor VIII) and cryoprecipitate should only be given in emergencies when factor VIII/vWF is unavailable.

N.B: No response to desmopressin in type 2 and even worsen the condition in type 3.

Liver disease

- ✓ Produces a complicated bleeding disorder with:
 - Decreased synthesis of clotting factors.
 - Decreased absorption of vitamin K.
 - Abnormalities of platelet function.
 - Functional abnormality of fibrinogen (dysfibrinogenemia).



Vitamin K deficiency

- Fat-soluble vitamin K is obtained from green vegetables and *bacterial synthesis in the gut*.
- Vit. K is necessary for *coagulation factor II, VII, IX, X and protein C & S*.
- **Deficiency of vitamin K is caused by:**
 - Newborn
 - Malabsorption e.g. obstructive jaundice
 - Vitamin K antagonist drugs such as warfarin

Hemorrhagic disease of the newborn

- Vitamin K-dependent factors are low at birth and fall further in breast-fed infants in the first few days of life.
- *Liver cell immaturity, lack of gut bacterial synthesis of the vitamin and low quantities in breast milk* may all contribute to a deficiency which causes hemorrhage, usually on the *second to fourth day of life*, but occasionally during *the first 2 months*.
- The PT and APTT are both abnormal. The platelet count and fibrinogen are normal with absent fibrin degradation products.
- **Prophylaxis:** For many years *vitamin K has been given to all newborn babies as a single intramuscular injection of 1 mg*.
- **In bleeding infants:** *Vitamin K 1 mg intramuscularly* is given every 6 hours with, initially, prothrombin complex concentrate if hemorrhage is severe.
- Deficiency resulting from obstructive jaundice, pancreatic or small bowel disease (Malabsorption) leading to less uptake of vitamin K (needed for synthesis of factors II, VII, IX, and X).
- **Diagnosis:**
 - Both PT and APTT are prolonged.
 - There are low plasma levels of factors II, VII, IX and X.



Disseminated intravascular coagulation (DIC)

- Systemic thrombo-hemorrhagic disorder

Characteristic features:

- Activation of coagulation system → thrombosis
- Consumption of clotting factors → bleeding
- Platelets aggregation → thrombocytopenia → bleeding
- Activation of fibrinolytic system → ↑FDPs & bleeding

C/P: *Thrombosis, Bleeding, hemodynamic complications and shock.*

- The mortality ranges from 30% to >80% depending on the underlying disease, severity of the DIC and age of the patient.

Associated clinical conditions:

- Sepsis & severe infections → the most common
- Trauma
- Serious tissue injury
- Fat embolism
- Cancer
- Obstetric complications (*Abruptio placentae, amniotic fluid embolism, fetal death in utero, septic abortion*).
- Vascular
 - Giant hemangioma * Aortic aneurysm
- Reaction to toxins
- Immunological disorders
 - Hemolytic transfusion reaction * Transplant rejection

Diagnosis

- **CBC:** Normocytic anemia, thrombocytopenia & reticulocytosis
- **Blood film:** schistocytes (fragmented RBCs)
- **Evidence of hemolysis:** increased serum bilirubin, mainly indirect bilirubin & increased LDH
- **Prolonged PT & APTT**
- **Serum fibrinogen:** decreased
- **D-dimer & fibrin degradation products (FDPs):** increased



Treatment

- **Treatment of the underlying cause** e.g. antibiotics in severe infections & sepsis.
- **Fresh frozen plasma & cryoprecipitate** in patients with active bleeding & hypofibrinogenemia
- **Platelets transfusion** in patients with severe thrombocytopenia $<20,000$, patients with active bleeding, and in patients at high risk for bleeding e.g. postoperative or if an invasive procedure is planned.
- **Anticoagulation:** Heparin is indicated in patients with acral cyanosis, digital ischemia, purpura fulminans, and chronic DIC.

Approach to patients with bleeding tendency

- Is there a generalized hemostatic defect? bleeding from *multiple sites*, *spontaneous bleeding* & *extensive bleeding after injury*.
- Is the defect *inherited* or *acquired*? (*Family history*)
- *Past or family history* of excess bleeding, e.g. during trauma, dentistry, surgery?
- Is there a secondary cause, e.g. drugs (warfarin), alcohol, liver disease, sepsis?
- Is the pattern of bleeding indicative of vascular, platelet or coagulation problem?
- Vascular/platelet bleeding is characterized by *easy bruising and spontaneous bleeding* from small vessels. There is often bleeding into the skin (*Purpura*).
- Coagulation disorders are typically associated with *hemarthrosis and muscle hematomas* and *bleeding after injury or surgery*.

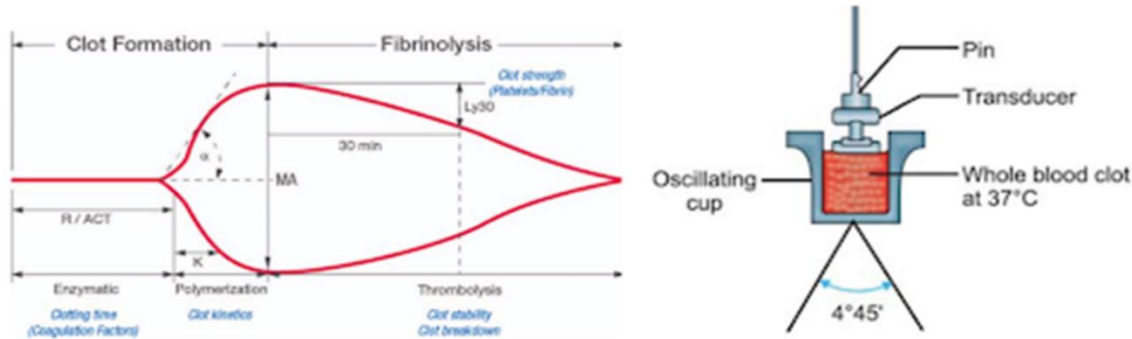
Investigations

- **CBC & blood film:**
 - Show the number and morphology of platelets and any blood disorder such as leukemia or lymphoma.
- **Prothrombin time (PT):**
 - Thromboplastin is added to test the *extrinsic system*.
 - *The normal PT is 16-18 seconds.*



- Is expressed as a ratio compared to control international normalized ratio- (INR), *normal range = 0.9–1.2*. It tests for abnormalities in factors I, II, V, VII, X (*but not factor VII*).
- Prolonged by: *Warfarin, vitamin K deficiency, liver disease, and DIC.*
- **Activated partial thromboplastin time (APTT):**
 - Tests for abnormalities (intrinsic pathway) in factor I, II, V, VIII, IX, X, XI, XII. (*but not factor VII*).
 - *Normal range 35–45seconds.*
 - Prolonged by: *Heparin treatment, hemophilia, DIC, liver disease.*
- **Thrombin (clotting) time:**
 - Thrombin is added to plasma to convert fibrinogen to fibrin.
 - *Normal range: 10–15s*
 - Prolonged by: *Heparin treatment, DIC, dysfibrinogenemia.*
- **Bleeding time:**
 - Measures platelets plug formation in vivo.
 - Two (1 mm deep, 1 cm long) incisions are made in the forearm. Each wound is blotted every 30 s and the time taken for bleeding to stop is recorded, *normally 3-10 minutes.*
 - *Prolonged bleeding times are found in patients with platelet function and number defects.*
 - *N.B: Bleeding time, a barbaric and unreliable test (the clue is in the name), is no longer used. Amongst the range of techniques to replicate the clotting process in vitro is thromboelastography (TEG)*
- **Thromboelastography (TEG)**
 - TEG permits rapid and more precise assays of clotting function under massive transfusion situations (e.g. major surgery, especially trauma). In particular, advances in this field have been driven by recent military usage.
 - Typical patterns showing results in *fibrinolysis, hypercoagulability, hemophilia and thrombocytopenia.*

Thromboelastography



- D-dimer: is a fibrin degradation product, released from cross-linked fibrin during fibrinolysis. This occurs during DIC, or in the presence of venous thromboembolism—deep vein thrombosis (DVT) or pulmonary embolism (PE). D-dimers may also be raised in inflammation, e.g. with infection or malignancy.
- Factor assays: used to confirm coagulation defects, especially when a single inherited disorder is suspected.
- Platelet function tests: such as platelet aggregation.

Hypercoagulable states (Thrombophilia)

Definition

- Patients with hereditary or acquired hypercoagulable states are more likely to develop clots, venous thrombosis, and arterial thrombosis, than healthy individuals.

Risk factors

1- Inherited:

- Factor V Leiden (activated protein C resistance) → The most common inherited hypercoagulable state
- Protein C deficiency → warfarin-induced skin necrosis
- Protein S deficiency
- Anti-thrombin III deficiency → resistance to heparin therapy
- Prothrombin gene mutation
- Homocysteinemia

2- Acquired:

- Antiphospholipid syndrome
- Advanced age
- Immobilization
- Obesity
- Diabetes mellitus
- Hyperlipidemia
- Pregnancy
- Drugs e.g. oral contraceptive pills
- Malignancy
- Myeloproliferative disorders e.g. polycythemia vera, ET, and myelofibrosis
- Paroxysmal nocturnal hemoglobinuria (PNH)
- Nephrotic syndrome

Clinical presentation

✓ Hereditary thrombophilia should be suspected in patients with a history of any of the following:

- Recurrent venous thromboembolism (VTE)
- Venous thrombosis before age of 40 years
- A family history of venous thromboembolism
- Thrombosis in unusual sites e.g. mesenteric, renal, and cerebral vein thrombosis and Budd Chiari syndrome
- Arterial thrombosis in absence of arterial disease
- Recurrent superficial thrombophlebitis

Work up

- CBC e.g. for polycythemia vera & essential thrombocytosis
- Blood smear & hemolysis markers (if anemic)
- Flow cytometry of peripheral blood RBCs (if suspected PNH) → absence of CD 55 & CD 59 on RBCs surface

- Factor V Leiden gene mutation assay
 - Prothrombin gene mutation assay
 - Protein C & S assay
 - Antithrombin III level
 - Homocysteine level
 - ANA, Anti-ds DNA & ANCA (P & C)
 - JAK 2 gene mutation assay by PCR
 - MPL & CALR gene mutation assay by PCR
 - Urine analysis: for proteinuria in nephrotic syndrome
 - Investigations for antiphospholipid syndrome →
lupus anticoagulant, anticardiolipin antibody, anti-B2 glycoprotein 1 antibody & prolonged APTT
 - Screening for malignancy e.g. tumor markers & CT scan
 - Duplex study to detect the site of venous or arterial thrombosis
- ✓ Protein C & S, antithrombin III, lupus anticoagulant & anticardiolipin antibody levels are all change during acute thrombosis. So, work up for hereditary thrombophilia must be done 3 months after acute thrombotic event to avoid misdiagnosis.

Treatment

- In patients with hereditary thrombophilia, the occurrence of a single thrombotic event is an indication for **prophylactic lifelong anticoagulation** to avoid recurrent thromboembolism.
- Indications for primary prophylactic anticoagulation for patients at risk, include prolonged hospitalization, postoperative status, immobilization, certain orthopedic disorders, and active cancers.
- **Approach to treatment of venous thromboembolism (VTE):**
 - ❑ **Initial treatment:**
 - Once VTE has been diagnosed, immediate anticoagulation therapy should be initiated.
 - Options for immediate anticoagulation include:
 - Low molecular weight heparin (LMWH)
 - Administered SC, as effective as unfractionated heparin (UFH) with a lower bleeding risk.
 - Advantages: predictable dose response, fixed dose schedule, lab. monitoring not required, safe & effective, low incidence of heparin induced thrombocytopenia (HIT)
 - Disadvantages: only partially reversible by protamine, long-term use associated with osteoporosis.
 - May require dose adjustment in patients with renal impairment.
 - Unfractionated heparin (UFH)
 - Advantages: rapidly reversible by protamine



- Disadvantages: must be monitored by APTT with dose adjustment to reach therapeutic level & high risk of HIT.
- Direct thrombin inhibitors e.g. dabigatran, argatroban, hirudin, lepirudin
- Direct factor Xa inhibitors e.g. apixaban & rivaroxaban
- Thrombolytic drugs e.g. streptokinase & tPA: reserved for acute life-threatening thrombosis with low bleeding risk.

❑ Long-term management:

- Warfarin:
 - Should be initiated with bridging overlap heparin therapy for 5 days to avoid risk of warfarin-induced skin necrosis, and then discontinue heparin after the patient has been fully anticoagulated with warfarin (INR >2 for 2 consecutive days).
 - It must be monitored by INR & its dose should be adjusted to maintain INR at 2-3
 - Disadvantages: high bleeding risk, frequent lab. monitoring by INR, multiple drug-drug and drug-food interactions.
- Direct oral anticoagulants (DOACs)
 - direct factor Xa inhibitors e.g. apixaban & rivaroxaban and direct thrombin inhibitors e.g. dabigatran
 - advantages over warfarin: less bleeding risk, no need for monitoring due to predictable pharmacokinetics, no or limited interactions with other drugs & food.
 - Contraindications: prosthetic heart valves, severe renal impairment, antiphospholipid syndrome, pregnancy, and lactation.

Duration of anticoagulant treatment:

- Provoked VTE with transient reversible risk factor: 3-6 months
 - Provoked VTE with ongoing risk factor: indefinite anticoagulation
 - Unprovoked VTE: consider indefinite anticoagulation therapy
 - Recurrent VTE: indefinite anticoagulation
 - Cancer-associated VTE: at least 3-6 months, longer if continued evidence of cancer
- ❖ In cancer patients → LMWH is more effective than warfarin to treat & prevent recurrence of cancer-associated VTE.
- ❖ In pregnancy → treat with LMWH during pregnancy, then LMWH or warfarin for 6 weeks postpartum (minimum total anticoagulation period is 3-6 months but must include 6 weeks postpartum as this a high-risk period. Avoid warfarin in pregnancy due to teratogenicity and avoid DOACs in pregnancy due to lack of data.

Antiphospholipid syndrome

✓ **Diagnostic criteria:** at least 1 clinical plus 1 laboratory criteria:

▪ **Clinical criteria:**

- **Vascular thrombosis:** one or more clinical episodes of venous or arterial thrombosis e.g. DVT & pulmonary embolism.
- **Pregnancy morbidity:** Recurrent ≥ 3 consecutive spontaneous abortions at <10 weeks gestation or ≥ 1 pregnancy loss at >10 weeks gestation after exclusion of other causes of pregnancy loss)

▪ **Laboratory criteria:**

- Antibodies must be positive on 2 occasions, at least 12 weeks apart
- Antibodies include: lupus anticoagulant, anticardiolipin antibody, and anti-beta 2 glycoprotein 1 antibody.

✓ **Treatment**

- Lifelong anticoagulation with warfarin with a target INR of 2-3 for the first thrombotic event and a target INR of 3-4 for recurrent thrombotic events.
- In pregnancy → aspirin and LMWH should be initiated.

Endocrinology



Contents

No.	Lectures Titles
1	Introduction and hypothalamus (diabetes insipidus and SIADH)
2	Growth hormone (acromegaly, gigantism and short stature)
3	Pan-hypopituitarism
4	Hyperthyroidism
5	Hypothyroidism
6	Disorders of parathyroid glands (Hyperparathyroidism and Hypoparathyroidism)
7	Diseases of the adrenal gland (Cushing, Addison and clinical use of corticosteroids)
8	Diseases of the adrenal gland (hyperaldosteronism and pheochromocytoma)

1

Introduction

❑ The endocrine system includes the following glands:

Pituitary gland

Thyroid gland

Parathyroid glands

Pancreas

Suprarenal glands

Gonads
(testes or ovaries)

❑ Endocrinal Diseases:-

Aetiology

Hyperfunction

• e.g. due to tumour or hyperplasia.

Hypofunction

• e.g. due to surgical removal or irradiation.

Clinical picture

Investigations

- Assay of hormonal level.
- Suppression tests: for diagnosis of gland hyperfunction.
- Stimulation tests: for diagnosis of gland hypofunction.
- Imaging of the gland.

Treatment

❑ **Hyperfunction:**

- Treatment of the cause.
- Symptomatic treatment.

❑ **Hypofunction:**

- - Replacement therapy.
- - Treatment of the cause,
- - Symptomatic treatment.

1)

PITUITARY GLAND

Anatomy:

- The pituitary gland lies inside the sella turcica in the base of the skull.
 - It is separated from the middle intracranial fossa by the diaphragma sellae which is pierced by the pituitary stalk connecting the pituitary to hypothalamus.

Relations:

- **Superiorly** : optic chiasma & hypothalamus.
- **Inferiorly** : sphenoidal air sinus.
- **Posteriorly** : upper part of brain stem,
- **Laterally** : cavernous sinuses & optic tracts.

Divisions:

The pituitary gland is composed of anterior & posterior lobes.

Anterior Pituitary (Adenohypophysis)

It contains three types of cells:

- **Acidophil cells** : secreting GH and prolactin
- **Basophil cells** : secreting ACTH, TSH, FSH, LH and MSH.
- **Chromophobe cells.**

• **Functions of Anterior Pituitary Hormones**

1) *Growth hormone (GH):*

- -Anabolic effect: increases protein anabolism with increased growth of bones, muscles, cartilage & viscera, This action is mediated by release of growth factors by the liver e.g. insulin-like growth factor-1 (IGF-1),
- -Diabetogenic effect increases gluconeogenesis.
- -Increases Ca, P & Na

2) *Prolactin:*

- Stimulates production of milk

3) *Adrenocorticotrophic hormone (ACTH):*

- Increases size of adrenal cortex. Increases function of adrenal cortex: stimulates secretion of glucocorticoids & a lesser effect on androgens & a minor effect on aldosterone.

4) *Thyroid stimulating hormone (TSH):*

- Increases size and vascularity of thyroid gland.
- Increases function of thyroid gland: stimulates all steps of thyroid hormones production.

5) Follicle stimulating hormone (FSH):

- **In females:** stimulates maturation of Graffian follicles with secretion of estrogen.
- **In males:** stimulates spermatogenesis.

6) Leutinizing hormone (LH):

- **In females:** stimulates rupture of Graffian follicles leading to ovulation & formation of corpus luteum with secretion of estrogen and progesterone.
- **In males:** stimulates Leyding cells leading to testosterone secretion. Also called "interstitial cell stimulating hormone (ICSH)".

7. Melanocyte stimulating hormone (MSI):

- Stimulates skin pigmentation.

• Control of the anterior pituitary hormones

- **Release factors (RF)**
- **Release inhibiting factors (RIF)**
 - are hormones secreted by the hypothalamus & transmitted through the hypothalamo-hypophyseal portal circulation to control secretion of hormones of anterior pituitary.

I- Release factors: e.g.

- 1- Thyrotropin-releasing hormone (TRH).
- 2- Corticotropin-releasing factor (CRF).
- 3- Gonadotrophines-releasing hormone (Gn-RH).
- 4- Growth hormone-releasing factor (GH-RF).

II-Release inhibiting factors: e.g.

- 1- Somatostatin (GH-RIF).
- 2- Dopamine (prolactin-RIF).

Release factors & release-inhibiting factors are controlled by:

1. Negative feedback control:

- Long feedback loop:

RF & RIF are regulated by the level of target gland hormones e.g. thyroxine.

-Short feedback loop:

RF & RIF are regulated by the level of anterior pituitary hormones e.g. TSH.

2. CNS:

which controls the hypothalamic & consequently pituitary functions.

Posterior Pituitary (Neurohypophysis)

The posterior pituitary secretes 2 hormones:

1. Antidiuretic hormone (ADH) or vasopressin:

- Retention of water by increasing reabsorption in collecting ducts
- Vasoconstriction.

2. Oxytocin:

- Stimulation of uterine contraction.
- Stimulation of milk ejection.

Note

These hormones are synthesized in the cell bodies of neurons of the supraoptic & paraventricular nuclei of the hypothalamus & pass down the axons to be stored in their terminal swellings within the posterior pituitary.

Control of ADH

- Stimulation of osmoreceptors of hypothalamus by increase plasma osmotic pressure: leading to secretion of ADH.
- Stimulation of volume receptors by decreased plasma volume: leading to secretion of ADH.
- Stressful conditions may modify secretion of ADH.

Disorders of Pituitary Gland

Disorders of anterior pituitary)

1. Hyperpituitarism:

- **Increased GH** : presents with gigantism or acromegaly,
- Increased prolactin : presents with hyperprolactinaemia,
- **Increased ACTH** : presents with Cushing's disease.
- **Increased TSH** : presents with hyperthyroidism
- **Increased gonadotrophines (Gn)**: presents with precocious puberty.

2. Hypopituitarism:

A. During adulthood:

1- Panhypopituitarism (Simmond's disease).

2- Isolated hormone deficiency: e.g.

- *Decreased TSH*: leading to secondary hypothyroidism.
- *Decreased ACTH*: leading to secondary hypocorticism.
- *Decreased gonadotrophines*: leading to secondary Hypogonadism

B. During childhood:

1- *Pituitary dwarfism* (Levi-Lorain syndrome).

2- *Frohlich's syndrome*.

Disorders of posterior pituitary

- 1- **Increased ADH:** leading to syndrome of inappropriate ADH secretion (SIADH)
- 2- **Decreased ADH:** leading to diabetes insipidus.

Diabetes insipidus

- Deficiency of ADH

Physiology :

- ADH is synthesized in hypothalamus & then transported along axons & stored in the posterior Pituitary.
- **Function of the hormone :**
 - ↑↑ water reabsorption by collecting tubules.
 - Large dose → vasoconstriction of blood vessels.
- **Regulation of this hormone :**
 - ↑↑ **ADH by :**
 - 1 . ↑ osmolality.
 - 2. Hypovolaemia & hypotension.
 - 3. Drugs : nicotine , barbiturates.
 - ↓↓ **ADH by :**
 - osmolality, Hypertension & Cold weather.

Etiology :

- A. **Central DI :** (damage of hypothalmo hypophyseal axis)
 - Idiopathic : Hereditary (autosomal dominant)
 - Injury: after hypophysectomy.
 - Infection : T.B.
 - Infiltration : sarcoidosis
 - Infarction: Sheehan syndrome.
 - pituitary tumor.
 - Pregnancy : deficiency of ADH can result from increased metabolism by a placental enzyme (vasopressinase enzyme). It resolves within a few weeks after delivery.
- B. **Familial DI (Wolfram syndrome, DIDMOAD):** hereditary condition in which there is a defect in osmo-receptors (*NB: Osmoreceptors are present in anterior hypothalamus*).
 - Diabetes Insipidus.
 - Diabetes Mellitus.
 - Optic atrophy.
 - Deafness.

C. Nephrogenic DI (Renal tubules not responding to ADH)

- Hereditary (X-linked recessive so, males are predominantly affected)
- Acquired
 - Chronic renal failure.
 - Renal tubular acidosis
- 2. Kidney amyloidosis.
- 3. Polycystic kidney & Pyelonephritis.
- 4. Hypercalcemia & Hypokalemia.
- 5. Drugs : Lithium , Cholchicine, Demeclocycline.
- 6. Heavy metal poisoning

Clinical Picture:

- a. **C/P of the cause:** e.g. history of hypophysectomy.
- b. **C /P of the hormone :**
 - Polyuria (up to 20 L/day) & polydipsia.
 - Severe dehydration -+ weakness , weight loss & fever.
 - Hypovitaminosis : of water soluble vitamins.
 - Complications: shock & death.

Investigation :

- a. **Investigations for the cause: Pituitary Imaging: X-ray , CT & MRI**
- b. **Assay of the hormone**
- c. **Investigations for the function of ADH :**
 1. **Urine analysis :**
 - Polyuria with low Specific gravity.
 - No pathological constituents.
 - After fluid deprivation : Polyuria persists.
 2. **Plasma osmolality :** ↑ due to loss of free water.
- d. **Stimulation tests :**
 - i. **Test the hypothalamus: nicotine test (1-3 mg nicotine)**
 - 1. normal : oliguria due to stimulation of ADH.
 - 2. central DI : -ve (no stimulation)
 - ii. **Test osmo-receptors : IV hypertonic saline (NaCl 2.5 %)**
 - normal : oliguria.
 - osmo-receptors defects : -ve (no oliguria)
 - iii. **Test kidney (vasopressin test) :** to differentiate between central & nephrogenic DI
 - 1 . in Central DI : oliguria (Concentrated urine)
 - 2. in Nephrogenic DI : No change.
 - iv. **Water deprivation test:**
 - monitor serum and urine osmolality, urine volume and body weight hourly for 8 hours.
 - Proceed to desmopressin administration i.m. and monitor the response.

Treatment :

- **Replacement therapy:** synthetic ADH (Desmopressine) 20 µg/d intra nasal
- **Diet :** ↓ Na , ↑ vitamins.
- **Drugs**
 - Cholpropamide : for central DI.
 - Carbamazepine : in nephrogenic DI.
 - Thiazide may be used in nephrogenic DI.
 - Amiloride (potassium-sparing diuretic) : is the drug of choice in lithium toxicity induced DI.

Differential diagnosis of polyuria

Urine output > 3L/d in adult

1. Physiological :

- Winter months.
- Excess coffee or tea.
- Diuretics.

2. Psychological : Hysterical polydipsia.

3. Pathological

- **Endocrinal :**

- DI : with its causes.
- DM

- **Renal**

- i. Nephrogenic DI.
- iii. Diuretic phase of acute renal failure.

Syndrome of inappropriate ADH secretion

Physiology :

- ❑ *It means an excessive release of ADH*

Etiology:

- **Tumors release ADH** : Oat cell carcinoma & lymphoma.
- **Pulmonary lesion** : Legionella pneumonia, TB.
- **CNS** : meningitis, encephalitis & head injuries.
- **Drugs** : cholropropamide, carbamazepine, cyclophosphamide.

Clinical Picture:

- Increased ADH causes water retention → dilutional hyponatremia
- (weight gain , weakness , confusion, convulsions & coma)
- No edema because of natriuresis.

Investigation:

- ↓Na (< 130 mEq /L)
- ↓ serum osmolality (< 270 mosmol/kg) .
- ↑ urine Na concentration (> 20 mEq/L) .

Treatment :

- a. Fluid restriction→ 0.8 -1 L / day.
- b. Demeclocycline:
 - 600 -1200 mg /day→ inhibit the action of ADH.
 - Given to patient unresponsive to fluid restriction.
- c. For sever hyponatremia : hypertonic saline(5%) IV slowly.

2

Growth hormone and Its Disorders

Growth hormone (GH)

- Also called somatotropin or human growth hormone
- peptide hormone secreted by the anterior lobe of the pituitary gland.
- It stimulates the growth of essentially all tissues of the body, including bone.
- GH is synthesized and secreted by anterior pituitary cells called somatotrophs, which release between one and two milligrams of the hormone each day.
- GH is vital for normal physical growth in children; its levels rise progressively during childhood and peak during the growth spurt that occurs in puberty.
- After puberty the concentrations of IGF-1 gradually decrease with age, as do GH concentrations.

Function

- **CHO** : hyperglycemia.(It antagonizes the action of insulin).
 - Decreases glucose uptake by the cells.
 - Increases the insulin resistance.
 - **Fat** : lipolysis.
 - **Protein**: anabolic (bone, muscle & viscera)
 - **Minerals**: ↑ (Na, K, Ca & P04).
- ❑ GH may act directly on tissues, but much of its effect is mediated by stimulation of the **liver** and other tissues to produce and release **insulin-like growth factors**, primarily insulin-like growth factor 1 (**IGF-1**; formerly called **somatomedin**).
- ❑ The term insulin-like growth factor is derived from the ability of high concentrations of these factors to mimic the action of insulin

Regulation

- GH secretion is **stimulated by growth hormone-releasing hormone (GHRH)** and is **inhibited by somatostatin**.
- GH secretion is pulsatile, with surges after the onset of deep sleep (especially prominent at the time of puberty).
- GH secretion increases in response to decreased food intake and to physiological stresses
- GH secretion decreases in response to food ingestion.

Pituitary Gigantism

Aetiology:

Increased secretion of GH before epiphyseal closure due to:

- 1- Pituitary hyperplasia.
- 2- Pituitary adenoma (rare).

Clinical picture

A- Endocrinal manifestations:

- 1) Disproportionate gigantism:
 - Span > height.
 - Lower segment > upper segment.
- 2) Later on features of acromegaly may develop.
- 3) Later on features of hypopituitarism may develop (rare).

B- Headache: (rarely occur in cases of pituitary tumours)

It typically passes through three stages:

- 1) At first, there is bitemporal headache due to increased intrasellar pressure.
- 2) Then, headache disappears due to rupture of sella.
- 3) Later, headache reappears & become generalized due to increased ICT.

C- Pressure manifestations: (rarely occur in cases of pituitary tumors)

1) Anteriorly:

- Compression of optic chiasma: bitemporal hemianopia.
- Compression of olfactory tract: ipsilateral anosmia.

2) Posteriorly:

- Compression of the upper part of brain stem: bilateral pyramidal affection & ophthalmoplegia.

3) Laterally:

- Compression of cavernous sinus: 3rd 4th & 6th cranial nerve paralysis & sensory loss in area supplied by ophthalmic division of 5th nerve,
- Compression of optic tract; contralateral homonymous hemianopia,

4) Superiorly:

- Compression of hypothalamus leading to hypothalamic syndrome e.g. diabetes insipidus, hypersomnia, obesity, autonomic epilepsy & hypopituitarism.

Investigations

1. Skull X-ray: may show features of intrasellar tumours (rare).

2. MRI & CT of skull:

- For accurate diagnosis of pituitary tumors.
- For detection of suprasellar extension.

3. Assay of GH level:

- Elevated (Normal level: < 20 ng/ml).

N.B. There is elevated level of IGF-1.

4- Glucose suppression test: administration of glucose fails to decrease GH level.

5- Blood chemistry: increased Ca, P & glucose.

N.B. Examination of field of vision may show bitemporal hemianopia

Treatment

1- Medical treatment:

- Bromocriptine: (5-20 mg/day) Decreases level of GH & may cause shrinkage of tumour.
- Octreotide (somatostatin analogue): 100-200 ug/8 h SC.

2- Surgical hypophysectomy:

- Indicated in patients having pituitary tumors with suprasellar extension
- Hypophysectomy may be transsphenoidal (most commonly used or transfrontal.

3- Pituitary irradiation:

Indicated:

- In patients with no suprasellar extension.
- If surgery is contraindicated or refused.

Methods:

- Transsphenoidal implantation of yttrium 90.
- External irradiation.

4- Symptomatic treatment.

Causes of Tall Stature

defined as height significantly above the average for age and sex (typically >2 standard deviations above the mean or above the 97th percentile), can result from a variety of causes. These range from normal variants (familial/genetic) to pathological conditions. Below is an organized overview:

I. Normal Variants (Non-Pathological)

- **Familial (Constitutional) Tall Stature**
 - Most common cause. Child's height aligns with genetic potential (tall parents).
 - Growth velocity and bone age are normal, with proportional body features.
- **Constitutional Advanced Growth (Early Maturation)**
 - Early puberty leads to a growth spurt, but final adult height is normal. Bone age is slightly advanced.

II. Endocrine Causes

- **Growth Hormone (GH) Excess**
 - Gigantism: Caused by a pituitary adenoma secreting excess GH before growth plate closure (in children). Leads to rapid, disproportionate growth.
 - Untreated cases can progress to acromegaly in adulthood.
- **Hyperthyroidism**
 - Accelerates growth rate temporarily but may advance bone age, leading to shorter final height if untreated.
- **Precocious Puberty**
 - Early puberty triggers a growth spurt, but premature closure of growth plates may result in shorter adult height. Rarely causes persistent tall stature.
- **Exogenous Obesity**
 - Childhood obesity may cause accelerated growth due to hormonal effects (e.g., insulin/leptin), but final height is typically normal.

III. Genetic Syndromes

- **Marfan Syndrome**
 - Autosomal dominant disorder (FBN1 gene mutation). Tall, slender build, long limbs (arm span > height), aortic dilation, lens dislocation.
- **Homocystinuria**
 - Similar to Marfan but with intellectual disability, thromboembolic risk, and homocysteine elevation.
- **Sotos Syndrome (Cerebral Gigantism)**
 - Overgrowth in childhood (large head, advanced bone age), developmental delays, and distinctive facial features.
- **Beckwith-Wiedemann Syndrome**
 - Prenatal/childhood overgrowth, macroglossia, abdominal wall defects, and increased cancer risk.
- **Klinefelter Syndrome (47,XXY)**
 - Males with tall stature, long limbs, hypogonadism, and infertility.
- **Fragile X Syndrome**
 - Intellectual disability, macrocephaly, and tall stature in some cases.
- **XXY Syndrome**
 - Males with mild developmental delays and above-average height.

IV. Other Causes

- **Aromatase Deficiency**
 - Impaired estrogen production delays growth plate closure, leading to continued linear growth into adulthood.
- **Exogenous Hormone Exposure**
 - Excessive GH or androgen use during childhood.
- **Chromosomal Abnormalities**
 - e.g., Duplications in SHOX gene (short stature homeobox-containing gene) or other growth-related genes.
- **Overnutrition**
 - Rarely, excessive caloric intake in early childhood may accelerate growth.

Evaluation of Tall

Red Flags prompting investigation:

- Height significantly discordant with parental height.
- Rapid growth velocity.
- Dysmorphic features (e.g., long fingers, heart defects).
- Signs of endocrine dysfunction (e.g., headaches, vision changes in gigantism).

Tests: Bone age X-ray, GH/IGF-1 levels, thyroid function, karyotyping, and genetic testing for syndromes.

Acromegaly

Aetiology:

Increased secretion of GH after epiphyseal closure due to:

- 1- Pituitary adenoma.
- 2- Pituitary hyperplasia due to ectopic secretion of GH-RF (very rare).

Clinical picture

- -Age usually starts in the third or fourth decade
- -The onset is insidious.
- -The course is slowly progressive

A. Headache

B. Pressure manifestations

C. Endocrinal manifestations:

1. Face(Ape- like facies):

- -Big skull with long face
- -Prominent frontal bosses, supra orbital ridges & zygomatic arches
- -Prognathism with separation of teeth
- -Enlargement of ears, nose, lips & tongue
- -Skin of forehead is thickened wrinkled greasy & sweaty

2-Hands & feet

- -Enlarged hands with blunt fingers (spade - like).
- -Enlarged feet with increase thickness of heel pad

3. Organomegaly

- -Hepatosplenomegaly.
- -Cardiomegaly and may be cardiomyopathy leading to heart failure
- -Hypertrophy of the larynx causing thick voice

4.Muscular manifestations:

- Pseudohypertrophic myopathy

5.Skin manifestations:

- The skin is thickened, wrinkled greasy & sweaty

6. Skeletal manifestations:

- Osteoarthritis & kyphosis may occur.

7- Endocrinal & metabolic manifestations:

- -Diabetes mellitus in 30% of cases.
- - Hypertension (due to increased body Na).
- - Galactorrhea may occur.
- -Later on features of hypopituitarism may develop.

8- Late complications:

- -Increased risk of atherosclerosis.
- -Increased incidence of colonic cancer.

Investigations

1- Skull X-ray:

- -Thick cortex
- -Wide frontal sinuses.
- -Prognathism with separation of teeth.
- -Features of intrasellar tumors.

2- Hands X-ray:

- -Broadening of phalanges.
- Tufting of terminal phalanges (mushroom-like).

3- MRI & CT of skull:

- - For accurate diagnosis of pituitary tumors.
- - For detection of suprasellar extension.

4- Assay of GH level:

- - Elevated (Normal level: < 20 ng/ml).
- N.B. There is elevated level of IGF-1.

5. Glucose suppression test: administration of glucose fails to decrease GH level.

6- Assay of prolactin level: elevated in 30% of cases.

7- Blood chemistry: increased Ca, P & glucose.

- N.B. Examination of field of vision may show bitemporal hemianopia.

Treatment

1- Surgical hypophysectomy:

- - Indicated in patients having pituitary tumours with suprasellar extension.
- - Surgical hypophysectomy may be transsphenoidal (most commonly used) or transfrontal.

2- Pituitary irradiation:

Indicated:

- In patients with no suprasellar extension.
- If surgery is contraindicated or refused
- If there is a residual tumour after surgery

Methods:

- - Transsphenoidal implantation of yttrium⁹⁰
- - External irradiation.

3-Medical treatment:

- ✓ **Bromocriptine: (5-20 mg/day)**
 - Decreases level of GH & may cause shrinkage of tumor.
- ✓ **Octreotide (somatostatin analogue):** 100-200 ug/8 h SC.

4- Symptomatic treatment: e.g. for diabetes mellitus, hypertension.

Pituitary Dwarfism Pituitary Infantilism (Levi-Lorain syndrome)

Causes of Pituitary Dwarfism

1. Congenital Causes (Present at Birth):

- Genetic mutations affecting the pituitary gland or growth hormone production
- Birth trauma or hypoxia affecting the pituitary gland.

2. Acquired Causes (Develop Later in Life):

- Brain tumors (e.g., craniopharyngioma, pituitary adenoma).
- Head trauma or injury to the pituitary gland.
- Infections or inflammation affecting the brain (e.g., meningitis, encephalitis).
- Radiation therapy to the brain or pituitary region.
- Autoimmune diseases affecting the pituitary gland.

3. Idiopathic Causes:

- In some cases, the cause of growth hormone deficiency is unknown.

Symptoms of Pituitary Dwarfism

- Short stature: Height significantly below the average for age and sex.
- Slow growth rate: Growth velocity is slower than expected for age.
- Delayed puberty: Absence or delay in the development of secondary sexual characteristics.
- Childlike facial features: The face may appear younger than the child's actual age.
- Increased fat around the waist: Body proportions may appear abnormal.
- Hypoglycemia: Low blood sugar, especially in infants and young children.
- Delayed bone age: Bone maturation is slower than chronological age.
- Other pituitary hormone deficiencies: If the pituitary gland is affected more broadly, deficiencies in hormones like thyroid-stimulating hormone (TSH), adrenocorticotrophic hormone (ACTH), or gonadotropins may occur.

Diagnosis of Pituitary Dwarfism

1. Clinical Evaluation:

- Assessment of growth patterns using growth charts.
- Physical examination to evaluate body proportions and signs of delayed puberty.

2. Laboratory Tests:

- **Growth hormone stimulation tests:** Measure GH levels after administering substances that normally stimulate GH release (e.g., insulin, arginine, clonidine).
- **Insulin-like growth factor 1 (IGF-1) and IGFBP-3 levels:** These are often low in children with GHD.

3. Imaging Studies:

- **MRI of the brain:** To evaluate the structure of the pituitary gland and hypothalamus and rule out tumors or structural abnormalities.
- **X-rays of the hand and wrist:** To assess bone age.

4. Other Hormonal Tests:

- To check for deficiencies in other pituitary hormones (e.g., TSH, ACTH, LH, FSH).

Treatment of Pituitary Dwarfism

1. Growth Hormone Replacement Therapy:

- Synthetic growth hormone (somatropin) is administered daily via subcutaneous injections.
- Treatment is most effective when started early and continued until growth is complete.

2. Management of Other Hormonal Deficiencies:

- If other pituitary hormones are deficient, replacement therapy may be needed (e.g., thyroid hormone, cortisol, or sex hormones).

3. Regular Monitoring:

- Growth progress, bone age, and hormone levels are monitored regularly to adjust treatment.

4. Psychological Support:

- Counseling or support groups may help children and families cope with the emotional and social challenges of short stature.

Frohlich's syndrome

Aetiology: Hypothalamic-pituitary lesions e.g. craniopharyngioma, encephalitis.

C/P:

- Trunkal obesity.
- Genu valgum
- Dwarfism.
- May be hypogonadism
- Diabetes insipidus.
- Polyphagia.
- Hypersomnia.

Short Stature

Aetiology

1. Normal Variants (Non-Pathological Causes)

- **Familial Short Stature:** A child inherits short stature from parents who are also short. Growth velocity and bone age are normal.
- **Constitutional Growth Delay (CGD):** A child is shorter than peers during childhood but has a delayed bone age and will eventually catch up in height during late adolescence. Often associated with a family history of "late bloomers."

2. Endocrine Causes

- **Growth Hormone Deficiency (GHD):** Insufficient production of growth hormone by the pituitary gland.
- **Hypothyroidism:** Low levels of thyroid hormone, which is essential for normal growth.
- **Cushing's Syndrome:** Excess cortisol levels, which can suppress growth.
- **Precocious Puberty:** Early puberty can lead to an initial growth spurt but premature closure of growth plates, resulting in short adult height.

3. Genetic and Chromosomal Disorders

- **Turner Syndrome (45,X):** A chromosomal disorder in females that leads to short stature and other physical features.
- **Down Syndrome (Trisomy 21):** A chromosomal disorder associated with short stature and developmental delays.
- **Noonan Syndrome:** A genetic disorder causing short stature, heart defects, and distinctive facial features.
- **SHOX Gene Mutations:** Mutations in the SHOX gene can cause short stature, as seen in Léri-Weill dyschondrosteosis.

4. Skeletal Dysplasias (Bone Disorders)

- **Achondroplasia:** The most common form of dwarfism, caused by a mutation in the FGFR3 gene.
- **Hypochondroplasia:** A milder form of skeletal dysplasia.
- **Osteogenesis Imperfecta:** A disorder characterized by brittle bones and short stature.

5. Intrauterine Growth Restriction (IUGR)

- Poor growth in the womb due to placental insufficiency, maternal malnutrition, or infections during pregnancy. Babies born small for gestational age (SGA) may not catch up in height.

6. Chronic Diseases

- **Malnutrition:** Inadequate caloric or protein intake can impair growth.
- **Celiac Disease:** Malabsorption of nutrients due to gluten intolerance.
- **Chronic Kidney Disease:** Affects growth due to metabolic imbalances and poor nutrition.
- **Inflammatory Bowel Disease (IBD):** Conditions like Crohn's disease or ulcerative colitis can impair growth.
- **Congenital Heart Disease:** Can lead to poor growth due to increased energy demands and reduced oxygen delivery.

7. Idiopathic Short Stature (ISS)

- Short stature with no identifiable cause after thorough evaluation. These children are otherwise healthy and have normal growth hormone levels.

Investigations

- Exclude systemic disease
- Thyroid function tests - serum **TSH** and **free T4** to exclude hypothyroidism.
- GH status. Basal levels are of little value, though urinary GH measurements may prove to be of some value in screening.
- Dynamic tests include the GH response to insulin.

Test should only be performed in centers experienced in their use and interpretation.

Assessment of bone age. Non-dominant hand and wrist X-rays allow assessment of bone age by comparison with standard charts.

Treatment

- Systemic illness should be treated and primary hypothyroidism replaced with thyroxine.
- For GH insufficiency, recombinant GH (somatropin) is given as nightly injections in doses of 0.17-0.35 mg/kg per week.
- Treatment is expensive and should be supervised in expert centers.
- GH treatment in so -called 'short normal' children has not been shown to produce any worthwhile increase in final height.

Summary

- **Growth hormone (GH), or somatotropin**
- **Peptide hormone secreted by the anterior lobe of the pituitary gland.**
- **It promotes growth of bone and other body tissues by stimulating protein synthesis and fat breakdown (for energy).**
- **Excessive production causes gigantism, acromegaly, or other malformations**
- **Deficient production results in dwarfism, dramatically relieved if GH is given before puberty.**
- **Genetic engineering techniques now permit large-scale production of adequate amounts of GH for that purpose.**

3

Panhypopituitarism

Definition

- Loss of anterior pituitary function

Etiology

1. Destruction of pituitary gland by:

- Surgery.
- Irradiation.
- Tuberculosis, sarcoidosis, hemochromatosis.

2. Sheehan syndrome : pituitary infarction after sever postpartum hemorrhage

3. Pituitary apoplexy : Hemorrhage in pituitary tumor.

4. Genetic: Isolated GH or GnRH deficiency.

5. Autoimmune.

Clinical picture

1. C / p of the cause: Surgery, Sever post-partum hemorrhage.

2. C / p of the hormones :

- According to the frequency of appearance : GH & Gondotrophins (FSH , LH) are lost early then TSH then ACTH deficiency.

a. Gonadal deficiency :

- ✓ **Females:** amenorrhea, ! libido, infertility, loss of pubic & axillary hair.
- ✓ **Males:** impotence,! libido, infertility, loss of pubic & axillary hair.

b. GH deficiency :

- ✓ **Adult:** Fasting hypoglycemia, osteoporosis & fine wrinkles.
- ✓ **Children:**
 - **GH deficiency only** : short stature with proportionate dwarfism.
 - **GH with FSH & LH deficiency** : dwarfism with infantile proportions (upper> lower segment & span shorter than height.

3. TSH deficiency : result in 2ry hypothyroidism.

- In contrast to 1ry hypothyroidism hypercholesterolemia is rare.
- Skin is thin with wrinkling.
- no menorrhagia in 2ry hypothyroidism.

4. ACTH deficiency : Results in 2ry adrenal insufficiency,

- simulates Addison's disease but :
 1. No pigmentation (absent ACTH)
 2. No marked hypotension , due to normal aldosterone.

5. Prolactin : Low with failure of lactation in **Sheehan syndrome**.

- It may be high in lesions causing pituitary stalk compression.

6. Coma : due to

- Hypoglycemic coma due to ↓↓ GH & cortisone.
- Myxedema coma.
- Pressure in cases of pituitary tumors.
 - NB : If the patient has associated DI: the problem is at the hypothalamus.

Differential diagnosis

- From 1 ry Addison's disease & 1 ry hypothyroidism
- From anorexia nervosa :
 - Normal hair & breast.
 - Aggressive attitude.
 - Normal cortisol & High GH due to hypoglycemia.

Investigation

❑ 1-Investigations for the cause:

- Pituitary imaging :X-ray, CT & MRI

❑ 2. Assay of the hormones level :

- ↓ GH [n : 1 - s n.gm/ml] & ! IGF-1.
- ↓ FSH, ! LH & ! sex hormones.
- ↓ TSH, ! T3 & ! T4.
- ↓ACTH & ! cortisol.
- Prolactin : Low or high

❑ 3. Investigations of the function of the hormone : e.g.

- Hypoglycemia.
- Hyponatremia.

❑ 4. Stimulatory test :

✓ Insulin tolerance test

- It is a gold standard for diagnosis of GH deficiency.
- After overnight fasting, regular insulin (0.1 U/Kg) is given :
 1. Normal : +ve (↑GH > 7 ng/ml)
 2. In pan-hypopituitarism : -ve (no stimulation) .

Treatment

1. Treatment of the cause.

2. Replacement therapy: Multiple hormones must be replaced.

- Hydrocortisone : 30 mg/ d. (20 mg a.m. & 10 mg p.m.)
- Gonadotrophin deficiency
 - ✓ *If fertility is not a problem :*
 - In males: testosterone.
 - In females: estrogen/ progesterone.
 - ✓ *If fertility is a problem :* (both males & females)
 - LH : Human chorionic gonadotropin (hCG) .
 - FSH : human menopausal gonadotropin.
 - GnRH : may be occasionally used.

3. TSH deficiency:

- Oral Levo-thyroxine, **should be started after steroid replacement, to avoid acute adrenal failure.**

4. ADH deficiency:

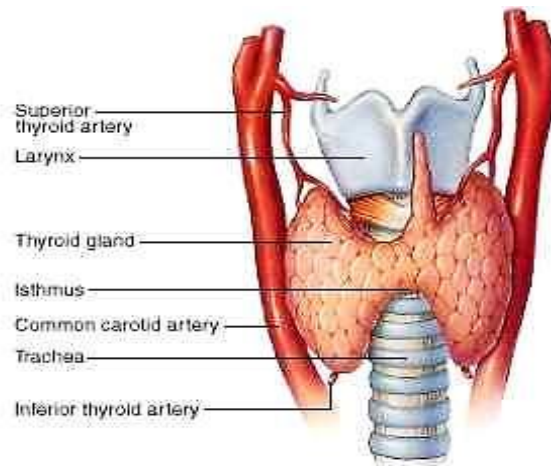
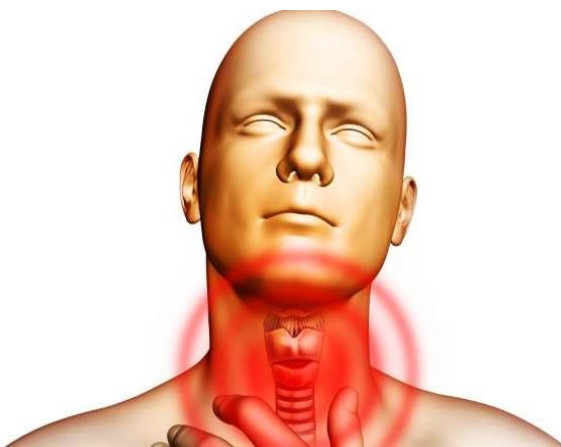
- Desmopressin (intranasal, SC, oral)

5. Recently :

1. Purified pituitary hormones.
2. Hypothalamic releasing factors

4

Thyroid gland & Diseases Of Thyroid gland



Thyroid Gland
is **UNIQUE** among other endocrine
glands in 3 ways

1. PHYSICAL

Only gland which
can be seen &
palpated.

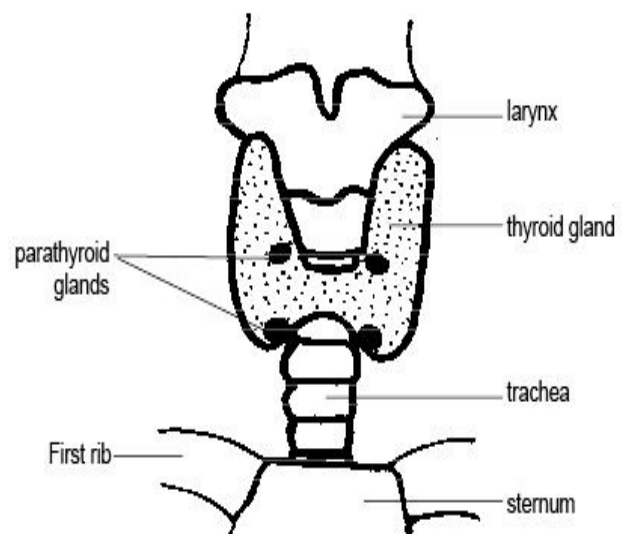
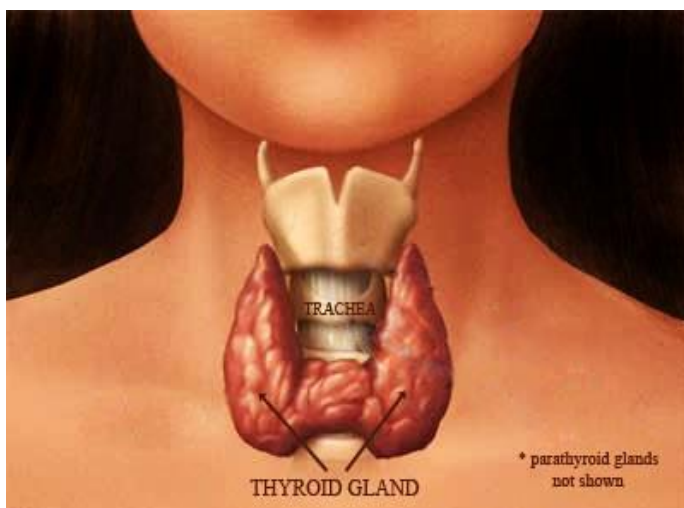
2. BIOCHEMICAL

Only hormone that
needs a trace
element Iodine for
its production

3. PHYSIOLOGY

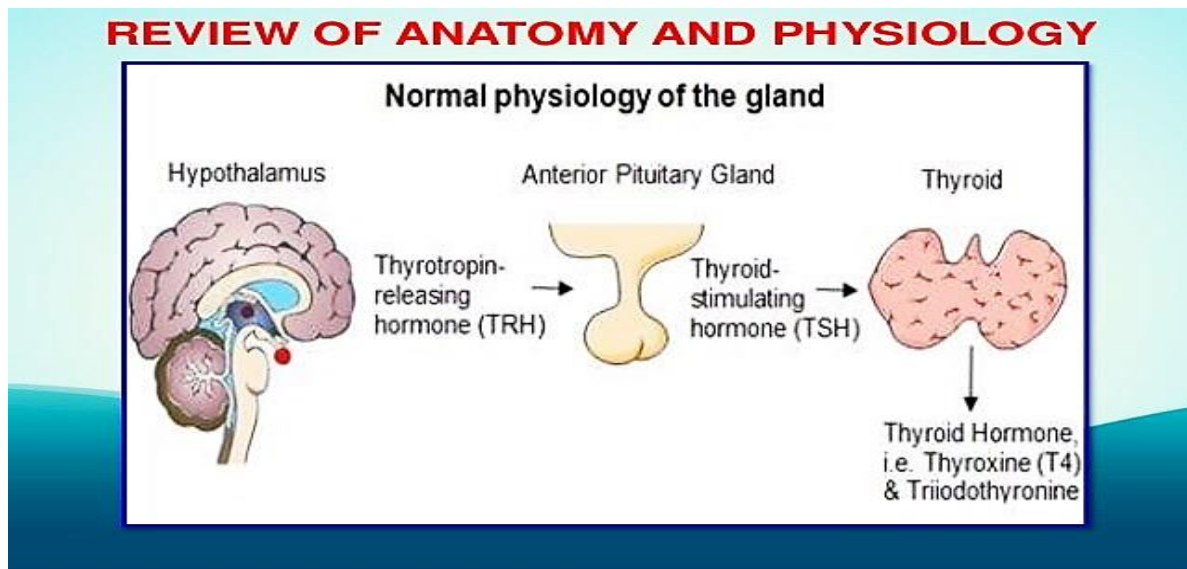
Hormone is stored
in an extracellular
site in the **Thyroid
Colloid**.

LOCATION OF THE THYROID GLAND

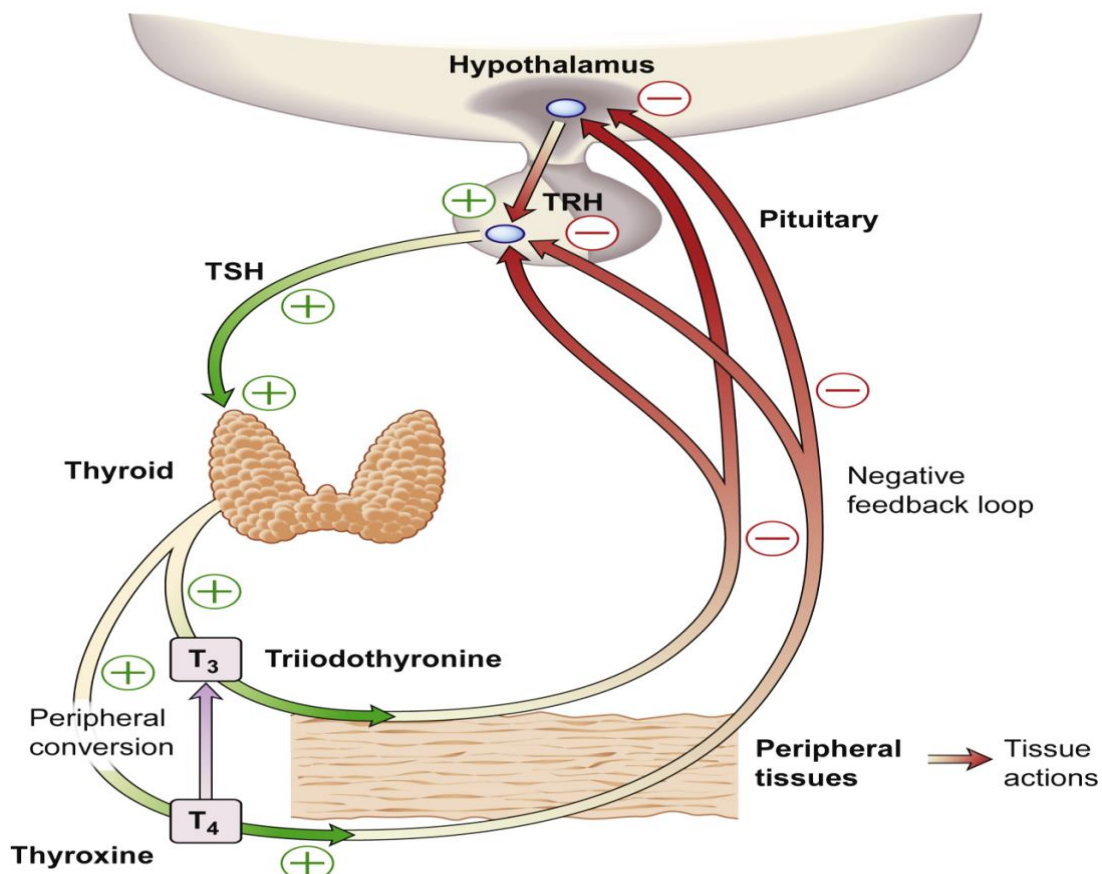


REVIEW OF ANATOMY AND PHYSIOLOGY

- -The thyroid gland is a butterfly-shaped gland located in the front of the neck. The thyroid gland itself is regulated by the pituitary gland in the brain, and the pituitary gland is regulated by the hypothalamus, another gland in the brain.
- It consists of two lateral lobes connected by an isthmus. The gland is about 5 cm long, 3 cm wide and weight about 30 g.
- The thyroid gland produces three hormones (T₄), triiodothyroxine (T₃), and calcitonin



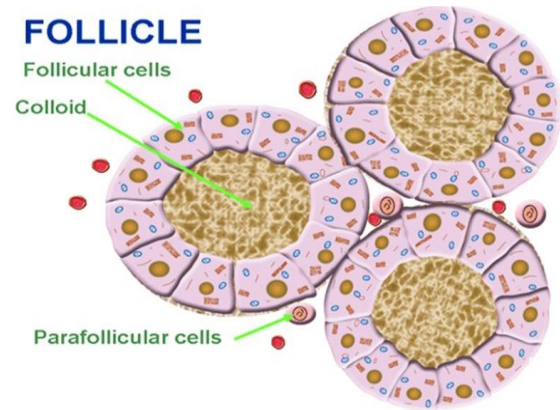
CONTROL & FEEDBACK



THYROID GLAND

The thyroid gland consists of 2 types of cells:

1. Follicular cells: These are more abundant, and the major secretory cells. They secrete Thyroid hormone.
2. Parafoallicular cells or C-cells: These are fewer in number & interspersed. They secrete Calcitonin.



THYROID HORMONES

- The Thyroid gland secretes 3 major hormones
- 1. Thyroxine or T4 : having 4 atoms of Iodine. (secreted in largest amount)
- 2. Triiodothyronine or T3 : having 3 atoms of Iodine (secreted in lesser amount)
- 3. Reverse T3 also called RT3. (secreted in the least amount)
- 4. Calcitonin: which is an important hormone of calcium metabolism.

LABORATORY EVALUATION AND IMAGING STUDIES

- Serum T4
- Serum T3
- TSH
- Anti-thyroid antibodies
- Thyroid stimulating Immunoglobulins
- Thyroid uptake and scan
- Thyroid Ultrasound

Serum Thyroxine (T4)

- **Measure free T4, not total T4**
- Only free T4 is biologically active
- Conditions that alter TBG alter total T4 but not free T4
- Pregnancy raises total T4
- Chronic liver failure lowers total T4
- **High in hyperthyroidism**
- **Low in hypothyroidism**

Serum Triiodothyronine (T3)

- Measurement of free T3 is preferable to total T3.
- High in hyperthyroidism
- Low in hypothyroidism
- But generally not worth measuring in hypothyroidism because T3 is less sensitive and less specific than the decrease in free T4

Serum Thyrotropin (Thyroid Stimulating Hormone; TSH)

- TSH is the most sensitive screening test for hyperthyroidism and primary hypothyroidism
- TSH within the normal range excludes these diagnoses
- **TSH is LOW in hyperthyroidism**
- **TSH is HIGH in hypothyroidism**

Antithyroid Antibodies

1. Antimicrosomal antibodies, thyroid peroxidase antibodies & Anti-thyroglobulin antibodies Present in ~95% of Hashimoto's and ~60% of Graves' patients at the time of diagnosis
2. Thyroid Stimulating Immunoglobulins (TSI) Is present in Graves' disease

Imaging studies

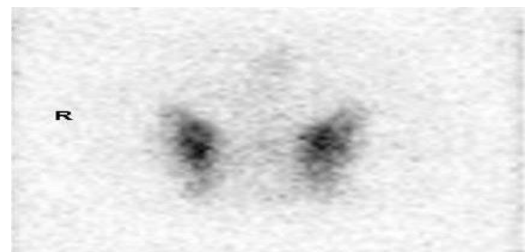
- Thyroid uptake and scan
- Thyroid US
- Neck CT

Thyroid uptake and scan

- I-123
- I-131
- Technetium 99

*Radiotracer:

- **Injectable IV:** Technetium (15min later: scan)
- **Oral:** I 131 and 123 (24 h later: scan/uptake)
- **Scan:** structure
- **Uptake:** function
- Obtain pregnancy test before the test

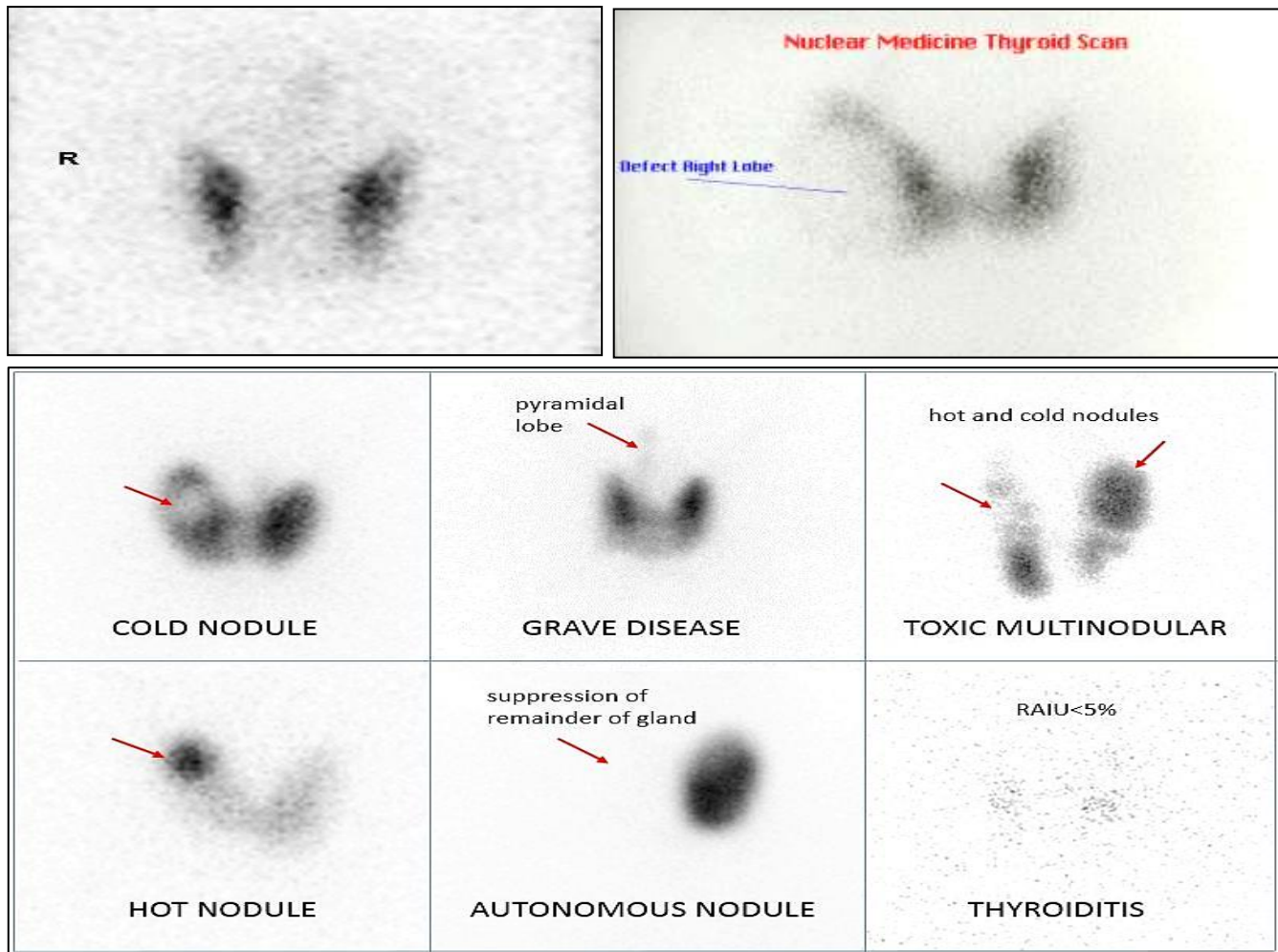


Radioiodine Uptake

- **Used to evaluate the cause of hyperthyroidism**
- High if the thyroid is hyper-functioning, e.g. Graves'
- Low if thyroid hormone is leaking out of damaged thyroid cells (subacute thyroiditis) or the patient is taking excess exogenous thyroid hormone
- Expressed as a NUMBER (e.g., 35%)
- Used to calculate the dose of I-131 to treat hyper-functioning thyroid tissue or cancer.

Thyroid Scan (nuclear Scan)

- Primary use is to determine whether palpated nodules are functional or non-functional.
- **“Hot”** nodules concentrate the radionuclide and are essentially always benign.
- **“Cold”** nodules are usually benign but are sometimes malignant.



THYROID ULTRASONOGRAPHY

- Painless, quick, no contrast material, no radiation
- Can be used in pregnancy,
- Can detect thyroid nodules as small as 2-3 mm and provide guidance for FNA biopsy

Thyroid disorders

- **Hypothyroidism**
 - Underactive thyroid
- **Thyroid enlargement**
 - Goiter
- **Hyperthyroidism**
 - Overactive thyroid

1) Thyrotoxicosis

DEFINITION

Hyperthyroidism is the term for overactive tissue within the thyroid gland, resulting in overproduction and thus an excess of circulating free thyroid hormones: thyroxine (T4), triiodothyronine (T3) or both.

Hyperthyroidism is hyperactivity of the thyroid gland with sustain increase in synthesis and release of hormones.



- **Thyrotoxicosis**
 - Refers to the manifestations of excessive quantities of circulating thyroid hormone .
- **Hyperthyroidism**
 - Means that the thyroid gland is functioning more than normal.
 - Therefore ,a hyperthyroid patient is thyrotoxic , but a thyrotoxic patient need not have an overactive thyroid and is therefore not actually hyperthyroid.
- **Thyrotoxicosis:**
 - High(free T4) and suppressed (TSH).
- **T3 toxicosis:**
 - free T4 normal, high (freeT3) and TSH is low **subclinical thyrotoxicosis**
 - both free hormones are normal, but TSH is low

Causes of hyperthyroidism

1-Grave's disease "Diffuse toxic goiter":

- Caused by a generalized over activity of the thyroid gland
- Most common cause
- Gland has lost the ability to respond to the normal control by TSH.
- Common among women more than men.
- Triggered by stress, smoking, radiation to the neck, medications, and infectious organisms such as viruses
- It may be associated with Graves' ophthalmopathy & dermatopathy.

2-Toxic nodular goiter "Multi-nodular goiter":

- A condition in which one or more nodules of the thyroid become overactive. The overactive nodules actually act as benign thyroid tumors.
- Symptoms of toxic nodular goiter do not include bulging eyes or skin problems.

3- Thyroiditis:

- Temporary hyperthyroidism then hypothyroidism.
- 3 types:
 - 1- Hashimoto's Thyroiditis.
 - 2- Subacute Granulomatous Thyroiditis: associated with fever & sore throat. It is tender to touch.
 - 3- Silent Lymphocytic Thyroiditis: associated with accumulation of lymphocytes.

4-Excessive iodine intake:

- by medications " amiodarone" - which is used for treatment of heart problems - or in diet.

5- Excessive secretion of TSH:

- due to a tumor in the pituitary gland. This condition is very rare and can be associated with other abnormalities of the pituitary gland.

CLINICAL TYPES

- Primary thyrotoxicosis (Grave's disease)
- Toxic multinodular goitre
- Solitary toxic nodule
- Hyperthyroidism due to rarer causes

Grave's Disease

Characteristics

- ˆ The most common cause of thyrotoxicosis (60-80%)
- ˆ It is an auto-immune disease
- ˆ The most important autoantibody is
 - ❖ Thyroid Stimulating Immunoglobulin(TSI)or TSA
 - ❖ Others(anti-TPO) (anti-TG)
- ˆ Symmetrical enlargement of the thyroid
- ˆ Hyper secretion of thyroid hormones
- ˆ Patients tend to be young women

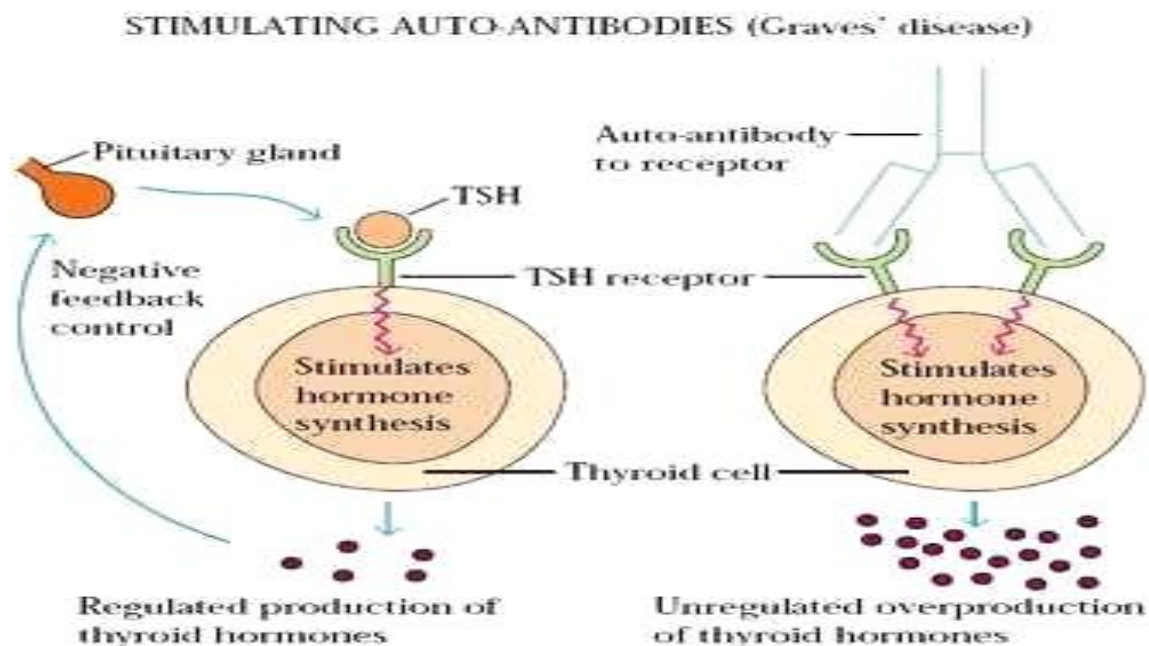
Caused by a generalized over activity of the thyroid gland

-Gland has lost the ability to respond to the normal control by TSH.

-Triggered by stress, smoking ,radiation to the neck ,medications ,and infectious organisms such as viruses

-It may be associated with Graves'

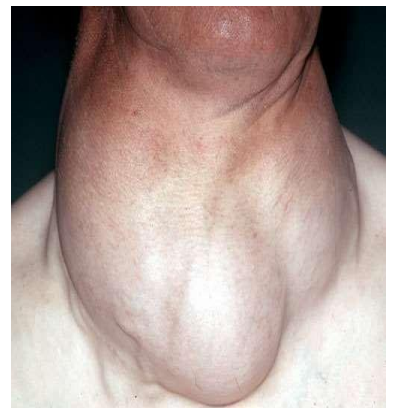
Etiology ,Pathogenesis ,and Pathology



Toxic Multinodular Goiter(TMg)

Characteristics

- TMG is the next most common cause of hyperthyroidism
- Caused by focal regions of hyper functioning independent of TSH
- Due to mutation of the TSH receptor Or Can be the result of chronic iodine deficiency
- Excessive TSH stimulation induces
 - ❖ Focal hyperplasia
 - ❖ Subsequent necrosis and hemorrhage
 - ❖ Nodule formation
 - ❖ <CV manifestations predominate



Toxic Single Adenoma (TSA)

Characteristics

- TSA is a single hyper functioning follicular thyroid adenoma.
- Benign monoclonal tumor that usually is larger than 2.5 cm
- Nuclear Scintigraphy scan shows only a single hot nodule
- 'TSH is suppressed by excess of thyroxines ,So the rest of the thyroid gland is suppressed

- Temporary hyperthyroidism then hypothyroidism.

3 types:

1. Hashimoto's Thyroiditis.
2. Subacute Granulomatous Thyroiditis :ALSO KNOWN AS De Quervain's thyroiditis . associated with fever & sore throat .It is tender to touch.
3. Silent Lymphocytic Thyroiditis :associated with accumulation of lymphocytes.

Hashimoto's thyroiditis

- also known as autoimmune or chronic lymphocytic thyroiditis Or Hashitoxicosis
- the most common form of thyroiditis
- Biochemically and clinically ,there is an initial period of thyrotoxicosis secondary to the release of thyroid hormones from the inflamed gland.
- thyrotoxic phase ranged from 31 to 168 days This is followed by the development of hypothyroidism or recovery.
- Treatment of these disorders is symptomatic

De Quervain 's thyroiditis

- This is transient hyperthyroidism from an acute inflammatory process ,probably viral in origin.
- There is usually fever ,malaise and pain in the neck with tachycardia and local thyroid tenderness.
- Thyroid function tests show initial hyperthyroidism ,the(ESR) and plasma viscosity are raised.
- Hypothyroidism ,usually transient.
- Treatment of the acute phase is with aspirin ,using short term prednisolone in severely symptomatic cases.

CLINICAL FEATURES OF HYPERTHYROIDISM

❑ Those related to hyperthyroidism

Symptoms	Signs
Palpitations	Tachycardia
Nervousness	Goiter
Easy fatigability	Weight loss
Hyperkinesias	Tremor
Diarrhea	Atrial fibrillation
Excessive sweating	Muscle weakness
Heat intolerance	Stare
Increased appetite	Lid lag

❑ Specific to Grave's Disease

1- Diffuse painless and firm enlargement of thyroid gland

2- Ophthalmopathy – Eye manifestations – 50 % of cases

Eye Signs in Toxic Goitre

- In early stages, may be unilateral but later may become bilateral.
- Order of appearance of signs
 - **Stellwag's sign** : Absence of normal blinking—so staring look.
 - **Von Graefe's sign** : Upper eye lid lags behind the eye ball as the patient is asked to look downwards.
 - **Dalrymple's sign** : Upper sclera is visible due to retraction of upper eye lid.
 - **Joffroy's sign** : Absence wrinkling in the forehead on looking upwards with inclined downwards.
 - **Moebius sign** : Inability or failure to converge the eye balls
 - **Gifford's sign** : Difficulty in eversion of the upper lid.



3- Thyroid dermopathy

- consists of thickening of the skin, particularly over the lower tibia ,due to accumulation of glycosaminoglycans
- Is usually bilateral

4- Thyroid Acropachy

- Thyroid acropachy is clubbing of fingers and toes.

Pretibial myxoedema

- B/L ,non pitting edema $\pm$,a/w clubbing
- Always a/w exophthalmos
- **Cause** : deposition of myxomatous tissue(GAG's) mainly in pretibial region due to stimulation of fibroblasts by thyroid immunoglobulin
- Skin – dry coarse and swelling due to obliterated lymphatics by mucin



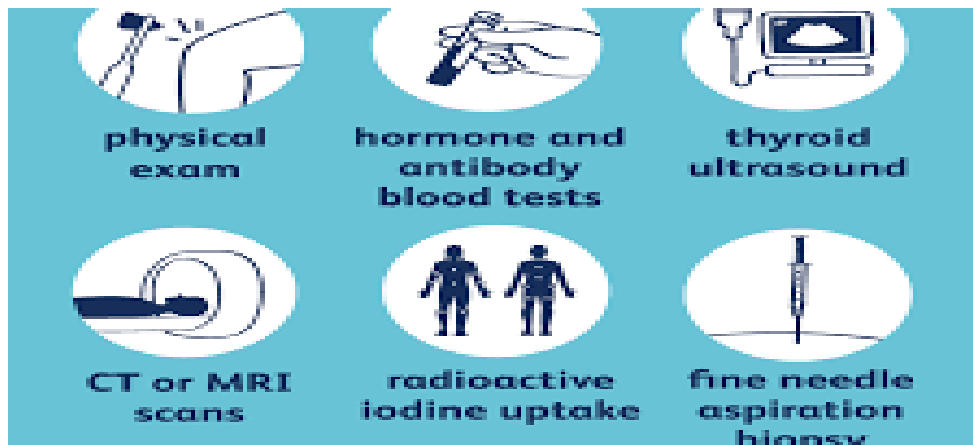
Diagnosis of Hyperthyroidism

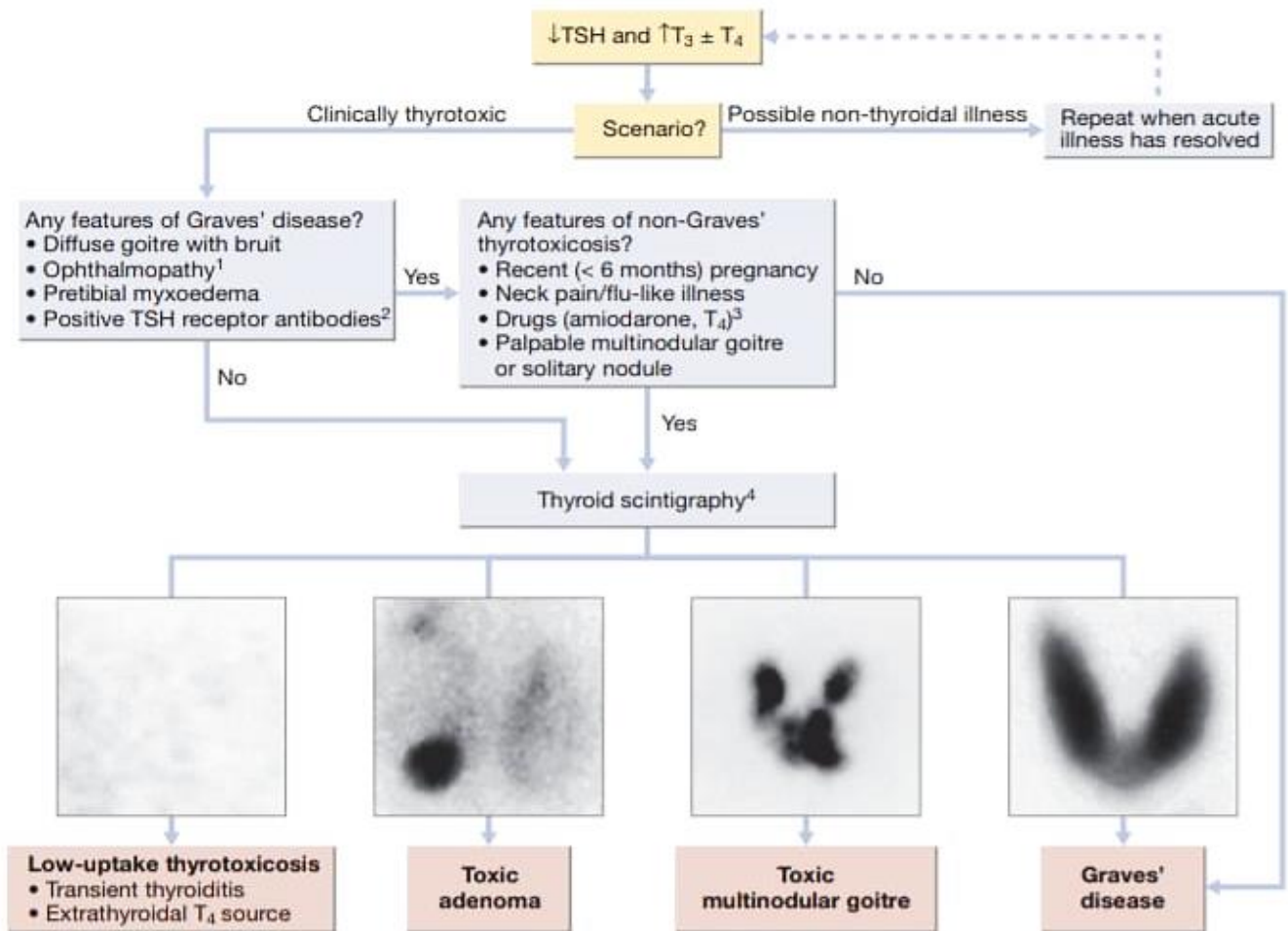
- **Hyperthyroidism can be suspected in patients with:**
 - Tremors
 - Excessive Sweating
 - Tachycardia and AF
 - Enlarged thyroid gland.
 - Puffiness around the eye & characteristic stare.
- However ,**In all cases, a blood test is needed to confirm the diagnosis.**
- The main tool for detection of hyperthyroidism is measurement of the blood TSH level
- If there is obvious involvement of the eyes, **a diagnosis of Graves' disease is almost certain.**

Differential diagnosis

- Hyperthyroidism is often clinically obvious but treatment should never be instituted without biochemical confirmation.
- Differentiation of the mild case from anxiety states may be difficult ;useful positive clinical markers are eye signs ,a diffuse goiter ,proximal myopathy and wasting.
- Weight loss despite a normal or increased appetite is a very useful clinical symptom of hyperthyroidism.
- The hyperdynamic circulation with warm peripheries seen with hyperthyroidism can be contrasted with the clammy hands of anxiety.

Diagnosis





	Graves' disease	Solitary or multinodular	Thyroiditis
History	Symptoms of thyrotoxicosis	Symptoms of thyrotoxicosis	Symptoms of thyrotoxicosis Recent viral infection Recent delivery of child Painful neck swelling
Clinical examination	Signs of thyrotoxicosis Diffuse goitre Extrathyroidal manifestations including ophthalmopathy	Signs of thyrotoxicosis Palpable solitary nodule or multinodular goitre	Signs of thyrotoxicosis Diffuse goitre Tender on palpation
Biochemical tests	TSH suppressed FT4 and FT3 elevated	TSH suppressed FT4 and FT3 elevated (only FT3 elevated in T3 toxicosis)	TSH suppressed FT4 and FT3 elevated Raised inflammatory markers (ESR, CRP)
Immunological tests	TSH receptor Antibody raised TPO Ab raised		TPO (Ab) may be raised
Thyroid isotope uptake scan (Scintiscan ⁹⁹Tm)	Diffuse uptake of tracer	Focal uptake of tracer in nodules with reduced uptake in rest of thyroid gland	Minimal uptake of tracer

Treatment of Hyperthyroidism

• The options for treating hyperthyroidism include:

- Treating the Symptoms
- Anti-thyroid Drugs
- Radioactive Iodine
- Surgery

Treating the symptoms

❑ Beta blockers

- Inhibit adrenergic effects

• Indications

- ❖ Prompt control of symptoms; but DO NOT affect the thyroid hormone levels in the blood.
- ❖ treatment of choice for thyroiditis;
- ❖ first-line therapy before surgery ,radioactive iodine, and antithyroid drugs;

Contraindications

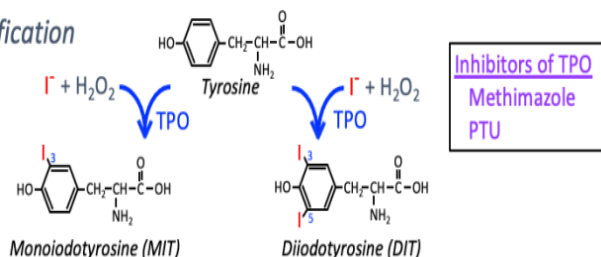
- ❖ Use with caution in older patients and in patients with pre -existing heart disease , COPD ,or bronchial asthma

Propranolol is the most commonly prescribed medication in doses of about 20 to 40 mg four times daily

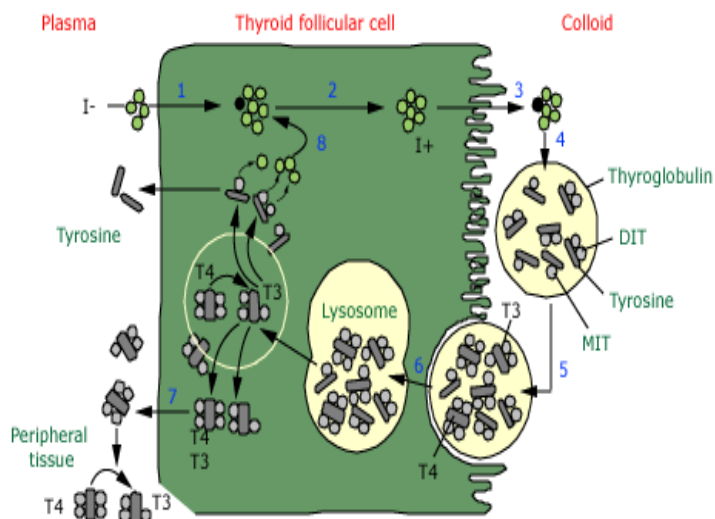
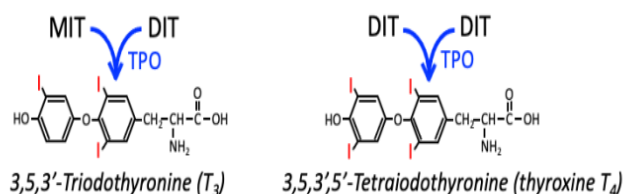
❑ Antithyroid drugs

- The drugs commonly used are propylthiouracil(PTU) and methimazole. Both drugs reduce thyroid hormone production by inhibiting the organic binding of iodine and the coupling of iodotyrosines) mediated by thyroid peroxidase TPO).
- ❖ PTU also inhibits the peripheral conversion of T4 toT3 making it useful for the treatment of thyroid storm.
- ❖ PTU has a lower risk of transplacental transfer so can be used during pregnancy.

Organification



Coupling





Side effects

Include reversible **granulocytopenia** ,skin rashes ,fever, peripheral neuritis, polyarteritis , vasculitis,and rarely ,agranulocytosis and aplastic anemia

❑ Radioactive Iodine:

- Radioactive iodine is given orally on a one-time basis to ablate a hyperactive gland.
- The isotope iodine 131 is used in treatment, while for a routine scan, iodine 123 is used.
- It is picked up by the active cells in the thyroid only and destroys them without any side effects.
- **It can't be used during pregnancy or breast-feeding**
- It is the treatment of choice for *recurring Graves' disease*
- More than 80% of patients are cured with a single dose of radioactive iodine.
- **Permanent hypothyroidism** is the major complication “after treatment”

❑ Surgical Treatment

Surgical treatment is reserved

- ❖ patient preference
- ❖ Pregnant women who can't tolerate ATD
- ❖ child or adolescent intolerant of ATDs
- ❖ large goiter ,with or without compressive symptoms
- ❖ Severe Graves'ophthalmopathy
- ❖ the presence of suspicious nodules

DEFINITION

goiter, is a swelling in the neck resulting from an enlarged thyroid gland. ... A **goitre** can be associated with a thyroid that is not functioning ...

CLASSIFICATION OF GOITRE

☐ **Simple goiter (euthyroid)**

- Diffuse hyperplastic
 - ❖ Physiological
 - ❖ Pubertal
 - ❖ Pregnancy

☐ **Multinodular Goiter**

☐ **Toxic**

- Diffuse
 - ❖ Graves' disease
- Multinodular
- Toxic adenoma

☐ **Neoplastic**

- Benign
- Malignant

☐ **Inflammatory**

- ❖ Autoimmune
- ❖ Chronic lymphocytic thyroiditis
- ❖ Hashimoto's disease
- ❖ Granulomatous
- ❖ De Quervain's thyroiditis
- ❖ Fibrosing
- ❖ Riedel's thyroiditis

☐ **Infective**

- ❖ Acute (bacterial thyroiditis, viral thyroiditis, 'subacute thyroiditis')
- ❖ Chronic (tuberculous, syphilitic)



THYROID:

a historical perspective

Goitre & mountains:

- Goitre was first seen in inhabitants of Alps. Initially they did not know that goitre is enlargement of thyroid gland. It was then documented that consuming sea weeds caused remarkable reduction of the swelling in these patients. It was later discovered that sea weeds contain large amounts of iodine.
- 1619:
Thyroid enlargement was first described as a cause of swelling on the front of the neck. It was also clearly noted that this swelling moved up and down when the patient attempts to swallow

Primary and secondary gland failure

‘Primary’ gland failure due to a disease process in the endocrine end-organ (thyroid, adrenal or gonad) will lead to deficiency of the hormone produced by that gland and a loss of negative feedback and subsequent elevation in the corresponding anterior pituitary hormone.

- In ‘secondary gland failure’, there are low or ‘inappropriately normal’ levels of the pituitary trophic hormone in the face of a low end-organ hormone level. For example, if a patient has low circulating free T₃ (fT₃) and T₄ levels in the context of a low TSH, pituitary disease should be suspected.

Hypothyroidism

Definition

- A clinical and biochemical syndrome that results from deficiency in thyroid hormone secretion from thyroid gland or in the action
- The ranges from subclinical hypothyroidism to primary and secondary hypothyroidism and the extreme medical emergency, myxedema coma.

Classifications

Hypothyroidism & Myxedema:-

1. Time of Onset:
 - a. Congenital (Cretinism)
 - b. Acquired
2. Level of endocrine dysfunction:
 - a. *Primary*
 - b. *Secondary*
3. Severity:
 - a. Overt (Clinical)
 - b. Mild (Subclinical)



Etiology:

PRIMARY

1- Congenital

- Agenesis
- Ectopic thyroid remnants

2- Defects of hormone synthesis

- Iodine deficiency
- Dyshormonogenesis

3- Antithyroid drugs, other drugs

- (e.g. lithium, amiodarone, interferon)

4- Autoimmune

- Atrophic thyroiditis
- Hashimoto's thyroiditis
- Postpartum thyroiditis

5- Infective

- Post-subacute thyroiditis

6- Post-surgery

7- Post-irradiation

- Radioactive iodine therapy
- External neck irradiation

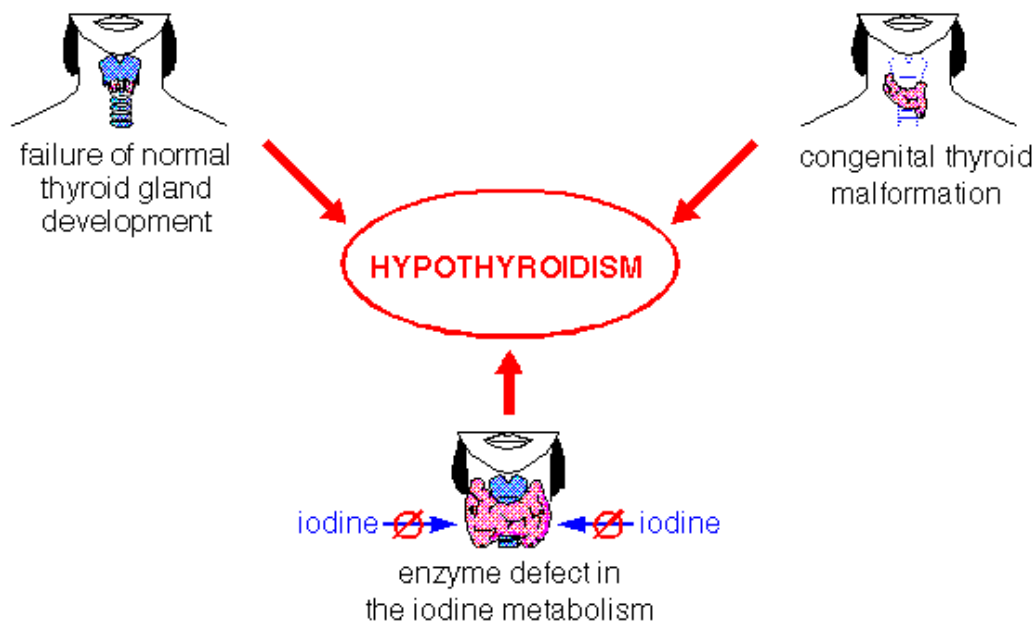
8- Infiltration

- Tumor

SECONDARY

- *Hypopituitarism*
- *Isolated TSH deficiency*
- *Peripheral resistance to thyroid hormone*

Causes of congenital hypothyroidism



© EMI Pharma

A Congenital Hypothyroidism (Cretinism)

Causes:

- a) Maternal iodine deficiency
- b) Fetal thyroid dysgenesis
- c) Maternal thyroid antibodies crossing the placenta

Cretinism

- ❖ It results in impaired brain development.
- ❖ Suggestive features are :-
 - postmaturity, hypothermia and delayed passage of meconium.
 - Decreased activity
 - Large anterior fontanelle
 - Poor feeding and weight gain
 - Hoarse cry
 - Later on retardation of physical and mental development.
 - Child has protuberant abdomen, dry skin, dry hair and delayed dentition
- ❖ Can be prevented by thyroid hormone therapy before 2 ms of age.

Defects of hormone synthesis

— Iodine deficiency

Dietary iodine deficiency still exists as 'endemic goitre' in some areas where goitre, occasionally massive, is common. The patients may be euthyroid or hypothyroid, depending on the severity of iodine deficiency. The mechanism is thought to be borderline hypothyroidism leading to TSH stimulation and thyroid enlargement in the face of continuing iodine deficiency. Efforts to prevent deficiency by providing iodine in salt continue worldwide but often with incomplete success.

— Dyshormonogenesis

This rare condition is due to genetic defects in the synthesis of thyroid hormones; patients develop hypothyroidism with goitre

B Autoimmune hypothyroid disorders

A-Atrophic (autoimmune) hypothyroidism

- This is the most common cause of hypothyroidism and is associated with anti-thyroid autoantibodies, leading to lymphoid infiltration of the gland and eventual atrophy and fibrosis.
- The condition is associated with other autoimmune diseases, & other endocrine deficiencies. Occasionally, intermittent hypothyroidism occurs with subsequent recovery; antibodies that block the TSH receptor may sometimes be involved in the aetiology.

B- Hashimoto's thyroiditis

- This form of autoimmune thyroiditis, again more common in women and most common in late middle age, produces atrophic changes with regeneration, leading to goitre formation.
- The gland is usually firm and rubbery but may range from soft to hard.
- TPO antibodies are present, often in very high titres (>1000 IU/L). Patients may be hypothyroid or euthyroid, though they may go through an initial toxic phase, 'Hashi-toxicity'.

C-Postpartum thyroiditis:-

- This is usually a transient phenomenon observed following pregnancy and may involve hyperthyroidism, hypothyroidism or the two sequentially.
- It is believed to result from the modifications to the immune system necessary in pregnancy.
- The process is normally self-limiting, but when conventional antibodies are found there is a high chance of this proceeding to permanent hypothyroidism.

Risk factors for autoimmune hypothyroidism

- ❖ Family history.
- ❖ Late middle age.
- ❖ Female.
- ❖ Having another autoimmune disorder: Type 1 DM ,hemolytic anemia, thrombocytopenia, vitiligo

Who should be screened for hypothyroidism?

- . Newborns
- . Early in pregnancy or preconception
- . Strong family history of thyroid disease
- . Postpartum with symptoms
- . Other autoimmune diseases
- . Presence of a goitre

Clinical picture:-

- Hypothyroidism produces many symptoms .The alternative term 'myxedema' refers to the accumulation of mucopolysaccharide in subcutaneous tissues.
- The classic picture of the slow, dry-haired, thick-skinned, deep-voiced patient with weight gain, cold intolerance, bradycardia and constipation makes the diagnosis easy.
- Milder symptoms are, however, more common and hard to distinguish from other causes of non-specific tiredness. Many cases are detected on biochemical screening.

1-General manifestations:

- Easy fatigability
- Weight gain
- Cold intolerance

2-Cutaneous manifestations:

- Dry skin
- Myxedema (accumulation of mucopolysaccharides in subcutaneous tissues)
- Dry brittle hair
- Brittle nail
- Alopecia
- Periorbital oedema

3- cardiovascular manifestations

- Bradycardia
- Ischemic heart disease
- Heart failure
- Hypertension
- Hypercholesterolemia
- Pericardial effusion

4- Gastrointestinal:

- Constipation
- Poor appetite
- Decreased gastrointestinal motility
- Autoimmune gastritis with failure of absorption of vit. B12

5- Neuromuscular manifestations:

- Myalgia
- Carpal tunnel syndrome
- Hoarseness of voice
- Delayed relaxation of tendon reflexes
- Disturbance in memory, concentration and intellect
- Slow thought and slow speech
- Increase need for sleep
- Myotonia
- Cerebellar ataxia
- Depression
- Psychosis
- Myxedema madness
- Coma

6- reproductive system:

- Infertility
- Menorrhagia or oligomenorrhoea
- Galactorrhoea
- Increased abortion and stillbirth

7- Respiratory manifestations:

- Obstructive sleep apnea
- Upper airway obstruction

8- Hematological manifestations:

- Anemia (normocytic-microcytic-macrocytic)
- usually normochromic and normocytic in type but may be macrocytic (sometimes this is due to associated pernicious anaemia) or microcytic (in women, due to menorrhagia)
- Bleeding (abnormal platelets function)

Diagnosing hypothyroidism

© EIM Pharma



physical examination and medical history



measurement of TSH (thyroid stimulating hormone) and T4 in the blood



testing for the presence of certain thyroid antibodies in the blood

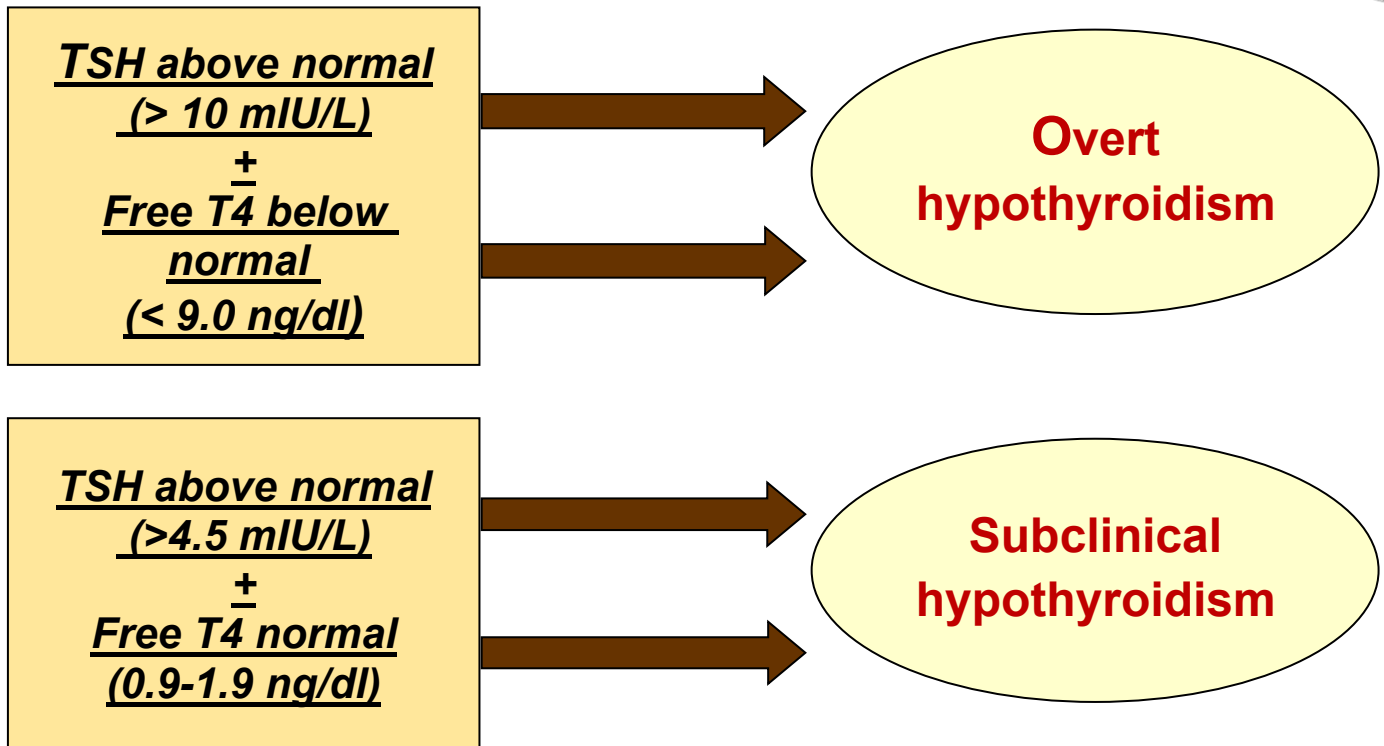


ultrasound (sonography) in selected patients with an enlarged thyroid gland

Mild thyroid failure (subclinical hypothyroidism)

- Increased TSH & Normal free T4& T3
- Symptoms may be present or absent
- Same causes as overt hypothyroidism
- there is a risk of progression to overt thyroid failure, particularly if antibodies to thyroid peroxidase are present or if the TSH rises above 10 mIU/L

Often diagnosed through routine screening or during evaluation of nonspecific symptoms or hypercholesterolemia



Treatment:

- Treatment is with **levothyroxine replacement**. It is customary to start with a low dose of 50 µg per day for 3 weeks, increasing thereafter to 100 µg per day for a further 3 weeks & finally to a maintenance dose of 100–150 µg / day.
- In younger patients, it is safe to initiate levothyroxine at a higher dose (e.g. 100 µg per day), to allow a more rapid normalisation of hormone levels.
- The dose of levothyroxine should be adjusted to maintain serum TSH within the reference range.
- at least 6 weeks should pass before repeating thyroid function tests (as TSH takes several weeks to reach a steady state) and adjusting the dose.
-

Transient hypothyroidism

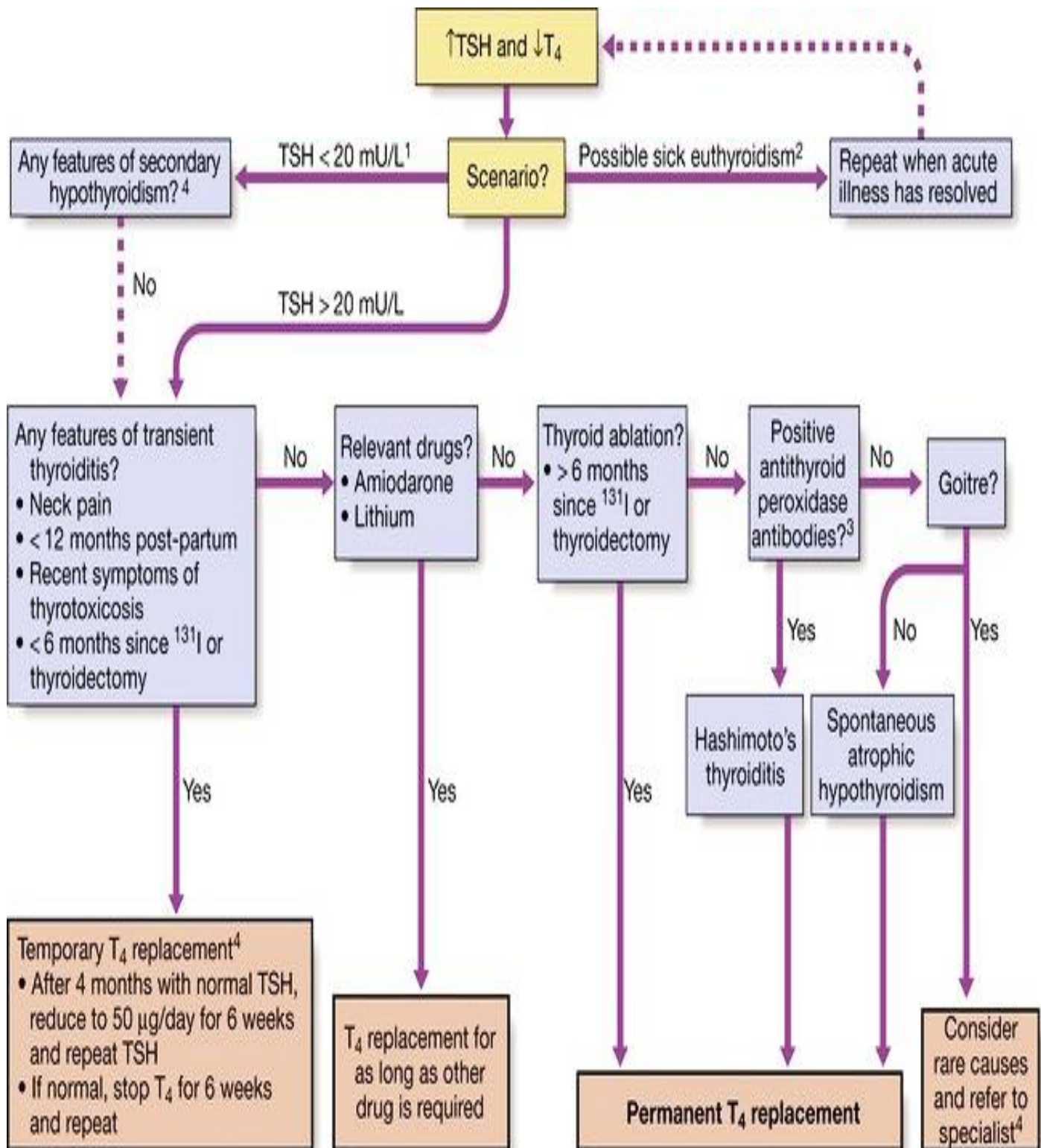
Care must be taken to identify patients with transient hypothyroidism, in whom life-long levothyroxine therapy is inappropriate. This is often observed during

- the first 6 months after thyroidectomy or ¹³¹I treatment of Graves' disease,
- in the post-thyrotoxic phase of subacute thyroiditis
- and in post-partum thyroiditis.

In these conditions, levothyroxine treatment is not always necessary, as the patient may be asymptomatic during the short period of thyroid failure

Levothyroxine replacement in ischaemic heart disease

- People with ischaemic heart disease require even lower initial doses, It is recommended to begin with 25 µg daily and perform serial electrocardiograms (ECGs), increasing the dose at 3- to 4-week intervals if angina does not occur or worsen, and the ECG does not deteriorate



Hypothyroidism in pregnancy

- ❖ Women with hypothyroidism usually require an increased dose of levothyroxine in pregnancy.
- ❖ Inadequately treated hypothyroidism in pregnancy has been associated with impaired cognitive development in the foetus

Treatment of Subclinical hypothyroidism

- In patients with non-specific symptoms, a trial of levothyroxine therapy may be appropriate.
- In those with positive autoantibodies or a TSH greater than 20 mIU/L, it is better to treat the thyroid failure early rather than risk loss to follow-up and subsequent presentation with profound hypothyroidism. Levothyroxine should be given in a dose sufficient to restore the serum TSH concentration to normal.

Monitoring

- The aim is to restore TSH to the normal range.
- Adequacy of replacement is assessed clinically and by thyroid function tests after at least 6 weeks on a steady dose.
- If serum TSH remains high, the dose of T_4 should be increased in increments of 25–50 μg , and the tests repeated at 6–8-week intervals until TSH becomes normal. Clinical improvement on T_4 may not begin for 2 weeks or more, and full resolution of symptoms may take 6 months.

Myxedema coma

© EMI Pharma

Causes

Myxedema coma is a rare, life-threatening form of hypothyroidism. It can occur in patients with undiagnosed, untreated hypothyroidism or in diagnosed hypothyroid patients who do not take their prescribed thyroid hormone replacement for months.

Symptoms



The signs and symptoms of myxedema coma consist of a progressive worsening of the signs and symptoms of hypothyroidism, eventually leading to respiratory failure and a loss of consciousness.

Treatment



Patients with myxedema coma are hospitalized in an Intensive Care Unit. Treatment includes intravenous levothyroxine and, frequently, life support with a respirator.

Treatment of myxedema coma:

ICU Management

- 1-Warming
- 2-Adequate oxygenation
- 3-Dextrose infusion for hypoglycemia
- 4- T_3 orally or intravenously in doses of 5-20 mcg bolus dose then 2.5-10 mcg every 8 hours
- 5-Hydrocortisone 100mg/ 8 hrs ,until exclusion of adrenal insufficiency
- 6-Antibiotic if there is infection

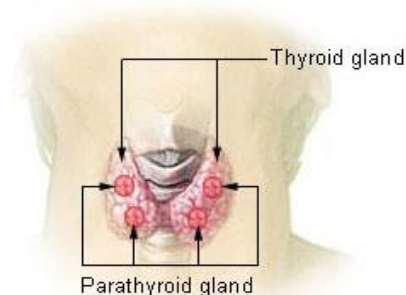
6

Parathyroid glands

The **parathyroid glands** are small endocrine glands located on the posterior surface of the thyroid gland. Despite their small size, they play a crucial role in maintaining calcium and phosphorus homeostasis in the body.

Structure and Location

- **Number and Size:** Typically, there are four parathyroid glands, each about the size of a grain of rice.
- **Location:** Embedded in the posterior surface of the thyroid gland, though their exact position can vary.



Hormone Produced: Parathyroid Hormone (PTH)

- **PTH:** A peptide hormone composed of 84 amino acids.
- **Secretion:** Released in response to low blood calcium levels (hypocalcemia).

Regulation of PTH Secretion

- **Calcium-Sensing Receptor (CaSR):**
 - Located on the surface of parathyroid cells.
 - Detects changes in blood calcium levels.
 - When calcium levels drop, CaSR signaling decreases, leading to increased PTH secretion.
 - When calcium levels rise, CaSR signaling increases, suppressing PTH release.

Actions of Parathyroid Hormone (PTH)

PTH acts on three primary target organs to increase blood calcium levels:

a. Bones

- **Stimulates Osteoclast Activity:**
 - PTH promotes bone resorption by activating osteoclasts, which break down bone tissue, releasing calcium into the bloodstream.
- **Net Effect:** Increases blood calcium levels.

b. Kidneys

- **Increases Calcium Reabsorption:**
 - PTH enhances the reabsorption of calcium in the renal tubules, reducing calcium loss in urine.
- **Decreases Phosphate Reabsorption:**
 - PTH reduces phosphate reabsorption, leading to increased phosphate excretion in urine.
- **Activates Vitamin D:**
 - PTH stimulates the kidneys to convert inactive vitamin D (25-hydroxyvitamin D) into its active form (1,25-dihydroxyvitamin D), which enhances intestinal calcium absorption.

c. Intestines (Indirectly via Vitamin D)

- **Enhances Calcium Absorption:**

Active vitamin D (calcitriol) increases the absorption of dietary calcium in the intestines.

Calcium Homeostasis

(normal 8.5 to 10.2 milligrams/deciliter (2.15 to 2.55 millimoles/liter))

- **Negative Feedback Loop:**
 - When blood calcium levels rise, PTH secretion is suppressed, reducing calcium release from bones, reabsorption in kidneys, and activation of vitamin D.
 - This feedback mechanism ensures tight regulation of blood calcium levels.

Role of Phosphorus

- PTH also regulates phosphorus levels by promoting its excretion in the urine, preventing hyperphosphatemia, which can lead to complications like soft tissue calcification.

Clinical Relevance

- **Hyperparathyroidism:** Excessive PTH secretion leads to hypercalcemia, bone demineralization, and kidney stones.
- **Hypoparathyroidism:** Insufficient PTH secretion results in hypocalcemia, causing muscle cramps, tetany, and neurological symptoms.

Hypercalcemia

- **Hypercalcemia** is a condition characterized by elevated levels of calcium in the blood (typically serum calcium >10.5 mg/dL or 2.6 mmol/L). It can result from a variety of causes, ranging from benign to life-threatening conditions. Below is a detailed breakdown of the **causes of hypercalcemia**:

Category	Specific Causes
Primary Hyperparathyroidism	Parathyroid adenoma, hyperplasia, carcinoma
Malignancy	Humoral hypercalcemia of malignancy, osteolytic metastases, ectopic PTH secretion
Vitamin D-Related	Excessive vitamin D intake, granulomatous diseases, lymphoma
Medications	Thiazides, lithium, vitamin A toxicity, calcium supplements
Endocrine Disorders	Hyperthyroidism, adrenal insufficiency, pheochromocytoma
Immobilization	Prolonged bed rest or immobilization
Renal Failure	Tertiary hyperparathyroidism, calcium-containing phosphate binders
Genetic Disorders	Familial hypocalciuric hypercalcemia, Jansen's metaphyseal chondrodysplasia
Other	Milk-alkali syndrome, Paget's disease of bone

Hyperparathyroidism

Hyperparathyroidism is a condition characterized by excessive production of **parathyroid hormone (PTH)** by the parathyroid glands. PTH plays a key role in regulating calcium and phosphorus levels in the body. When PTH is overproduced, it disrupts calcium homeostasis, leading to a range of clinical manifestations.

Types of Hyperparathyroidism

1. Primary Hyperparathyroidism:

- **Definition:** Overactivity of the parathyroid glands due to intrinsic glandular abnormalities.
- **Causes:**
 - **Parathyroid Adenoma:** A benign tumor of one gland (80-85% of cases).
 - **Parathyroid Hyperplasia:** Enlargement of all four glands (15-20% of cases).
 - **Parathyroid Carcinoma:** A rare malignant tumor (<1% of cases).
- **Mechanism:** Excessive PTH secretion leads to increased calcium levels (hypercalcemia) and decreased phosphorus levels (hypophosphatemia).

2. Secondary Hyperparathyroidism:

- **Definition:** Overproduction of PTH in response to chronic low calcium levels, often due to an underlying condition.
- **Causes:**
 - **Chronic Kidney Disease (CKD):** Reduced kidney function leads to phosphate retention and impaired vitamin D activation, causing hypocalcemia.
 - **Vitamin D Deficiency:** Inadequate vitamin D reduces calcium absorption, triggering PTH release.
 - **Malabsorption Syndromes:** Conditions like celiac disease or Crohn's disease impair calcium and vitamin D absorption.
- **Mechanism:** PTH levels rise to compensate for low calcium, but calcium levels remain low or normal.

3. Tertiary Hyperparathyroidism:

- **Definition:** Autonomous PTH secretion after long-standing secondary hyperparathyroidism, even after the underlying cause (e.g., CKD) is corrected.
- **Mechanism:** The parathyroid glands become unresponsive to calcium feedback and continue to overproduce PTH.

Symptoms of Hyperparathyroidism

Symptoms vary depending on the severity and duration of the condition. Many patients are asymptomatic, but common symptoms include:

1. Hypercalcemia-Related Symptoms:

- **General:** Fatigue, weakness, and depression.
- **Renal:** Kidney stones, polyuria (excessive urination), and polydipsia (excessive thirst).
- **Gastrointestinal:** Nausea, vomiting, constipation, and peptic ulcers.
- **Neurological:** Cognitive impairment, memory loss, and confusion.
- **Musculoskeletal:** Bone pain, osteoporosis, and fractures.

2. Chronic Complications:

- **Osteitis Fibrosa Cystica:** Severe bone demineralization with cystic changes.
- **Cardiovascular:** Hypertension and calcification of blood vessels.

Diagnosis of Hyperparathyroidism

1. Laboratory Tests:

- **Serum Calcium:** Elevated in primary hyperparathyroidism.
- **PTH Levels:** High in primary and tertiary hyperparathyroidism; elevated or inappropriately normal in secondary hyperparathyroidism.
- **Phosphorus:** Low in primary hyperparathyroidism; high in secondary hyperparathyroidism (due to CKD).
- **Vitamin D Levels:** Often low in secondary hyperparathyroidism.
- **Renal Function Tests:** To assess kidney function (e.g., creatinine, GFR).

2. Imaging:

- **Ultrasound:** To locate enlarged parathyroid glands.
- **Sestamibi Scan:** A nuclear medicine scan to identify parathyroid adenomas.
- **Bone Density Scan (DEXA):** To assess bone loss and osteoporosis.

Treatment of Hyperparathyroidism

1. Primary Hyperparathyroidism:

- **Surgical Treatment:**
 - **Parathyroidectomy:** Removal of the affected gland(s) is the definitive treatment for symptomatic patients or those with complications (e.g., kidney stones, osteoporosis).
- **Medical Management:**
 - **Cinacalcet:** A calcimimetic drug that reduces PTH secretion.
 - **Bisphosphonates:** To prevent bone loss.
 - **Vitamin D Supplementation:** If vitamin D levels are low.

2. Secondary Hyperparathyroidism:

- **Treat Underlying Cause:**
 - **Vitamin D Supplementation:** For vitamin D deficiency.
 - **Phosphate Binders:** For CKD patients to reduce phosphate levels.
 - **Calcium Supplements:** To correct hypocalcemia.
- **Medications:**
 - **Cinacalcet:** To lower PTH levels.
 - **Active Vitamin D Analogs:** Calcitriol or paricalcitol for CKD patients.

3. Tertiary Hyperparathyroidism:

- **Parathyroidectomy:** Often required due to autonomous PTH secretion.

Complications of Untreated Hyperparathyroidism

- **Renal:** Chronic kidney disease, nephrolithiasis (kidney stones).
- **Skeletal:** Osteoporosis, fractures, and bone deformities.
- **Cardiovascular:** Hypertension, vascular calcification, and increased risk of cardiovascular events.

Malignant hypercalcemia

Is a common and serious complication of cancer, occurring in up to **20-30% of cancer patients** at some point during their disease. It is caused by tumor-induced mechanisms that disrupt calcium homeostasis, leading to elevated blood calcium levels (hypercalcemia).

Pathophysiology of Malignant Hypercalcemia

- **Humoral Hypercalcemia of Malignancy (HHM):**
 - Tumors secrete **PTH-related peptide (PTHrP)**, which mimics PTH and increases bone resorption and renal calcium reabsorption.
 - Common in **squamous cell carcinoma, breast cancer, and renal cell carcinoma.**
- **Local Osteolytic Hypercalcemia:**
 - Bone metastases (e.g., from **multiple myeloma, breast cancer, or lung cancer**) cause direct bone destruction, releasing calcium into the bloodstream.
- **Ectopic PTH Secretion:**
 - Rarely, tumors (e.g., ovarian or lung cancer) produce PTH directly.

Clinical Presentation

- **Symptoms:**
 - **General:** Fatigue, weakness, dehydration.
 - **Neurological:** Confusion, lethargy, coma.
 - **Gastrointestinal:** Nausea, vomiting, constipation, anorexia.
 - **Renal:** Polyuria, polydipsia, kidney stones, acute kidney injury.
 - **Cardiac:** Arrhythmias, QT shortening, hypertension.

Diagnosis

- **Laboratory Tests:**
 - **Serum Calcium:** Elevated total calcium (>10.5 mg/dL or 2.6 mmol/L) or ionized calcium.
 - **PTH Levels:** Low or suppressed (distinguishes it from primary hyperparathyroidism).
 - **PTHrP Levels:** Elevated in humoral hypercalcemia of malignancy.
 - **Phosphorus:** Often low due to increased renal excretion.
 - **Renal Function:** Assess for acute kidney injury (elevated creatinine).
- **Imaging:**
 - **X-rays or CT scans:** To detect bone metastases or lytic lesions.
 - **Bone Scan:** To evaluate skeletal involvement.

Management of Malignant Hypercalcemia

The management involves **acute treatment** to lower calcium levels and **long-term strategies** to address the underlying malignancy.

Acute Management

1. Hydration:

- **Normal Saline (0.9% NaCl):** Aggressive IV hydration is the first step to correct dehydration and promote calcium excretion.
 - **Dose:** 2-4 liters over 24 hours, adjusted based on renal and cardiac function.
- **Goal:** Restore intravascular volume and induce calciuresis (calcium excretion in urine).

2. Loop Diuretics:

- **Furosemide:** Used **after hydration** to enhance calcium excretion.
 - **Dose:** 20-40 mg IV, as needed.
- **Caution:** Avoid in hypovolemic patients, as it can worsen dehydration.

3. Bisphosphonates:

- **First-Line Therapy:** Inhibit osteoclast-mediated bone resorption.
- **Options:**
 - **Zoledronic Acid:** Most potent; 4 mg IV over 15-30 minutes.
 - **Pamidronate:** 60-90 mg IV over 2-4 hours.
- **Onset:** Ca levels begin to drop within 24-48 hours, with maximal effect in 4-7 days.
- **Duration:** Effects last 2-4 weeks.

4. Denosumab:

- **Mechanism:** A monoclonal antibody that inhibits RANK ligand, reducing osteoclast activity.
- **Use:** For patients with **resistant hypercalcemia** or **renal impairment** (bisphosphonates are contraindicated in severe renal failure).
- **Dose:** 120 mg subcutaneously weekly for 4 weeks, then monthly.
- **Onset:** Rapid, within 1-2 days.

5. Calcitonin:

- **Mechanism:** Rapidly inhibits bone resorption and increases renal calcium excretion.
- **Dose:** 4 IU/kg subcutaneously or intramuscularly every 12 hours.
- **Onset:** Works within 4-6 hours, but effects are short-lived (24-48 hours).
- **Use:** Adjunct therapy for severe hypercalcemia while waiting for bisphosphonates to take effect.

6. Glucocorticoids:

- **Mechanism:** Effective in hypercalcemia caused by **lymphomas** or **multiple myeloma** (tumors sensitive to steroids).
- **Dose:** Prednisone 20-40 mg/day orally or IV.
- **Onset:** Calcium levels decrease within 2-5 days.

7. Dialysis:

- **Indication:** For severe, refractory hypercalcemia or patients with renal failure.
- **Modality:** Hemodialysis with low-calcium dialysate.

Long-Term Management

1. Treat the Underlying Malignancy:

- **Chemotherapy, Radiation, or Surgery:** To control tumor growth and reduce PTHrP production or bone destruction.

2. Maintenance Therapy:

- **Bisphosphonates or Denosumab:** For ongoing prevention of hypercalcemia in patients with bone metastases.

3. Hydration and Monitoring:

- Encourage oral hydration and monitor calcium levels regularly.

Monitoring and Follow-Up

- **Serum Calcium:** Monitor daily during acute treatment, then weekly or as needed.
- **Renal Function:** Assess creatinine and urine output.
- **Electrolytes:** Monitor potassium, magnesium, and phosphate levels (may be affected by treatment).

Prognosis

- Malignant hypercalcemia is often a sign of advanced cancer and is associated with a poor prognosis (median survival: **2-3 months**).
- Early recognition and aggressive treatment can improve quality of life and prevent complications.

Hypocalcemia

Hypocalcemia is a condition characterized by abnormally low levels of calcium in the blood (typically serum calcium <8.5 mg/dL or 2.1 mmol/L). Calcium is essential for numerous physiological processes, including muscle contraction, nerve function, and bone health. Hypocalcemia can range from asymptomatic to life-threatening, depending on the severity and rate of onset.

Causes of Hypocalcemia

A. Parathyroid-Related Causes

1. Hypoparathyroidism:

- **Post-Surgical:** Most common cause, often after thyroid or parathyroid surgery.
- **Autoimmune:** Destruction of parathyroid glands (e.g., autoimmune polyglandular syndrome).
- **Genetic:** Congenital absence or dysfunction of parathyroid glands (e.g., DiGeorge syndrome).

2. Pseudohypoparathyroidism:

- Resistance to parathyroid hormone (PTH) due to genetic mutations.

B. Vitamin D Deficiency or Resistance

1. Nutritional Deficiency:

- Inadequate dietary intake or lack of sunlight exposure.

2. Malabsorption:

- Conditions like celiac disease, Crohn's disease, or bariatric surgery.

3. Chronic Kidney Disease (CKD):

- Impaired conversion of 25-hydroxyvitamin D to active 1,25-dihydroxyvitamin D.

4. Vitamin D Resistance:

- Rare genetic disorders affect vitamin D receptors.

C. Magnesium Abnormalities

1. Hypomagnesemia:

- Low magnesium levels impair PTH secretion and action.
- Causes include alcoholism, malnutrition, and diuretic use.

D. Other Causes

1. Acute Pancreatitis:

- Calcium binds to free fatty acids released during pancreatic inflammation.

2. Alkalosis:

- Increased binding of calcium to albumin, reducing ionized calcium levels.

3. Medications:

- **Bisphosphonates:** Inhibit bone resorption.
- **Calcitonin:** Reduces calcium levels.
- **Anticonvulsants:** Interfere with vitamin D metabolism.

4. Massive Blood Transfusion:

- Citrate in transfused blood binds calcium.

Hypoparathyroidism

Hypoparathyroidism is a rare endocrine disorder characterized by insufficient production or secretion of **parathyroid hormone (PTH)** by the parathyroid glands. PTH plays a critical role in regulating calcium and phosphorus levels in the body. Hypoparathyroidism leads to **hypocalcemia** (low calcium levels) and **hyperphosphatemia** (high phosphate levels), resulting in a range of clinical manifestations.

Causes of Hypoparathyroidism

Hypoparathyroidism can be classified into **acquired** and **congenital** forms:

A. Acquired Hypoparathyroidism

1. Post-Surgical Hypoparathyroidism:

- **Most Common Cause:** Damage or removal of the parathyroid glands during neck surgery (e.g., thyroidectomy, parathyroidectomy).
- **Incidence:** Occurs in 1-5% of thyroid surgeries.

2. Autoimmune Hypoparathyroidism:

- Destruction of parathyroid glands by autoantibodies (e.g., autoimmune polyglandular syndrome type 1).

3. Radiation-Induced Hypoparathyroidism:

- Radiation therapy to the neck or head can damage the parathyroid glands.

4. Infiltrative Diseases:

- Conditions like hemochromatosis, Wilson's disease, or metastatic cancer can infiltrate and damage the glands.

B. Congenital Hypoparathyroidism

1. **Genetic Disorders:**
 - **DiGeorge Syndrome (22q11.2 Deletion Syndrome):** Developmental defect causing absence or hypoplasia of the parathyroid glands.
 - **Familial Isolated Hypoparathyroidism:** Mutations in the *PTH* or *GCM2* genes.
2. **Autoimmune Polyglandular Syndrome Type 1:**
 - Associated with hypoparathyroidism, adrenal insufficiency, and chronic mucocutaneous candidiasis.

C. Functional Hypoparathyroidism

1. **Hypomagnesemia:**
 - Low magnesium levels impair PTH secretion and action.
2. **Activating Mutations in the Calcium-Sensing Receptor (CaSR):**
 - Mimics hypercalcemia, suppressing PTH secretion.

Symptoms of Hypoparathyroidism

Symptoms are primarily due to **hypocalcemia** and can range from mild to severe:

Neuromuscular Symptoms

- **Paresthesia:** Tingling or numbness in the fingers, toes, or around the mouth.
- **Muscle Cramps:** Painful spasms, especially in the hands and feet.
- **Tetany:** Severe muscle spasms, carpopedal spasm (wrist and foot flexion), and laryngospasm.
- **Chvostek's Sign:** Facial twitching when the facial nerve is tapped.
- **Trousseau's Sign:** Carpal spasm induced by inflating a blood pressure cuff above systolic pressure.

Neurological Symptoms

- **Seizures:** Due to increased neuronal excitability.
- **Cognitive Impairment:** Memory loss, confusion, or depression.

Cardiovascular Symptoms

- **Prolonged QT Interval:** Can lead to arrhythmias (e.g., torsades de pointes).
- **Heart Failure:** Rare, due to decreased cardiac contractility.

Chronic Symptoms

- **Basal Ganglia Calcification:** Can cause movement disorders.
- **Dental Abnormalities:** Enamel hypoplasia or delayed tooth eruption.
- **Cataracts:** Due to long-term hypocalcemia.

Diagnosis of Hypoparathyroidism

1. **Laboratory Tests:**
 - **Serum Calcium:** Low total calcium (<8.5 mg/dL or 2.1 mmol/L) and ionized calcium.
 - **Phosphorus:** Elevated (>4.5 mg/dL or 1.45 mmol/L).
 - **PTH Levels:** Low or inappropriately normal despite hypocalcemia.
 - **Magnesium:** To rule out hypomagnesemia.
 - **Vitamin D Levels:** 25-hydroxyvitamin D and 1,25-dihydroxyvitamin D.
2. **Electrocardiogram (ECG):** Prolonged QT interval, arrhythmias.
3. **Imaging: Brain CT or MRI:** To detect basal ganglia calcification.

Management of Hypoparathyroidism

The goal of treatment is to **normalize calcium and phosphorus levels** and **alleviate symptoms**.

A. Acute Management (Symptomatic Hypocalcemia)

1. Intravenous Calcium:

- **Calcium Gluconate:** Preferred due to lower risk of tissue necrosis.
 - **Dose:** 1-2 g (10-20 mL of 10% solution) IV over 10-20 minutes, followed by infusion (e.g., 1 g in 1 liter of saline over 4-6 hours).
- **Calcium Chloride:** Reserved for severe cases (contains more elemental calcium but is more irritating to veins).
 - **Dose:** 1 g (10 mL of 10% solution) IV over 10 minutes.
- **Monitoring:** Check serum calcium every 4-6 hours and adjust infusion rate.

2. Magnesium Replacement:

- If hypomagnesemia is present, administer magnesium sulfate (1-2 g IV over 10-20 minutes).

B. Chronic Management

1. Oral Calcium Supplements:

- **Calcium Carbonate:** 1-3 g/day in divided doses (best absorbed with meals).
- **Calcium Citrate:** Preferred in patients with achlorhydria or on proton pump inhibitors.

2. Active Vitamin D (Calcitriol):

- **Dose:** 0.25-2 mcg/day, adjusted based on calcium levels.
- **Mechanism:** Enhances intestinal calcium absorption.

3. Thiazide Diuretics:

- Reduce urinary calcium excretion in patients with chronic hypocalcemia.

4. Recombinant PTH (rhPTH):

- **Teriparatide:** For patients with refractory hypoparathyroidism.
- **Dose:** 20 mcg subcutaneously twice daily.

5. Dietary Modifications:

- Increase intake of calcium-rich foods (e.g., dairy products, leafy greens).
- Limit phosphate-rich foods (e.g., processed foods, carbonated drinks).

Monitoring and Follow-Up

- **Serum Calcium:** Monitor regularly to adjust treatment.
- **Phosphorus and Magnesium:** Ensure levels are within normal range.
- **24-Hour Urine Calcium:** To assess risk of nephrolithiasis.
- **Renal Function:** Especially in patients on calcium and vitamin D supplements.

Pseudohypoparathyroidism (PHP)

Pseudohypoparathyroidism (PHP) is a rare genetic disorder characterized by **resistance to parathyroid hormone (PTH)**. Unlike hypoparathyroidism, where PTH levels are low, patients with PHP have **normal or elevated PTH levels** but exhibit hypocalcemia and hyperphosphatemia due to end-organ resistance to PTH. PHP is often associated with distinctive physical features and is classified into several subtypes based on genetic and clinical characteristics.

Types of Pseudohypoparathyroidism

PHP is classified into several subtypes based on the underlying genetic defect and clinical presentation:

A. PHP Type 1a

- **Features:**
 - **Albright Hereditary Osteodystrophy (AHO):** Short stature, round face, obesity, brachydactyly (shortened fingers and toes), and subcutaneous ossifications.
 - **Multihormone Resistance:** Resistance to other hormones that act via $G_s\alpha$ (e.g., TSH, gonadotropins).
 - **Hypocalcemia and Hyperphosphatemia:** Due to PTH resistance.

B. PHP Type 1b

- **Features:**
 - **No AHO Phenotype:** Patients lack the physical features of AHO.
 - **Isolated PTH Resistance:** Hypocalcemia and hyperphosphatemia without multihormone resistance.

C. PHP Type 1c

- **Features:**
 - **AHO Phenotype:** Similar to PHP type 1a.
 - **Multihormone Resistance:** Resistance to multiple hormones.

D. PHP Type 2

- **Features:**
 - **No AHO Phenotype:** Patients lack the physical features of AHO.

Symptoms of Pseudohypoparathyroidism

Symptoms are primarily due to **hypocalcemia** and **hyperphosphatemia** and can include:

Management of Pseudohypoparathyroidism

- The goal of treatment is to normalize calcium and phosphorus levels and alleviate symptoms.
- **Pseudopseudohypoparathyroidism (PPHP)** is a rare genetic disorder that is closely related to **pseudohypoparathyroidism (PHP)**. However, unlike PHP, patients with PPHP do not exhibit **hormone resistance** or **biochemical abnormalities** such as hypocalcemia and hyperphosphatemia. Instead, PPHP is characterized by the **physical features of Albright Hereditary Osteodystrophy (AHO)** without the endocrine dysfunction seen in PHP.

7

Adrenal Gland Disorders

The *adrenal glands* are the pair endocrine gland, located above the superior kidneys' poles. They are against the interspace between the first lumbar vertebra and eleventh thoracic vertebra. They have triangular form.

The weight of both adrenal glands is 6 to 12 g, the length is 4-6 cm, the width is 2-3 cm, and the thickness is to 1 cm.

The external portion of the adrenal gland is surrounded by the connective tissue capsule.

The adrenal gland consists of outer *cortex* and an inner *adrenal medulla*.

Functional Anatomy and Physiology

■ Adrenal cortex

1. Zona glomerulosa ----- Aldosterone
2. Zona fasciculate ----- Cortisol
3. Zona reticularis ----- Adrenal androgens

■ Adrenal Medulla

Component of the sympathetic nervous system ----- Catecholamines.

■ Glucocorticoids

- Major one in human is cortisol.
- Diurnal pattern of secretion via ACTH stimulation.
- Rise dramatically during stress.
- 95% bound to cortisol-binding globulin.
- Has a minimal mineralocorticoid activity.

Principal functions:

1. Regulation of carbohydrate metabolism.
2. Increase protein catabolism.
3. Immunomodulation.
4. CV regulation

■ Mineralocorticoids

- Major one is Aldosterone.
- It is the most important sodium-retaining hormone.
- The principle stimulus is Angiotensin-II (renin-angiotensin system).

Principal functions:

4. Sodium retention.
5. Potassium excretion

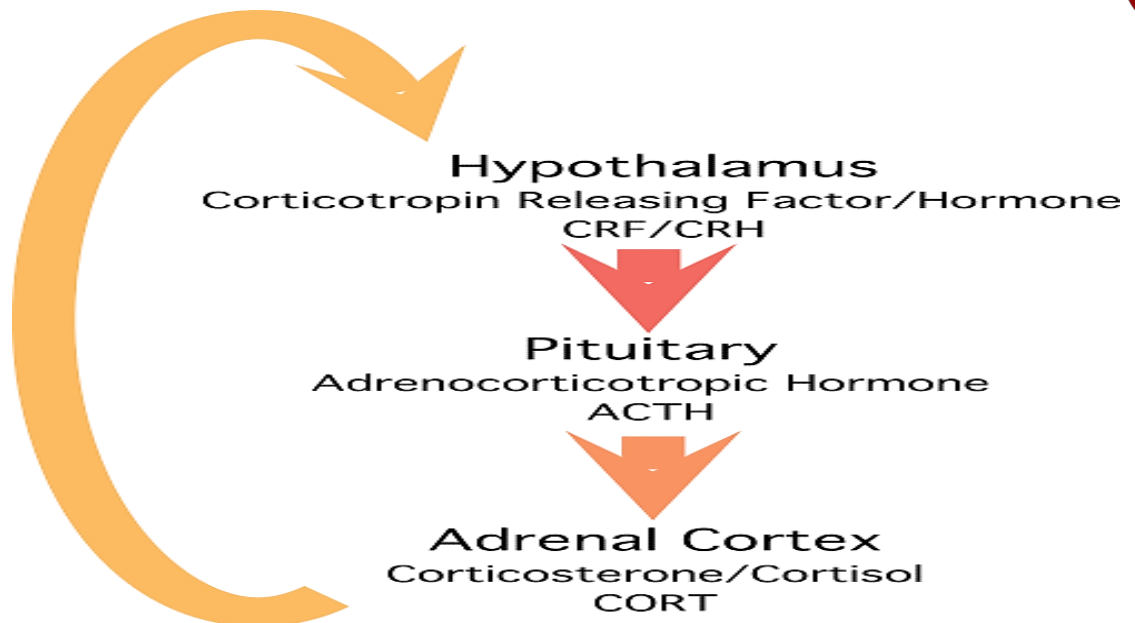
■ Catecholamines

- Man, small proportion of circulating noradrenaline is secreted from adrenal medulla, much more is released from the nerve endings.
- Conversion of noradrenaline to adrenaline is induced by glucocorticoids.

Principal functions:

6. Increase heart rate.
7. Regulation of vascular tone.
8. Antagonise insulin action

HPA axis



Classification

- **Adrenal hyper function :-**
 - Cushing syndrome “excess cortisol”
 - Hyperaldosteronism
 - Adrenogenital syndrome “excess androgen
- **Adrenal hypofunction :**
 - Primary adrenal insufficiency “ Addison disease
 - “Secondary adrenal insufficiency

A Cushing syndrome

Definition

- Cushing’s syndrome :constellation of symptoms associated with cortisol excess.
- Cushing's syndrome develops when the level of a glucocorticoid in the body is too high over a long period of time.
- Too much glucocorticoid can occur from an exogenous or endogenous source
- Described by Harvey Cushing in 1932

Etiological classification

1. **ACTH dependent:**
 - Pituitary- dependent (Cushing’s disease).
 - Ectopic ACTH secretion.
2. **Non-ACTH-dependent:**
 - Adrenal hyperplasia
 - Adenoma or
 - Carcinoma.

Cushing's syndrome; Clinical features

1. Weight gain (central obesity), moon face, dorsal fat pad.
2. Plethora, acne, Hirsutism, scalp hair-thinning, frontal balding.
3. Depression/ psychosis, insomnia.
4. Skin; thin skin, infections, easy bruising, striae, pigmentations, poor wound healing.
5. Musculoskeletal; back pain, osteoporosis, kyphosis, muscular weakness, proximal muscle wasting, pathological fractures.
6. Menstrual irregularities, decreased libido.
7. Hyperglycemia, hypertension.

Diagnosis

- The first step in diagnosing Cushing's is to determine whether the patient has high levels of cortisol.
- Cortisol levels vary throughout the day, making testing more difficult. The normal "diurnal rhythm" for cortisol secretion is that cortisol and ACTH levels are the highest in the morning and the lowest at 11PM to midnight.
- The second step in the diagnosis of Cushing's is to determine whether the cortisol production is dependent on ACTH (pituitary or ectopic sources) or ACTH independent (adrenal tumors). This is termed the "differential diagnosis."

1. Investigations to confirm the diagnosis:

- 24-h urinary cortisol level: This test is considered the gold standard diagnostic test.
- 2- Diurnal rhythm of plasma cortisol: In patients with Cushing syndrome this circadian rhythm is lost, and cortisol levels at midnight and 8 A.M. are usually comparable.

□ **Low dose Dexamethasone suppression tests:**

- helps in the initial diagnosis of Cushing syndrome, as a screening or a confirmatory test. It can be performed either as an overnight or a two-day test.
 - **Overnight, 1 mg test:** Due to ease and convenience, the overnight test is the most commonly used screening test. Dexamethasone 1 mg is administered orally between 11 PM and midnight. Serum cortisol levels are drawn the next morning between 8 and 9 AM
 - **Two-day, 2 mg test:** Dexamethasone 0.5 mg is administered orally every 6 hours (9 AM, 3 PM, 9 PM, 3 AM) for two days (total dose 4 mg). Serum cortisol level is drawn 6 hours (9 AM) after the last administered dose.
- Normal response: A normal result is decrease in cortisol levels upon administration of low- dose dexamethasone.
- Cushing's disease involve no change in cortisol on low-dose dexamethasone

2. Investigations for the cause:

1. Plasma ACTH level

- Once the diagnosis of Cushing syndrome is confirmed, the next step is to categorize ACTH-independent vs. ACTH-dependent Cushing syndrome, by checking the plasma ACTH levels.
- In ACTH-independent Cushing syndrome, the plasma ACTH is low or undetectable, indicating an adrenal etiology (causing pituitary suppression of ACTH).
- While, in ACTH-dependent Cushing syndrome, the plasma ACTH is inappropriately normal or high, suggesting either pituitary or an ectopic source.

2- High-dose dexamethasone-suppression test;

- HDDST can help distinguish pituitary (i.e., Cushing disease) from an ectopic source of ACTH overproduction.

High-dose dexamethasone-suppression test; 2mg/6h for 2 days;

- ✓ in pituitary dependent disease; plasma cortisol on day 2 is less than 50% that on day 0
- ✓ failure of suppression suggests ectopic source.

3. Others tests;

- ❖ Effects of hypercortisolism include:
 - Hyperglycemia
 - Hyperlipidemia
 - Hypokalemia(excrete K⁺)
 - Metabolic alkalosis(excrete H⁺)

4. Imaging

- ❖ Adrenal CT or MRI
- ❖ Brain CT or MRI
- ❖ Chest x-ray or CT

Cushing's syndrome; Treatment

➤ Medical treatment

- ❖ **Steroidogenic inhibition:** Metayrapone; 11-B hydroxylase blocker.
- ❖ **Ketoconazole (200mg 3times/day);** synergistic to metyrapone to reduce its dose.

➤ Surgical treatment

❖ Cushing's disease

- Trans sphenoidal microadenomectomy
- Pituitary radiation
- Bilateral total adrenalectomy

❖ Adrenal adenoma and carcinoma

- Surgical removal

❖ Ectopic ACTH Syndrome

- Surgical removal of the ectopic tumor
- Radiotherapy

B

Adrenal Insufficiency

- ✓ Arises when cortisol levels are not sufficient to meet the needs of the body
- ✓ May be primary or secondary
- ✓ May be congenital or acquired
- ✓ Can be fatal if left untreated

Primary adrenal insufficiency (high ACTH) (Addison's disease)

- Primary Adrenal Insufficiency is also known as Addison's Disease in honor of Dr. Thomas Addison
- Addison disease (or Addison's disease) is adrenocortical insufficiency due to the destruction or dysfunction of the entire adrenal cortex. It affects glucocorticoid and mineralocorticoid function.
- The onset of disease usually occurs when 90% or more of both adrenal cortices are dysfunctional or destroyed .

Epidemiology

- Addison's disease is a rare and chronic disease.
- 6-110 cases diagnosed per 100,000 in the world per year.
- 1.4 million deaths per year around the world.
- Usually effects 30-50 year-olds, but can be seen in all ages

Etiology

- Autoimmune:
 - **Isolated autoimmune adrenalitis (30- 40%)
 - **Polyglandular syndrome 1 &2 (60- 70%)
- Infection:TB, HIV, CMV, cryptococcosis, histoplasmosis, coccidioidomycosis
- Bilateral adrenal hemorrhage
- Deposition Diseases: Hemochromatosis, amyloidosis, Sarcoidosis
- Congenital Adrenal Hyperplasias
- Metastases
- Bilateral adrenalectomy

Secondary adrenal insufficiency

Etiology

- ❖ Hypothalamic or pituitary disease:
 - Pituitary tumors (active and inactive adenomas, carcinoma)
 - Mass lesions affecting the hypothalamic-pituitary region: Craniopharyngioma, meningioma, metastases
 - Pituitary irradiation
 - Autoimmune hypophysitis
 - Pituitary apoplexy/hemorrhage
 - Pituitary infiltration (TB, actinomycosis, sarcoidosis, histiocytosis X, Wegener's granulomatosis, metastases)
- ❖ Exogenous Glucocorticoid Therapy

Addison's disease; clinical features (low ACTH)

- Chronic presentation
- Acute presentation
- Acute on top of chronic

Chronic presentation

- Non-specific; malaise, fatigue, anorexia, nausea, vomiting, weakness.
- Postural hypotension.
- Hypoglycemia.
- Pigmentation. Increased skin pigmentation can be seen with primary adrenal insufficiency secondary to melanocyte stimulating activity associated with ACTH
- Decrease in body hairs.

Addisonian crisis (acute presentation)

- Severely low blood pressure (shock)
- Hyperkalemia
- Hyponatremia
- Hypoglycemia
- Hypercalcemia
- Unexplained fever, diarrhea, vomiting
- Coma and death
- Precipitated by infection, surgery or intercurrent disease

Investigation

- **In the blood analysis:** lymphocytosis, eosinophilia, erythrocyte sedimentation rate is decreased, when the active tuberculosis is present this rate is increased;
- **The electrolyte:** hyponatremia, hypochloremia, hyperkaliemia; The glucose tolerance test – flat with marked hypoglycemic phase in three hours;
- **Thyroid function test :-**
 - Decreased thyroid hormones and high TSH, due to associated autoimmune hypothyroidism
- **Serum Cortisol and ACTH levels :-**
 - The content of ACTH is increased;
 - The content of cortisol is decreased;
- **Rapid ACTH stimulation test:**
 - -Blood is drawn in 2 separate tubes for baseline cortisol and aldosterone values
 - -Synthetic ACTH (1-24 amino acid sequence) in a dose of 250 mcg (0.25 mg) is given IM or IV.
 - -30 or 60 minutes after the ACTH injection, 2 more blood samples are drawn; one for cortisol and one for aldosterone.

❖ Interpreting rapid ACTH stimulation test:

Two criteria are necessary for diagnosis:

- (1) an increase in the baseline cortisol value of 7 mcg/dL or more and (2) the value must rise to 20 mcg/dL or more in 30 or 60 minutes, establishing normal adrenal glucocorticoid function.
- In patients with Addison disease, both cortisol and aldosterone show minimal or no change in response to ACTH.

❖ **To differentiate between primary and secondary adrenal insufficiency :**

- **Plasma ACTH** :high in primary and low or undetectable in secondary adrenal insufficiency.
- **Plasma renin activity** :high plasma renin activity and low plasma aldosterone in primary adrenal insufficiency.
- **Prolonged ACTH stimulation test** : synthetic ACTH,250mcg is given as an iv infusion over 8 hrs for 48hrs , 24 hr urine 17 hydroxy corticosteroids is measured at baseline and daily during the test .
 - In secondary adrenal insufficiency :2-5 fold increase in 17-OHCS occur compared to baseline 17OHCS. In primary adrenal insufficiency the is no response.

Treatment

- Replacement therapy (always need glucocorticoids + usually mineralcorticoid therapy)
 - Hydrocortison orally 15 mg at morning and 5 mg at evening
- Doses change according to lifestyle:
 - ❖ doubling the routine oral dose in the case of intercurrent illness with fever
 - ❖ IV hydrocortisone injection at a daily dose of 100 mg in cases of prolonged vomiting, surgery, or trauma
- Have to carry emergency injection of hydrocortisone and card/bracelet identifying their condition
 - Mineralocorticoid replacement:-
 - Aldosterone is not available, fluodrocortisone is used 0.5-1 mg daily.
 - ❖ Adequacy of replacement is assessed by measurement of s. electrolytes, blood pressure and plasma renin activity. Also by blood picture.
 - Adrenal androgen replacement is an option in patients with lack of energy, and in women with loss of libido.
 - ❖ It can be achieved by once-daily administration of 25–50 mg DHEA. Treatment is monitored by measurement of DHEAS, androstenedione, testosterone.
 - Initiation of thyroxin replacement when there is hypothyroidism prior to treatment of adrenal insufficiency can trigger an adrenal crisis , as thyroxin will speed up the inactivation of residual cortisol. So re-evaluation is required after adrenal insufficiency has been appropriately replaced for a few weeks .

Management of Addisonian crisis:- It is a medical emergency

- IV fluid (normal saline 1 L/h with continuous cardiac monitoring and 10% dextrose)
- Hydrocortisone 100 mg bolus followed by 100–200 mg hydrocortisone over 24 h infusion or i.v doses until GIT symptoms improve then start oral therapy
- Mineralocorticoid replacement can be initiated once the daily hydrocortisone dose has been reduced to <50 mg
- Treat precipitating cause

8

Diseases of the adrenal gland (hyperaldosteronism and pheochromocytoma)

Hyperaldosteronism

- ☐ common form of secondary hypertension.
- ☐ Hypertension associated with low renin and increased aldosterone levels that are not suppressed by appropriate testing
- ☐ **Primary hyperaldosteronism**
 - Causes**
 1. Adrenal adenoma (Conn's syndrome); 60% often very small, occur in young females.
 2. Bilateral adrenal hyperplasia; more common in males, rare before 40ys.
 - ☐ **Secondary Hyperaldosteronism**
 - Causes**
 1. Accelerated hypertension
 2. Renal artery stenosis.
 3. Heart failure.
 4. Liver cirrhosis

Primary Hyperaldosteronism

☐ Clinical features

- Primary hyperaldosteronism may be asymptomatic, particularly in its early stages.
- When symptoms are present, they may be related to hypertension (if severe-----hypokalemia, or both).

■ Hypertension :

- Headaches, Facial flushing
- If hypertension is severe, weakness, visual impairment, impaired consciousness, and seizures (hypertensive encephalopathy) almost invariable but it is a rare cause of secondary hypertension (<1%).
- Hyponatremia and increased potassium excretion. Hypokalemia muscle weakness, polyuria, secondary to tubular damage (nephrogenic diabetes insipidus).
- Peripheral edema may occur.

Endocrine causes hypertension

- Pheochromocytoma
- Adrenal adenoma (Conn syndrome)
- Acromegaly
- Cushing syndrome
- Thyrotoxicosis
- Myxedema

❑ Investigations

- Beta-blockers and other drugs may interfere with renin activity, and spironolactone, angiotensin-converting enzyme (ACE) inhibitors and angiotensin II receptor antagonists will all affect results; all should be discontinued, if possible.
- Hypokalemia, increased urinary potassium excretion.
- Increased K daily excretion in urine
- Increased plasma aldosterone/ renin ratio.
- High plasma aldosterone not suppressed by 0.9% saline infusion or fludrocortisone injection.
- Investigation for the cause; adrenal CT or MRI, venous catheterization of the adrenal glands for aldosterone level

❑ Treatment

- ❑ GENERAL PRINCIPLES — Establishing the correct diagnosis is essential since the treatment of primary aldosteronism is based upon whether the adrenal aldosterone hypersecretion is unilateral (adenoma, unilateral hyperplasia, or carcinoma) or bilateral (idiopathic adrenal hyperplasia or glucocorticoid-remediable aldosteronism).
- ❑ Surgery is curative only in patients with unilateral disease. As adenoma .
- ❑ Patients with bilateral idiopathic adrenal hyperplasia are treated with a mineralocorticoid antagonist. ; spironolactone 100-400 mg/day. Side effects; skin rash, nausea, gynecomastia.
- ❑ Aldosterone receptor antagonist eplerenone can be used if there is side effects of spironolactone.

Pheochromocytoma

❑ Catecholamine excess: pheochromocytoma

- Rare tumor of chromaffin tissue which secretes catecholamines.
- Responsible for <0.1% of cases of hypertension.
- 10% are malignant.
- 10% are extra-adrenal.
- 10% are familial.

■ Some are associated with:

- MEN 2 syndrome.
- Tuberous sclerosis
- Von Hippel Lindau disease

Clinical features

- Hypertension (usually paroxysmal, sometimes orthostatic hypotension).
- Cyanosis and Raynaud syndrome due to vasospasm.
- CVS: palpitation, tachycardia, arrhythmias, chest pain, acute coronary syndrome, cardiomyopathy, and heart failure.
- Abdominal pain, nausea or vomiting, constipation or diarrhea and intestinal ischemia.

- Glucose intolerance Or DM .
- Occasionally; complications of hypertension as stroke, myocardial infarction, retinopathy
- Weight loss
- Headache, sweating, pallor or flushing .
- Attacks of anxiety or even panic attacks
- CNS: Headache, paresthesias, numbness, dizziness, TIA, hemiplegia, seizures, hemorrhagic stroke.
- Multisystem crisis: Severe hypertension/ hypotension, fever, encephalopathy, renal failure, ARDS, death

❖ Causes of death in patients with unsuspected pheochromocytomas

- Myocardial infarction
- Cerebrovascular accident
- Arrhythmias
- Irreversible shock
- Renal failure
- Dissecting aortic aneurysm
- Acute respiratory distress syndrome

Investigations

- Direct measurement of plasma catecholamines.
- Measurement of urinary metabolite (metanephrines), 24 h urinary collection is more useful as the secretion is usually paroxysmal.
- Clonidine suppression test.
- Plasma chromogranin A (a storage vesicle protein) is raised.
- Localization of the tumor; abdominal CT or MRI.
- Extra-abdominal sites can be detected by scintigraphy using meta-iodobenzyl guanidine (MIBG) labelled with radioactive iodine.
- Positron emission tomography (PET) is also used by some centers in the localization of pheochromocytomas
- Genetic testing should be performed in all people with confirmed pheochromocytoma or paraganglioma.

Treatment

1. Surgical removal (5-year survival is about 95% for non-malignant Tumors).
2. Medical therapy for preparation of the patient for a minimum 6 weeks; complete alpha and beta blockade by phenoxybenzamine (20-80 mg daily) then propranolol (120-240 mg daily)
3. sodium nitroprusside and phentolamine (a rapid-acting α -blocker) should be available in case sudden severe hypertension develops.

Multiple endocrine neoplasia (MEN)

1) MEN 1	2) MEN 2
• Parathyroid	• Adrenal
• Pituitary	• Thyroid
• Pancreas	• Parathyroid
• Adrenal	
• Thyroid	

List of references

1- Text Books:

- Kumar & Clark's Clinical Medicine, 10th edition 2021
- -Davidson's Principles and Practice of Medicine 23rd edition
- Internal Medicine Correlations and Clinical Scenarios for USMLE Step
- Goldman-Cecil Textbook of Medicine; 26th edition, 2019.
- Harrison's principals of internal medicine , 21 st edition, 2022.

2- Periodicals and Web Sites of Medicine.

-
-